OECD ECONOMIC SURVEYS

SWEDEN

1997

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ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF SWEDEN

THE LAND

Area, (1 000 sq. km)	450	Inhabitants in major cities, including suburbs	
Lakes (1 000 sq. km)	39	(end of 1994) thousands:	
Arable area (1 000 sq. km)	30	Stockholm	1 554
Woodland (1 000 sq. km)	228	Göteborg	761
		Malmö	498

THE PEOPLE

Population (end 1995), thousands	8 838	Net natural increase per 1 000 inhabitants	
Number of inhabitants per sq. km	20	(average 1991-95)	2.5
Net natural increase (average 1991-95), thousands	21.5	Net migration (average 1991-95), thousands	27.9

PRODUCTION

Gross domestic product in 1995 (Kr billion)	1 634.9	Gross fixed capital formation in 1995:	
GDP per head, US$	25 944	Per cent of GDP	14.5
		Per head, US$	3 774
		Employment, 1995:	
		Total civilian, thousands	3 986
		Per cent of total:	
		Agriculture, forestry, fishing	3.8
		Industry	23.0
		Other	73.2

THE GOVERNMENT

Per cent of GDP in 1995:		Composition of Parliament (number of seats):	
Public consumption	26	Social Democrats	161
General government current revenue	58	Conservatives	80
Public gross fixed capital formation	2	Centre	27
		Liberals	26
		Left Party	22
		Environmental party	18
		Christian Democrats	15
			349

Last general election: September 1994
Next general election: September 1998

FOREIGN TRADE

Exports of goods and services, 1995		Imports of goods and services	
as per cent of GDP (average 1991-95)	33.1	as per cent of GDP (average 1991-95)	30.1
Main exports in 1995 (per cent		Main imports in 1995 (per cent	
of total merchandise exports):		of total merchandise imports):	
Wood, pulp and paper	6.2	Food, beverages, tobacco	6.4
Metal and metal products	10.9	Fuel and energy	10.4
Machinery	30.8	Machinery and transport equipment	40.8
Transport equipment	14.8	Other manufactured goods	26.9

THE CURRENCY

Monetary unit: Krona	Currency units per US$, average of daily figures:	
	Year 1995	7.13
	October 1996	6.60

Note: An international comparison of certain basic statistics is given in an Annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Sweden by the Economic and Development Review Committee on 14th October 1996.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 28th October 1996.

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The previous Survey of Sweden was issued in August 1995.

Assessment and recommendations

Better balance is making for sustained growth

The imbalances that characterised the Swedish economy before it moved out of recession in early 1994 are gradually being unwound. The recovery, which was initially export-led, has enabled a restructuring of balance sheets to take place both in the private and the public sector. The process is still continuing among households, where consumption has been depressed for several years, but the corporate sector has been the driving force of the upturn, first as exports were boosted by a highly competitive cost level and higher market growth; and second as the strong expansion of fixed investment in manufacturing spread to sectors catering to the domestic economy. On present intentions, business fixed capital formation can be expected to increase by 10 per cent in 1996 and 5 per cent in 1997, allowing a gradual build-up of the productive capacity of the economy. The economy should thus be well-placed to respond to a more vigorous expansion of export markets projected from the second half of 1996. Growth in private consumption could remain sluggish even though disposable income growth should recover as the negative effect of the fiscal consolidation wears off, but with the inventory adjustment completed economic growth should increase from 1¾ per cent in 1996 to above 2 per cent in 1997 and 1998.

Unemployment and wage growth are too high...

Higher average hours and a temporary reduction in the scope of active labour market programmes (ALMPs) blunted the impact of higher labour demand on unemployment in 1995, and the slowing of growth in early 1996 pushed unemployment up again. More than 12 per cent of the labour force were either unemployed or engaged in ALMPs in 1995. Nevertheless, the high level of unemployment failed to restrain wage settlements in the 1995 bargaining round, as awards in the more buoyant export industries, especially paper and pulp, were copied by other sectors. Moreover, as wage agreements were concluded for two or three years in all sectors, annual average wage growth well in excess of productivity gains has been embedded in the economy up to 1997/98. The long-term effects that this increase in labour costs is likely to set in train are already in evidence, both in a declining growth in export shares and in investment, which is highly slanted towards capital-deepening. If the next round of wage bargaining gives a similar result to the last, a prolonged weakness of the economy is likely to occur, which would leave producers with few options but to shed labour to contain costs. So far, however, wage increases are being absorbed by higher productivity and lower profit margins in business, with both producer prices and consumer prices net of indirect taxes lower than one year ago.

... but inflation is on target and monetary policy has gained credibility

With inflation receding rapidly, both official price projections and inflation expectations have become consistent with the central bank's target. Based on bringing the average increase in consumer prices in 1995 down to 2 per cent and maintaining inflation at that level thereafter, with a tolerance margin of 1 per cent, the target was under threat in early 1995. Inflation was then running above 3 per cent, and with wage increments being predicated upon continued high price expectations, the central bank signalled its

preference for the currency to strengthen. Policy-controlled interest rates were kept high throughout the second half of 1995. As inflation prospects improved, the central bank allowed its key signal rate to fall by a cumulative 4 percentage points between January and October 1996. In the summer of 1996, headline inflation came down below 1 per cent. By establishing the credibility of its overriding objective, monetary policy has been able to provide support to the economy during 1996, and – with the krona strengthening – monetary conditions have changed in favour of domestically-oriented sectors at the expense of the exposed sectors.

Long-term interest rates have also declined substantially, as a stability-oriented monetary policy has been underpinned by increasing financial-market confidence in the fiscal consolidation programme. But there is still a significant premium on Swedish interest rates and bond yields are higher than the inflation expectations of domestic agents would seem to warrant, probably attributable to a perceived exchange rate risk. According to its budget and inflation performance, the Swedish economy is well placed to take part in Economic and Monetary Union from its inception on the criteria laid down in the Maastricht Treaty, and its financial market structures should be well-suited to the operation of a common monetary policy. The remaining risk premium would seem to reflect *inter alia*, doubts among financial markets as to whether Sweden will participate in the first round of monetary union, as the decision about participation will be taken by the Parliament in late 1997. Irrespective of this decision, however, the requirements flowing from inflation targeting and an irrevocably fixed exchange rate are identical over the medium term. Either arrangement would be critically dependent on the creation of a wage formation process that reflects underlying productivity and employment prospects.

Fiscal consolidation is progressing in accordance with medium-term objectives...

During the recession of the early 1990s, the general government budget deficit underwent an acute deterioration. The subsequent consolidation effort has been among the most substantial undertaken by any OECD economy. Reasserting control over public finances has entailed the setting out of medium-term objectives, underpinned by a detailed specification of expenditure cuts and tax increases which have now been adopted by Parliament. The programme covers the years 1995-98, the objective being to achieve a balanced budget by 1998, compared with a peak deficit of 12.3 per cent of GDP for 1993. Avoiding a spiralling of public debt has necessitated a front-loading of the programme, which has ensured that around seventy per cent of the savings will show up in the budget numbers for 1996. A follow-up procedure has been introduced to make sure the consolidation programme is on track, based on six-monthly reviews of overall budget developments. After a minor reinforcement in 1996, made according to the review procedure, the programme amounts to an overall improvement in the structural budget balance of 8 per cent of GDP by 1998. Given a strict commitment to the objectives set gross debt should continue to fall relative to GDP from its peak in 1994. Although recent reforms have made the public finances less sensitive to fluctuations of economic activity, high tax levels and generous transfer schemes mean that the budget remains vulnerable to the economic cycle. The goal should thus be to achieve a surplus over the cycle, as is the recently-established objective. To achieve this goal, while minimising the need for further discretionary measures, structural reform needs to progress far enough for it to be able to support longer-term economic and employment growth.

...underpinned by stricter budgetary control processes

Wide-ranging changes in the budget process have been major factors underpinning the commitment to sound public finances, insofar as the Riksdag (parliament) has

formally tied its own hands with respect to the overall expenditure level over the years 1997-1999. The central government expenditure-to-GDP ratio has been set on a declining trend, supported by mechanisms to secure better control of expenditure growth in statutory programmes. Cost overruns in such programmes are to be fully covered within three years, either by legislative changes to secure the savings required or by cuts in related expenditures. Since overall expenditure growth will now be derived from underlying fiscal objectives and not from programme commitments, the expenditure/tax spiral prevalent in previous years should be avoided. However, the new framework has still to be fully tested and its success is dependent on a continued commitment to overall macroeconomic balance in the face of high unemployment, which is fuelling demand for increased public outlays. Continued improvement in the public finances is also dependent on maintaining current policies *vis-à-vis* the municipalities, which commit them to medium-term budget balance while discouraging tax increases.

... but the structure of public finances needs to be changed

An underlying principle behind the fiscal consolidation programme has been to balance expenditure cuts and tax increases in such a way that sacrifices are shared among all sections of the population. This has been crucial to its wide acceptance and speedy implementation. Expenditure cuts have acted to improve incentives, but tax increases have had the opposite effect. By international comparison, Sweden stands out as having a high expenditure level, reflecting both extensive public transfers and a wide range of provision of services taken on by the public sector. It has a correspondingly high aggregate tax level. The amount of transfers to the working-age population is surpassed by only a few countries, and the level of services-in-kind in the social area by none. High transfers are motivated by redistributive concerns, but two-thirds of transfers reflect

redistribution over the life-cycle of the individual, rather than between individuals. A greater reliance on insurance-based systems, *inter alia* occupational pensions, and on private savings to meet foreseeable events over the life-cycle, could thus reduce pressures on public spending without significant sacrifice of distributional goals. With the social security system projected to move into substantial deficit in the next century, such a reform would contribute to the sustainability of the public finances without having to resort to higher taxation. The objective of the old-age pension reform initiated in 1994, and still not fully detailed, is to link future pension outlays to the growth of resources by relating future payments to demographic and real income trends, and introducing more actuarially-correct inducements to labour supply and private savings. However, the effective retirement age will have to increase substantially if the costs of the social security system are to be sustainable.

Another area requiring action is tax expenditures, which are an important element of support to businesses and households. While the tax system has benefited in simplicity and transparency from reforms to broaden the tax base and lower marginal tax rates, a recent survey of tax expenditures indicated scope for further reform in this direction, insofar as these expenditures constitute over 20 per cent of the tax base for indirect taxes and over 15 per cent of the income tax base. Since such tax expenditures are still highly selective in character, a broadening of the tax base would be a move towards greater fiscal neutrality and would serve to improve efficiency as well as the public finances.

Extensive scope for improving labour market flexibility...

Swedish labour market performance has deteriorated significantly over the 1990s, with total unemployment above 12 per cent over the past four years if those engaged in active labour market schemes (ALMPs) are included.

Improving the job-generating capacity of the economy should thus be one of the main priorities of economic policy, requiring wide-ranging measures embodying short- and long-term effects and relating to the labour market proper, to product markets, and to enterprise dynamism. Applied to the Swedish case, the *OECD Jobs Strategy* would call, in particular, for: *i)* increased aggregate wage flexibility to improve the balance between demand and supply in the labour market; *ii)* increased relative wage flexibility to ensure that wage differentials align with productivity and skill differentials to a greater extent, thereby reducing differences in unemployment risk and improving incentives to human capital formation; *iii)* reforms to tax and transfer schemes to lower reservation wages and encourage job search and work effort; *iv)* a redirection of the resources spent on active labour-market policies to lower long-term joblessness while using educational spending more effectively to ease school-to-work transition; *v)* an alleviation of features of government regulations relating to employment protection and working-time flexibility which contribute to insider-outsider mechanisms, low capital utilisation and low enterprise flexibility; and *vi)* an enhancement of product market competition, particularly with respect to contestability in the provision of services. Achieving these objectives will also depend on the nurturing of an entrepreneurial climate which favours enterprise creation and technology diffusion. Most immediately, it requires reforms in the areas of tax and transfer systems, employment security and industrial relations legislation, active labour market policies, education and training, and competition-enhancing regulatory reform. Many of these have already been set in train, both by the previous administration and in the strategy outlined and proposals contained in the Employment Bill in the spring of 1996. Though the current government has not maintained all the reforms introduced by its predecessor, the current

programme aims at improving overall labour market flexibility, *inter alia* by enhancing incentives for job search and to accept job offers, while increasing the capacity of the education system and strengthening the role of small and medium-sized enterprises in the economy.

... in particular with respect to wage formation

Aggregate real wage flexibility in Sweden has traditionally derived from the effects of variations in price rather than wage inflation, and has proved highly inadequate in the disinflationary climate of the 1990s. Strong wage-wage links prevent nominal wage growth from responding to high unemployment, since the present dominant industry-based wage-bargaining mode tends to weaken the perceived links between real wage increases and unemployment as compared with bargains struck either at enterprise level or the economy-wide level. Nominal wages are currently increasing significantly faster than the average European rate in spite of historically high unemployment. Given the commitment to price stability, high wage increases in upcoming wage rounds would be a major impediment to employment growth. Major changes in wage formation are required to achieve lower unemployment, and the Government's invitation to the social partners to present a joint proposal on how the wage formation process should be changed is an important move in this direction.

Evidence from the last decade indicates that the willingness to abide by centrally-imposed wage ceilings is very limited. But a further move towards a more decentralised wage system would be needed to reduce the strong wage-wage links across sectors. The present bargaining system could, nevertheless, produce outcomes more conducive to higher employment if the financing of unemployment benefits and ALMPs were shifted from government to unemployment insurance funds organised at industrial level; the unemployment cost of excessive wage agreements could then be internalised in the bargaining process.

Wage compression has economic costs...

The compression of wage differentials which has resulted from the present set of bargaining institutions and income replacement schemes reflect deeply-embedded equity concerns in Swedish society, but it has served to create wide differences in unemployment risk, as highly-educated employees have maintained their overall labour market attachment to a higher extent than other groups during the 1990s. A further decentralisation of wage bargaining would be beneficial insofar as it would allow locally-set wages to respond to productivity and skill differentials, enterprise earnings and to the balance in the local labour market. This would allow the unemployed and those at risk from unemployment the opportunity to price themselves into jobs to a greater extent than today. Positive effects on life-time employment and earnings would follow from on-the-job experience, reflecting the high upward earnings mobility of those having gained employment in the Swedish labour market. Indeed, a wider income dispersion would probably increase such mobility, since wages act not only as current remuneration of labour, but also as indicators of the prospective returns of investment in human capital; from this perspective the compression of wage differentials has resulted in a fall in the returns to higher education in Sweden with consequent effects on the supply of highly-qualified labour. While enrolment rates in tertiary education have recently picked up in response to higher unemployment, and the number of places available is currently a limiting factor, educational wage premia might still be too small to sustain this increase.

... while high replacement rates have disincentive effects

While the main responsibility for initiatives affecting both wage levels and differentials rests with the social partners, the policy response to the unemployment problem needs to focus on the framework for wage-setting given, *inter alia*, by the formulation of tax and transfer systems with respect to their effects on the work incentives of job seekers, and

the returns to individuals from investing in education and training. Poverty traps are less prevalent in Sweden than in many other countries, reflecting a lower degree of means-testing in transfer schemes and only small progressivity in the income tax schedule over the majority of income-earners. But with a consequent high transfer expenditure level to finance, indirect taxes and in particular payroll taxes combine with income taxes to impose an overall high tax wedge on labour income. Moreover, impediments to effective job search are still strongly embedded in the unemployment benefit scheme, not only as a result of still highly generous replacement rates for lower incomes, but also because of their long effective duration. An upper limit to the duration of unemployment benefits and a further tightening of eligibility criteria are pending. This will allow the Public Employment Service (PES) to focus more on reducing long-term unemployment by intensified job broking and more targeted use of ALMPs. However, a reduction in the overall taxation of labour, combined with less generous transfer schemes with respect to replacement rates and duration, would act to improve the quality and quantity of labour, while increasing incentives to job search and, by allowing a wider wage dispersion, enabling low-skilled workers to compete more effectively for jobs.

Active labour market schemes need re-orienting...

ALMPs have been used extensively to maintain and upgrade the qualifications of the unemployed, both in response to the effect of narrow wage differentials on demand for low-skilled workers and in the face of weak labour demand in general. The consequence has been an over-reliance on this policy instrument, with declining or even in some cases negative returns on the resources allocated. Consequently, a major change of emphasis is taking place, with ALMPs being scaled down to provide scope for intensified efforts in adult and tertiary education. In the process a targeted and more selective use of ALMPs, to

support the placement and reintegration activities of the PES and thereby reduce long-term joblessness, is evolving and this trend is to be welcomed. Moreover, the practice of allowing unemployed to enter an ALMP at a fairly early stage of an unemployment spell has acted as an impediment to their undertaking effective job search on their own and diverted the resources and attention of PES officers from the hard-to-place. In these respects, a concentration and reorientation of PES activities in line with the recommendations given in the OECD thematic review (*Enhancing the Effectiveness of Active Labour Market Policies*) could bring substantial gains. This could include an integration of the administration of unemployment benefits with the PES which would improve the enforcement of job availability provisions.

... and the focus of the formal education sector needs changing

Long-term employment growth is dependent on a supply of highly qualified entrants from the formal education system, along with a rapid pace of skill acquisition in the workplace. Sweden has been in the forefront in the latter aspect, but has gradually fallen behind with respect to the formal education sector. This reflects not only a low demand for higher education because of insufficient incentives, but also deficiencies on the supply side. There has been a fairly high drop-out or non-completion rate in upper secondary and tertiary education, affecting the bottom end of the skill distribution. At the higher end, a shorter tertiary education than in many other countries and a curriculum directed towards the requirements of the public sector have put Swedish industry at a disadvantage, particularly in the area of scientific and engineering personnel. The present strengthening of adult and tertiary education may have positive short-term effects in this regard, but a further shift in the balance towards tertiary education will only be beneficial for long-term growth if individual choices as to the skills required are those appropriate to a competitive, enterprise-based economy.

Enterprise flexibility should be strengthened

The issue of enterprise flexibility versus job security has taken on greater significance as the economy has moved out of recession. Swedish employment protection legislation is on the stricter side by European standards and seems to add to the problems of non-experienced workers in gaining a foothold in the labour market, by adding hiring and firing costs to an already high entry wage. In this respect, the most effective approach from a longer-term perspective would be to leave provisions in this area to be agreed at the local level as an integral part of a decentralised wage-setting system, with government regulations establishing the necessary framework. It would be of particular benefit to small and medium-sized enterprises, which have relatively high compliance costs. Restrictions on employment adjustment are compounded by working-time regulations and industrial relations provisions, which allow secondary action, imposing wages and conditions inconsistent with local employment possibilities, and by restricted access to outsourcing and the hiring of self-employed. It would appear that this framework is more restrictive than in other countries and needs to be reviewed.

... and the environment for small and medium-sized enterprises improved

In respect of its dynamism and efficiency of its larger enterprises, the Swedish business sector is relatively well-performing, a situation attributable in part to the focus of company law on active ownership, which has been promoted by a system which allows majority control to be wielded by groupings of shareholders. This concentration of control, often around family names in industry and commerce, can, in principle, reduce the influence of wide groups of shareholders, particularly smaller-shareholder interests, but in the Swedish case majority owners are still putting up a substantial capital of their own. Moreover, returns on equity among larger Swedish enterprises are high and equity supply to quoted enterprises has been above that of other small countries. Although the influx of capital

through foreign portfolio investment over the past years, together with added emphasis on individual equity savings in the future, may make owner-based controlling shareholdings more difficult to sustain, the system has so far achieved a relatively effective marriage of ownership and control for listed companies. It is, however, less clear that the present framework ensures an efficient allocation of capital for both established and newer, smaller firms.

The regulatory and legislative framework applying in the large corporations (often multinational, which have dominated Swedish industry) has, in general, been applied to all enterprises, regardless of size and kind of activity. It would appear that this has affected negatively the survival rate, and more particularly the incentives for small and medium-sized enterprises (SMEs) to expand. With respect to start-ups this segment of the economy performs relatively well, but it would appear that employment creation is inhibited by the existing regulatory framework. Employment patterns are dominated by large enterprises to a far larger degree than elsewhere and establishment size and net job creation are inversely correlated. Government intervention aimed at offsetting some of the disadvantages of SMEs, especially with respect to technology development and supply of risk capital, has been extensive in its range, but has added to their information problems by its complexity. Converting existing support into a more favourable tax treatment of capital income in unquoted enterprises, both for entrepreneurs and local investors, together with a simplification of the tax code, would be more effective. An example of the former is the proposed relief of the double taxation of dividends, which has impeded individual participation in the stock market. The proliferation of agencies involved in support and counselling of SMEs extends also to R&D expenditure. R&D spending is high in Sweden, reflecting the influence of the large multinationals, but the ensuing concentration on basic research in a few product areas may have imparted only few spin-offs to other enterprises and

has not facilitated technology diffusion. A shift in the balance of government support from university-based institutions to cofinanced science parks may prove more beneficial, while the process of technology diffusion to SMEs would be enhanced if there were some rationalisation of the currently dispersed responsibilities for new technologies, market opportunities, training and capital.

Enhancing the competitive climate

A competitive market is the most effective way of ensuring that the newest technology is embodied and that enterprises can grow. The manufacturing sector has been subject to a competitive international trade regime over several decades, with ensuing positive effects on productivity. A similar pressure for structural change has not been present for major utilities, nor for the non-manufacturing sectors in general, where food distribution and construction display particular problems arising from inadequate competition. The adoption in 1993 of the prohibition principle as the basis for competition regulations has signalled a far stronger pro-competitive and activist stance in the sheltered (non-tradeable) sectors. However, thus far incumbents have usually managed to hold on to their dominant position, major privatised utilities in transportation and communications having sufficient leverage to fend off competitors. The underlying issues in this case relate to the inadequacy of the regulatory regime applying to natural monopolies, to cross-subsidisation from sheltered to exposed (tradeable) activities and to the access to and pricing of interconnected networks. Such issues have clearly not been fully resolved and need to be dealt with, as for example in the telecommunications sector, postal services and railways.

The wide scope of the public sector in general has an important bearing on the development of the private services sector. While the production of public services has been increasingly separated from their provision, the public sector retains a dominant role in producing a wide range of services, creating barriers to entry and hampering

competition. Moreover, local governments are majority owners of 1 500 companies with a turnover of SKr 115 billion and they have wide scope to engage in activities on a par with private enterprises. The present lack of clear guidelines and procedures in this field provides latitude for distorting competition, insofar as such enterprises can draw on public resources – explicit or implicit by cross-subsidisation, guarantees or capital cost reductions – in their bids for contracts, both in private and public tenders. This gives scope for them to pursue predatory pricing strategies, a danger amplified by the conflict of interest arising from their double role as in-house advisor in public tenders. The overall competitive climate may thus be inhibiting private enterprise creation, and would be improved by a redefinition of the scope of activities that municipally-owned enterprises may engage in. A set of regulations preventing cross-subsidisation between monopoly and competitive activities, with full-cost accounting rules and with closer monitoring of fully specified public tenders, would be especially desirable.

Summing up

The macroeconomic policies pursued since 1994 have achieved a remarkable improvement in the public finances and in the credibility of monetary policy. Budget consolidation is proceeding in accordance with overall objectives, while price developments have been brought into line with targets. Sweden has thus gone a long way towards securing a more stable economic environment and the economy is well-placed to take advantage of an upturn in world trade. However, laying the foundation for balanced growth and increased employment over the medium and long term – which would facilitate the achievement of the targeted budget surplus without additional taxes or expenditure cuts – is equally dependent on the progress to be made in improving labour market structures and raising the potential for growth in the market-based sectors of the economy. In

this respect pressing structural problems remain with respect to unemployment, particularly in the area of wage formation but also with respect to the failure of small and medium-sized firms in Sweden to expand. The recent programme set out in the Employment Bill needs to be carried out without delay and extended to cover all dimensions of labour and product market flexibility, drawing in this respect on the wide-ranging recommendations embodied in the *OECD Jobs Strategy*. Assisted by initiatives to provide markets with a greater role in the allocation of resources, and underpinned by ongoing fiscal consolidation, such a programme would allow for a more auspicious climate for overall economic development.

I. Economic trends and prospects

Having accelerated strongly from mid-1993, as a result of stronger foreign demand, economic activity became more broadly based during 1994 and early 1995, as business investment responded to emerging capacity constraints in the tradeable sector (Table 1). Growth then slowed in the latter part of 1995 but picked up in the first half of 1996 when exports showed renewed strength. However, private consumption increased only slowly as households continued to restructure their financial balances, an adjustment prolonged by higher precautionary savings in the face of high unemployment (Figure 1). Price inflation, measured by the consumer price index, was moving above the central bank's inflation target in early 1995, but high interest rates and increased confidence in the fiscal consolidation programme led to a substantial appreciation of the krona

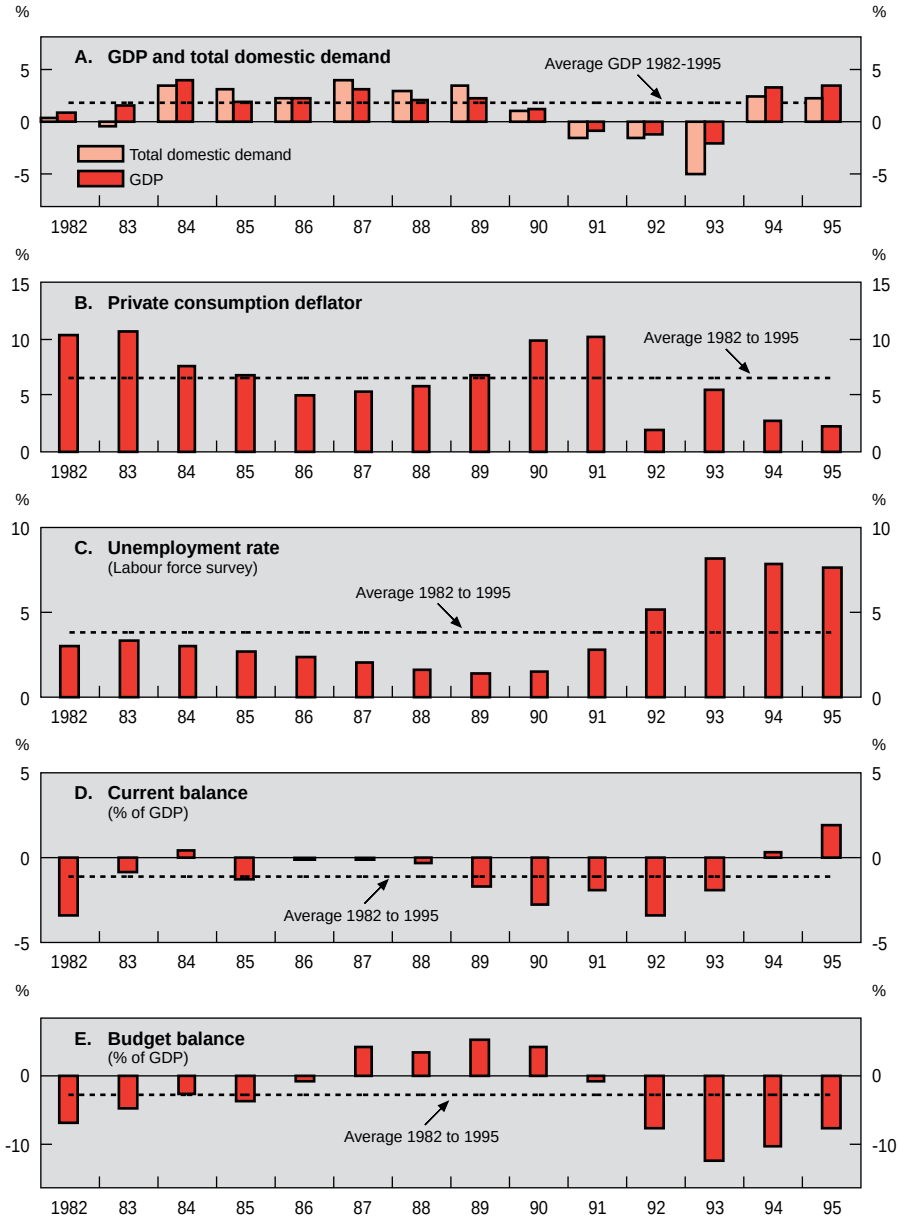
Table 1. **Main demand components**

Seasonally adjusted growth at annual rate, per cent

	1993		1994		1995		1996
	I	II	I	II	I	II	I
Private consumption	–5.9	2.1	1.7	1.7	0.1	1.3	0.6
Public consumption	3.1	–2.5	0.7	–1.7	–1.3	0.2	–6.1
Gross capital formation	–15.5	–20.3	15.5	2.2	14.3	12.9	12.8
Housing	–37.2	–37.2	–35.0	–33.5	–29.1	–17.9	..
Business	–10.9	–9.7	39.3	12.2	34.8	17.8	..
Public	30.6	–21.8	31.3	11.8	–10.2	18.2	–24.5
Change in stocks[1]	–1.0	0.8	1.4	2.0	0.1	–0.4	–1.5
Exports	4.5	23.1	8.6	15.9	15.1	5.0	5.6
Total demand	–3.7	3.5	5.8	6.3	5.5	3.1	1.0
Imports	–8.0	7.5	15.6	13.5	11.7	5.0	0.8
GDP	–2.8	1.9	3.4	4.0	4.4	3.0	0.5

1. Contribution to GDP.
Source: OECD, Statistiska Centralbyrån (SCB).

Figure 1. **MACROECONOMIC PERFORMANCE**



Source: OECD.

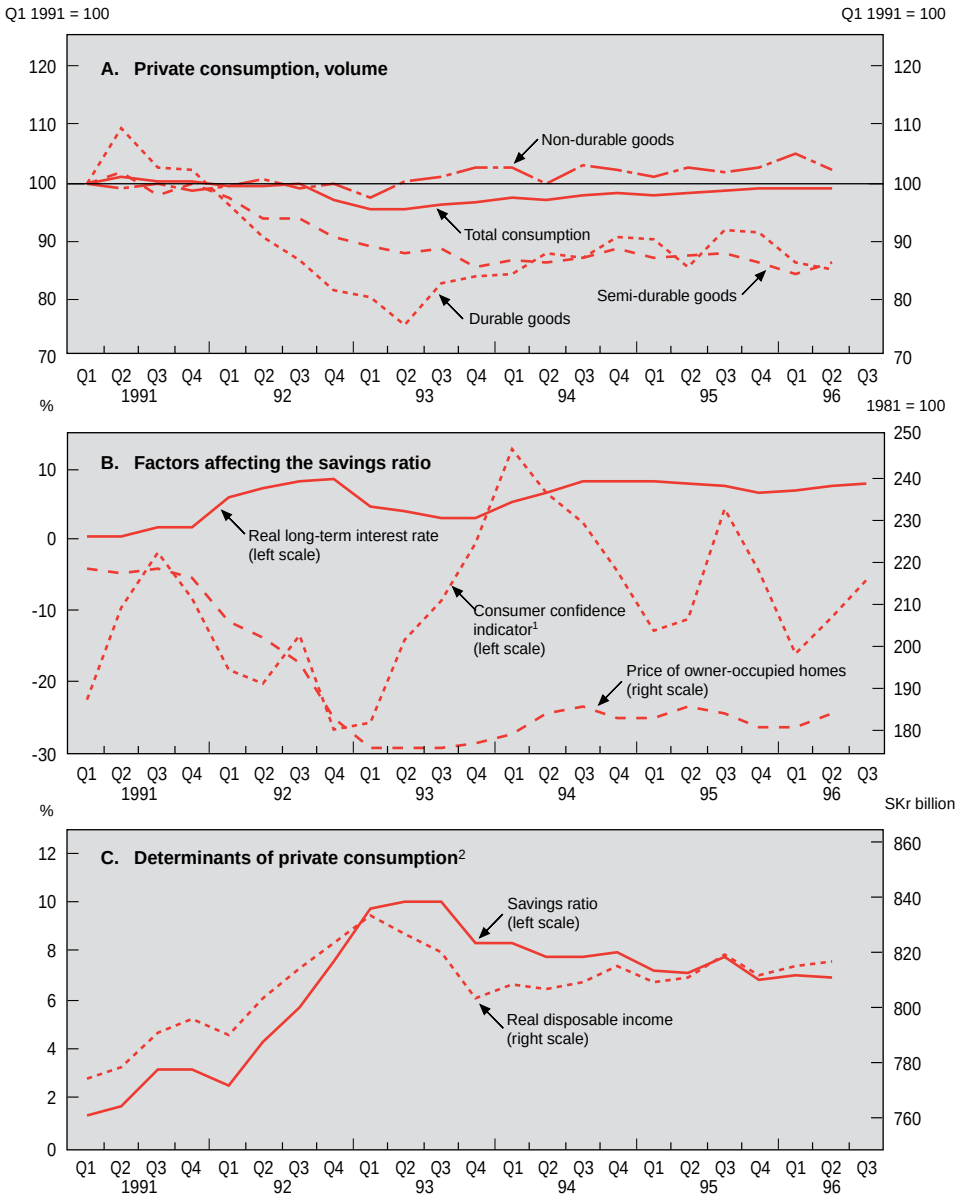
in late 1995 and to a subsequent deceleration of inflation. This has since allowed the central bank to lower short-term interest rates, helping to offset the negative impact of the higher krona on activity. The growth pause is ending as the positive effects of the lower short-term interest rates feed into the economy, thereby broadening the upswing, and as the front-loading in 1995-96 of the fiscal consolidation programme is coming to an end. Moreover, the upswing should be further supported by a strengthening of export market growth, setting the framework for a sustained upswing over the medium term. Employment could rise as a result, but with strong wage-wage links leading to average wage increases well in excess of productivity gains, unemployment in the economy is likely to remain high.

A modest recovery

Weak domestic demand

Private consumption has been weak throughout the 1990s (Figure 2, Panel A). From 1990 to 1994 this was due to a sharp increase in the savings ratio from slightly negative in 1990 to 8.0 per cent in 1994, which more than offset the increase in disposable income following, *inter alia*, from the under-financed tax reform. The saving rate has since come down somewhat, but the fall has been restrained by continued high unemployment, high real interest rates and households' continued efforts to make up for the wealth loss associated with the fall in house prices up to early 1993 (Panel B). The weak employment situation and fiscal consolidation[1] have also depressed real disposable income growth since 1994 (Panel C). Consumer spending has thus expanded only modestly. Consumer confidence remains subdued and highly volatile (Panel B), reflecting concern over prospects for the Swedish economy and the risk of unemployment, as well as negative sentiments concerning households' economic situation. On the other hand, spending plans for durable goods have been supported by replacement demand for ageing consumer durables. Indeed, there has been a 30 per cent increase in new car registrations from their low in 1993 to mid-1996, supported by the fall in interest rates seen over the past few years. Other indicators, such as retail sales, convey an impression of continued retrenchment in consumer spending. A comparison across Nordic countries of financial consolidation in households and its relation to the fall in household property wealth is given in Box 1.

Figure 2. **PRIVATE CONSUMPTION**



1. Breaks in the series in end-1992 and in end-1995.
2. Seasonally adjusted, 4 quarter moving average.

Source: OECD; Statistiska Centralbyrån (SCB).

Residential prices fell 20 per cent in nominal terms from their peak in 1991 to the beginning of 1993 (Figure 2, Panel B). Prices have since been stable, despite a significant increase in the taxation of owner-occupied dwellings. At the same time, the cost of constructing new houses has not followed house prices down (Figure 4), leading to an almost 20 per cent fall in house prices relative to construction costs over the period 1991-95. Consequently, housing starts are only one fifth of their 1991 level and overall housing investment is just below half its level of 1990. Within this total, maintenance spending grew from less than a third

Box 1. **Households' financial savings and house price falls in the Nordic countries**

Swedish house prices have come down on average by around 17 per cent since their peak in 1991, which is the equivalent of an estimated wealth loss of close to SKr 270 billion* (or 36 per cent of disposal income in 1991) for the household sector. Since 1991 households have increased their net financial position by an estimated SKr 370 billion (50 per cent of the 1991 disposable income level), primarily through a reduction in their bank loans. But relative to disposable income total net wealth is still 27 per cent (SKr 200 billion) below 1991. Hence, the household sector still has some way to go before it has restored its wealth position. Moreover, the increase in real interest rates over the past five years, before as well as after tax, suggests that the composition of wealth should move further towards financial assets at the expense of real assets.

The wealth loss to Swedish households, at 36 per cent of disposable income from the peak year, compares with a loss in Denmark of 24 per cent and a 40 per cent loss in Norway, while Finland has seen a much larger loss of 65 per cent (Figure 3). Households in the three other countries responded by increasing their financial savings, primarily through a reduction in debt as in Sweden. The improvement in the net financial position of households in Denmark, Norway and Finland has been, respectively, 81 per cent, 34 per cent and 19 per cent of the disposable income level in the peak year, while relative to disposable income, total wealth in Denmark and Norway, 64 per cent and 38 per cent respectively above the level in the peak years, had more than recovered from the losses imposed by the house price falls – even before their respective property markets rebounded. In both countries this was one of the key factors behind the revival of growth in private consumption, and if this experience is repeated in Sweden, households there may still have to increase their financial assets before private consumption can be expected to recover, a process that in the other countries has been long drawn-out.

* Estimates of housing wealth are based upon Barot and Kanis (1993).

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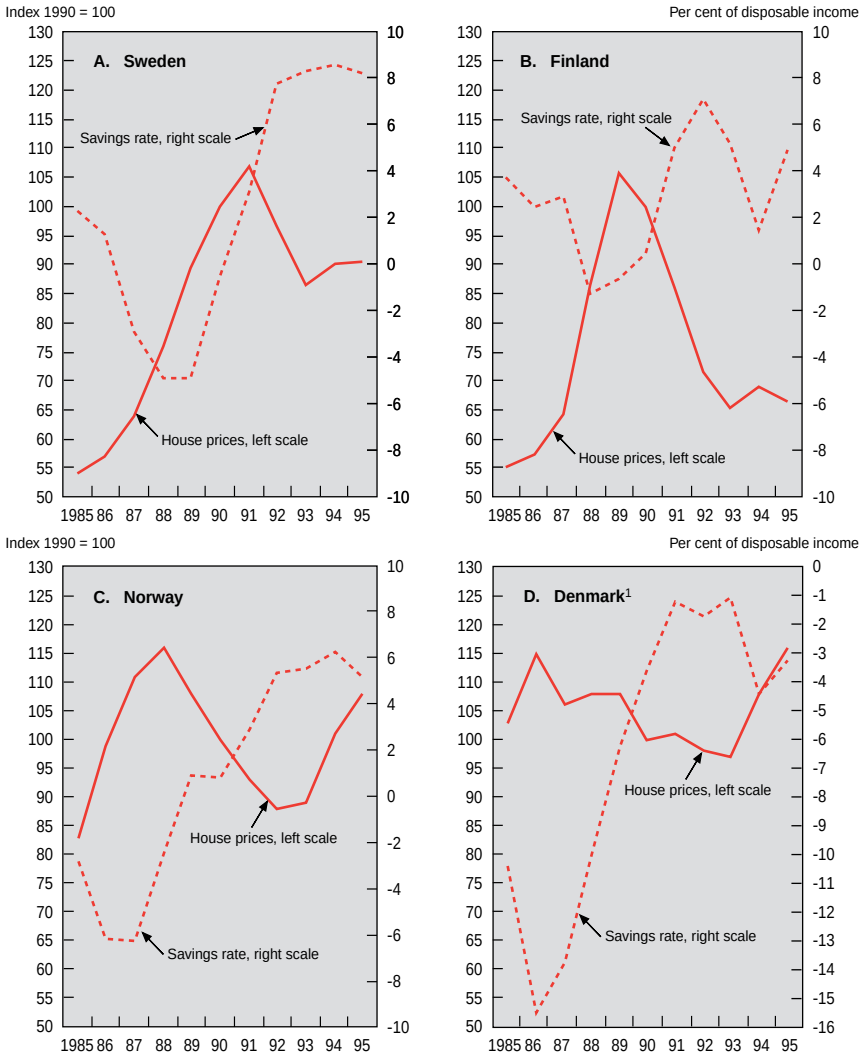
(continued)

Figure 3. **HOUSE PRICES AND HOUSEHOLD SAVINGS RATES IN THE NORDIC COUNTRIES**

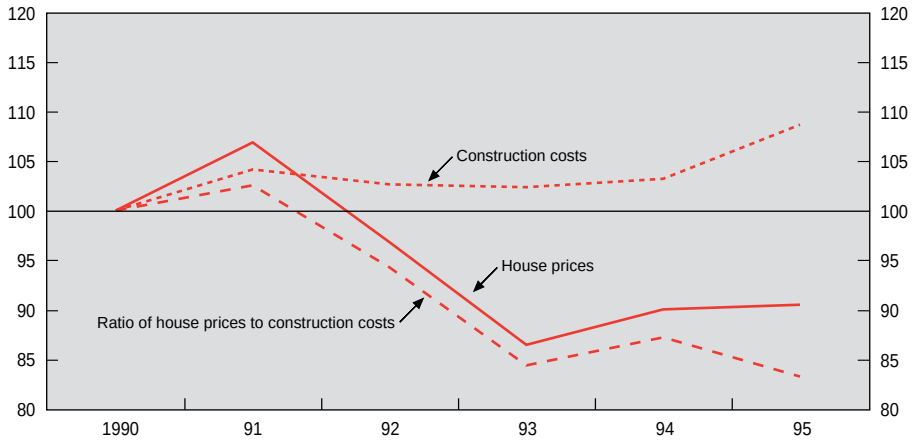


1. OECD estimate for 1995.

Source: OECD; National central banks and statistical institutes.

Figure 4. **HOUSE PRICES AND CONSTRUCTION COSTS**

Index 1990 = 100

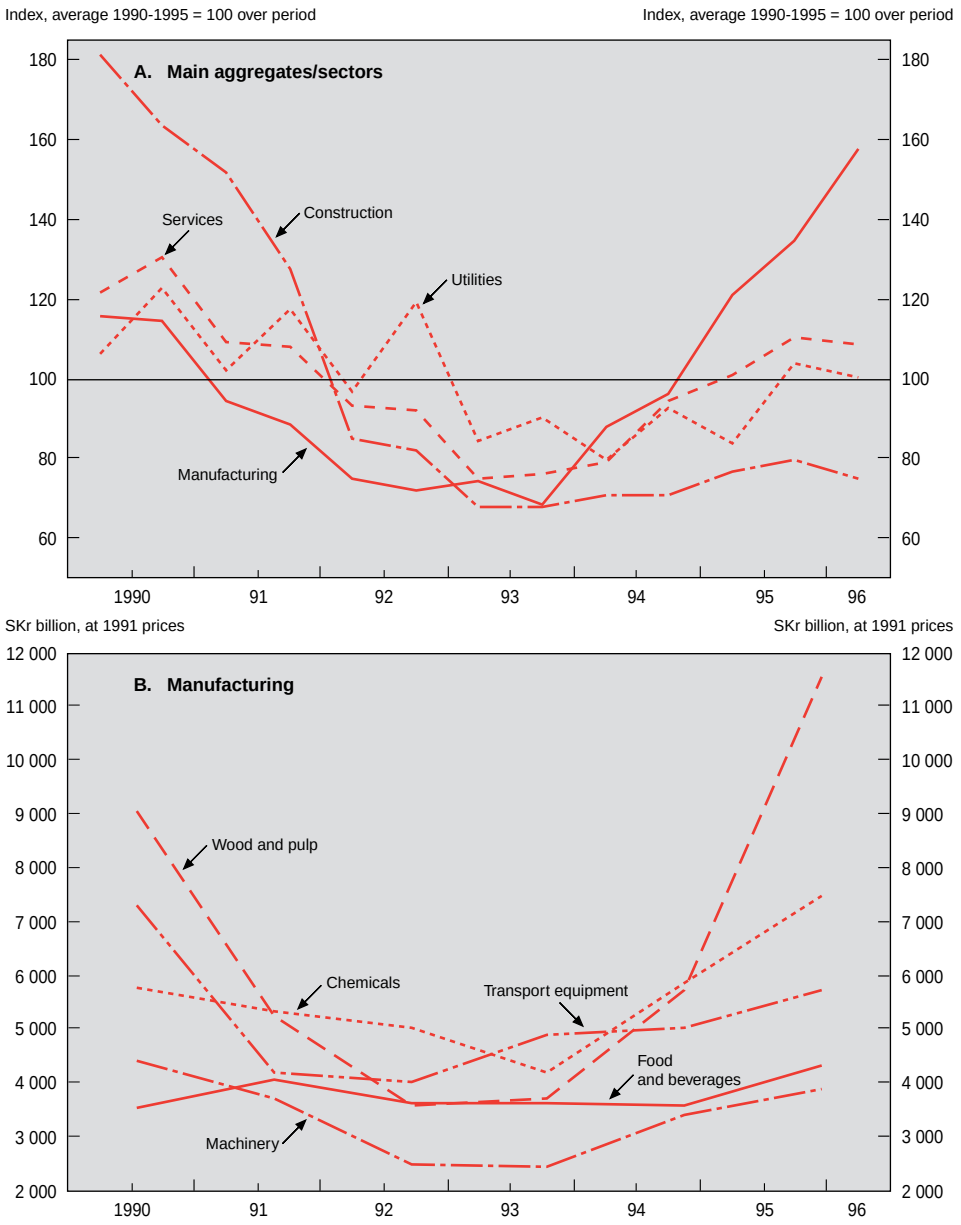


Source: National Institute of Economic Research (NIER).

of total construction investment in 1991 to more than half in 1995, partly due to temporary tax subsidies of the order of SKr 5 billion for maintenance in 1995. However, in mid-1996 house prices started to increase in the larger cities and in expanding areas with low vacancies. The number of apartments under construction increased by 40 per cent during the first half of 1996, and housing investment – although starting from an historically low level – is estimated to have increased by 20 per cent in 1996.

Business sector investment in Sweden fell by more than one third from 1990 to 1993, lowering the ratio of total investment to GDP from 12½ per cent to 8 per cent. This decline reflected both the need for a restructuring of financial balance sheets and the impact of high real interest rates. Subsequently, a revival in business sector investment has taken place, originating in the manufacturing sector (with an investment increase of 120 per cent over the period 1994-96) and spreading to the service sectors and utilities, where increases of 35 to 40 per cent have occurred over the same period (Figure 5, Panel A), taking the overall investment ratio above 11 per cent. The present investment upswing has been driven by increasing capacity utilisation following the export-led upswing.

Figure 5. **GROSS FIXED INVESTMENT, BY SECTOR**



Source: SCB and NIER.

Scale-based industries, such as wood and pulp, chemicals, and transport equipment have recorded particularly large increases in investment (Figure 5, Panel B), while industries with a lesser reliance on price competitiveness have seen smaller investment growth. Overall, investment was not sufficient to offset the depreciation of the capital stock in 1993 and 1994, and just sufficient to maintain the capital stock in 1995. Moreover, the period up to 1993 was characterised by an increase in the capital/labour ratio, as labour-shedding took place on a large scale. Recent investment trends, with the growth in machinery investment exceeding the growth of investment in structures by a factor of three, indicate that capital-deepening is continuing.

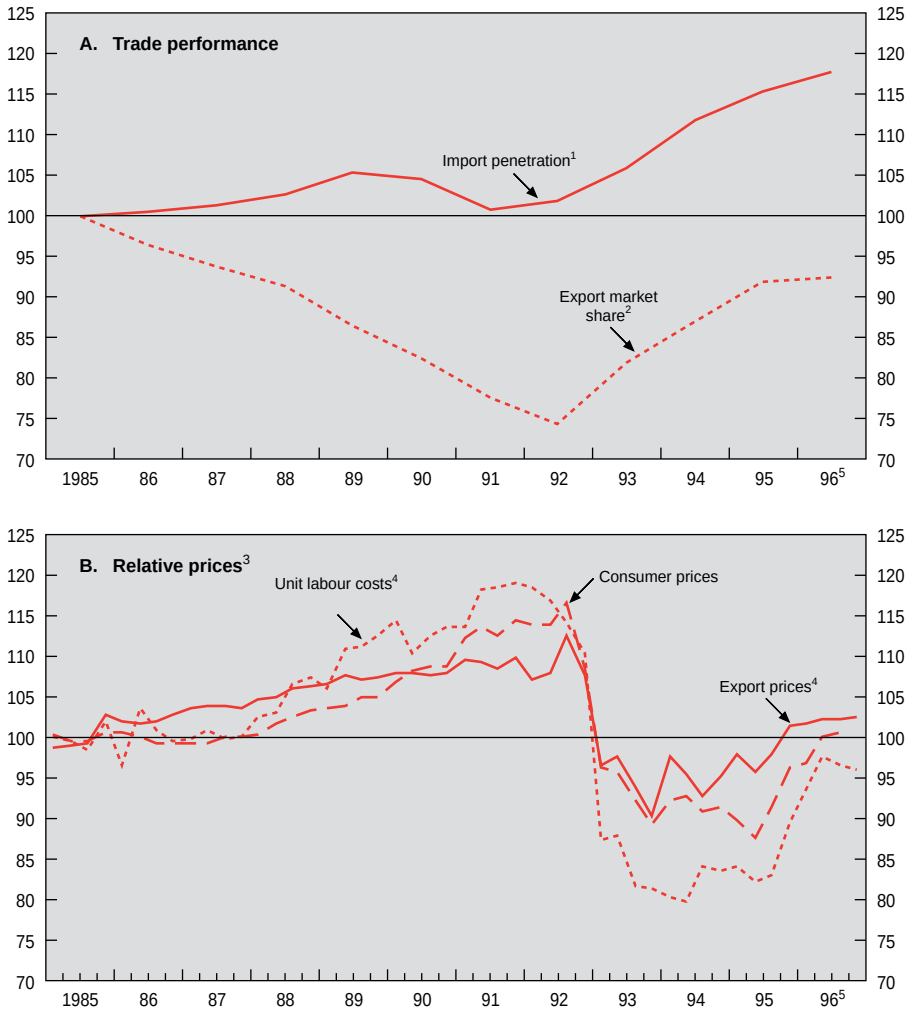
Trade performance and the external sector

The growth in exports exceeded market growth over the 1993-95 period and Sweden's market share increased by more than 20 per cent (Figure 6, Panel A). Instrumental in achieving this was the substantial improvement in unit labour costs following from the depreciation of the exchange rate in late 1992 and the large efficiency gains in manufacturing since then (Figure 6, Panel B). The slowing of growth in continental Europe in the second half of 1995 and the appreciation of the Swedish krona that started in the middle of the year dented export growth, but export growth showed renewed strength in mid-1996, with market shares for Swedish manufacturing continuing to increase (Table 2, Panel A). Gains in market shares have been geographically broadly based (Table 2, Panel B), but growth has been unevenly distributed among export categories (Table 3, Panel A). Export volumes and prices of paper products have been held back by weak international demand and competition from other depreciating countries, while more cyclically sensitive products such as metal products and machinery and equipment saw growth of around 25 per cent volume in 1994. The large price increases in 1995 for iron and metal and paper products were partly the result of substantial world market price increases at the beginning of the year, which were largely reversed over the course of the year. Other export products were subject to modest price increases, but overall export prices rose significantly (Table 3, Panel A).

The ratio of imports to GDP increased by more than 8 percentage points from 1992 to 1995 to above 34 per cent. Five percentage points of this increase can be attributed to the shift in composition of demand associated, *inter alia*, with

Figure 6. **RELATIVE PRICES AND TRADE PERFORMANCE**

Index 1985 = 100



1. Import of goods relative to import-weighted final demand (using 1985 input-output cœfficients).
2. Manufactured goods, at fixed 1991 prices.
3. Sweden relative to an export-weighted average of other OECD-countries; measured in common currency.
4. Manufacturing industry.
5. Estimates.

Source: OECD.

Table 2. **Export performance**

A. Export growth

	Seasonally adjusted growth at annual rate, per cent						
	1993	1994	1995	1994		1995	
				I	II	I	II
Manufactured products							
Relative export prices	–14.8	1.0	6.0	4.4	4.2	13.7	2.3
Export volume growth	9.8	18.6	15.5	13.8	22.3	21.0	0.0
Market growth	–0.4	11.8	9.4	13.8	14.3	9.2	5.3
Export performance	10.2	6.0	5.6	0.0	7.1	10.8	–5.0
Total goods							
Export volume growth	10.5	15.8	14.3	8.6	19.7	19.5	0.3
Market growth	–0.6	11.4	8.6	13.2	13.2	8.4	4.6
Export performance[1]	11.0	4.7	5.5	–2.8	6.5	10.2	–4.1

Source: OECD.

B. Swedish export market share in selected countries

	Per cent				
	Weight 1993	1992	1993	1994	1995
Germany	14.4	2.1	2.1	2.2	2.3
United Kingdom	10.3	2.5	2.5	2.8	2.8
United States	8.4	0.9	0.7	0.7	0.8
Norway	8.1	18.2	16.9	18.3	18.1
Denmark	6.6	11.4	10.8	12.6	11.8
France	5.4	1.4	1.3	1.4	1.6
Netherlands	5.1	2.4	2.2	2.7	2.8
Finland	4.6	13.7	12.7	12.6	13.3
BLEU	4.1	1.9	2.0	2.6	2.4

1. Goods exports, current prices.
Source: SCB.

a higher share of investment in machinery. This has left the import share three percentage points higher than would be implied by conjunctural developments on the basis of unchanged supply patterns. Increased international specialisation should lead to increased import and export shares over time, but the substantial increase in imports of, for example, wood, paper and pulp (Table 3, Panel B), for which there exist domestic substitutes, may reflect emerging capacity constraints in some sectors. The slowing of economic growth in late 1995 eased capacity constraints, and capacity utilisation for most manufacturing sectors continued to

Table 3. **Commodity composition of goods exports and imports**

Per cent growth

A. Exports

	Share in 1992	Volume			Prices[1]		
		1993	1994	1995[2]	1993	1994	1995[2]
Total	100.0	9.8	17.0	10.6	8.2	3.8	8.9
of which:							
Manufactured products[3]	96.8	9.1	18.0	11.5	8.2	4.0	8.9
Food and beverages	1.7	25.0	23.1	25.4	5.9	5.8	–0.4
Wood products	3.6	22.7	11.4	1.1	3.7	15.6	3.1
Paper products	9.3	2.9	7.1	–10.6	4.4	3.2	34.9
Iron and steel	5.0	9.8	25.8	–2.9	9.5	6.6	21.5
Machinery and equipment	48.6	7.5	24.2	21.5	9.1	2.1	2.0

1. Paasche price index.
2. Estimates.
3. Excluding ships and aircraft.

B. Imports

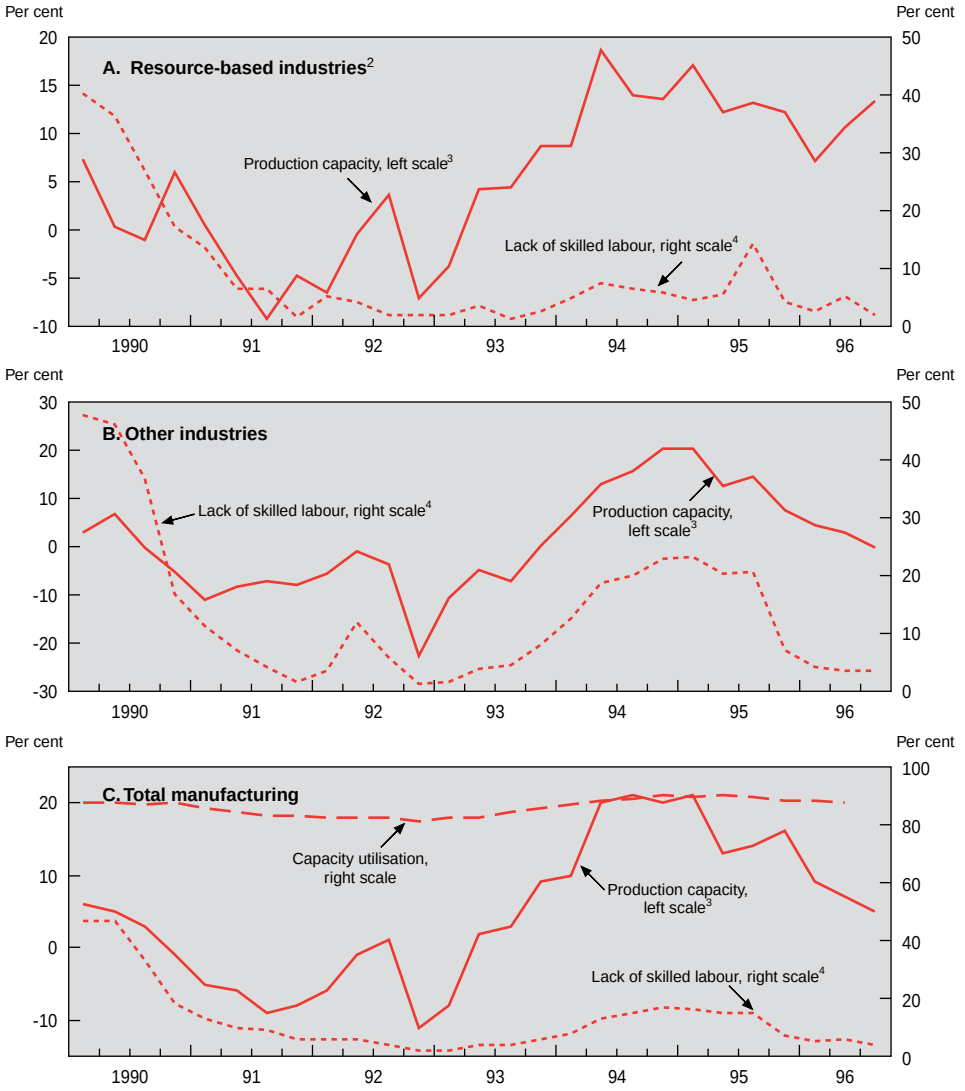
	Share in 1992	Volume			Prices[1]		
		1993	1994	1995[2]	1993	1994	1995[2]
Total	100.0	2.5	15.2	9.2	12.1	3.6	5.0
of which:							
Agricultural products	3.5	–7.4	10.4	1.8	15.8	15.1	8.3
Mineral products[3]	6.0	6.0	1.2	–5.3	16.9	–3.6	2.5
Manufactured products[4]	89.3	3.2	16.1	10.8	11.7	3.6	5.0
Food and beverages	5.3	2.3	15.5	–8.4	17.4	5.1	5.8
Wood, paper and pulp	1.0	3.0	31.8	8.2	–1.3	10.6	22.8
Oil products	3.9	–6.1	7.9	–4.1	23.4	–5.0	–0.6
Iron and steel	2.3	12.9	35.3	2.3	8.9	10.5	20.3
Machinery and equipment	43.1	3.0	20.9	23.1	13.1	1.8	0.8
Other manufactures	31.6	3.7	8.1	–0.5	8.1	5.3	10.1

1. Paasche price index.
2. Estimates.
3. Including crude oil.
4. Excluding ships and aircraft.

Source: *Konjunkturinstitutet* (National Institute of Economic Research – NIER), *Analysunderlag*, June 1996.

fall during 1996. Taking into account the new capacity coming on line as ongoing investment projects are completed, capacity should be adequate to meet an increase in export and domestic orders. Skilled workers have in general been in abundant supply in the manufacturing sector (Figure 7), but bottlenecks have emerged for some highly specialised groups, particularly in the service area (Figure 8).

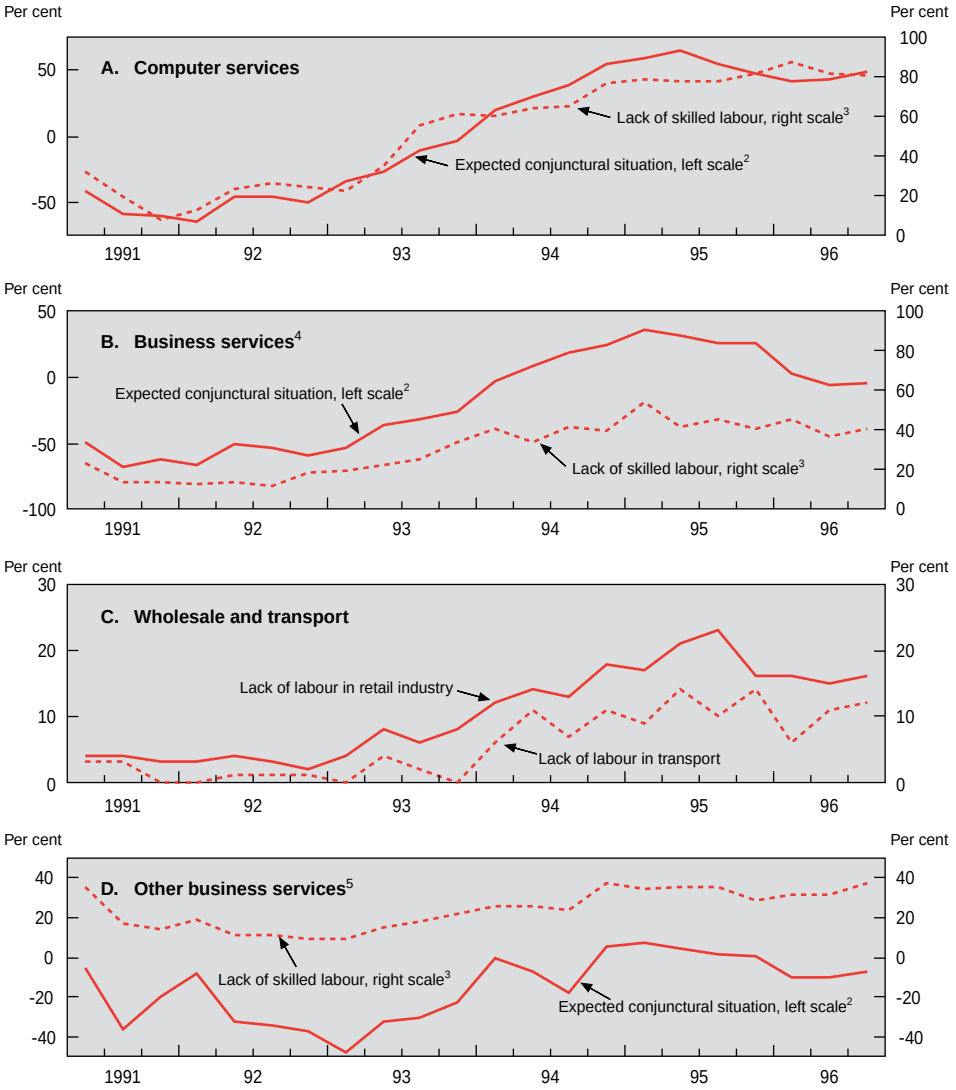
Figure 7. **RESOURCE USE IN THE MANUFACTURING SECTOR**[1]



1. Business survey data.
2. Wood, paper and pulp, minerals, iron and steel.
3. Balance between firms reporting increases (+) or decreases (-) in production capacity.
4. The share of firms reporting a shortage of skilled labour.

Source: NIER, SCB, OECD.

Figure 8. **RESOURCE USE IN THE SERVICE SECTOR**[1]



1. Business survey data.
2. Balance between firms' optimistic (+) and pessimistic (-) answers regarding the conjunctural situation in the forthcoming semester.
3. The share of firms reporting a shortage of skilled labour.
4. Marketing, organisation, and advertising services.
5. Consulting, accounting and legal services.

Source: NIER.

The current account returned to surplus in 1994 after nine years of deficit, and the improvement continued into 1995 and 1996 (Table 4). A substantial and increasing surplus in trade of goods has been only partially offset by higher interest payments on external debt (presently at 40 per cent of GDP), larger transfers arising from EU-membership and a declining surplus on the services balance. The surplus on the trade balance is now above 6 per cent of GDP. This reflects not only Sweden's favourable competitive position but also slower growth in domestic demand than its trading partners, which in turn has forced domestic-oriented producers to seek outlets for their production abroad.[2] The capital account indicates substantial portfolio changes; an important merger between a Swedish and a US pharmaceutical company alone involved gross capital flows of SKr 60 billion. High capital inflows resulting from net foreign direct investment were more than offset by outflows associated with portfolio investment in shares and by Swedish acquisitions of foreign bonds, while foreign acquisitions of Swedish bonds were mainly financed by borrowing in Swedish banks and thus neutral from a balance-of-payments point of view.

Table 4. **Balance of payments**

SKr billion

	1993	1994	1995	1996 I
Trade balance	53.1	67.1	105.6	71.1
Net investment income	–66.9	–44.6	–46.0	–31.9
Unrequited transfers[1]	–15.4	–16.1	–25.8	–9.2
Current account	**–29.2**	**6.4**	**33.8**	**30.0**
Direct investment	18.1	–3.3	24.9	–3.2
Trade in shares	32.2	37.4	–37.7	2.6
Loans/bonds in foreign currency	–35.5	–22.7	5.2	–15.1
Loans/bonds in SKr	42.7	2.6	44.4	–9.4
Changes in official reserves	–75.5	–18.1	9.9	25.3
Others[2]	47.2	–2.2	–80.4	–30.3
Capital account	**29.2**	**–6.4**	**–33.8**	**–30.0**

1. Including wages and salaries.
2. Including financial derivatives, other capital transactions and net errors and omissions.

Source: Sveriges Riksbank.

The labour market and inflation

Continuing labour shake-out

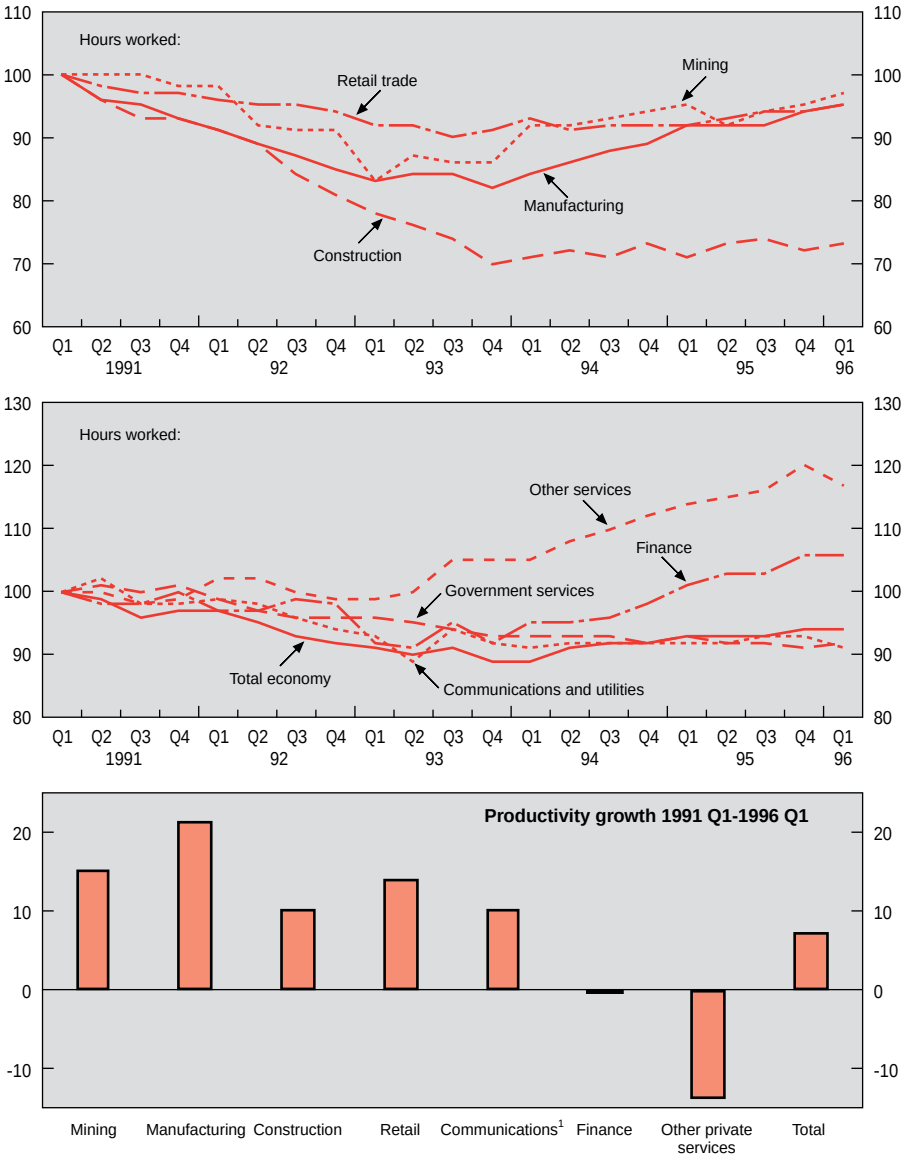
Since the onset of the recession, there has been a considerable labour shake out. From the first quarter of 1991 to the first quarter of 1994 the number of hours worked fell by 10.6 per cent for the total economy. Over the same period, total employment fell even more, as average hours worked per employee increased by 3 per cent in total and a full 8 per cent in the manufacturing sector.[3] During the present upswing, total hours worked in manufacturing have increased by more than 11 per cent over the three years 1994-96, whereas the government sector continued to contract (Figure 9, Panel A). In total, hours worked increased by 3½ cent in 1994-95 and just above 1 per cent year-on-year in the first half of 1996. Increased labour demand has been evident in a number of the service sectors, such as wholesale and retail trade, the financial sector and the category ''other services'', comprising such services as professional and household services. The financial sector and the category ''other services'' are currently the only sectors with more hours worked than in 1991. The construction sector has seen the greatest contraction, but since the end of 1993 hours worked appear to have stabilised at around three quarters of their 1991 level.

Productivity per hour has been increasing in most sectors (Figure 9, Panel B), particularly in the manufacturing and mining industries, while the service sectors have experienced somewhat smaller gains. The expansion of the economy, however, was sufficiently fast for employment to expand by 60 000 persons in 1995 before decreasing again in the first half of 1996. Variations in labour demand during the first half of the 1990s mainly affected dependent employment (Figure 9, Panel C) as self-employment has continued its expansion at similar rates both during the recession and the subsequent period of increasing labour demand.

Labour supply responded with a delay to the increase in labour demand and began to grow only as from 1995 (Figure 10, Panel C). From the beginning of 1991 until mid-1994 close to 350 000 persons left the labour force. Labour supply subsequently increased by 70 000 persons, but it fell during 1996. The variation in recorded labour supply in the last couple of years can to some extent be explained by the use of active labour market policies in Sweden, with just over 200 000 currently either in training or in subsidised work programmes (Figure 10, Panel A). Participants in training schemes are not available for the

Figure 9. **HOURS WORKED AND PRODUCTIVITY GROWTH**

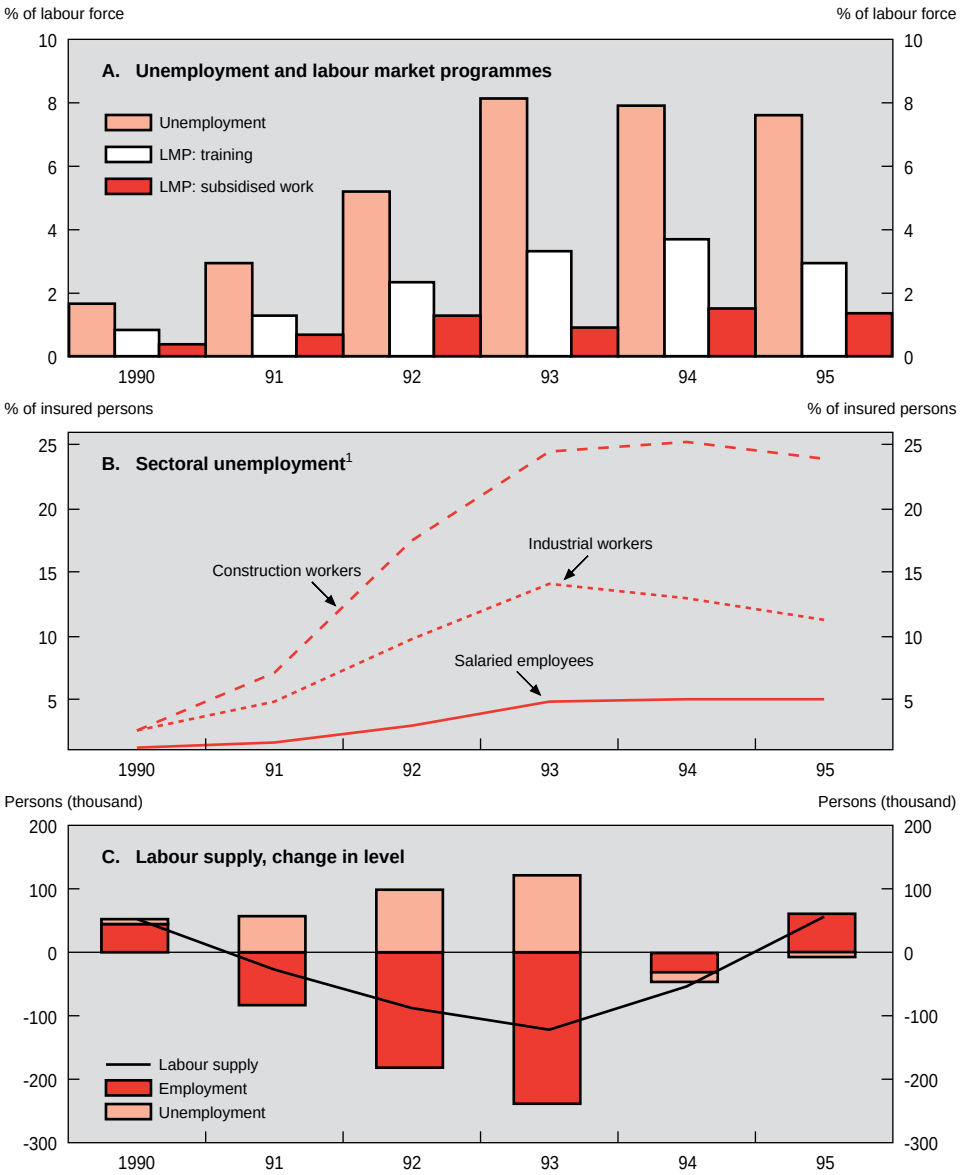
Index 1991 Q1 = 100



1. Including utilities.

Source: SCB.

Figure 10. **UNEMPLOYMENT, LABOUR MARKET PROGRAMMES (LMP) AND LABOUR SUPPLY**



1. Insured unemployed persons as percentage of all insured persons in the relevant UI-fund.
Source: SCB and NIER.

labour market and as such are treated as outside the labour force. Adjusted for a contraction in training schemes in 1995, which is being reversed in 1996, the underlying growth in the labour force can be estimated at 0.5 per cent in 1995 and close to 0 in 1996. Open unemployment nevertheless fell from 8.0 per cent in 1994 to 7.7 per cent in 1995, but increased somewhat in 1996.

The total of open unemployment and ALMP participants was 13.2 per cent in 1994 and just over 12 per cent in 1995. Moreover, it should be noted that discouraged-worker effects have been strong during the 1990s, with the number of students increasing by 100 000 up to 1994 and a further 25 000 over the past two years. The number of discouraged workers not in the labour force, but who would like to work, has followed labour market developments quite closely with substantial increases until 1994, but then falling by 15 per cent in 1995. However, this development was reversed in the first half of 1996.

Faster wage growth, but subdued inflation

Real wage growth was rather muted from 1991 to 1994, as the increase in unemployment restrained excessive wage demands. In 1994 emerging capacity constraints in the export-oriented sector led to an acceleration in wages and salaries within this sector (Table 5). Wage growth in the domestically-oriented sector was more subdued and this continued into 1995, although wage growth picked up somewhat in a number of areas. Wage drift increased slightly from 1994 to 1995 and was in the range of 1½ per cent economy-wide, explaining about 40 per cent of observed wage increases since 1994. As the wage negotiations were finalised rather late in the year recorded wage growth was artificially low in 1995 with a corresponding increase in early 1996. Most of the present agreements are running for two to three years and are estimated to yield on average wage increases of close to 5.5 per cent for 1996 and 4.5 per cent the following year, the slowing reflecting, *inter alia*, front-loading of wage increases in 1996. The impact was evident in the data for the first half of 1996, which registered wage increases of about 5¾ per cent year-on-year.

The rather high outcome of the negotiations in face of the high unemployment should be seen in the light of calculations made ahead of the bargaining round that wage increases should not exceed 3½ per cent if external competitiveness were to be maintained. However, as inflation expectations in the labour market were running between 3½ and 4 per cent at the time, this ‘‘wage room’’

Table 5. **Wages and salaries in the private sector**[1]

Per cent

	1994		1995		1996
	I	II	I	II	I
Hourly earnings					
Production sector[2]	3.7	4.3	4.5	5.3	6.7
of which:					
Metal industry	3.4	4.5	5.9	6.1	5.3
Construction	0.4	0.9	3.3	4.9	4.3
Trade and services	1.5	2.7	4.1	5.2	5.5
Transport	3.0	3.5	4.6	5.3	2.4
Financial sector	2.4	2.0	0.6	1.2	7.5
Total	2.5	3.4	4.1	5.1	5.7
Monthly salaries[3]					
Production sector	3.3	3.9	3.8	3.8	7.2
of which:					
Metal industry	4.5	5.3	2.7	2.7	7.3
Construction	1.8	2.5	3.0	3.0	4.0
Trade and services	2.4	3.4	2.7	2.6	5.0
Transport	1.5	2.9	3.2	2.5	6.0
Financial sector	3.9	2.3	2.6	2.7	5.1
Total	2.1	1.9	2.7	2.9	5.9
Memorandum item:					
Consumer price index	2.1	2.6	3.1	2.7	1.4

1. Year-on-year changes, excluding retroactive supplements.
2. Mining, quarrying, manufacturing and utilities.
3. Including commissions and fees.

Source: SCB.

effectively established the floor rather than the ceiling in all areas of negotiations. The resulting strong wage-wage links counteracted the effects of the high unemployment on wage growth. Moreover, the employers' willingness to accept the outcome of the negotiations can partly be attributed to the depreciation of the krona into 1995 and the strong expansion of the economy at the time, which led to a substantial increase in the private sector's net profit rate from 18½ per cent in 1991 to 28½ per cent in 1995. Annual productivity increases have amounted to 5 per cent since 1991 in the manufacturing sector and close to 2 per cent in a number of domestically-oriented sectors. The hike in profit margins enables producers to absorb wage increments above productivity growth without a significant short-run impact on consumer price inflation. The scope for a lowering of profits is, however, limited over the medium term by the need to maintain an internationally competitive rate of return on invested capital.

Swedish consumer price inflation was below those of its main European trading partners in early 1994, but started to rise above them later in the year as indirect taxes were increased and underlying inflation tended upwards. The price impulses from the 1992 depreciation were initially modest, as importers squeezed their margins during the recession, but a further weakening of the exchange rate in early 1995 resulted in import and export price increases of 14 and 10 per cent respectively, foreshadowing a further acceleration of domestic inflation. The central bank reacted with interest rate increases in the second half of 1994 and these continued in the first half of 1995 as consumer price inflation moved outside the bank's tolerance range for inflation of 2 per cent ±1 per cent and peaked at 3.3 per cent in April. The appreciation of the krona, about 18 per cent over the period April to December 1995, reined in price inflation, which fell to 0.2 per cent year-on-year in September 1996, while consumer prices net of indirect taxes fell by 1 per cent. Moreover, import and export prices were 4½ and 8½ per cent respectively below their level a year ago, and producer prices were also declining. Hence, the outlook is for continued low increases in prices.

The short-term outlook

Economic policy assumptions and the external environment

The fiscal consolidation programme, discussed in greater detail in Chapter II, should move the primary balance into surplus during 1997, the programme being heavily front-loaded. Seventy per cent of the tightening is concentrated in 1995 and 1996, with the negative impact on real income and domestic demand diminishing over time (Table 6). The fall in consumer price inflation has allowed an easing of monetary policy, with Swedish short rates narrowing the spread with similar German rates to about 150 basis points, and inflation prospects pointing to some scope for further narrowing of the differential over the projection period. Long-term interest rates have fallen by 3½ percentage points since last year to around 7 to 7¼ per cent, with the differential relative to Germany presently around 1¼ percentage points. German long rates are not expected to fall much further, but sustained progress in fiscal consolidation would mean that the differential could continue to fall. The exchange rate is, in effective terms, almost 12 per cent above its 1995 average and the projections are based on the technical assumption that this will be maintained throughout the projection period.

Table 6. **Economic policy assumptions and external environment**

	1995	1996	1997	1998
Fiscal policy				
General government financial balance (per cent of GDP)	–7.9	–3.8	–2.5	–0.5
Change in general government financial balance (per cent of GDP)	2.4	4.1	1.3	2.0
of which:				
Cyclically adjusted	1.1	4.1	1.1	2.0
Interest and exchange rates				
Effective exchange rate (1991 = 100)	81.8	90.2	91.7	91.9
Kronor per DM	5.0	4.5	4.4	4.4
Three-month German rates	4.5	3.3	3.1	3.7
Three-month domestic rates	8.7	5.9	4.6	4.9
Ten-year domestic bond yields	10.2	8.1	7.2	7.1
Export-market growth for total goods (per cent)				
Total OECD	7.0	5.1	6.2	6.9
Germany	2.1	2.1	4.0	6.5
United Kingdom	4.0	7.3	6.9	6.7
Norway	8.1	8.1	5.6	3.7
United States	0.9	6.7	6.1	6.2
Finland	11.7	5.5	7.2	7.0
Unit labour cost growth in competitor countries (per cent)[1]				
Germany	0.9	0.6	0.6	0.8
United Kingdom	0.9	0.8	1.1	1.8
Norway	2.4	1.9	3.1	4.1
United States	3.1	2.7	2.3	2.4
Finland	–0.3	1.4	0.7	1.5

1. Business sector unit labour costs.
Source: OECD.

Economic activity gained strength in mid-1996 and is projected to continue its upward trend as export market prospects improve (Table 7). The quickening of economic growth in Europe will stimulate Swedish exports, but the gain in market shares should start to reverse as the high wage growth following the agreements made in 1995 coincides with a slowing of productivity growth, accentuating the negative effects of the higher nominal effective exchange rate. Growth will be supported by buoyant investment activity, driven by the export-oriented sector which is facing a good profit situation and falling interest rates. Import growth will be stimulated given the high, and possibly increasing, import content of export and investment goods, leaving a strongly declining contribution from net exports to growth over the projection period.

Table 7. **Short-term outlook**

Percentage changes (1991 prices)

	1993 current prices		1995	1996	1997	1998
	SKr billion	Per cent of GDP				
A. Demand and output						
Private consumption	796.3	55.1	0.8	0.8	1.5	1.8
Government consumption	406.1	28.1	–1.0	–1.5	–0.2	–1.0
Gross fixed investment	205.8	14.2	10.9	9.8	4.1	6.1
Final domestic demand	1 408.2	97.4	1.8	1.7	1.5	1.9
Stockbuilding[1]	–13.6	–0.9	0.5	–1.4	0.0	0.2
Total domestic demand	1 394.6	96.4	2.3	0.2	1.5	2.1
Exports of goods and services	472.9	32.7	12.6	6.0	6.0	4.5
Imports of goods and services	421.4	29.1	10.3	2.5	5.0	5.0
Foreign balance[1]	51.6	3.6	1.4	1.5	0.8	0.2
GDP at constant prices			3.6	1.7	2.2	2.1
GDP at current prices	1 446.2	100.0	7.4	3.3	4.7	4.7
GDP price deflator			3.7	1.6	2.5	2.6
B. Memorandum items						
Private consumption deflator			2.4	1.5	2.1	2.2
Total employment			1.6	–0.5	0.5	0.4
Unemployment rate			7.7	7.9	7.4	7.2
Breakdown of gross fixed investment						
Public sector	34.3	2.4	–0.5	–5.0	1.4	3.5
Private sector residential	56.1	3.9	–28.0	15.0	2.0	10.0
Private sector non-residential	115.2	7.9	25.1	12.5	5.0	6.0
Short-term interest rate			8.7	5.9	4.6	4.9
Long-term interest rate			10.2	8.1	7.2	7.1

1. Changes expressed as percentage of GDP in the previous period.
Source: OECD.

Short-term outlook and risks

The domestic-oriented sectors could see some improvement in demand, as high real wage growth and falling interest rates induce consumers to lower their saving ratio, with an accompanying increase in private consumption. Overall GDP growth is estimated at 1¾ per cent in 1996, and projected to rise above 2 per cent in 1997 and 1998. Growth will be sustained mainly by exports and business fixed investment, while national accounts-based consumer price inflation may increase from 1½ per cent in 1996 to 2¼ per cent in 1998, as embedded wage increases outstrip productivity growth. Short-run employment prospects

should nevertheless improve as production picks up, the unemployment rate being projected to come down to close to 7 per cent by the end of the projection period. Participation in labour market programmes is estimated to increase from 4½ per cent of the labour force in 1995 to 5¼ per cent in 1998, leaving the total of unemployed and ALMP participants at 12½ per cent of the work force.

The main downward risk to the projection stems from the effects of the high wage increases agreed for 1996-97 and an eventual continuation of these. Not only is capital-deepening gathering pace as producers react to the high price of labour, but a prolonged slowdown in the economy would also force them to shed labour to contain costs, with adverse short-term effects on employment and on progress towards fiscal consolidation. Moreover, there are limits to the extent that producers can absorb high wage growth even under a more balanced development in demand. A pick-up in consumer price expectations may thus occur, especially if wage growth appears unlikely to subside when present agreements come up for renewal during 1997 and 1998; long-term interest rates may then increase and renewed exchange rate instability could follow. On the upside, private consumption may be stronger than expected as households' financial consolidation comes to an end, particularly if underpinned by an improved balance in the housing market. Moreover, with a build-up of capacity in the exposed sectors Sweden should be in a good position to respond if export market growth is higher than expected.

II. Macroeconomic policies

Benefiting from the austerity measures adopted during 1994 and 1995, the budget deficit has been declining rapidly: from a peak of 12.3 per cent in 1993, the general government deficit fell to 7.9 per cent of GDP in 1995 and is estimated to have fallen below 4 per cent in 1996. The primary deficit is nearing a level at which net debt stabilises, and government gross debt has already been set on a declining path, assisted by a substantial privatisation programme. This improvement in fundamentals has served to restore the confidence of financial markets, allowing bond yields to fall by close to 2 percentage points over the year to September 1996, and has been underpinned by monetary policy. Assisted by an appreciation of the krona from mid-1995, the central bank was able to rein in consumer prices, which had exceeded the target range in the first half of 1995, and with inflationary pressures abating strongly during 1996, short-term interest rates have been able to provide support to the economy.

On present trends with respect to budget deficits and price performance the Swedish economy will meet the requirements for participation in the Economic and Monetary Union of the European Union from its inception, as laid down in the Maastricht Treaty. The Swedish authorities have stated that the question of EMU participation will be assessed carefully over the next year and thereafter be a matter of decision in the Swedish parliament in the autumn of 1997. In the meantime, the operational background to monetary policy remains inflation targeting, the longer-term requirements of which are virtually identical to those flowing from a fixed exchange rate. Indeed, the uncertainty about the decision on the EMU gives added emphasis to the need to implement fiscal and monetary objectives in full in order to sustain external confidence, and the key to achieving those objectives – within or outside a monetary union – lies in reducing the structural budget deficit and unemployment. In this respect, the fiscal

consolidation implemented over the past few years, and discussed below, needs to be accompanied by structural reforms both with respect to public transfers and to the implementation in full of the planned reform of the old-age pension system.

The chapter proceeds as follows: the first section reviews the implementation of monetary policy during 1995 and 1996 before moving on to a discussion of the main influences on long-term interest rates and the alternatives for the monetary policy framework in the years ahead. The fiscal policy section starts with a review of the development of government finances over the past few years and the concomitant changes in the budget process. Securing control over budget developments over the medium term requires also a change in the structure of public finances and this is discussed in the remainder of the fiscal policy section, with emphasis on public transfers, tax expenditures and the on-going pension reform.

Monetary policy

Implementation of monetary policy 1995-96

Following the floating of the Swedish krona in late 1992, the central bank announced that it was adopting a price inflation target. The target was defined as 2 per cent growth in the unadjusted Consumer Price Index, with a tolerance margin of 1 per cent on either side. This was to be achieved by 1995. During 1994, price developments remained favourable, as the expected large price impulse from the depreciation in 1992 failed to show up, and monetary policy responded with a generally supportive stance. However, by August 1994 it was evident that inflationary pressures had started to build up, and the central bank increased the repo rate – its main signalling rate – in a series of steps during the second half of 1994 and first half of 1995. By early 1995 the target variable was moving out of the stated range; it reached 3.3 per cent in April, reflecting the inflationary impact of a depreciating currency and the extensive use of indirect tax increases to strengthen public finances. Surveys of inflation expectations indicated a widespread scepticism about the overall objective of monetary policy both over the short and medium term. As the wage-bargaining round proceeded it became evident that expectations of continued high price increases, both

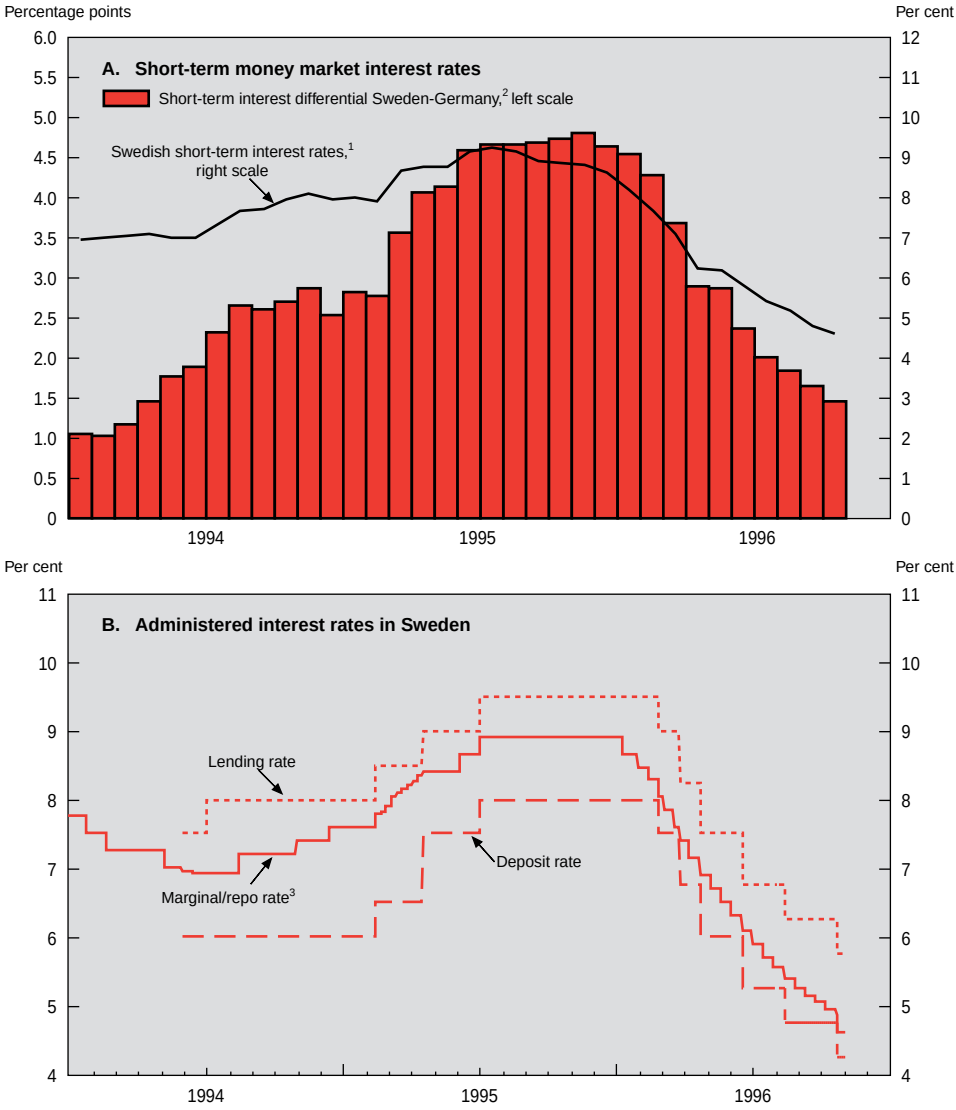
domestically and for externally traded goods and services, were embedded in the wage increments awarded. Consequently, the focus of monetary policy has been on establishing the non-inflationary credentials of the central bank, support to the economy being countenanced only if price developments could be expected to remain within the tolerance range over a full two year horizon.

Over the year to June 1995 the repo rate had been increased by 1¾ percentage points. When this proved insufficient, the central bank declared its intention of further tightening monetary policy in order to bring price developments into line. One complicating factor in this situation was the relatively weak immediate effect of interest rates on domestic demand (lending to a large degree being conducted at fixed interest rates) and the two to three-year horizon of wage contracts, making an exchange rate appreciation the most effective way of a rapid dampening of price increases. The repo rate was raised by another ¼ percentage point in July to reach 8.9 per cent (Figure 11) and maintaining the key rate at this level during the second half of 1995 proved to be sufficient, with increasing recognition by the financial markets of the fiscal consolidation effort, to set off an appreciation of the krona against the Deutschemark by 7 per cent from early September to the end of the year (Figure 12).

By the beginning of 1996, price inflation had been brought back into the upper part of the tolerance range, where it was expected to remain through 1996 and 1997. With growth prospects being revised downwards, the repo rate was lowered in a series of small steps from early January to late October, by a cumulative 4¼ percentage points. This reduced the differential with short-term German rates by 3 percentage points to 1½ percentage points while the krona appreciated by another 4 per cent against the Deutschemark. Consumer price inflation fell sharply in the process; driven partly by the effects of monetary policy on import prices and lower housing costs, it fell below the lower end of the tolerance range of 1 per cent in the summer of 1996. Although some pick-up in inflation is expected as these effects peter out, consumer price inflation now appears to be securely embedded within the tolerance range over a two-year horizon.

In contrast with other monetary policy frameworks, which rely on intermediate targets such as the exchange rate or the money supply, inflation targeting is not associated with a well-defined intermediate target. But in order to assess the impact of monetary policy on economic developments, the monetary authorities

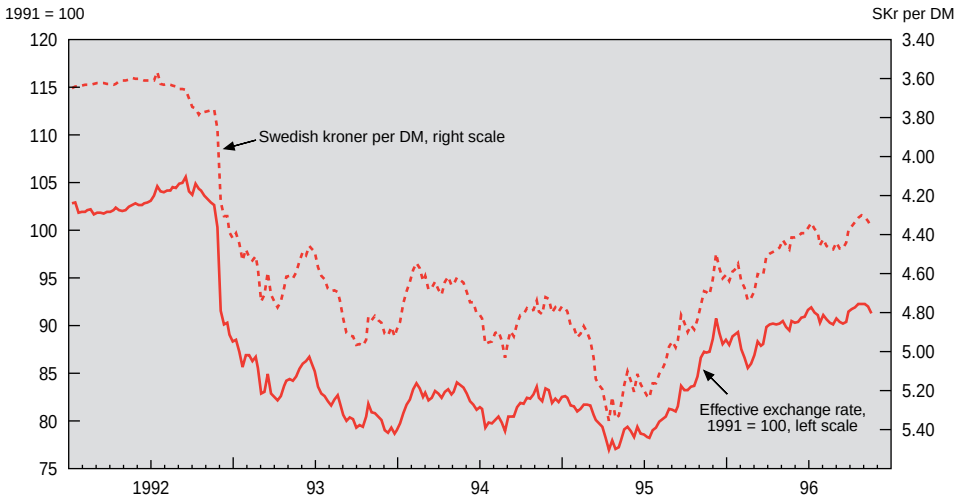
Figure 11. **SHORT-TERM INTEREST RATE DEVELOPMENTS**



1. 3-month Treasury discount notes.
2. 3-month Fibor.
3. Sweden introduced a system with a deposit rate and a lending rate as respectively floor and ceiling for money market rates on June 1, 1994. The repo rate replaced the marginal lending rate as the key signal rate.

Source : OECD, *Main Economic Indicators*; Sveriges Riksbank.

Figure 12. **EXCHANGE RATE DEVELOPMENTS**

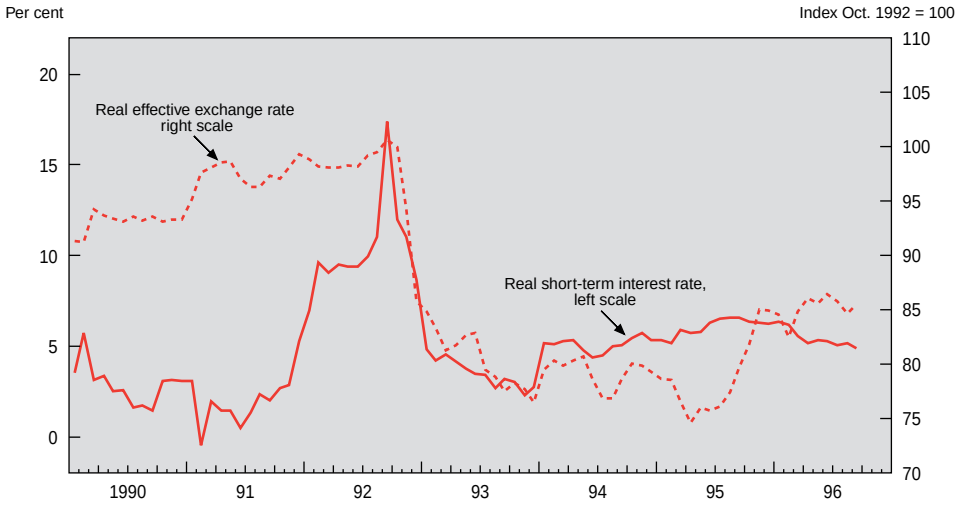


Source: OECD.

have *inter alia* used a monetary conditions index, which weights changes in short-term interest rates and the exchange rate according to their likely effects on the output gap of the economy one year ahead. This index provides some guidance for the current direction of monetary policy.[4] It indicates that a significant overall tightening occurred in the second half of 1995 which has been partially reversed with the fall in interest rates in 1996. This has served to restrain demand and, together with the direct price effects of the appreciation of the krona, price developments through 1996, but should have an expansionary effect through 1997.

During 1993 and 1994 the Swedish economy was characterised by a strong duality, as the exposed sectors benefited from the strong depreciation of the currency while high interest rates and financial restructuring in the private sector acted to depress domestic-oriented sectors. This feature was even more pronounced in the first half of 1995, as short and long-term interest rates increased further, but this was offset by a further depreciation of the currency (Figure 13). Monetary developments during the second half of 1995 and 1996 have to a large

Figure 13. **DEMAND IMPULSES GIVEN BY MONETARY DEVELOPMENTS**

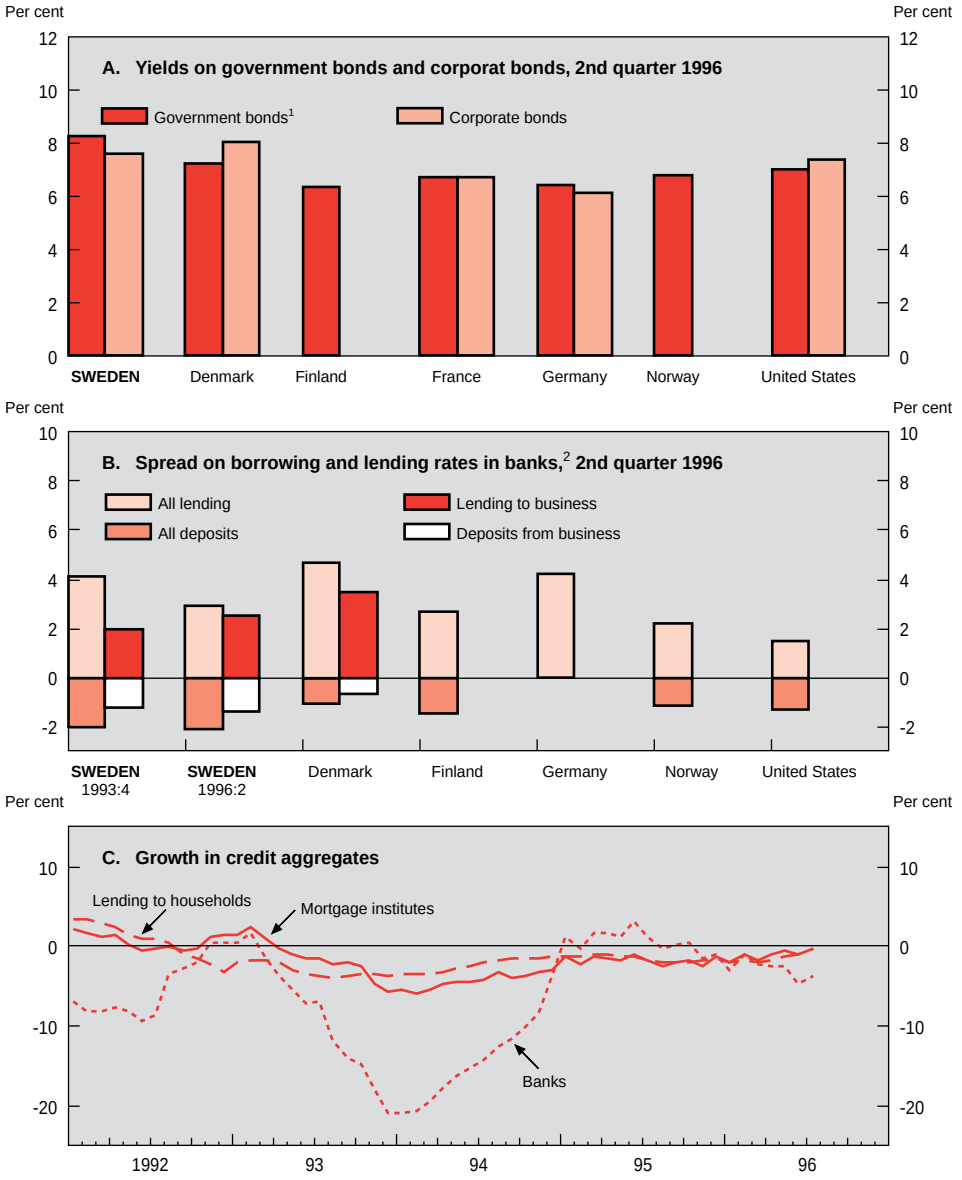


Source: OECD, *Main Economic Indicators* .

extent reversed this duality of the economy, with real interest rates coming down and the exchange rate appreciating. Nevertheless, real long-term rates are fairly high, both in an international context (Figure 14, Panel A) and – following the 1990/91 tax reform which lowered marginal tax rates and hence the value of tax deductions – relative to the 1980s. Investment carried out under present interest rate and price constellations has to satisfy a high rate of return. In spite of a pick-up in investment during 1995 and 1996 – both in manufacturing and sectors catering to the domestic market – credit aggregates have remained weak, and mortgage and bank lending have declined over the last year by ½ and 4 per cent respectively (Figure 14, Panel C). Investment has thus been financed out of retained profits, as financial consolidation has continued both in households and enterprises. In the process assets contained in broad money (M3) have grown, partly at the expense of securities.

Effects of supply-side developments on credit growth are hard to quantify, even during the years of the 1991-93 banking crisis, but since early 1995 they would have been positive (Figure 14, Panel B). The earlier wide spread between

Figure 14. **INTEREST RATES AND CREDIT AGGREGATES**



1. 10-year government bond rates.
2. A 3-month money market rate is used as the reference rate.

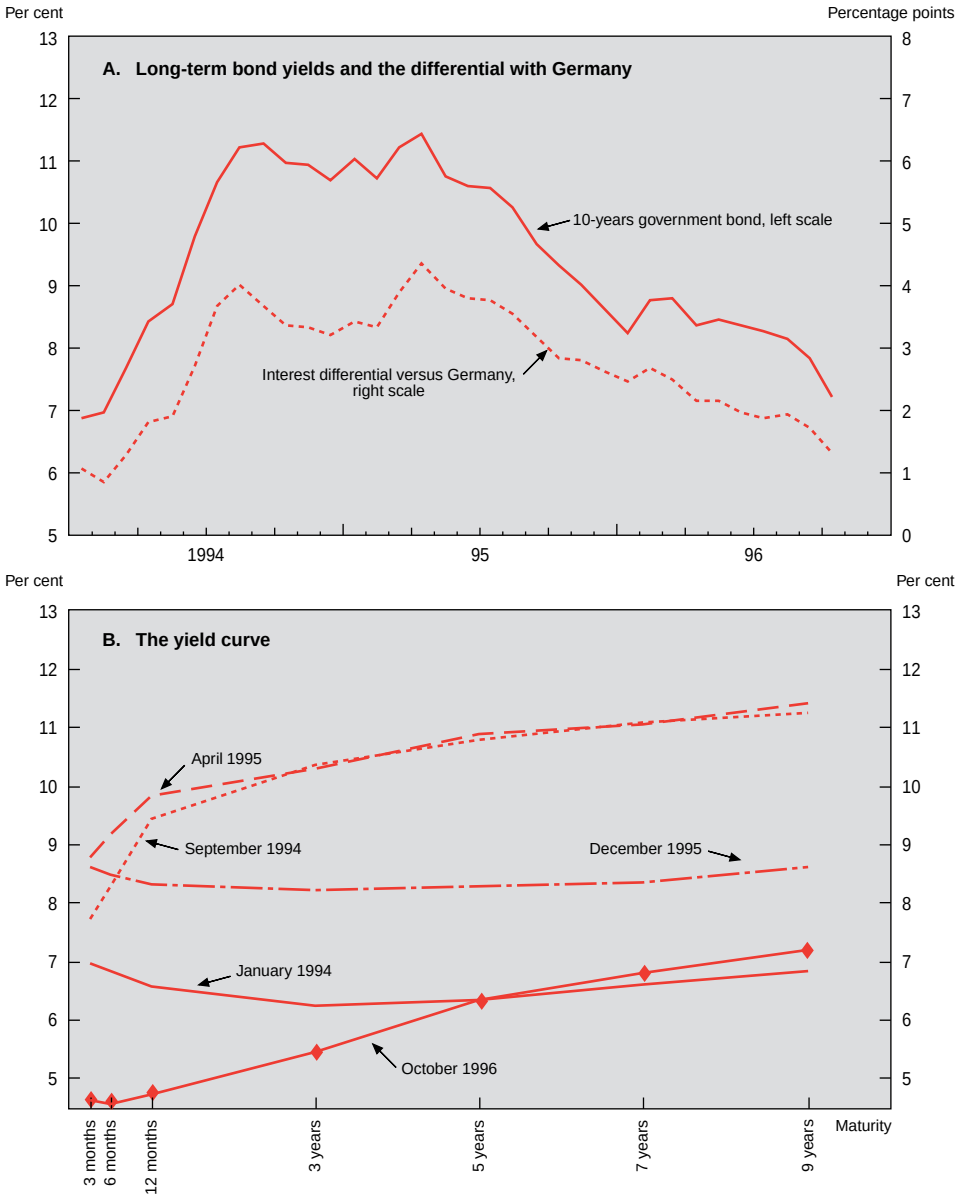
Source: OECD; Sveriges Riksbank; National sources.

deposit and lending rates of banks, which reflected the need to build up their capital, fell from 5.8 percentage points in the second half of 1994 to 4.9 percentage points in the first half of 1996. The capital base of banks and mortgage institutions has been strengthened considerably and capital ratios are now among the highest of OECD economies and far in excess of the 8 per cent risk adjusted capital adequacy requirement following from EU directives. As a consequence of this, the government has been able to withdraw its unlimited guarantee of banks' commitments towards creditors and depositors with effect from 1 July 1996. A deposit insurance scheme, funded by premium payments from financial institutions, went into force on 1 January 1996 and now constitutes the safety net for depositors, covering all deposits up to SKr 250 000. The last remnants of the banking crisis may be seen in the separate property-holding companies, both in government hands (Securum and Venantius) and under bank ownership.

Declining bond yields

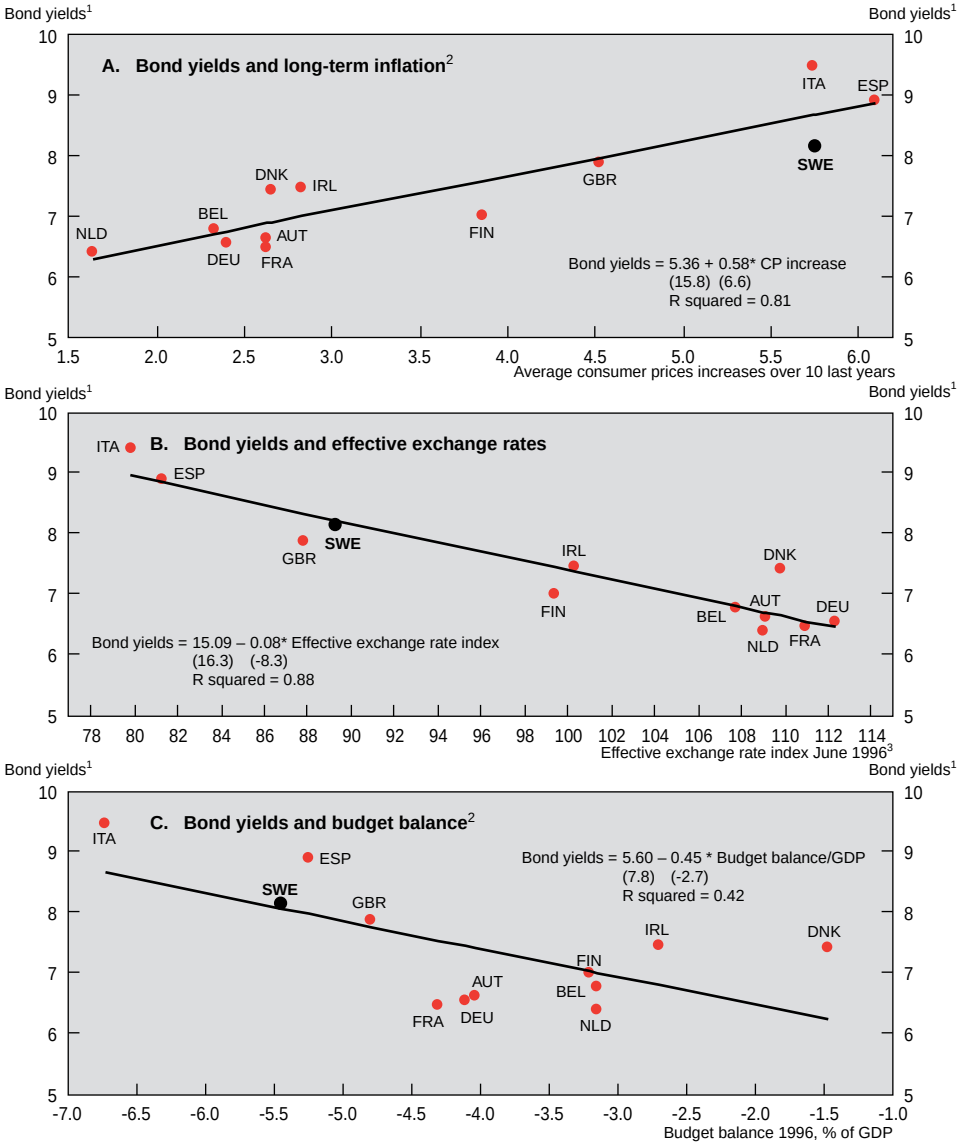
Swedish bond yields shifted dramatically upwards from early 1994 (Figure 15), as long-term interest rates increased in most countries. However, the upward shift in Sweden was particularly pronounced, as long-term interest rates moved from 7 per cent at the start of 1994 to above 11 per cent in late 1994, where they remained until mid-1995. They fell below 9 per cent in late 1995 and were between 7 and 7½ per cent in October 1996, taking the differential against Germany down from 3¼ percentage points in June 1995 to 1¼ percentage points. Swedish rates still exceed those prevailing two and a half years ago. However, just as it would appear that Swedish interest rates were slow to respond to worsening fundamentals in 1994, the full effect of the fiscal consolidation programme appears not yet to be fully incorporated in interest-rate differentials. Further gains might thus be expected as the fiscal consolidation programme continues to show up in declining deficits and lower debt ratios. But, from international evidence, inflation and exchange rate history are probably more powerful in explaining longer-run differences in bond yields than current budget imbalances (Figure 16), and in this context the long history of high inflation and substantial depreciations of the exchange rate may continue to give rise to a premium on Swedish bonds, even though present price performance is among the best of OECD countries.

Figure 15. **BOND YIELDS**



1. Effective yields on 3-6-12 months Treasury bills and 3-5-7-9 years government bonds.
Source: OECD, *Main Economic Indicators;* Sveriges Riksbank.

Figure 16. **BOND YIELDS, INFLATION PERFORMANCE AND EXCHANGE RATES**



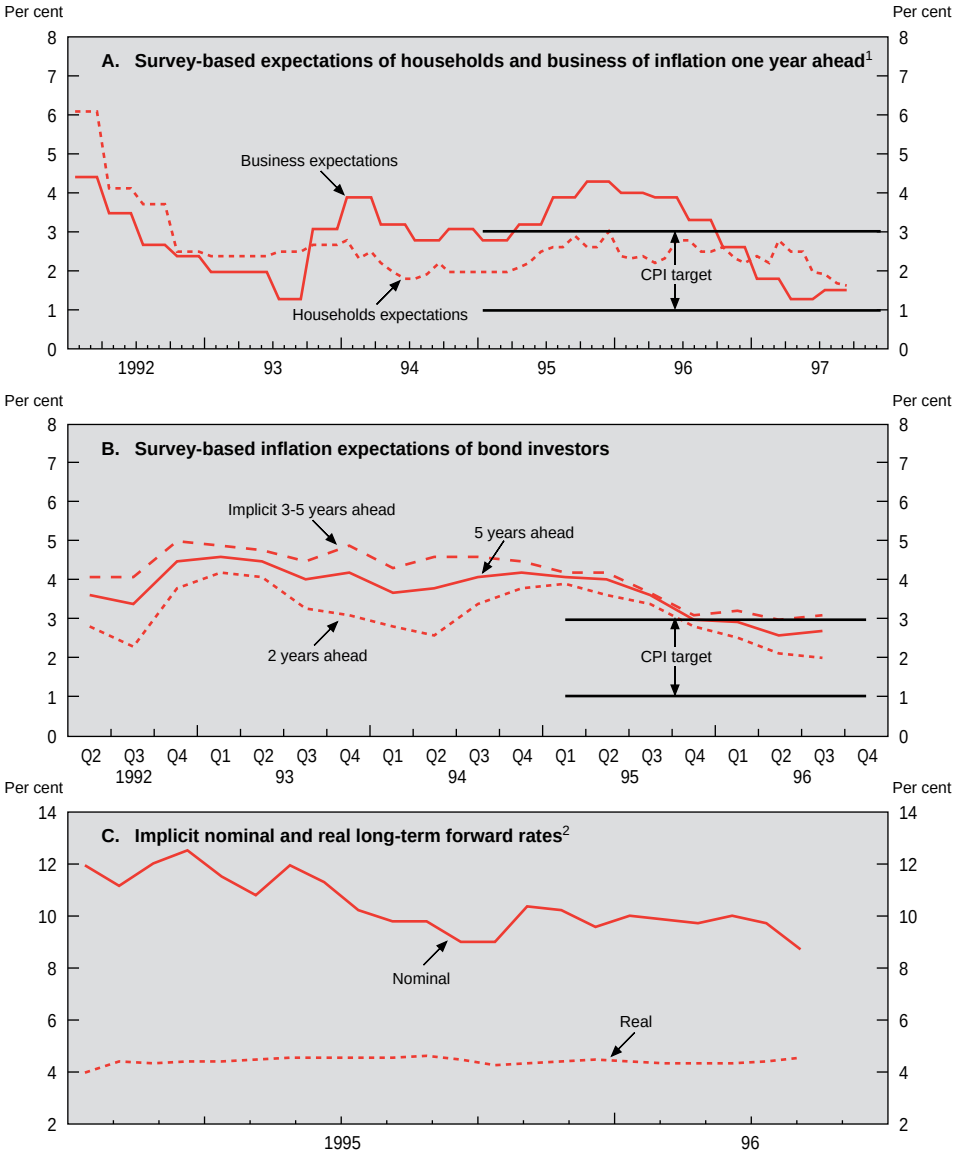
1. The yield on 28 June 1996.
2. Based on *OECD Economic Outlook* 59.
3. January 1992 = 100.

Source: OECD.

The steep slope of the yield curve, while obviously reflecting expectations of higher nominal interest rates in the future, would seem to contrast with inflation expectations among the labour market partners, industry, households and investors in the bond market which have all come down substantially over the last year and are now within the inflation range targeted by the central bank (Figure 17). The differential between implicit nominal and real forward interest rates[5] is now around 4¼ per cent, but still higher than observed inflation expectations warrant, reflecting *inter alia* a risk premium against European currencies which indicates uncertainty as to whether Sweden will participate in the first round of the European Economic and Monetary Union (EMU).[6] Economic policy in Sweden has over the past two decades been characterised by a search for an anchor which could stabilise wage and price developments, and inflation targeting has now been operative for more than three years. So far price developments have been in line with the target set. But this has been partly due to the substantial output gap that has persisted through the period and an active monetary policy has been necessary to secure compliance with the target, even in the face of a substantial fiscal consolidation. While price expectations have come down, it remains to be seen how wage drift and longer-run wage-bargaining behaviour will respond to the requirements set by a low inflation monetary policy framework.

On present economic trends with respect to budget deficits and price performance Sweden will probably be in a position to take part in the Economic and Monetary Union from its inception.[7] EMU participation raises a number of issues with respect to compatibility of economic structures. Participation in a currency union is assumed to require a similar economic structure between countries taking part, insofar as asymmetries in cyclical developments and different impacts of changes in short-term interest rates will render a common monetary policy less appropriate. Depending on the methodology applied, quite opposing assessments have been made of the similarity between Swedish economic structures and those of other European countries.[8] More generally, it may be argued that historical asymmetries are of little significance, partly because participation in a currency union may itself change economic structures.[9] Moreover, the EU internal market is still in the process of completion and may impart substantial changes in the industrial structure of member countries, most probably in the direction of greater specialisation as efficiency gains are assumed to accrue from

Figure 17. **INFLATION EXPECTATIONS**



1. Graphs have been shifted 12 months forward to coincide with the point in time to which expectations refer.
2. 10 years. Real rates are calculated from indexed bonds.

Source: Sveriges Riksbank.

realising economies of scale. With regard to the impact of monetary policy, Sweden already shares one important element of the interest rate transmission channel with continental European countries insofar as household and business loans in Sweden are to a large degree extended at fixed rates,[10] and the similarity may increase over time if capital income taxation continues to move in the same direction in EU countries.

Labour market institutions have been slow to adapt to a strictly non-inflationary environment. Within the present inflation targeting framework, higher wage growth than in other countries would need to be offset by an appreciation of the krona, thus squeezing profits and reducing employment. Over the economic cycle an irrevocably fixed exchange rate and an inflation target impose the same conditions on economic developments if the credibility of the inflation target is to be retained. In either case the key to a balanced development lies in a wage formation process that reflects underlying productivity and employment prospects, and an exchange rate which is consistent with long-term equilibrium considerations, reflecting both long-term trends in productivity growth, the terms of trade and the external debt position. A weak productivity performance and increased external indebtedness has shifted the real exchange rate for Sweden downwards over time but the real exchange rate of the krona is now estimated to be fairly close to its long-term equilibrium rate.[11]

Fiscal policy

Setting fiscal policy on a sustainable path

As described in the previous *OECD Economic Survey of Sweden*, the 1990-93 period saw a shift from budget surplus in 1990 to a deficit in 1993 of 12.3 per cent of GDP, partly reflecting a temporary increase in government outlays due to the banking crisis.[12] Reasserting control over public finances has been the main priority of economic policy over the past three years (Table 8). The improvement in public finances in 1994 was modest, even with the cessation of banking support, and it was only when medium-term objectives were underpinned by the implementation of the corrective measures, specified in detail, needed to address the imbalances that confidence in the fiscal consolidation was established. This was achieved by mid-1995, by which time the specified

Table 8. **Budget plans and outcomes**[1]

General government, per cent of GDP

	1993		1994		1995		1996		1997
	Budget	Outcome	Budget	Outcome	Budget	Outcome	Budget	Estimated outcome	Budget
Net lending	–12.0	–12.3	–11.1	–10.3	–10.2	–7.9	–4.6	–4.0	–2.6
Central government	–15.4	–15.2	–13.1	–11.7	–10.9	–8.8	–5.8	–4.7	–2.8
Social security	2.3	2.2	1.6	1.6	1.0	1.2	1.2	0.9	0.9
Municipalities	1.1	0.7	0.4	–0.2	–0.3	–0.2	0.1	–0.2	–0.7
Memorandum item:									
Net lending (SKr billion)	–172	–177	–165	–158	–161	–129	–79	–67	–47
Revenues and expenditures									
Direct taxes	21.7	21.2	21.7	22.1	22.7	23.3	23.4	24.3	24.1
Indirect taxes	14.6	14.9	13.7	14.3	13.2	12.8	13.9	14.6	14.0
Social security contributions	14.4	14.0	13.7	13.3	14.4	13.9	14.6	14.7	14.5
Transfers	34.6	37.4	33.5	33.0	32.6	31.9	29.7	31.1	29.4
Consumption and investment	30.8	29.2	29.9	30.2	30.2	28.8	28.8	28.7	28.3
Other items, net	2.7	4.2	3.2	3.2	2.3	2.8	2.0	2.2	2.5

1. Budget plans are based on assessments on a national accounts basis in November/December ahead of the calendar year. For estimated outcome 1996 and 1997 according to the Finance Bill of September 1996.

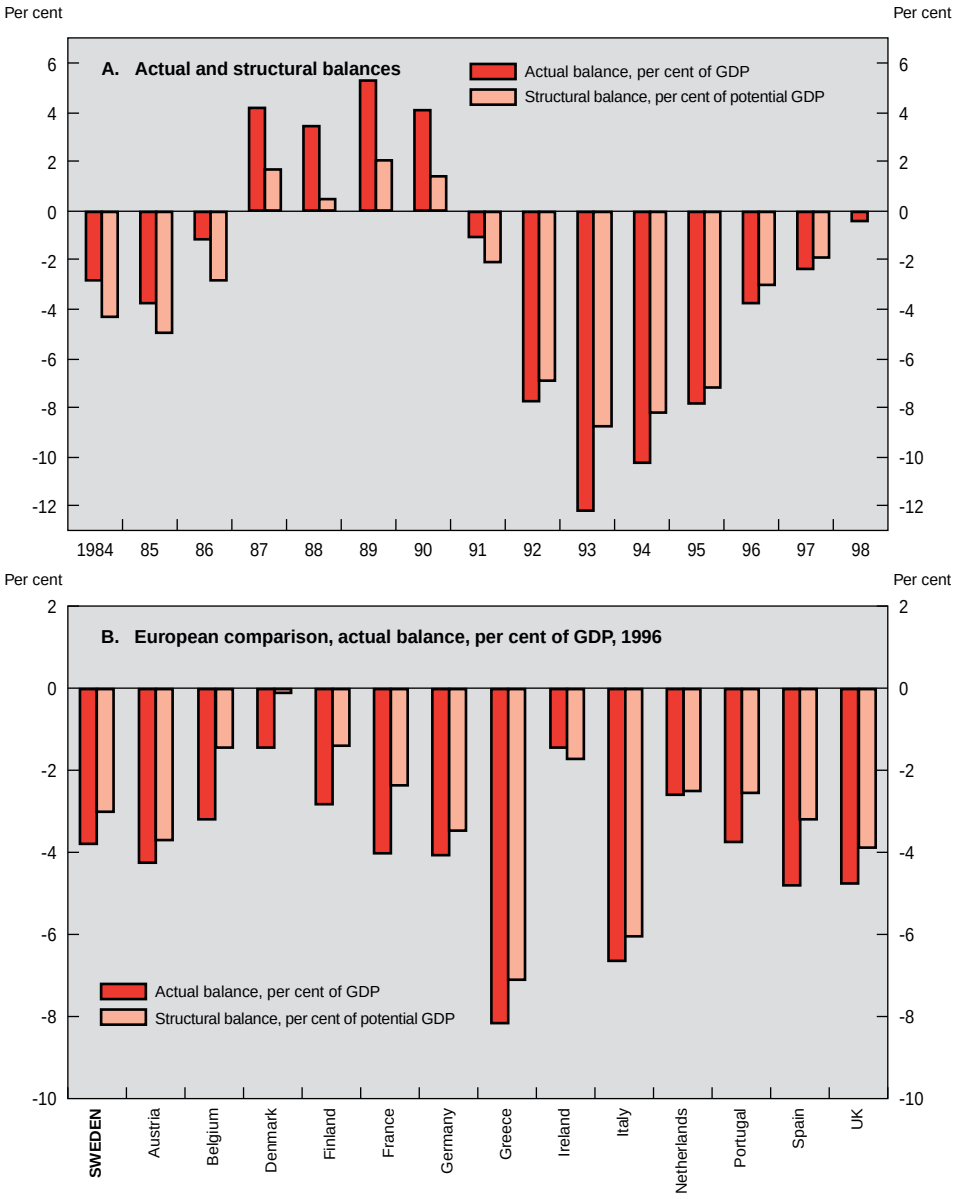
Source: Ministry of Finance, *The Swedish Economy* (various issues).

measures amounted to savings of 7 per cent of GDP, bringing the projected deficit for 1998 to below 1 per cent of GDP. The first Swedish convergence programme presented to the European Union in June 1995 stated the overall objective as eliminating the general government deficit by 1998, and on the official growth assumptions adopted (somewhat above the long-term growth rate of the economy) the measures adopted already came close to fulfilling this strategic aim of fiscal policy.

In order to assess compliance with objectives set, the government instigated at the same time half-yearly reviews of budget prospects, committing itself to additional measures to offset any shortfall. As growth prospects appeared to improve in 1995, forecasts for the budget balance correspondingly improved and no further discretionary measures were proposed upon the first follow-up of the convergence programme in October 1995. As the economy slowed in late 1995 and early 1996, budget projections presented in the spring of 1996 indicated a prospective failure to achieve the stated objectives. The 1994/95 fiscal consolidation programme of SKr 118 billion was thus supplemented by a permanent reinforcement of SKr 8 billion, with temporary reinforcements for 1997 and 1998 of another SKr 2 and 4 billion respectively (see Box 2). These were designed to bring the budget deficit down from a baseline of 5.1 per cent of GDP in 1996 to 2.6 per cent in 1997 and to zero by 1998. The supplementary package was fully specified and quickly adopted by the Parliament.[13] Since tax receipts turned out more favourably than projected in 1996, *inter alia* due to higher revenues from corporate taxes, it would appear that the deficit has fallen below 4 per cent (Figure 18).

By the end of 1994 the liabilities of the public sector exceeded assets by 22 per cent of GDP, and the consolidated gross debt of the public sector amounted to 79 per cent of GDP, having risen at a rate of some 9 percentage points of GDP a year in the 1991-94 period. Adopting the objective of eliminating the budget deficit by 1998 implied that the ratio of net government debt to GDP would peak in 1997, while the convergence programme of June 1995 added the objective of stabilising gross debt relative to GDP in 1996. This requires a primary budget surplus of 1½ per cent of GDP in 1998 and as a result of the overall budget developments described above, the improvement in the primary balance has proceeded to a point where net government debt should start to fall in

Figure 18. **GENERAL GOVERNMENT BUDGET BALANCES**



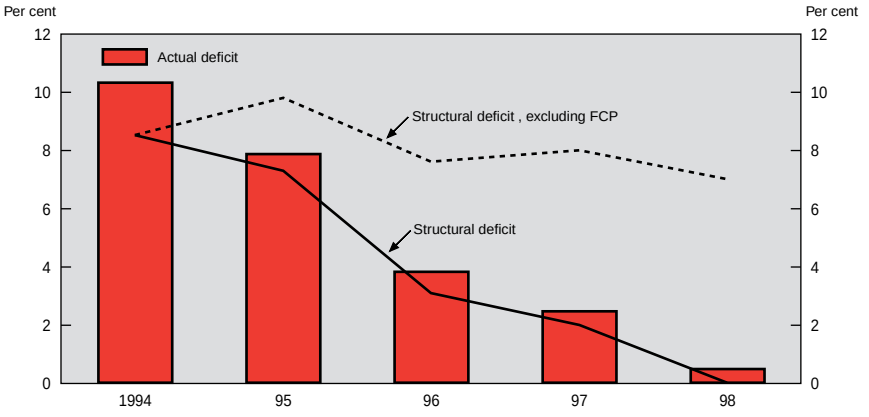
Source: OECD.

Box 2. **The fiscal consolidation programme: its profile, components and implementation**

Compared with a baseline of unchanged provisions and parameters in the tax and transfer system, the Swedish fiscal consolidation programme is calculated to result in savings to the central government budget of SKr 118 billion, later extended to SKr 126 billion. SKr 20 billion incorporated in the programme will not show up in an improved balance but will finance the Swedish EU contribution. The actual improvement in the general budget balance will be a result not only of the implementation of the programme but also of underlying cyclical developments, as recent growth rates have been above the historical trend of the economy. Local government finances may deteriorate in the short term, but are then set to improve as municipalities and counties are legally obliged to balance their books by 2000.

The consolidation programme will be fully effective by 1998, but close to half of the savings were effected by 1995 as revenue increases, in particular, were front-loaded (see item II below and Figure 19). The majority of measures were fully detailed at the outset, and the programme is now fully adopted by Parliament and largely implemented (item III below). The major steps in the policy process were as follows:

Figure 19. **THE IMPACT OF THE FISCAL CONSOLIDATION PROGRAMME (FCP)**
Per cent of GDP



Source: OECD; Ministry of Finance.

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I. The political process

Decisions taken prior to the autumn of 1994:	SKr 18.3 billion
November 1994 proposals:	SKr 56.4 billion
January 1995 proposals:	SKr 19.3 billion
Autumn 1994 – April 1995: Financing the EU contribution	SKr 20.0 billion
April 1995 proposals:	SKr 3.6 billion
Total:	SKr 117.7 billion
April 1996: Additional measures, permanent effect	SKr 7.8 billion
Total:	SKr 125.5 billion

II. Profile of savings effects (per cent of GDP)

	Annual impact	*Cumulative*	*Per cent of total*
1995:	3½ per cent	3½ per cent	44
1996:	2 per cent	5½ per cent	69
1997:	1½ per cent	7 per cent	86
1998:	1 per cent	8 per cent	100

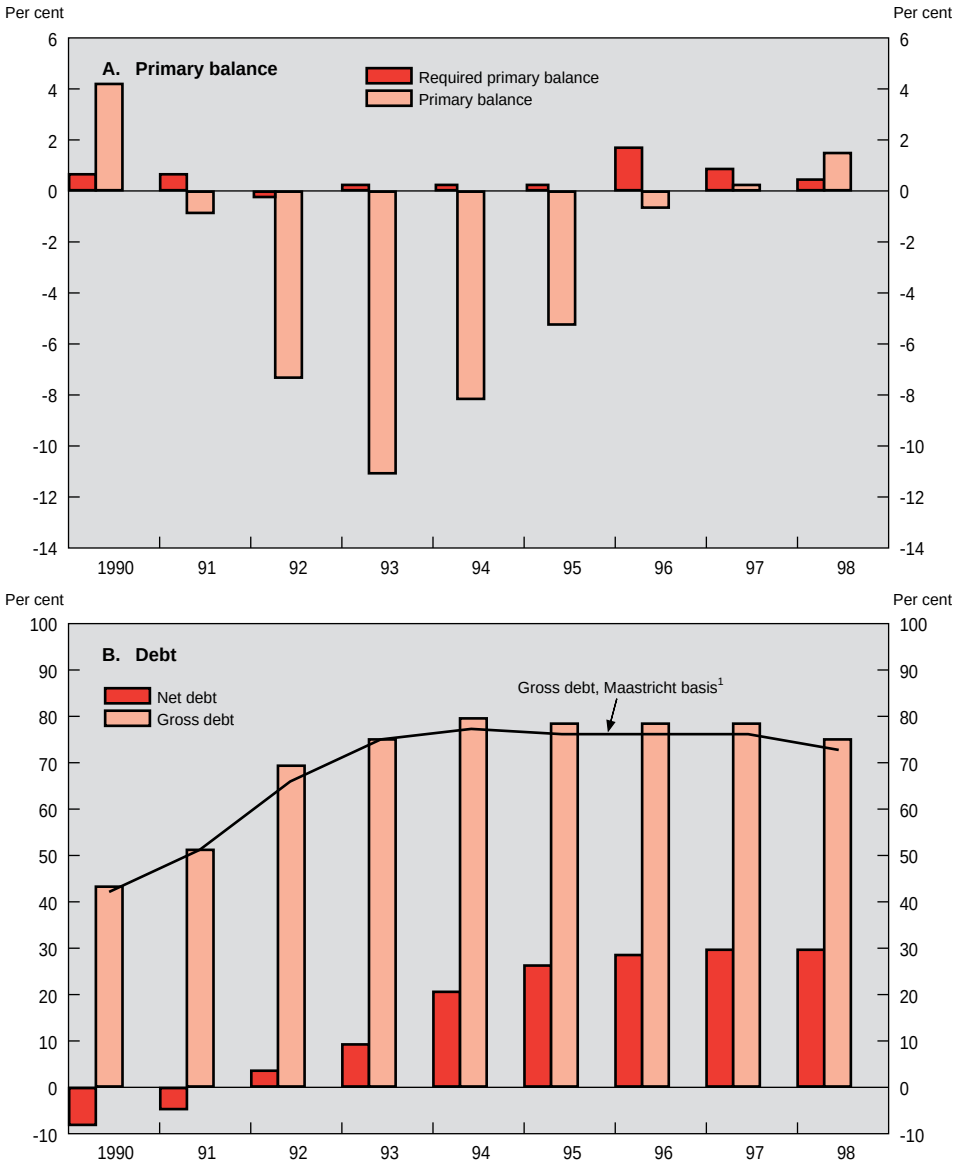
III. Components and state of implementation as of September 1996 (SKr billion)

	Revenue reinforcements	*Expenditure cuts*	*Total*
Original programme 1994-95			
Parliament decisions	47.9	54.6	102.6
Government proposals	9.4	5.1	14.4
To be formulated in legislation	0.0	0.7	0.7
Additional proposals 1996			
Parliament decisions	0.6		0.6
Government proposals	1.4	5.6	7.0
To be formulated in legislation		0.2	0.2
Total	59.3	66.2	125.5

line with earlier plans (Figure 20, Panel A). Assisted by a substantial privatisation programme amounting to SKr 50 billion over the period 1995-98, the strong appreciation of the krona, a reduction in deposits with the Central Bank and a simplification of borrowing arrangements with the National Debt Office, consolidated gross debt is now estimated to have peaked in 1994 at 79½ per cent of GDP (Figure 20, Panel B). Insofar as government assets – if needed – can be sold to redeem debt, gross government debt should continue to fall. The improvement

Figure 20. **GENERAL GOVERNMENT DEBT AND THE PRIMARY BALANCE**

Per cent of GDP



1. Data are provided by the Commission Services of the European Union for 1990 to 1994.

Source: OECD.

made so far in public finances in Sweden has made a controlled development of deficits and net debt, even in fairly adverse circumstances, the most likely outcome. High tax levels and extensive transfer schemes mean, however, that the budget remains vulnerable to the economic cycle, although recent reforms should have acted to reduce its sensitivity in this respect (see Box 3).

The combination of a strong political commitment to fiscal consolidation embodied in the biannual follow-up procedure, and better budgetary control (see below) should help to ensure that overall objectives are strictly adhered to and the consolidation measures already in place would lead to substantial budget surpluses and debt reductions over the five-year horizon up to 2000 (Table 9, Official projection). Improvements from 1996 on will be made more slowly as tax receipts which have increased markedly relative to GDP in 1996, reflecting *inter alia* the strong cyclicality of corporate taxes, may weaken somewhat as cyclical factors peter out. Moreover, official projections as they stand rely on economic growth ¼ to ½ percentage points a year above the long-term growth rate of the economy. Structural changes, particularly in wage formation, may not

Box 3. **The cyclical sensitivity of public finances in Sweden**

The impact of a shock to economic activity on public finances depends most importantly on *i)* the overall tax share of the economy, *ii)* the elasticity of taxes with respect to output, reflecting the structure of the tax system, *iii)* the impact on unemployment of changes in output (the reciprocal of the Okun coefficient) and the coverage and generosity of unemployment-related transfer schemes. Relative to other OECD countries Sweden has a high tax share and, on evidence up to the early 1990s, higher than average tax elasticities both for the personal income tax and social security contributions. A low impact on unemployment of shocks to GDP is more than offset by a generous unemployment benefit system and a relatively high share of such expenditures compared to other countries, but for all countries changes in revenue following from a change in GDP far outweigh changes in expenditures. Altogether, it would appear that the size of the automatic stabilisers in Sweden exceeds that of other countries (Figure 21), even those at the same level of taxation, although changes in the Swedish tax and transfer system undertaken over the past years should have acted to modify this conclusion somewhat.

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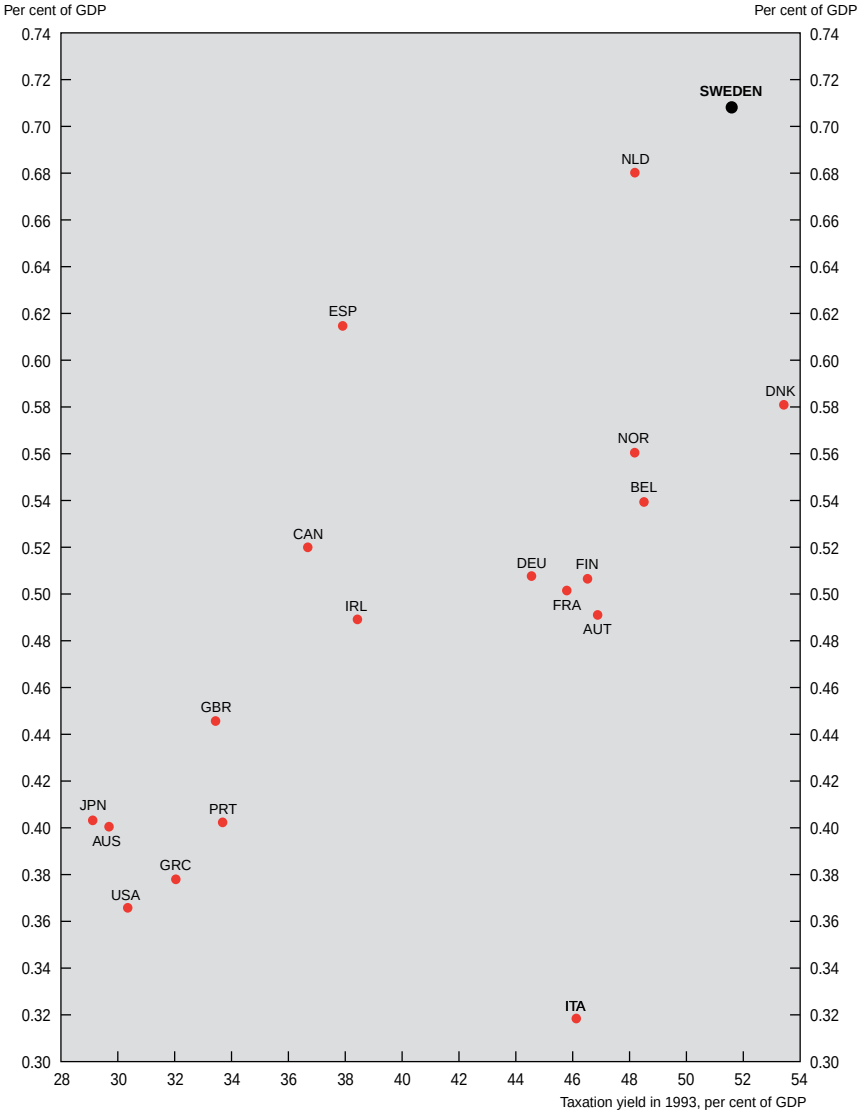
Figure 21. **AUTOMATIC STABILISERS AND GOVERNMENT BORROWING**[1]



1. The figure shows the effects on government borrowing, as per cent of GDP, of a 1 per cent change in GDP.
Source: Hœller *et al.* (1996).

Table 9. **Projections of public finances**

Per cent of GDP

	1995	1996	1997	1998	1999	2000	Average GDP growth 1996-2000
Official projection							
Baseline							2.2
Taxes	50.4	53.7	52.6	53.1	52.9	53.3	
Transfers	32.2	31.0	29.4	27.9	26.9	26.3	
Consumption, investment	28.9	28.7	28.3	27.6	27.0	26.7	
Net interest income	–1.4	–2.2	–1.6	–1.6	–2.1	–1.9	
Other income	4.0	4.3	4.1	4.0	3.9	3.9	
Financial balance	–8.1	–4.0	–2.6	0.0	0.9	2.3	
Net debt	27.8	30.5	31.7	30.6	28.5	25.5	
OECD illustrations							
Growth equal to long-term trend[1]							1.7
Taxes	50.4	53.7	52.0	52.3	52.2	52.4	
Transfers	32.2	31.0	29.8	28.9	28.3	27.8	
Consumption, investment	28.9	28.7	29.0	28.6	28.6	28.9	
Net interest income	–1.4	–2.2	–1.6	–1.8	–2.5	–2.6	
Other income	4.0	4.3	4.1	4.0	4.0	4.0	
Financial balance	–8.1	–4.0	–3.8	–2.4	–2.3	–1.9	
Net debt	27.8	30.5	33.1	34.5	35.6	36.5	
Higher interest rates[2]							1.7
Financial balance	–8.1	–4.0	–3.8	–2.5	–2.4	–2.1	

1. Economic growth is assumed to follow the long-term trend of the economy. Weaker private demand than in the official projections is offset by higher public demand, while higher unemployment necessitates a higher transfer level. Taxes are assumed to stabilise at a somewhat lower level relative to GDP than in official projections, reflecting an assumed strong cyclical impact in 1996.
2. Long-term interest rates are assumed to remain around 1 percentage point higher than in the other two scenarios.

Source: Ministry of Finance (1996*b*); OECD.

have progressed far enough to deliver such growth rates, while under the assumption of broadly neutral economic policies official projections which rely on private consumption growth being supported by a fall in the households' savings rate may also prove optimistic. More sluggish growth, lowering tax revenues and increasing transfers, and a traditional recourse to fiscal stimuli to offset weak employment creation in the private sector would slow further consolidation substantially and wider interest rate differentials could again reinforce imbalances (Table 9, OECD illustrations). Scenarios embodying such slippages carry a low probability on present performance, but further efforts may be required to secure compliance with objectives set.

Reforming the budget process

An integral part of fiscal consolidation has been the reform of the budget process. According to an evaluation undertaken in the early 1990s,[14] the Swedish budget process performed relatively badly by European comparison with respect to expenditure control. This lack of control reflected a weak role of the Ministry of Finance in relation to other ministries in the preparation of the budget; a fragmented budget process within the parliament; an absence of transparency and inadequate information content of the budget, and weak central control over the implementation of the adopted budget.[15] The result was heavy cost overruns that accumulated through the budget year, primarily on transfer programmes (Table 8). From this followed an upward tendency in tax levels as deficits had to be contained. Moreover, the underlying political priorities tended to disappear in such a process.

A major step to improving the Swedish budget process has been the lengthening of the parliamentary period from three to four years. This should reduce the impact of the electoral cycle on budgets and strengthen accountability by giving governments time to impose their budgetary priorities. The lack of control over expenditures can be traced back to the fact that 70 per cent of these were governed by statutory rules, with no obligation to find matching cuts if estimates were exceeded, or to implement legislative changes in order to bring expenditures in such programmes back to a baseline. This problem is now handled by imposing three-year ceilings for each of the 27 major expenditure areas (including social security transfers but excluding interest payments). Cost overruns for one programme have to be covered either by drawing down other programmes in the same area or by matching cost overruns one year by corresponding savings over the following two years. These expenditure ceilings, set in nominal terms, are the cornerstone of the new budget process, the total expenditure ceiling being derived *ex ante* from overall budgetary objectives (see above) and not from component commitments embodied in the programme set-ups. The ceilings for the period 1997-99 were decided upon by the parliament in the spring of 1996, with the ensuing detailed allocations to be presented and decided in the autumn. Forthcoming spring budget rounds should then, in principle, consist primarily of decisions as to the expenditure ceiling for the new year that is added to the three-year horizon. However, this part of process has not yet been tried out and is in the end dependent on a continued commitment to overall macroeconomic balance.

Table 10. **Public sector revenues and expenditures in selected OECD countries**[1]

Per cent of GDP at market prices

	Sweden	Denmark	Finland	France	Germany	Netherlands	Norway	United Kingdom	United States
Revenue									
Taxes on income, profits and capital gains	**21.6**	31.3	18.4	7.8	11.8	11.6	14.6	13.1	12.1
Personal[2]	**18.5**	28.0	16.5	6.1	10.7	8.3	10.7	9.7	9.7
Corporate[2]	**3.0**	2.1	1.9	1.7	1.1	3.3	3.8	3.4	2.4
Social security contributions	**14.4**	1.6	12.4	19.3	15.4	18.8	9.8	6.3	6.9
Taxes on property, etc.	**2.5**	2.1	1.2	5.3	1.1	2.0	1.2	3.8	3.3
Taxes on goods and services	**11.7**	16.7	13.8	12.2	10.9	12.0	16.0	12.2	4.9
Total taxes	**50.2**	51.7	45.8	44.5	39.1	44.4	41.5	35.3	27.2
Other revenue	**7.9**	8.9	6.3	4.3	6.8	3.2	9.0	2.3	3.5
Total revenue	**58.2**	60.6	52.1	48.8	45.9	47.6	50.5	37.6	30.7
Disbursements									
Final consumption expenditure	**25.9**	25.0	21.5	19.6	19.5	14.0	20.6	21.3	15.9
Interests and dividends	**7.2**	7.3	5.4	3.8	3.8	5.8	2.9	3.7	2.0
Subsidies	**5.2**	3.9	2.8	1.7	2.1	1.8	3.8	1.0	0.4
Other current transfers	**26.8**	24.7	26.1	25.5	21.3	27.1	18.4	14.9	13.7
Total current disbursements	**65.1**	60.8	55.8	50.5	46.7	48.8	45.6	41.0	32.0
Capital outlays	**1.2**	1.6	1.8	3.3	2.8	2.1	1.8	2.4	1.0
Memorandum items:									
Gross social expenditures[3]	**31.8**	25.4	28.5	21.5	21.8	23.4	22.5	17.7	9.8
Gross to net social expenditure adjustment[4]	**–8.3**	–7.2	n.a.	n.a.	–3.7	–8.1	n.a.	–1.9	1.0

1. 1995, except United States 1994.
2. Excluding taxes not entirely allocable to either sector.
3. 1993 excluding health, comprising both transfers and services in kind.
4. 1993. Includes effects of differences in mandatory social programmes, tax treatment of benefits, indirect taxes and tax expenditures for social purposes.

Source: OECD, *National Accounts and Revenue Statistics;* OECD, SOCX database; Adema *et al.* (1996).

The European experience indicates that the budget process has an important influence on public indebtedness,[16] and the budget reforms now implemented in Sweden would seem to establish a firmer control over central government finances than exists in most other countries. The total expenditure ceiling, which is scheduled to fall from 40.7 per cent of GDP in 1997 to 37.5 per cent in 1999, includes, on average, unallocated reserves amounting to 1 per cent of GDP, and the government may let expenditures fall below the adopted allocations without specific proposals to Parliament. A further fiscal tightening could thus be rapidly implemented if necessary. The framework of spending control has been strengthened by a comprehensive review of tax expenditures, and the assets and liabilities of government bodies are calculated annually to assess the net worth of the central government. The finances of local authorities, which are independent of the central government because of their constitutional right to set their own taxes, has to be influenced indirectly, by the allocation of block grants. So far, local government finances have remained close to balance since 1994, but a deterioration is expected in 1997. A binding requirement on the accounts of local government to balance, combined with requirements to fund their commitments in the area of occupational pensions, will apply from 2000. For the time being, local government finances are constrained by an arrangement which encourages municipalities and counties to maintain local tax levels unchanged insofar as block grants will be reduced to those municipalities which raise taxes, effectively withdrawing 50 per cent of any tax increase.

Restructuring public finances

At the same time as firmer control is being imposed on the budget process, renewed efforts need to be devoted to addressing the longer-term incentive effects of the present fiscal structure. In an international context Sweden stands out as having both high transfer and consumption levels relative to GDP, with the expenditure level for general government peaking at 70 per cent of GDP in 1993 before falling back to 67 per cent of GDP in 1995 (Table 10). The high level of taxes needed to finance this spending not only makes the budget balance highly sensitive to external and internal demand shocks, but also induces serious distortions in the economy and acts as a brake on job creation (see Chapter IV). A comparison of transfers and services-in-kind across countries indicates that social security expenditures in Sweden are around 5 percentage points higher relative to

GDP than in Denmark, a country with similar structural unemployment, and 10 percentage points higher than in continental countries like France and Germany. These differences may partly be attributed to different tax treatment of household transfers, with Sweden most consistently applying a gross transfer system which technically inflate both tax and transfer levels. Subject to this caveat, disaggregation (Figure 22) indicates that:

- Except for early retirement benefits, all types of income transfers in Sweden are above OECD averages. This applies in particular to disability and occupational injury benefits, but unemployment benefits and housing support are also above average. The overall transfer level to the working-age population is surpassed only by the Netherlands, Finland, and Denmark, and the share of the population having transfers as their main source of income has increased from 16 to 31 per cent over the period 1965-92.
- Services-in-kind in the social area, comprising active labour market programmes and services to families, elderly and disabled, are higher than in all other countries.

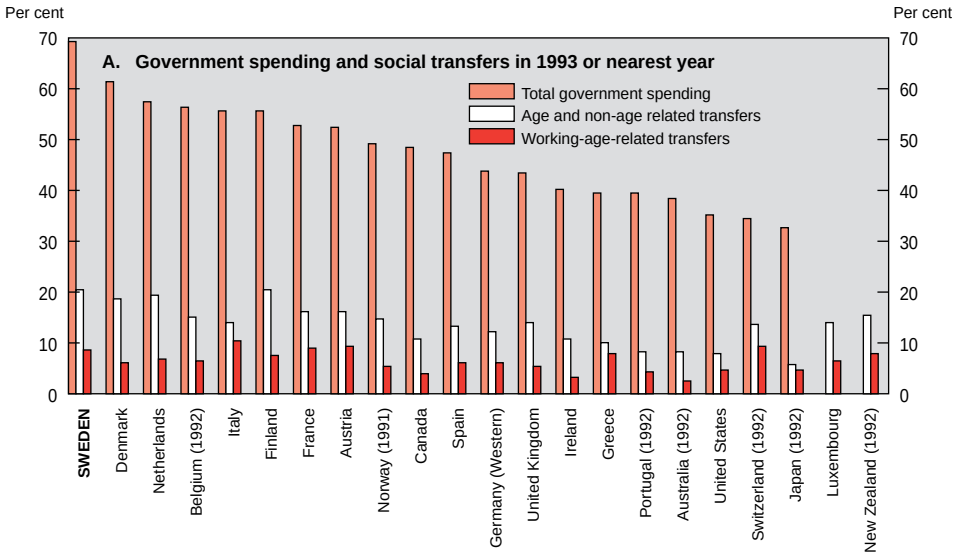
The interlinkage between these expenditure items should be noted. Generous transfer schemes combined with a compressed wage structure increases structural unemployment and necessitates in turn wide-ranging efforts to prevent human capital from decaying.

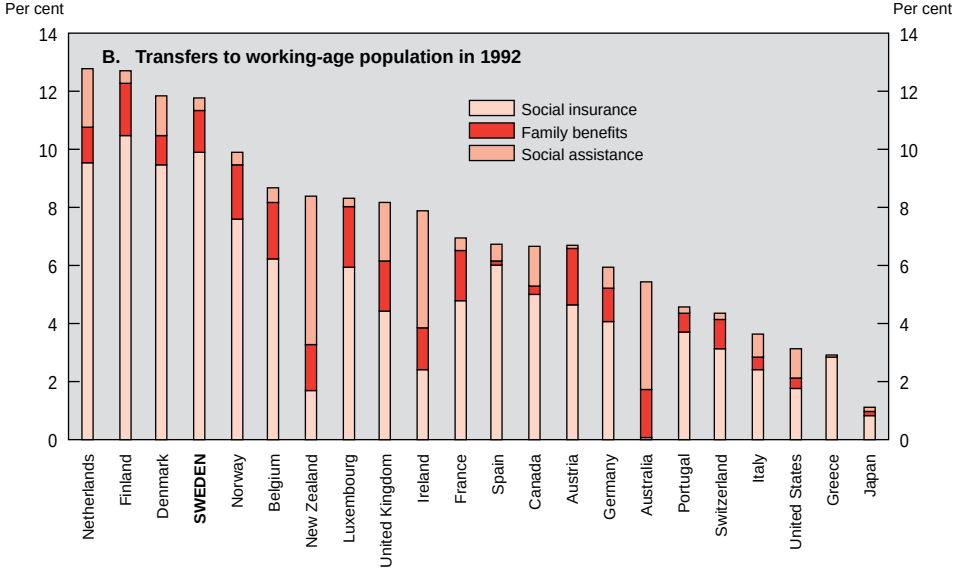
High public expenditure reflects an extensive government involvement in the economy, based on a far greater range of economic activities within the government sphere than is common in most other countries. While such involvement reflects both overall democratic choices and long-standing traditions, little consideration has been given to the efficiency outcomes or the ratio of benefits to cost. In this respect two considerations are of importance:

- first, introducing user fees to pay for services rendered by government bodies may serve not only to lower taxes but also to test the underlying demand for a given service, thus aligning production with the composition of demand;
- second, while government bodies may be the end-provider, competitive tenders or out-sourcing at all possible levels may secure the maximum of economy in the production of these services (see Chapter IV).

Figure 22. **THE STRUCTURE OF TRANSFERS**

Per cent of trend GDP





Source: OECD Social expenditure database.

It would appear that the scope for improving efficiency in the provision of public services is far greater than envisaged only a few years ago. Productivity in the public sector rebounded strongly in the early 1990s after declining over the previous two decades. Indeed, on available evidence the quality and output of important services have been maintained in the face of considerable shedding of manpower in the local government sector.

Reducing tax expenditures

The recent compilation of tax expenditures[17] provides a valuable supplement to the standard overview of transfers on the expenditure side. While tax expenditures are not yet treated on a par with transfers in the budget process, it is now possible to establish a comprehensive assessment of support provided through the fiscal budget. Tax expenditure estimates also indicate to what extent the tax system departs from a fully neutral system, insofar as collecting a given tax revenue while exempting parts of the tax base can be highly distorting. Overall tax expenditures for 1996 are calculated at 174 billion or 10 per cent of GDP. With respect to household transfers the compilation indicates that SKr 328 billion in transfers are supplemented by SKr 56 billion in tax expenditures. This indicates that, notwithstanding the gross transfer principle generally applied, a variety of support is still provided net of taxes.

Household tax expenditures can generally be thought of as neutral from a budget balance point of view, as their stated aim is to affect the disposable income of households in a prescribed way.[18] Even excluding these, however, tax expenditures still reduce the tax base by an amount close to SKr 115 billion. or 7 per cent of GDP,[19] indicating that a broadening of the tax base and sizeable reductions in marginal tax rates are still possible. Tax expenditures are particularly important in the case of the value added tax and excise taxes, with exemptions and tax rates below the standard rate reducing the corresponding tax bases by around 25 per cent. The income tax base is also reduced by 15 to 20 per cent, mainly as a result of accelerated depreciation schemes and tax reservation schemes for business and tax relief for pension savings. These compilations are particularly pertinent to decision-making in areas such as housing, where support is provided explicitly as transfers but also implicitly through a variety of decisions on the tax side, often with the aim of graduating support with the length of ownership (Table 11). It also appears that support to media such as newspapers

Table 11. **Tax expenditures**

SKr billion

A. By main categories, 1996

		Per cent of expenditure ceilings
Health and social insurance	72.7	21.1[1]
Labour market and education	9.8	13.1
Housing, regional development, culture	23.4	45.7
Business	35.2	86.0
General tax expenditures	32.8	. .
of which:		
To households	7.1	. .
Total	173.9	24.7

B. Housing support 1996[2]

	Tax expenditures	Tax sanctions	Transfers	Total	Support per SKr 1 000 of assessed value
Owner-occupied					
Recently built (no tax)	1.3		0.7	2.0	37
Medium age (half tax)	0.4	–0.2	0.5	0.8	45
Old building (full tax)	21.2	–15.3	3.0	9.0	10
Collective housing					
Recently built (no tax)	1.6	–0.8	6.8	7.5	159
Medium age (half tax)	0.8	–0.8	0.6	0.6	27
Old building (full tax)	3.2	–3.2	1.7	1.7	17
Multi-family dwellings					
Recently built (no tax)			14.0	14.0	151
Medium age (half tax)	–0.3		1.4	1.1	24
Old building (full tax)	–2.5		10.5	8.0	42
Total	25.7	–20.3	39.2	44.7	

1. Social insurance outside the central government budget is included in the ceiling.
2. Calculated net of taxes on transfers.

Source: Ministry of Finance (1996*a*).

and television is mainly achieved through favourable tax treatment. A reduction of business support – where tax expenditures play an important role – would serve to improve overall efficiency and government finances.

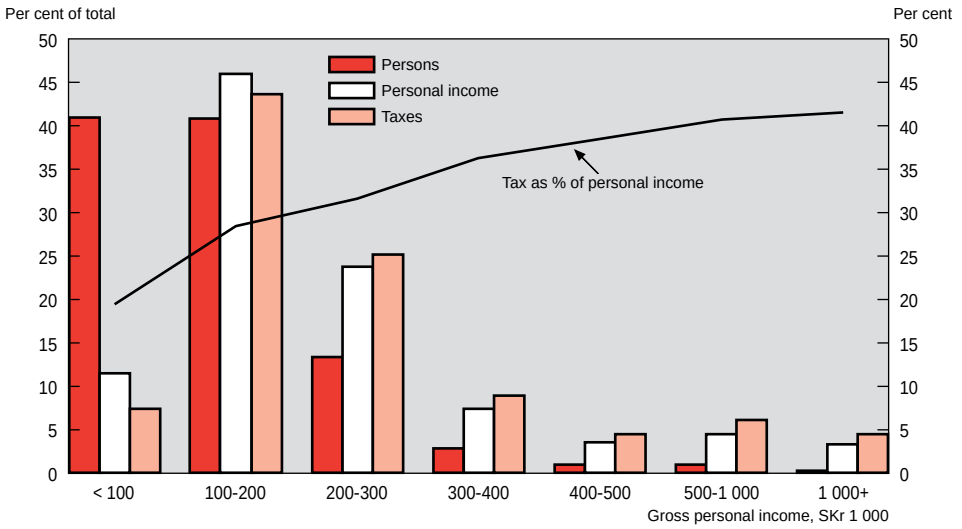
Achieving redistributive goals with fewer distortions

The high levels of government transfers and consumption have their origins in distribution concerns, which motivated the expansion of social insurance

systems during the 1960s and 1970s. Sweden opted at that stage for a comprehensive government-run old-age pension system whereas some other countries chose systems with larger private participation, partly through occupational pensions administered by the social partners and partly by individual pension savings to complement a basic publicly-provided pension. Unemployment, sickness and family insurance benefits were all integrated into the government-run system and financed out of general tax proceeds. The high tax level following from this was accompanied by a highly progressive system, although the effective redistribution of income was weakened by the fact that high-income earners could benefit from numerous deductions. The tax reform of 1990-91 greatly simplified the tax structure. Marginal income tax rates were brought down and the redistribution effected though the tax system was reduced in favour of redistribution effected on the expenditure side, via increased child benefits and housing allowances. The reform was neutral with respect to overall income equality.[20] As a consequence of the reform, the distribution of tax revenues by income group does not differ substantially from the underlying distribution of pre-tax incomes (Figure 23). Average taxes for the bulk of income earners are around 30 per cent, approaching 40 per cent only for very high incomes (1 per cent of income earners). But the combination of taxes on wage and entrepreneurial income, social security contributions and indirect taxes imposes a high tax wedge on labour.

The overall redistributive impact of the tax and transfer system depends not only the effects on the annual income distribution, but also on the distribution of life-time incomes. Recent estimates indicate that the life-time distribution of factor incomes is fairly even in Sweden, and far more even than the distribution of incomes for a given year. The level of transfers received over the life-cycle does not vary greatly between income groups, while taxes are broadly proportional across the income distribution,[21] so that the system acts to even the distribution. But about two-thirds of transfers reflect redistribution over the life-cycle of each individual whereas one-third reflects inter-personal redistribution. Even if the old-age pension is excluded, more than 50 per cent of unemployment, sickness and early retirement benefits pertain to redistribution over the life-cycle.[22] The extensive Swedish social safety net is thus supplying savings for contingencies, such as old-age pensions, which might, in principle, be left to individuals themselves. More reliance could, for example, be placed upon occupational pensions integrated into the collective bargaining system. Greater

Figure 23. **WHO PAYS THE TAXES?**



Source: SCB.

reliance on insurance premiums and on private savings to meet foreseeable events over the life-cycle would increase the robustness of government finance and social insurance systems.

Pension reform

Concern over the burden imposed by the pay-as-you-go system of financing social security prompted a broad cross-party agreement on a major reform of old-age pensions in 1994. The main features of the reform, which is to be phased in over a period of 20 to 25 years, include:

i) a combination of earnings-related and universal flat-rate (the guaranteed pension) components, the latter to be gradually clawed back as the earnings-related pension exceeds a certain threshold;

ii) the base for pension entitlements will be life-time income (with special provisions for time off work due to child-minding, military service and certain higher education) rather than the best 15 years as previously;

iii) pensions will be indexed to inflation, but with the indexing factor reduced to the extent that real growth is under a certain norm and increased to the extent that GDP growth exceeds that norm;

iv) the official retirement age will remain 65 years but people can retire from the age of 61 with an actuarial reduction in pension, or stay on until the age of 67 with an actuarial addition to pensions;

v) pensions will be linked to population-wide life expectancy at the age of retirement.

vi) the contribution rate is planned to be 18.5 per cent of which 2 percentage points will go into individual accounts, to be managed upon guidelines from the contributor.

A gradual implementation of the reform started with the introduction of an employee contribution in 1995. A number of issues are still being discussed, inter alia the assumed change-over from employers' to employees' contributions. The final proposal on the formulation and further implementation of the reform has not been put forward, more than two years after the initial agreement. In view of the challenges involved, an implementation in full of the planned reform is clearly warranted.

The blueprint for reform aimed at improving future public finances in several respects. The estimated benefit level will approach the present system only at a fairly high real income growth and there are mechanisms that adapt benefits to economic and demographic developments. The perceived tax wedge may be reduced as pension savings are partly individualised, and incentives to remain in the work force may improve. But the final income replacement ratio will still be higher than today since the present system is not fully mature. Demographic developments are also moving in an adverse direction, with the elderly dependency ratio increasing 15 percentage points over a 30-year period, and the system will come under substantial strain in the next century. Future pension expenditures in Sweden will thus be particularly high in an international context in the next three to four decades (Table 12). The pure ageing effect would, in the absence of offsetting measures, increase government debt by 115 per cent of GDP from 2000 to 2030.[23] A reversal of the fall in the effective retirement age would be highly effective in dampening the increase in social security expenditure, but would have to be supplemented by other cost-containing measures.

Table 12. **Net present value of social security payments**[1]

Percentage of 1994 GDP

	Baseline	Cost containment[2]	Later retirement[3]
United States	163	142	142
Japan	299	269	263
Germany	348	313	309
France	318	289	253
Italy	401	348	347
United Kingdom	142	147	122
Canada	204	174	166
Australia	97	78	79
Austria	298	268	246
Belgium	300	260	252
Denmark	235	211	179
Finland	384	341	304
Iceland	106	83	91
Ireland	107	106	90
Netherlands	214	181	179
New Zealand	213	178	171
Norway	230	189	201
Portugal	277	228	236
Spain	323	283	298
Sweden	**370**	**354**	**258**

1. The net present value refers to the value today of future payments. The calculation assumes a discount rate of 5 per cent per year over the period 1994-2070 and a productivity growth of 1.5 per cent.
2. Pension expenditures are kept unchanged relative to GDP from 2015.
3. Starting after 2005, the retirement age is raised by 0.5 year annually to reach 70 years.

Source: Roseveare *et al.* (1996).

A wider set of forces acting upon government finances is captured by the so-called generational accounts, which measure budget imbalances by the difference in implied net lifetime taxes (taxes paid minus transfers received) borne by present and future generations.[24] This gives an indication of the sustainability (and equity) of present policies. Such calculations are highly sensitive to the assumptions made about their starting point with respect to cyclical imbalances and corrections of fiscal imbalances over the medium term, which are critical assumptions in the Swedish case. If the actual 1995 budget is taken as a point of departure, it is evident that the net tax payments of present generations are not sufficient to cover public consumption, and with net government debt around 28 per cent of GDP, future generations will have to carry a higher tax burden than present generations: the imbalance, using standard assumptions about the

discount rate and economic growth, would on this basis be in the area of 150 to 250 per cent.[25] However, if total unemployment – open unemployment and ALMP participants – could be expected to fall to 5 per cent over the long run, the generational imbalance would be eliminated. In this respect, these calculations confirm the underlying message that a more flexible labour market is a key to a long-term improvement of the fiscal balance.

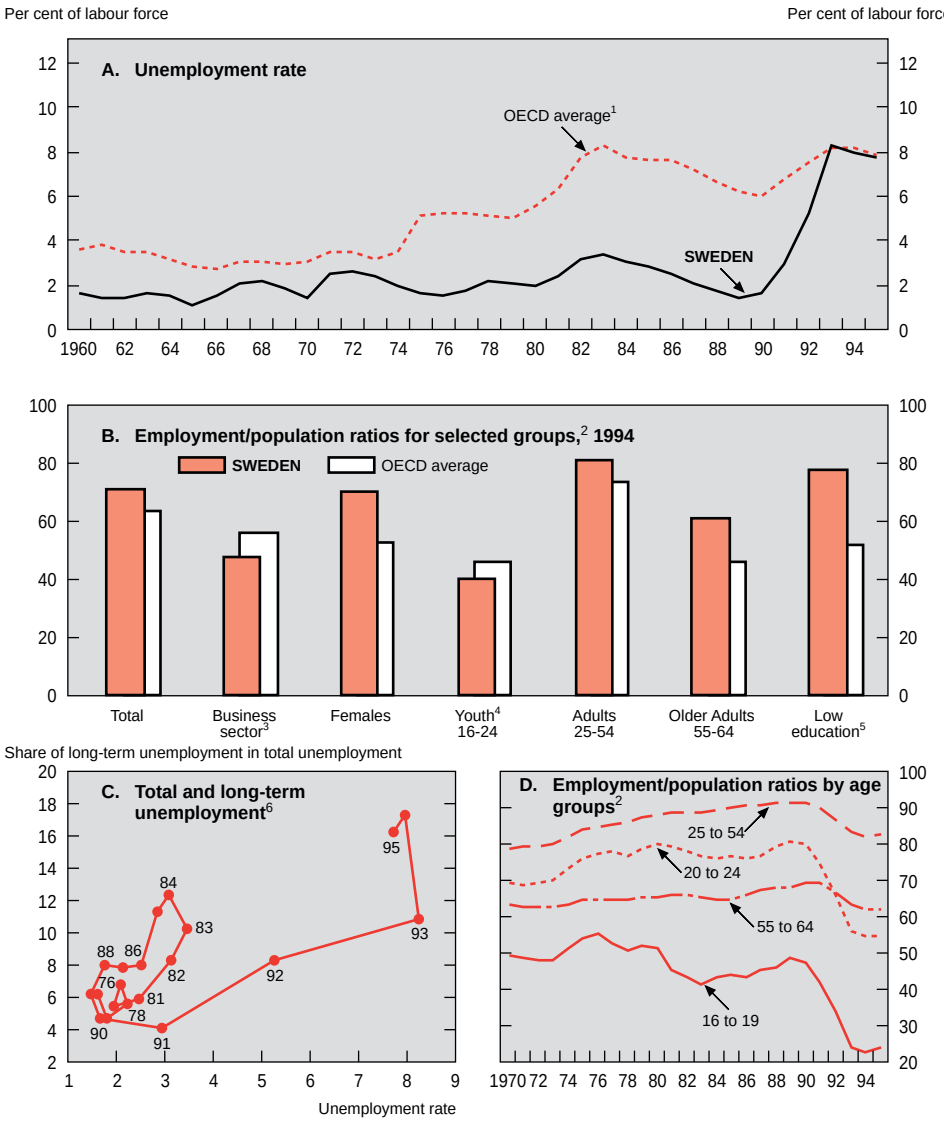
III. Implementing the OECD Jobs Strategy

Introduction

During the 1970s and 1980s the Swedish labour market was one of the best performing in the OECD area. The unemployment rate was very low compared with most other OECD countries and remained so even when external shocks hit the economy, while earnings were relatively evenly distributed. This successful performance was credited both to features of the labour market itself, particularly the macroeconomic flexibility associated with centralised wage bargaining, and to the combination of counter-cyclical fiscal policies, public-sector employment expansion and active labour-market measures. However, the Swedish ''model'' has come under intensified scrutiny during the 1990s as unemployment rapidly increased towards the OECD average during the recession at the start of the decade (Figure 24). With high tax rates and an excessive fiscal deficit signalling that the limits of public sector expansion have been reached, existing labour-market institutions and active labour market policies have so far proved ineffective in lowering unemployment. Indeed, insofar as the persistence of unemployment may be taken to reflect underlying impediments to labour market flexibility, there is a real danger that high unemployment will become entrenched, as has proved to be the case in other European countries. Avoiding this outcome calls for a concerted structural policy response.

The macroeconomic responses which have been tried to combat possible unemployment in the past seem to have contributed to current rigidities. The secular expansion of the public sector which emerged from past expansionary policies has caused a severe strain on public finances, leaving both tax rates and interest rates at levels inconsistent with the achievement of growth rates in the market-oriented sectors of the economy which could reverse Sweden's long-term decline. In this respect, a rebalancing of public and private sector resource claims

Figure 24. **LABOUR MARKET INDICATORS**



1. Commonly-used definition and for countries for which data are available.
2. Defined as the percentage of each population group that is employed.
3. Business sector employment divided by working-age population.
4. The minimum age for youth differs across countries (either 14, 15 or 16).
5. Completed less than upper secondary education.
6. Long-term unemployment is defined as individuals looking for work for one year or more.

Source: OECD.

is essential if sufficient employment opportunities are to be created and a broad-based welfare system is to be maintained. Fiscal retrenchment and structural adjustment are thus essential complements, a determined effort to remove obstacles to a well-functioning labour market being needed not only to allow further inroads into unemployment to be made, but to reach a surplus in government balances without heavy reliance on discretionary measures.

This chapter follows up the general structural policy recommendation of the OECD *Jobs Study* with specific recommendations for Sweden, drawing not only upon the general analysis in the Jobs Study, but also upon the special chapter on labour market institutions and policies in the *1995 OECD Economic Survey of Sweden*. The structure of the chapter is as follows. The first section contains an overview of the Swedish labour market and employment performance over the past two decades. The policy requirements to emerge from the Study are discussed in the second part, while a review of recent policy actions and an assessment of the scope for further action are given in the last section.

Labour market and employment performance

Employment was on a steady upward trend throughout the 1970s and 1980s, interrupted only by minor setbacks in conjunction with the international downturns in the mid-1970s and early 1980s. Annual employment growth averaged 0.8 per cent over the period 1973-1990, entailing at its peak an employment to working-age population ratio of over 80 per cent – the highest among OECD countries. Although this position could not be maintained given the adverse developments in the economy in the early 1990s, Sweden's employment-to-population ratio is still higher than in most other OECD countries (Figure 24). Nevertheless, since the early 1970s employment growth has reflected the continued expansion of the public sector, while employment in the business sector has stagnated over the period as a whole. Reaching a trough in 1984, business sector employment increased temporarily in the latter part of the 1980s, reflecting the boost given by a competitive currency and the credit-induced expansion of domestic demand during those years. But as the momentum associated with these factors petered out, to be replaced by a series of negative shocks to economic activity, employment in the private sector fell sharply from 1990 onwards, necessitating a correction in public employment as public-sector deficits reached crisis

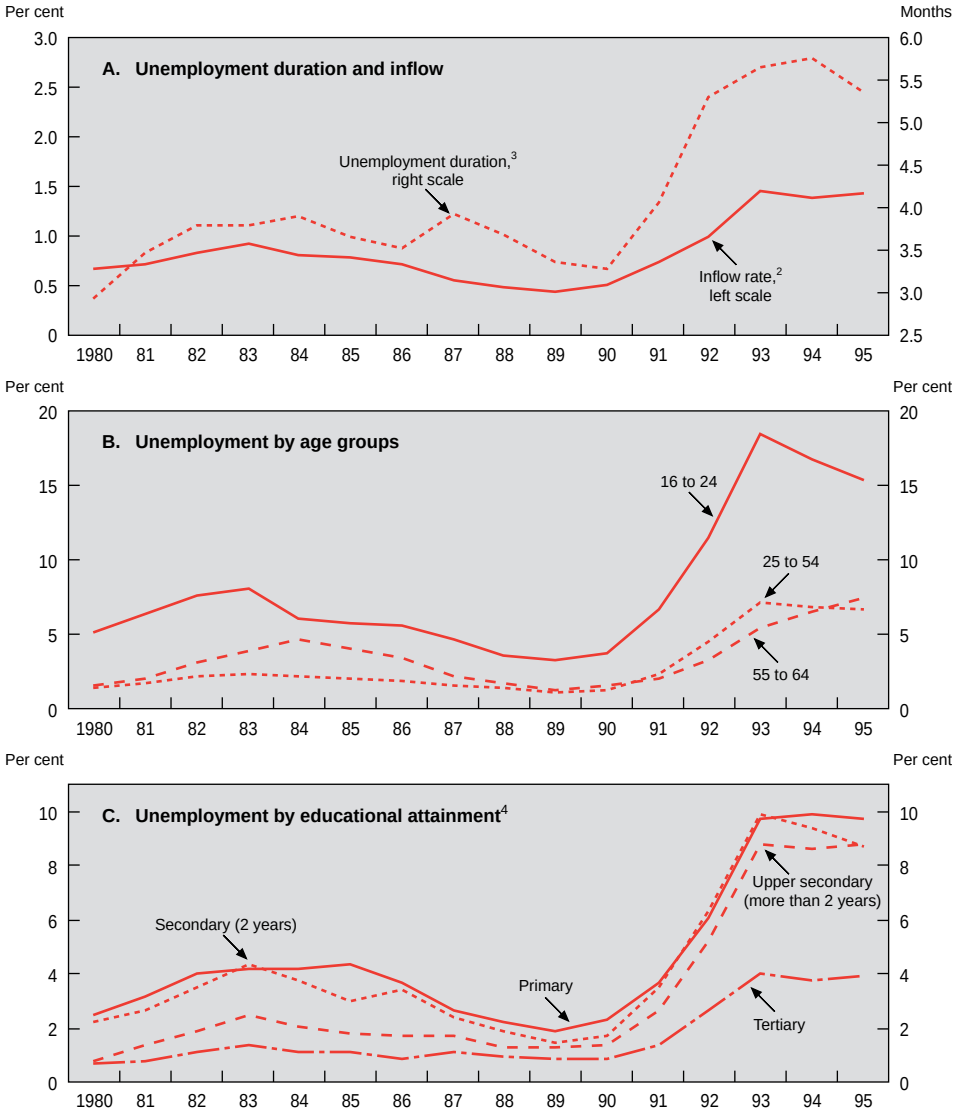
levels. Only in 1995 did private sector employment increase again – a short-lived expansion driven by the manufacturing sector, which benefited both from the strong depreciation of the Swedish currency and expanding international markets.

The concentration of employment growth in public services has had a counterpart in a continuous upward movement in the female participation rate, which increased from 63 per cent in 1973 to close to 80 per cent in 1990 and still remains above 75 per cent. Having peaked in 1975 at close to 90 per cent, the male participation rate has since fallen to 80 per cent. However, the downward trend was halted in the late 1980s and the labour market attachment of prime-age male workers (25 to 54 years of age) has remained strong. The fall in the male participation rate reflects, in particular, a withdrawal of older workers into early retirement and, more recently, a postponement of labour market entry by younger workers in favour of prolonged education. It should also be noted that from the early 1980s average working hours were on an upward trend, a strong decrease in average working hours having occurred during the 1970s associated with an increase in part-time employment, reductions in contractual weekly working hours, longer vacations and increased absence from work, as sickness and other welfare benefits became more generous.

While the longer-term tendency in Sweden has been for economic growth to have a high employment content, reflecting *inter alia* the increased weight of the public sector, the downturn in the 1990s has marked a trend towards more intensive use of labour and, through decreasing absenteeism, an upward shift in the growth of average working hours. The short run effect of the resulting decrease in employment growth has been to add to the pressure on active labour market policies (ALMPs). Having comprised on average 2 per cent of the labour force over the 1980s, the number covered by such measures amounted to 4½ per cent of the labour force over the years 1992-95, limiting the increase in ‘‘open’’ unemployment, which nevertheless rose from an average of 2½ per cent of the labour force in the 1980s to 7¼ per cent in the 1992-95 period.

ALMPs were very successful during the 1980s in keeping the average duration of unemployment spells, and hence long-term unemployment, low compared with other OECD countries. However, the strong increase in unemployment from 1990 onwards has reflected both an increased inflow into unemployment and a longer average duration of unemployment spells (Figure 25, Panel A). The share of long-term unemployed – those seeking a job for more than one

Figure 25. **UNEMPLOYMENT CHARACTERISTICS**[1]



1. Break in data in 1987.
2. Defined as the number of persons with unemployment duration less than 4 weeks as a percentage of the labour force.
3. Calculated as the unemployment rate divided by the inflow rate.
4. In per cent of educational group in the labour force.

Source: OECD; SCB.

year – in open unemployment has increased from around 5 per cent in the latter part of the 1980s to about 16 per cent in 1995. But since participation in ALMPs qualify for a new benefit period, long-term unemployment and the duration of unemployment spells are clearly understated.[26] Unemployment levels remained low in the 1980s for both youths and older workers, although labour market access for youths was relatively difficult even then. The 1990s have witnessed a rapid increase in unemployment rates for youths, even as their share in the overall labour force contracted sharply (Figure 25, Panel B). The upward trend in unemployed older workers has been cushioned by seniority rules.

The educational profiles of Swedish employment and unemployment differ in some respects from those in other OECD countries. Enrolment rates in education remain fairly high until 17/18 years of age, with 80 per cent of the age cohort completing upper secondary education; and Sweden has an internationally-high share of graduates with technical and vocational diplomas. Beyond this level, enrolment rates fall below those in neighbouring countries, particularly with regard to university education, but the proportion of mature students (aged above 30 years) in university education is higher in Sweden than in other countries. Indeed, insofar as the qualifications of the Swedish labour force compare well with those of other countries, this reflects the high share of older workers having achieved above-average qualifications. As in most other OECD countries, unemployment rates have shown significant differences according to educational attainment. Although such differentials are not, in general, pronounced for those with lower and secondary education, in some segments of the labour force workers with lower secondary education are up to twice as likely to be unemployed as those having completed upper secondary education. Moreover, only workers with tertiary education have been able to maintain overall low unemployment rates (Figure 25, Panel C).

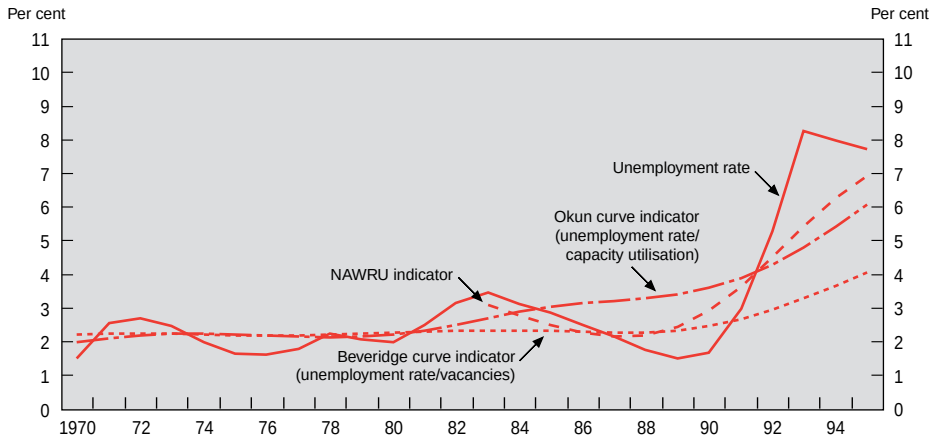
At the aggregate level, real wages in Sweden have traditionally responded relatively strongly to higher unemployment, thus facilitating rehiring of labour, at least up to the end of the 1980s.[27] However, the fall in real wages was most often effected via higher inflation associated with a depreciation of the Swedish currency and, in the disinflationary climate of the 1990s, real wages have been much more rigid. Indeed, following a slowdown in response to the strong increase in unemployment in the early 1990s, wage growth accelerated markedly in the wake

of the 1995 settlements in spite of continued high unemployment, as wage settlements in export industries, particularly in pulp and paper, were copied by other sectors.

The degree of mismatch in the labour market – as depicted by the relationship of vacancies to unemployment (the Beveridge curve) – was unchanged through the 1970s and 1980s, but the matching process between vacancies and job seekers may have begun to deteriorate in the 1990s.[28] In this connection, the short duration of unemployment in the past may be seen not simply as a result of a well-functioning market but also as an outcome of the rather strict practices of the public employment services in reinforcing job availability criteria, especially in confronting job seekers with job offers and placements in ALMP schemes. With higher overall unemployment, the low duration of unemployment spells has given way to longer spells as the strict practices applied by employment offices in the 1980s can no longer be consistently enforced. Persistent differences in unemployment rates by region and by educational attainment serve to reinforce the picture of underlying weakness in the functioning of the labour market.

At the aggregate level, OECD indicators point to a significant increase in structural labour market imbalances from the late 1980s to the mid 1990s and corroborate the growing evidence that unemployment in Sweden is now tending to exhibit strong persistence.[29] Structural unemployment – measured either as the open unemployment rate at which wage growth stabilises (NAWRU) or in terms of the relationship between open unemployment and capacity utilisation (Okun curve) – may well be in the area of 6 to 7 per cent (Figure 26). The principal implication of strong persistence effects in labour markets is that structural unemployment measured in this way will tend to follow movements in actual unemployment. This tendency will be particularly pronounced insofar as the influence of the unemployed on the wage bargaining process tends to diminish the longer they remain unemployed, either because of a reduction in their human capital or because they become ‘‘outsiders’’ to the wage-setting process. Such mechanisms may imply that wage growth can accelerate if unemployment declines quickly (and that unemployment therefore has fallen below the NAWRU even though it remains above some fundamental equilibrium level). The acceleration of wage growth in the 1995-96 round points to a possible malfunctioning of the labour market in this respect. But since the NAWRU can, in principle, decline in step with actual unemployment if the reduction towards an underlying

Figure 26. **STRUCTURAL ASPECTS OF THE LABOUR MARKET**



1. The trends of the three indicators are based on yearly estimates of unemployment rates which would have *a)* resulted in a normal vacancy rate, *b)* stabilised wage inflation or *c)* led to a normal rate of capacity utilisation. These estimates have subsequently been smoothed using a Hodrick-Prescott filter.

Source: OECD.

structural unemployment rate is gradual enough, significant inroads to unemployment may eventually be made, particularly if wage formation is adapted to a low inflation environment. Some positive evidence in this respect is provided by the fall in unemployment from 1994 to 1995, which was evenly spread across all duration categories of unemployed, indicating that the decay in human capital had not progressed too far. However, institutional features and weaknesses in policy implementation may cause imbalances to persist and these are the subject of the remainder of this chapter.

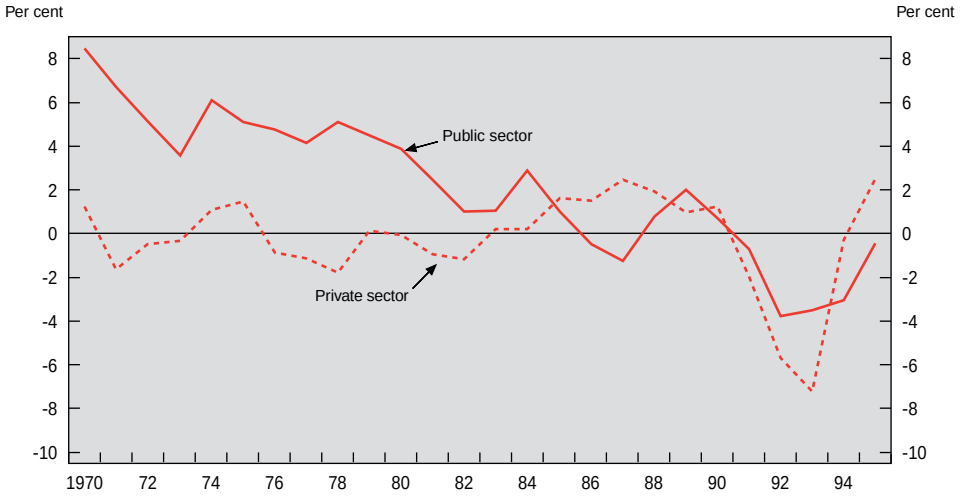
Policy requirements

Faced with the danger of high unemployment gradually being embedded in the Swedish economy and growth prospects remaining subdued, the traditional approach to labour market policies can no longer be considered sufficient to secure long-term employment goals. In particular, growth will need to depend on the private sector. A successful reorientation of labour market institutions and

policies is thus crucial and in this respect, two main concerns stand out. In the longer term, growth in employment and real wages needs to be predicated upon a high level of human capital formation, so that restoring the incentives to undertake such investment – both on an individual level and on an enterprise level – will be one of the foundations of labour market policy. This pillar would have to be supplemented by efforts to foster long-term structural change through innovation, technology diffusion and entrepreneurship. In the shorter run, labour market policies will need to ensure that overall wage growth is consistent with low inflation and overall productivity trends and facilitate the successful redeployment of labour from declining to expanding enterprises and the reintegration of the unemployed, in turn reducing the duration of unemployment spells and the deterioration of human capital. This requires well-functioning labour markets where wages and labour flows respond to skill differentials and relative shortages of workers without jeopardising moderate wage increases in the aggregate.

The long-term stagnation of employment in the private sector (Figure 27) reflects not only the strong claims on resources made by the public sector, but also a host of product- and labour market regulations impeding both competition

Figure 27. **EMPLOYMENT GROWTH BY SECTOR**



Source: OECD.

and the reallocation of labour; high non-wage labour costs and taxes, despite a tax reform which has removed the highest marginal income tax rates; a tax system which has locked in profits and discouraged the expansion of new businesses; an education system still catering to the demands of the government sector; wage differentials which have reduced the incentives to enter higher education; and high reservation wages, reflecting not only deeply-embedded equity concerns but also high replacements rates in social insurance and assistance schemes, in turn necessitating substantial efforts to enforce job availability criteria and a high level of public spending on ALMPs. In most of these areas the process of reform is already underway, but is as yet incomplete. In other areas, such as the need to increase relative wage flexibility, the government can wield only indirect influence on the social partners to allow reforms to take place. A synopsis of the recommendations made below can be found in Box 5.

Enhancing the ability to adjust and adapt

The wage formation system

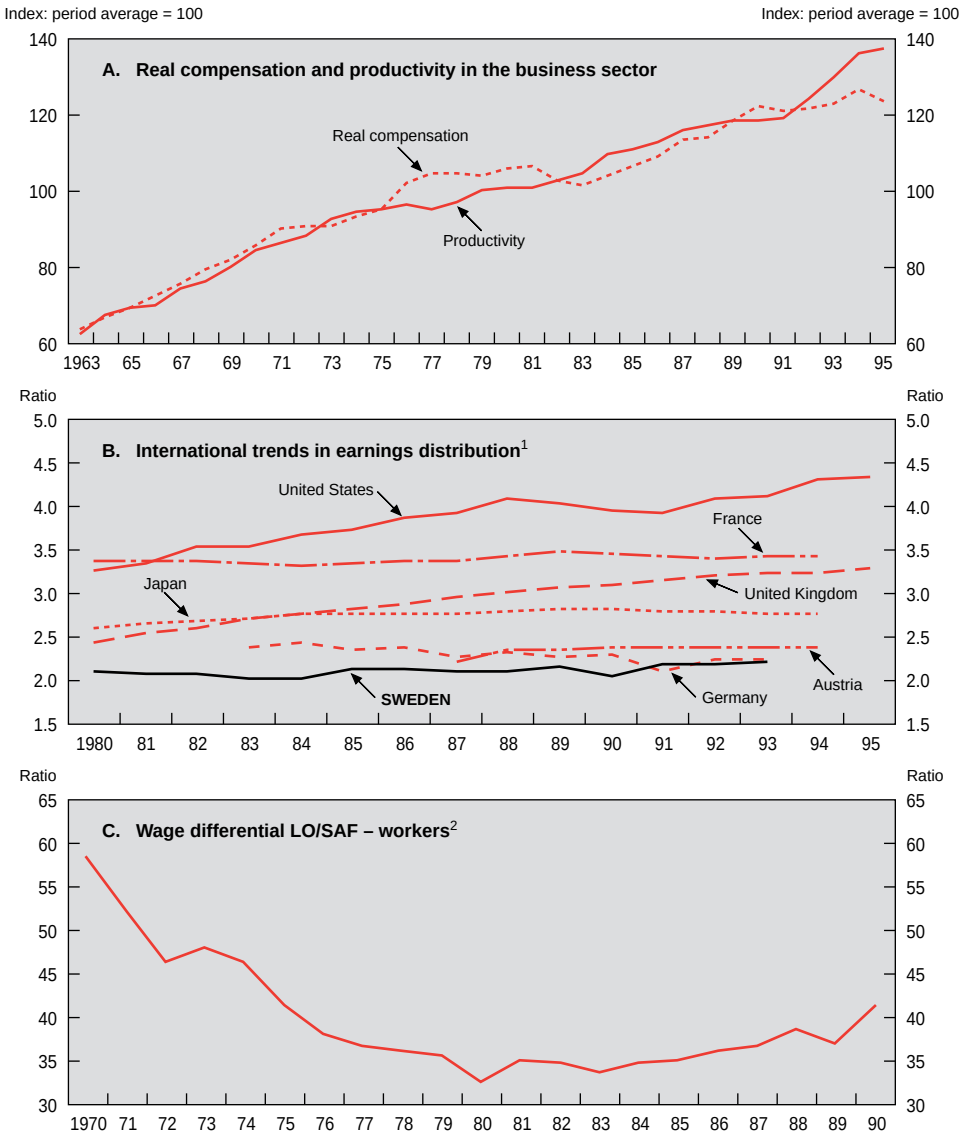
Trade union membership is high in Sweden, at around 83 per cent of the dependent workforce, reflecting *inter alia* the responsibility given to union-based insurance funds in the administration of unemployment insurance. The industrial unions affiliated to the Trade Union Congress (LO) are dominant on the blue-collar side, while two independent unions (TCO and SACO) are dominant for white-collar and professional workers. Of particular significance for long-term developments is the increasing influence of public sector unions, following the growth of the public sector. Together with their counterparts in the Swedish Employers' Confederation (SAF), the LO unions maintained a highly centralised wage bargaining process over the three decades up to the early 1980s. However, wages were still responsive to local conditions, with annual wage drift alone amounting to 3½ per cent, and in 1983 the employers and employees in the engineering trades broke away from the centralised arrangement. Although centralised agreements have occasionally been concluded in subsequent years and even though wage bargaining became temporarily subject to a strong degree of co-ordination in face of the serious macroeconomic imbalances in the early 1990s, the trend since 1983 has been fairly consistently towards bargaining at the industry and enterprise level.

Wage formation in Sweden can thus now be characterised as following an intermediate model, with bargaining at industrial level being dominant. This type of institutional configuration is usually associated with relatively unfavourable employment outcomes.[30] The high elasticity of wages with respect to unemployment which can be observed in the past can be traced in part to nominal wage stickiness in an inflationary environment. Overall, nominal wage growth in Sweden has generally outstripped that of trading partners, and several adjustments of the exchange rate were needed throughout the 1970s and 1980s to restore output profitability. Nevertheless, the rate of return on capital has been lower than in other countries, implying for the exposed sectors in particular a need to reduce costs, and thereby employment, in an effort to align profit levels with other countries (Figure 28, Panel A).

Within this framework and in the absence of any statutory minimum wage imposed by the government, the outcome in terms of the wage distribution is wholly the result of the bargaining process. At times, and for some bargaining areas, an explicit minimum wage applying without qualifications to all workers above the age of 18 has been an integral part of agreements, while in other cases income distribution concerns have been expressed in particular wage increments awarded to workers below a certain earnings threshold. Following a strong compression in wages over the 1960s and 1970s the overall distribution of wage rates remained fairly stable in the 1980s. The earnings distribution in Sweden has remained more compressed than in most other OECD countries, with wages in the ninth decile generally no higher than twice the wages in the first decile (Figure 28, Panel B). A particularly pronounced compression of differentials had taken place for blue-collar workers, where by 1983 wages in the ninetieth percentile were only 25 to 30 per cent above wages in the tenth (Figure 28, Panel C). The underlying philosophy of wage bargaining institutions up to the early 1980s argued that wage compression served to eliminate jobs in declining industries with low wages and to supply expanding and profitable industries with relatively cheap labour. Differentials in the LO/SAF-area have since widened and by 1993 were back to the level prevailing in the mid 1970s, reflecting the shift in the bargaining regime that took place over the 1980s. However, the legacy left behind by the traditional wage-compression model continues to negatively affect the Swedish labour market in several ways.

Firstly, the development of wage relativities is not conducive to an expansion of the market-based sectors of the economy. Wage developments through

Figure 28. **WAGE DEVELOPMENTS**



1. As measured by the ratio of the lower limit of earnings received by the top 10 per cent (9th decile) of all male workers relative to the upper limit of earnings received by the bottom 10 per cent (1st decile).
2. As measured by the ratio of wages in the 90th percentile to wages in the 10th percentile in the wage distribution.

Source: OECD (1996*a*); Edin and Topel (1995).

the 1970s and 1980s were characterised by a catch-up process whereby wages in the sheltered sectors gravitated towards manufacturing wages. This trend persisted in the first half of the 1990s in spite of a strong shift from domestic to external demand. Moreover, since the late 1980s wages in the government sector have started to increase relative to wages in manufacturing.[31] Furthermore, it would appear that the framework relating to industrial relations allows secondary action to a larger extent than in other countries, *inter alia* with respect to boycotts. Such action may under present circumstances prevent the expansion of small and medium-sized enterprises as unions try to impose wage rates not fully consistent with local employment possibilities. Also relating primarily to wage flexibility is the limitation of individual choice between self-employment and dependent employment insofar as this choice is not a personal one, but is obliged to follow objective criteria. Moreover, unions have the right to veto decisions to contract out work hitherto performed by an enterprise's own employees.

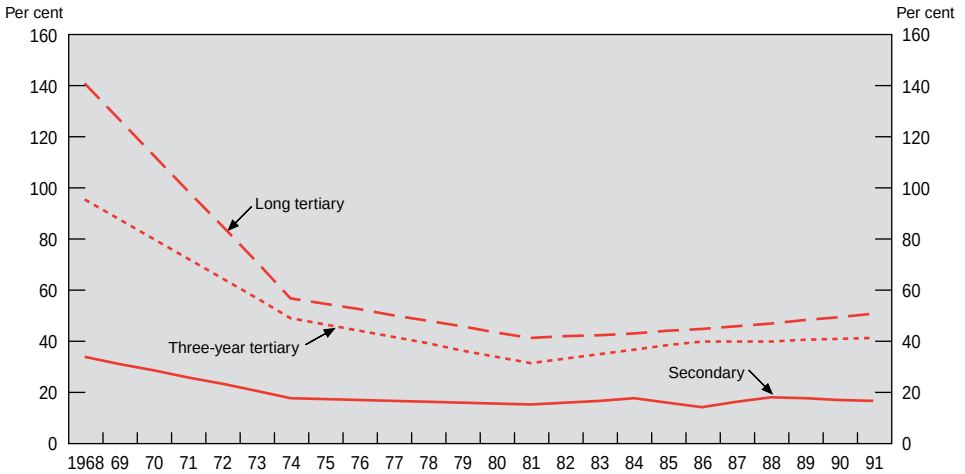
Secondly, the compression of wages imparts long-term costs to the extent that youths and low-skilled workers are put at a disadvantage by being priced out of ordinary jobs. Experience in other OECD countries indicates that long unemployment spells at the start of a working career can have lasting negative effects on earnings and job tenure. The rise in unemployment being relatively recent in Sweden, such effects are not yet ingrained in the Swedish labour market. But, as noted above, the unemployment rate for workers with basic and secondary education is twice the rate for workers with tertiary education, indicating that the unemployment risk is skewed towards low-skilled workers. Weak labour market prospects have led to an increase in enrolment in higher education over the 1990s and the acquisition of more qualifications may allow these cohorts to compete successfully once they enter the labour market. However, the size of these cohorts is such that a substantial adjustment of relative wages may be required if they are to find employment without displacing low-skilled workers. While historical experience over the 1970s and 1980s indicates that youth wages adjust quite strongly to changes in demand and supply conditions, distributional concerns have nevertheless prevented a full adjustment, the consequence being an overrepresentation of youths on ALMPs to avoid open unemployment.[32]

Thirdly, the strong compression of wage differentials has resulted in a fall in the returns to higher education. Such a shift was in part related to the increased supply of workers with upper secondary and tertiary education but, with strong

egalitarian forces at work, wage differentials have been driven lower than demand and supply considerations would have implied.[33] The effect of this can be seen in a fall in the rate of return to an additional year of schooling from 8 per cent in the late 1960s to 4 per cent in the early 1980s, a level still prevailing into the early 1990s. In a short-term perspective, the compression of wage differentials helps industry to recruit highly qualified labour at relatively low wages. But the longer-term effect will be a lack of skilled labour, as enrolment rates over time respond negatively to the change in incentives. These effects became evident by the early 1980s and were a reason for the reluctance of Swedish employers to continue with the highly centralised wage bargaining. In order to maintain the quality and competence of the labour force over the long run, the basic incentives for individuals to invest in further education and training need to be in place. In this context, the recent increase in enrolment rates is encouraging, but educational wage premia still seem small and may not be sufficient to sustain the higher enrolment rate in the longer term (Figure 29).

A wider wage dispersion is often seen as conflicting with overriding distributive goals, the alternative options being defined as low unemployment and a wide distribution of wages and earnings against a more equitable distribution of wages and earnings and higher unemployment compensated for by generous transfers. However, if a wide income dispersion is associated with a sufficiently strong upward earnings mobility, the trade-off between dispersion and equity will not exist. Those gaining their first work experience, either at low wages or as part-time workers, do not tend to stay on in that segment of the labour market. The Swedish and international experience indicates a significant upward mobility from the lowest part of the earnings distribution to other segments; over a seven-year horizon around 75 per cent of males and close to 60 per cent of females staying on in employment can be expected to move upwards by on average more than one quintile (Table 13).[34] Indeed, over a longer perspective, long-term non-employment constitutes the most important source of income inequality, as a result of lifelong effects on the earnings capacity of individuals, so that a narrow dispersion which prevents entry into employment can be counter-productive. Comparison across countries indicate that differences in the employment distribution are far more important than differences in the wage distribution in explaining differences in the distribution of earnings.[35]

Figure 29. **EDUCATIONAL ATTAINMENT AND WAGES**[1]



1. Percentage premium on life-time gross income relative to persons with primary education.
Source: Edin *et al.* (1994); SCB.

Increasing the flexibility of employment and working time

In the early 1970s, the focus of Swedish employment policies shifted from providing the assurance that work could be obtained, given a certain degree of occupational and geographical mobility, to providing greater security for those in work. Stronger employment protection was an integral part of this policy and the legislation enacted to provide this has largely remained in place, despite a less interventionist industrial policy, support to ailing enterprises being abandoned in the mid-1980s. Compared with other European countries the protection offered against unfair dismissals is fairly strong: notice periods are among the longest in the OECD area and the ‘‘last-in-first out’’ (LIFO)-principle has to be respected, implying up to and including 1996 a one-year-long rehiring obligation *vis-à-vis* former employees, now modified to nine months. On the other hand, enterprises can exercise substantial discretion in deciding when dismissals are necessary, and severance pay is not mandated. Although the general effect of employment protection legislation (EPL) is to reduce employment variations over the economic cycle, it has not prevented the labour market from adjusting to large

Table 13. **Alternative measures of five-years earnings mobility for the working-age population**

	Correlation coefficient	1986-91 transitions among quintiles[1]				
		Per cent				
		Stayed in lowest quintile	Stayed in same quintile	Moved up or down one quintile	Moved two or more quintiles	Weighted average move from lowest quintile
United States						
Males	0.72	28.2	50.5	32.7	16.8	1.4
Females	0.51	57.9	47.2	37.4	15.5	0.6
Germany						
Males	0.73	20.1	53.2	32.1	14.7	1.7
Females	0.65	48.3	49.0	38.0	13.0	0.8
France						
Males	0.76	21.3	55.0	29.4	15.6	1.7
Females	0.67	32.9	51.4	34.1	14.5	1.2
Italy						
Males	n.a.	29.8	56.2	30.0	13.8	1.4
Females	n.a.	41.2	50.5	35.2	14.3	1.0
Denmark						
Males	0.62	40.6	49.9	33.7	16.4	1.2
Females	0.48	44.0	44.0	38.0	18.0	0.9
Finland						
Males	0.64	45.3	48.0	30.8	21.2	1.1
Females	0.58	40.4	41.6	39.1	19.2	1.0
Sweden						
Males	**0.61**	**25.8**	**49.9**	**31.5**	**18.7**	**1.7**
Females	**0.30**	**42.9**	**38.5**	**42.2**	**19.3**	**0.9**

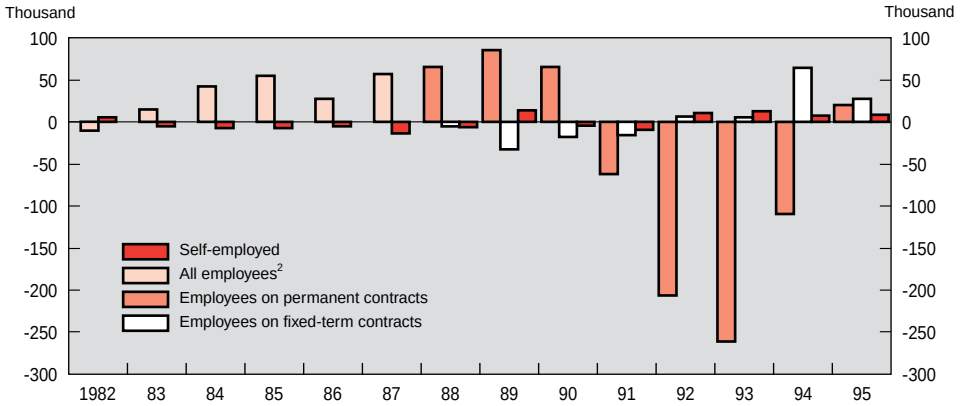
1. Covers only those individuals with positive earnings in both the origin and destination years and excludes the self-employed. The time period is 1987-92 for the United States, 1985-90 for Finland, and 1986-93 for Sweden.

Source: National sources, OECD.

structural shocks of the kind experienced in the early 1990s. However, in the aftermath of the substantial downsizing of the workforce then experienced, enterprises have been, in general, reluctant to take on new staff during the subsequent upturn – a reluctance reinforced by the direct cost of laying off workers and the rehiring obligation. During the 1995 upswing, after EPL had been tightened in late 1994 by the incoming government, increased labour requirements were to some extent met by use of overtime, indicating that EPL acted as a deterrent to

employment growth. Fixed-term contracts, most often used by youths to gain job experience, became more widespread in the face of uncertain unemployment prospects, thus reversing the decline in this form of employment in the late 1980s (Figure 30).[36]

Figure 30. **EMPLOYMENT PATTERNS**[1]
Annual changes, employed persons



1. New definition in 1986.
2. From 1988 total employees are split into two categories: those on permanent contracts and those on fixed-term contracts.

Source: SCB.

The short-term cost associated with EPL in the current economic cycle has to be assessed against the possible long-term gains accruing from a regime where long-term labour commitments induce enterprises and their workers to invest in firm-specific human capital, and where stability of employment may serve to avoid a process whereby unemployment with each cycle tends to be more strongly concentrated among less-productive workers.[37] However, these long-term effects may be achieved without EPL, insofar as seniority wages allow firms to internalise the benefits of investment in their employees' human capital. EPL may also entail longer-term costs if, by increasing hiring and firing costs, it prevents entrants to the labour market from obtaining work experience and on-the-job-training that may qualify them for permanent employment. It may also

add disproportionately to the cost of hiring older workers, leaving them at a disadvantage once they become unemployed.[38] The consequence will thus be a strengthening of the bargaining power of insiders relative to outsiders in the labour market. Unless this is offset by a fall in the relative wage of the outsiders, unemployment will eventually be concentrated among low-skilled and entrants who have little or no formal qualifications to prove their competences. EPL may thus be expected to affect negatively the employment of youths and other workers with a marginal attachment to the labour market, an inference borne out by Swedish data on teenage workers and by case studies for other EU countries.[39] In general, EPL has been identified as an important source of structural unemployment.[40] Adding to the costs in the Swedish case, EPL does not differentiate between large and small enterprises, whereas compliance costs for small enterprises are recognised to be relatively high[41] and such enterprises are therefore exempted from EPL in some countries. While the high degree of unionisation in Sweden may allow a flexible interpretation and implementation at the local level, the lack of dynamism in the small and medium-sized enterprise sector may *inter alia* reflect an overall negative impact of EPL.

Against this background some modifications to the present legislation are needed in order to reduce compliance costs for small enterprises and strengthen the bargaining power of outsiders in the labour market. Such changes could include a lengthening of probationary periods, shorter notice periods and a far more flexible interpretation of the last-in-first-out and rehiring rules, which would be beneficial in a long-term context and allow a faster employment response in the short run.[42] While the wide-ranging array of regulations in these areas may be – and often is – adjusted to circumstances prevailing at firm level, the negative effects on the incentives to set up new businesses and on the practices adopted at the workplace to avoid being in breach of the formal rules cannot be ignored. The most effective approach, which would allow labour market regulations to be adapted to the circumstances of the firm, would be basically to leave them to be agreed with the social partners, as an integral part of wage bargaining and underpinned by greater responsibility for the social partners in financing the cost of unemployment (see below).[43]

Government regulation of the labour market is generally not confined to job security provisions, but also comprises job safety provisions, with effective limits on the overall utilisation of labour. In most cases, these regulations originate in

provisions incorporated into collective agreements and cover such items as vacations, normal working hours and shift-time arrangements. While, in terms of persons employed, the overall utilisation of labour in Sweden has remained high as two-income families are the norm rather than the exception, one result of such provisions in the Swedish case is that average annual working hours per person employed are low by international comparison. Strong wage premia for overtime work – fixed in collective agreements and in the range of 50 to 100 per cent – do not appear to be an impediment to longer working hours during cyclical upturns (as at present). But the combined set of working-time provisions, the pay schedule for irregular work patterns and overtime premia do constrain effective working time in the long run and thereby also affect the utilisation of capital. In the latter respect, it would seem that the effective capital working time in Sweden is below the average in some European countries and the United States, in effect leaving capital idle for a substantial part of the week (Table 14). More flexible working-time regulations, supported in particular by lower premia for irregular working hours, would consequently increase the rate of return on investment, stimulating employment opportunities.

Table 14. **Capital working time and pay schedules for irregular work hours**

Manufacturing industry

	Capital working time per week[1]	Relative wages[2]		
		Daytime basic pay = 100		
	Hours	Discontinuous 2-shift system	Semi-continuous	Continuous
Sweden	**45.9**	**112**	**117**	**131**
France	48.1	112	106-130	111-140
Germany	49.7	110-145	110-150	n.a.
Norway	42.4	108	115	124
United Kingdom	52.4	118	122	126
United States	54.7	104	n.a.	n.a.

1. In 1989.
2. In 1990. Actual labour costs in Sweden, Norway and United States; for the other countries according to contractual arrangements.

Source: Anxo *et al.* (1995).

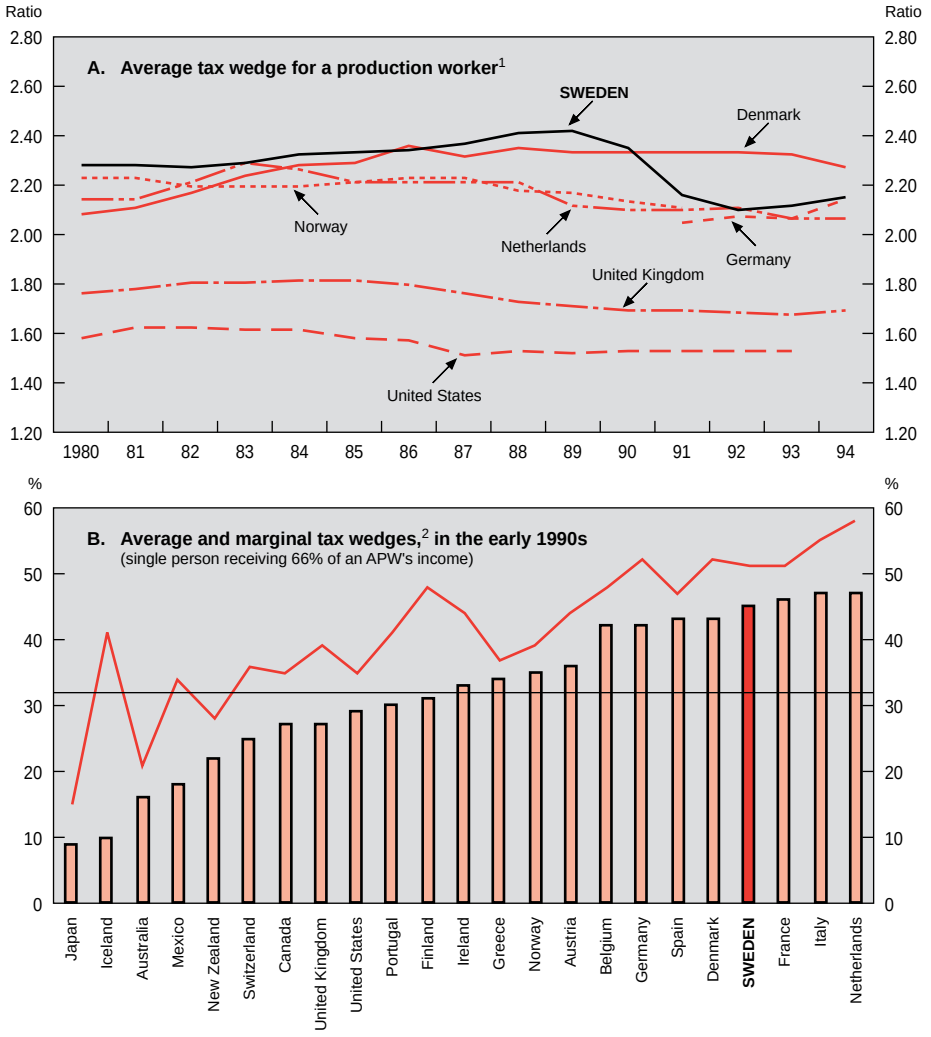
Taxes and transfers

The Swedish tax reform of 1991 was instrumental in bringing down taxation of labour towards average European levels (Figure 31), but the ratio of total labour cost to take-home pay is still above two. A comparison of the 1981 and 1991 tax and benefit system indicates that the main effect of the Swedish reforms over the past decade has been to lower marginal effective tax rates on higher incomes, while rates have remained unchanged for lower incomes. This type of reform would be expected to result in an increase in hours worked at the top end of the income scale, while labour force participation and working hours among lower income earners would not be affected. The 1991 reform is estimated to have increased working hours for married men by about 5 per cent in the long run, while labour supply for married women is estimated to have remained broadly unchanged.[44] This finding is in general supported by other studies, demonstrating that the reform had a positive effect on the supply of labour in addition to reducing the welfare loss associated with high marginal tax rates.

While the empirical evidence, though mixed,[45] suggests that the economic incidence of payroll taxes falls largely on wage earners in the longer run, hence affecting labour supply, some taxes on the use of labour may be borne by employers in the short-term, reducing labour demand. A general reduction in tax rates, if accompanied by expenditure cuts, could thus serve to reduce structural unemployment. In this context, a shift from taxation of labour to capital income might be thought to lower total labour costs, thus shifting the factor intensity of production towards use of more labour. But this would add to the cost of capital and depress investment in the short term, and would not prevent the ultimate burden falling on labour given free international capital movements.

It might also be argued that the tax system could be used to shift the composition of labour demand from better-qualified to low-skilled labour, thus offsetting the negative effects on demand for the latter group flowing from the narrow wage distribution. On the basis of previous experience, however, lowering social security contributions for sections of the labour force suffering from high unemployment could lead to stronger wage growth, limiting the improvement in wage-cost differentials. In any case, if public finances are to remain unchanged, such a restructuring of the tax system would imply fairly high tax increases on medium and high income earners. Their influence on wage formation is stronger than that of the low-skilled, with a higher wage pressure if they

Figure 31. **AVERAGE AND MARGINAL TAX WEDGES**



1. Total labour costs divided by real take-home pay for an average production worker, thus including effects of personal income taxes, social security contributions, indirect taxes and child benefits.
2. The average wedge is the sum of employees' and employers' social security contributions and personal income tax as a percentage of gross labour costs – gross wages plus employers's social security contributions. The marginal tax wedge is the percentage of any marginal increase in gross labour costs arising as a result of an increase in gross wages which is paid in employees' and employers' social security contributions and personal income tax.

Source: OECD, *The OECD Jobs Study, Implementing the Strategy*, 1995; OECD.

seek to shift higher taxes backwards in the short term. More fundamentally, the main reason for allowing wage differentials to widen is that this would allow wages at the enterprise level to respond to productivity developments, the financial situation of the enterprise and local employment conditions. In this respect, tax differentiation would be too blunt an instrument. The Swedish tax system has moved a long way towards becoming simpler and more transparent, and this process should continue.

With tax proceeds financing an array of transfers to persons of working age, the overall effects on labour supply need to take into account both taxes and transfers and their interaction. Indeed, if means-testing of transfers interacts with progressive tax systems to yield high composite (effective) rates of taxation, there may be severe disincentives for the unemployed to seek paid employment and for low-paid workers to either work harder or search for higher-paid jobs. The most important social insurance schemes set up to cover short-term income loss are unemployment and sickness benefits. Subject to caps on the level of income insured (now below the average production worker's wage), these schemes originally provided benefits of up to 90 per cent of wage income, a replacement rate which was gradually reduced through the 1990s to 75 per cent as of January 1996. Nevertheless, the schemes remain among the most generous in Europe. The duration of unemployment benefits is formally restricted to 60 weeks (90 weeks for workers above 55 years of age), but participation in ALMPs is sufficient to restore benefit rights if they are exhausted. For non-insured workers a parallel, but less generous scheme is operated by the Public Employment Service.

The social assistance scheme (SA) is available to any household with little or low income and thus in principle to unemployed or part-time workers, on the condition that the recipient is actively searching for (full-time) work. With benefits related to a subsistence or ''necessity'' norm, and being non-taxable, the replacement rate for lower incomes is fairly high, surpassing that of the unemployment benefit scheme in such cases, but access is restrained by a rather strong means test, requiring that capital assets have been realised. The average duration of social assistance in 1993 was 4½ months, and 7½ per cent of the population received cash assistance at some time during the year. The effectiveness of the work requirement with respect to improving reintegration in ordinary jobs is diminished by the fact that the corresponding job can be supplied by

municipalities and used by them to requalify recipients for unemployment benefits, which unlike SA are financed by the central government. In this respect the SA may work to sustain periods of joblessness, as it would seem that the SA recipients are not given high priority by the Public Employment Service. Incentives to search for jobs are further affected by housing benefits and payments for day care of children which are both income-related. Further details of income replacement schemes are provided in Box 4.

Replacement rates in the Swedish unemployment benefit and social assistance systems may be close to 100 per cent for some low-income workers, creating a potential unemployment trap (Figure 33).[46] With replacement rates falling as income increases, marginal effective tax rates will in general be fairly high for families with children, in the area of 50 to 60 per cent. However, compared with some other countries, the Swedish tax and benefit system seems to avoid poverty traps comprising wide segments of the workforce, reflecting the fact that means-testing is less widespread than in other countries and that access to transfer schemes is regulated through enforcement of strict eligibility criteria. The consequence, as noted, is a higher level of transfer expenditure than in most other countries.

Overall, while the nature of the Scandinavian welfare state results in high participation rates,[47] because a demonstrated work record and willingness to take available jobs are necessary for access to the social security safety net, the Swedish experience over the past decade indicates that changes to tax incentives can improve both the quantity and quality of labour resources. In this case, a further reduction of marginal tax rates, if combined with lower replacement rates in transfer schemes, could be expected to trigger additional efficiency gains, increasing both the return to education and strengthening incentives to active job search on the part of the unemployed. Further, as progressive tax rates still act as a deterrent to undertaking investment in higher education, a reduction of marginal tax rates reduces the overall effective tax on human capital.[48] In the process such a reduction would engender a closer alignment of the wage and skill distributions, while differences in unemployment risk – which are currently concentrated among lower income earners – would be gradually reduced. This would lower long-term unemployment, and imbalances in the labour market would be brought to bear more strongly upon aggregate wage growth, improving real wage flexibility.

Box 4. **Income replacement schemes**

Membership of government-financed, but trade union-administered *unemployment insurance* funds is open to all members of the labour force. Benefits may be claimed after 12 months of membership and 80 days of work within five months in this period, subject to a willingness-to-work requirement of 17 hours per week and a waiting period of five days. The replacement rate is 75 per cent of gross earnings, subject to a nominally fixed cap which used to be close to the earnings level of the average production worker but has fallen below 75 per cent of that wage. The formal duration of the benefit is 300 days, not necessarily consecutive, 450 days for workers 55 years or older, but benefits can be extended through participation in ALMPs. For other job seekers above 20 years of age with a minor work record or who have recently finished education, a parallel benefit (KAS) is available at a flat rate corresponding to about 30 per cent of APW gross earnings and for a duration of 150 days for workers below 55 years of age.

Income replacement in case of sickness is awarded after one waiting day for the first fortnight through a *sickness wage* paid by the employer (to be increased to four weeks from 1997) and for the remaining period through *sickness benefits* paid by social security. The replacement ratio is 75 per cent, but unlike the sickness wage, sickness benefits are capped at an income 15 to 20 per cent above the average industrial wage (7½ times the base amount in the social security system) and there is no upper limit on the length of the sickness period.

A partial early retirement pension is available for employees and self-employed 61 to 64 years of age who want to reduce their weekly working hours. The reduction cannot exceed ten hours per week and the pension amounts to 55 per cent of the estimated loss in earnings. However, the old-age pension is unaffected by this reduction in total income. This option will no longer be available when the planned pension reform enters into force (see Chapter II).

A disability pension is available for persons 16 to 64 years of age whose work capacity has been lowered by at least 25 per cent. The pension is graduated with respect to the loss in work capacity. An earlier option to supply this assessment with considerations with respect to age, local employment conditions etc. have been terminated, restricting the overall access to the scheme. The replacement rate is calculated on the same basis as for old-age pension with an assessment of the future income stream that would have accrued in the absence of illness.

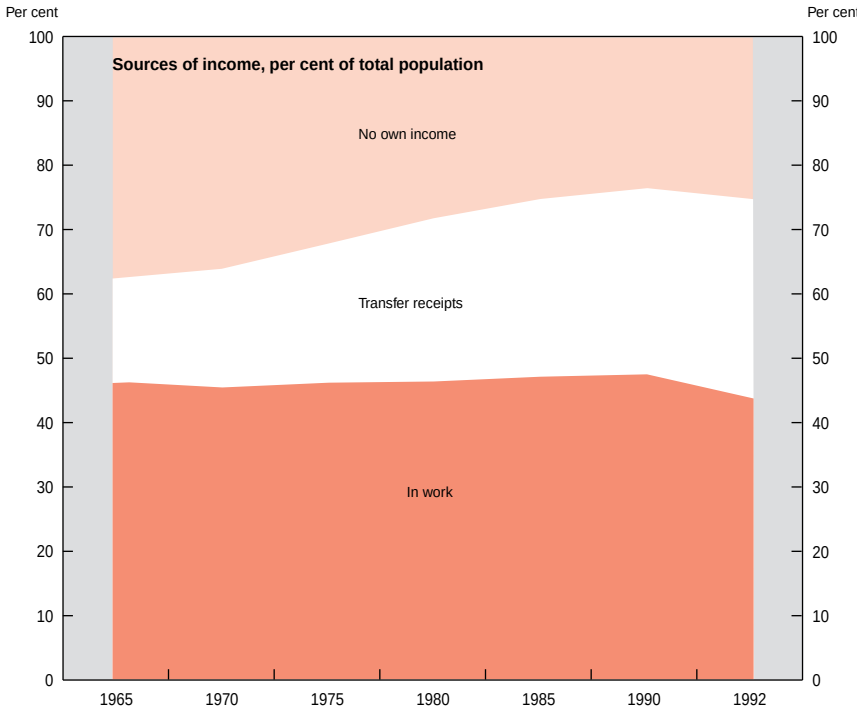
Social assistance benefits, administered by local authorities, are available to households who for reasons of low income are unable to obtain a reasonable living standard. Although centrally-provided norms are available, local adaptations take place with respect to design, content and levels of support, resulting in 25 per cent of municipalities adhering to the centrally-provided norms and the remainder adopting lower levels. In principle social assistance is not provided to households who have low earnings due to

(continued on next page)

(continued)

few hours of work; however, if they register as seeking full-time jobs, assistance will in practice be given on the proviso that allowances are reduced one-for-one with other income. Means-testing is also embedded into schemes for *housing benefits and payment for day care for children*, the former being phased out at a rate between 10 to 33 per cent (depending on family circumstances) of additional earnings for incomes above 60 to 70 per cent of an APW.

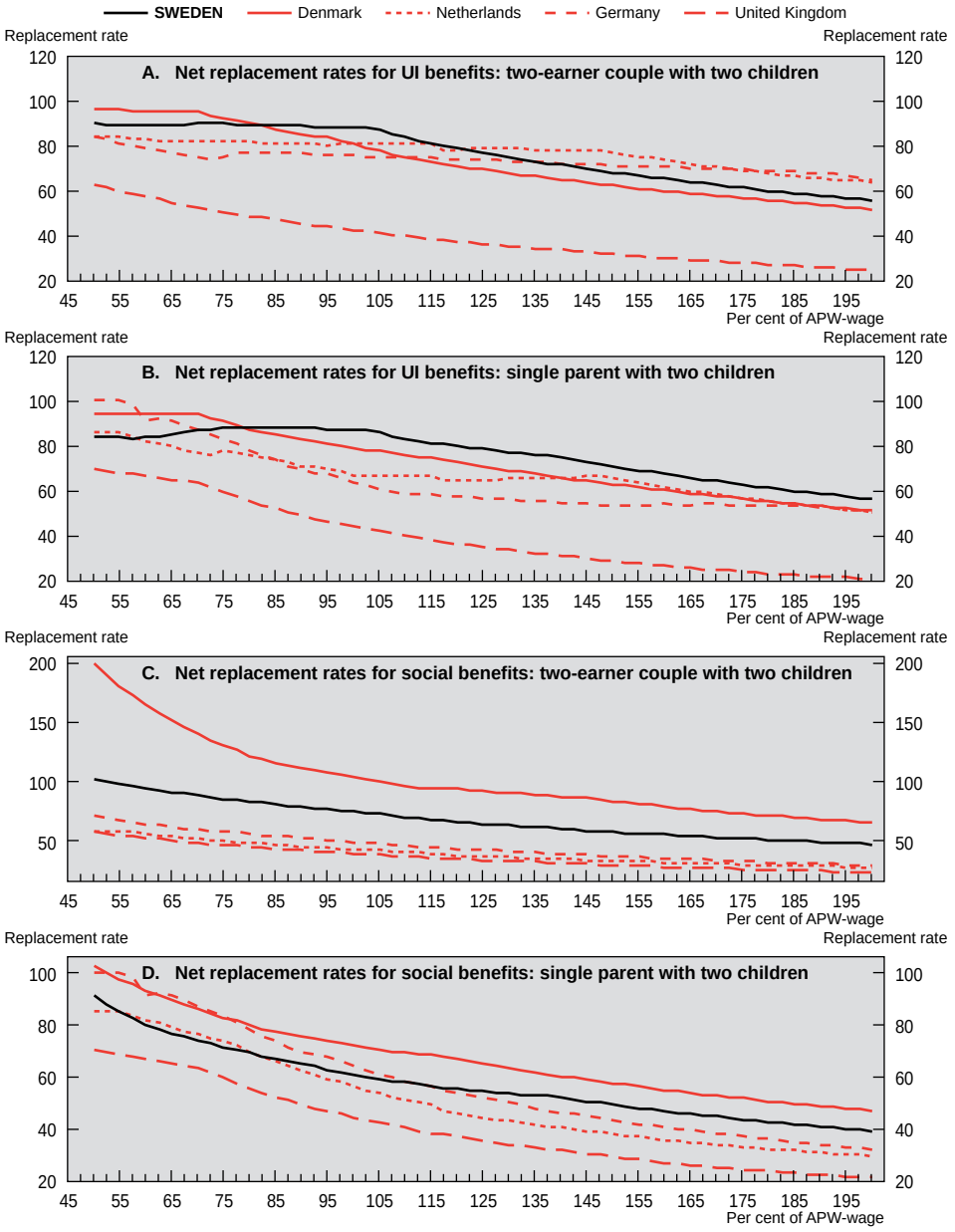
Figure 32. **RECIPIENTS OF INCOME REPLACEMENT BENEFITS**



Source: ESO Report, Ds 1994:133.

Most OECD social insurance schemes are financed out of general taxation and payments are generally unrelated to contributions made, either by employees or employers. While workers pay unemployment insurance premiums, they are

Figure 33. **NET REPLACEMENT RATES**[1]



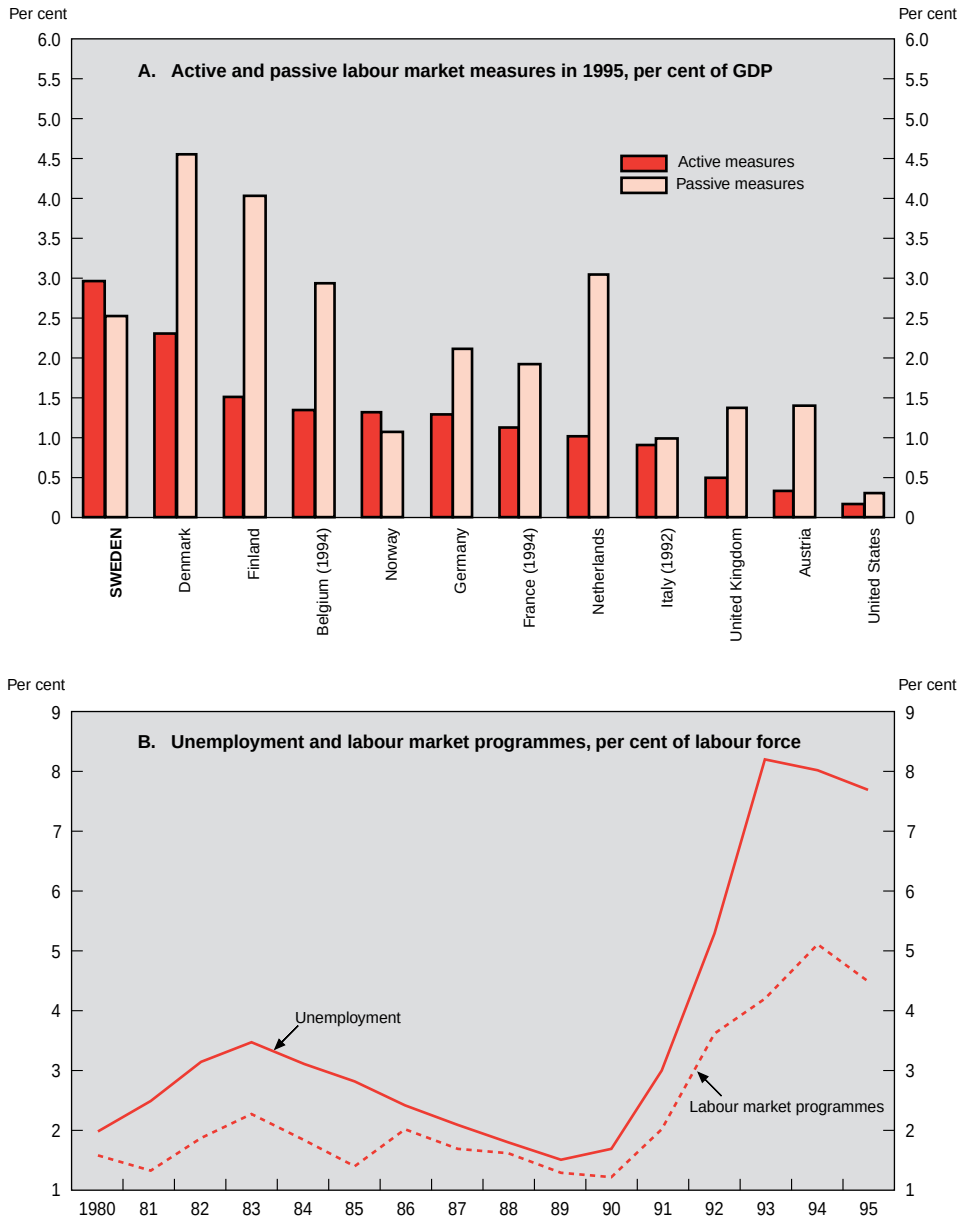
1. Based on rules in force in 1994, for the first year of unemployment.
Source: Ministry of Economic Affairs (1995).

based on a given level of unemployment and are not adjusted in line with changes in the unemployment level; any unemployment in excess of the given level is wholly financed by the central government. This type of system may engender risk-prone behaviour on the part of wage bargainers since social insurance expenditure arising from (for example) the unemployment consequences of excessive wage developments carries no direct financial penalty. An option would therefore be to establish a stronger direct link between social insurance expenditure and employee and employer contributions. In a Swedish context, where trade unions and employers' confederations at the industrial level still exert a substantial influence on wage outcomes, and UI funds are organised on an industrial basis, it may still be possible to establish links between contributions and wages in particular sectors, so that insurance contributions are more closely associated with conditions in those sectors. This relates not only to unemployment insurance, but also to active labour market policy expenditures.[49] One advantageous feature of such a scheme would be the strengthened incentive for UI funds to monitor the work availability of recipients of UI benefits.

Active labour market policies

Active labour market policies, comprising job broking, job subsidisation and training measures designed to stimulate labour demand or labour supply, and to improve the match between them, have been a central feature of Swedish economic policies since the 1950s. Expenditures on these programmes have consistently been both at a far higher level than in other OECD countries and higher than the level of resources going into passive measures (Figure 34). Originally intended as an integral part of a policy to speed up structural change, with a compression of the wage structure as the other pillar, the underlying philosophy has gradually shifted. Acknowledging that a compressed wage structure places high demands on low-skilled workers to keep up with structural change, the Swedish authorities use ALMPs, such as labour market training and subsidised jobs in the public and private sector, not only as alternatives to short-term income replacement, but also to upgrade and maintain the skills and competences of the unemployed, thereby securing a more equitable income distribution over the long run. The Swedish authorities emphasise in particular the contribution of ALMPs in strengthening the attachment of the unemployed to the labour market by limiting periods of passive cash support and their function as a test of job availability as ALMP participation is a prerequisite for continued benefit

Figure 34. **LABOUR MARKET MEASURES**



Source: OECD (1996*a*); SCB.

eligibility. For the long-term unemployed ALMPs are seen as an efficient way of restoring their employability and enabling them to complete successfully for jobs, as placement with private or public employers can supply them with references to be used for further job search.[50] ALMPs are seen both as an important short-term counter-cyclical instrument, more flexible than other fiscal policy levers, and as an efficient way of promoting labour market flexibility. In this respect, the Swedish authorities have stated their clear preference for stimulating labour market flexibility through education and training of the work force and a reluctance to use measures that entail weaker job protection and larger dispersion of incomes. An assessment of ALMPs thus needs to weigh their direct cost in terms of resources and their indirect costs in terms of a less flexible wage formation against their effectiveness in preventing unemployment persistence and increasing occupational mobility.[51]

Unemployment persistence may result where high replacement rates and long effective benefit duration weaken incentives to effective job search. Since the unemployed have the right to obtain an ALMP offer to restore benefit eligibility, the duration of unemployment benefits is in practice unlimited, thus making possible sustained periods of joblessness. With a relatively small work effort being required to access UI benefits (see Box 4), work attachment may be weakened for a fairly large category of workers, particularly since job offers carrying wages more than 35 per cent below previous earnings can be refused. Consequently, job search and job availability criteria need to be strictly enforced. In the generally favourable macroeconomic climate for employment growth over the 1980s, with its high level of vacancies, a virtuous circle of strict enforcement of job availability criteria, short unemployment spells and a low level of unemployment could be maintained. But once this circle was broken, as happened with the large negative demand shock in the early 1990s, ALMPs and intensified efforts in job broking can no longer be relied upon to offset adverse incentive structures. Indeed, if underlying structures are not changed, ALMPs will continue to weigh heavily on public finances.

With the cost of ALMPs amounting to 3 per cent of GDP and with 4 to 5 per cent of the labour force covered by such schemes, the efficiency of ALMPs in promoting the necessary labour market adjustments should be subject to continued scrutiny. Studies of *labour market training* in general fail to substantiate effects on wages and job prospects that indicate a significant rate of return.[52] A positive impact on prospects for workers with low incomes has been found, while

returns have been lowest in periods when programmes have been expanding and when they have been used to reinstate benefit eligibility. Overall, however, such programmes are subject to diminishing returns, suggesting that a targeting of low-skilled (but highly motivated) participants is warranted.[53] *Subsidised work programmes* are acknowledged to have a negative short-term effect on job search, resulting in a smaller number making the transition to ordinary jobs from these programmes than from open unemployment. The long-term effects, however, are generally positive, with programme participants more often securing permanent employment and suffering fewer relapses into unemployment than the unemployed who did not participate in such programmes.[54] A clear verdict on the effectiveness of such programmes is difficult because of the possibility that subsidies will go to workers who would have been hired anyway (''dead-weight loss'') or that the subsidies shift employment opportunities between workers and firms (''displacement effect''), as is likely in a stagnating or contracting labour market. In general, the net employment effect of these measures is estimated to be in the area of 30 to 50 per cent, indicating quite large dead-weight and displacement effects, but with some uncertainty as to the long-term effects.[55]

ALMP programmes may be successful even with high short-term displacement effects as long as workers with high unemployment risk gain at the expense of workers who have better prospects for eventual recruitment. In such a case, the result may be a reduction of the NAIRU. The Swedish experience, on the other hand, suggests that a shift from employment to ALMPs is associated with stronger wage growth than an equivalent shift into open unemployment, indicating that the availability of ALMPs as an alternative to normal unemployment reduces the flexibility of wages with respect to total unemployment. High bargained wages offered in subsidised work programmes – now mainly replaced by the work-for-benefit model – have had the same negative impact.[56] While the overall effect of the high level of ALMPs on macroeconomic performance in Sweden is open to dispute, the evidence suggests that they should be strictly targeted on those who are or have a high risk of becoming ''outsiders'', particularly on the long-term unemployed. This would allow a closer integration of ALMPs with job broking directed at this segment of the labour market, *inter alia* towards social assistance recipients. Moreover, in a labour market where displacement effects are large and where public finances are stretched, policy credibility will also require a clear recognition of the overall diminishing value for money of ALMPs, with a view to scaling down some of the programmes and reassessing the level of remuneration offered.

On recent evidence it would appear that compared with other countries, a relatively large part of the outflow from unemployment in Sweden is related to the take-up of ALMPs as a means of reinstating UI benefit eligibility. This diverts attention and resources from placement of job-seekers in ordinary jobs and explains the high monthly intake into ALMPs relative to the outflow from unemployment seen in Sweden.[57] Combined with other evidence that the return to participants from undertaking more than one ALMP programme is at best negligible and usually negative, due to negative signalling effects to employers, the present UI eligibility rules seem to be counterproductive. A well-formulated ALMP programme, undertaken as a part of a coherent plan to integrate the unemployed into an ordinary job, may improve job prospects substantially and reduce the probability of an unemployment spell evolving into continuous joblessness. In this respect, it is noteworthy that the outflow from unemployment to ordinary jobs increases substantially in the last four weeks preceding benefit termination.[58] The implication is that a shorter duration of unemployment benefits, combined with stricter eligibility rules and tight enforcement of job availability criteria, could spur stronger job search efforts, making for more realistic reservation wages.

An effect of the high level of ALMPs has been that less attention than envisaged has been paid to job-counselling and job-search assistance. In many cases, it would appear that PES officers refrain from calling on the unemployed for an interview if they do not have a specific job offer or ALMP placement available. The consequence has been long intervals between job interviews, which in many cases become a mere formality, and a weaker follow-up of the job availability criterion. It may also have led to an overuse of ALMPs, when job search alone might have been more effective. This would seem particularly pertinent during the first stage of an unemployment spell, and allowing the unemployed to enter an ALMP only after a few months should be avoided; at this stage chances of finding unsubsidised jobs could still be explored. It would also appear that, for several reasons, resources are often concentrated on job-broking and training for unemployed people who stand a good chance of finding a job on their own. In this respect, a concentration of efforts on the hard-to-place is warranted.[59] Moreover, the efforts of the PES officers to enforce job availability would be more effective if the administration of unemployment insurance were integrated with the PES.

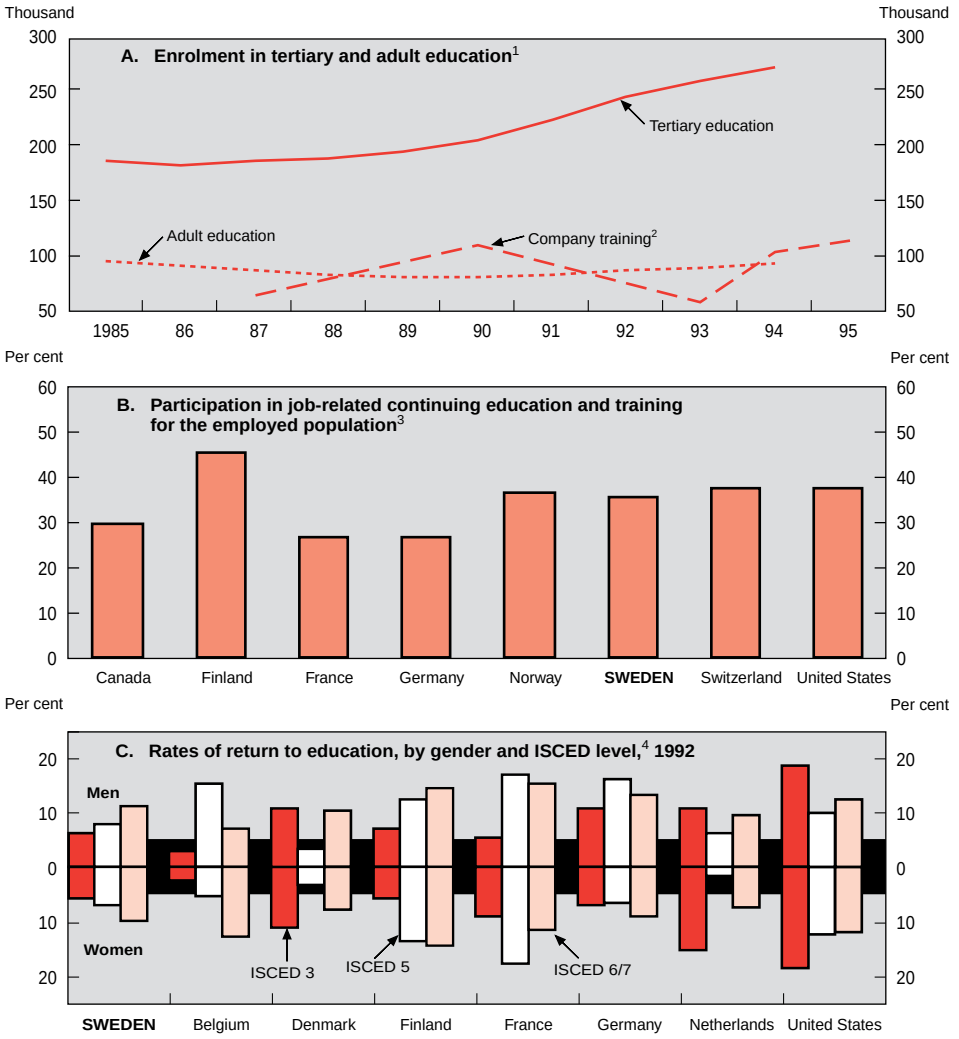
Increasing the knowledge base, efficiency and innovative capacity

Upgrading skills and competences

The educational attainment of the Swedish population is relatively high by international standards and the combination of relatively high spending on primary and secondary education and the Swedish emphasis on life-long learning has resulted in an above-average level of formal skills and a versatility of the labour force (Figure 35, Panels A and B). Indeed, international comparisons rank Sweden very high when it comes to maintaining the capacity of the workforce to adapt to new technology and new work methods.[60] Nevertheless, with respect to enrolment in tertiary education, Sweden gradually lagged behind comparable countries during the 1980s and the early 1990s, reflecting a response to the diminishing returns from undertaking tertiary education.[61] The share of the age-cohort undertaking university education declined from 14 per cent in 1950 to 7 per cent for cohorts born in the mid 1960s. This trend is in the process of being reversed, both as a response to improved incentives and to the depressed demand for labour in the early 1990s. However, enrolment rates in tertiary education in Sweden still remained below the OECD average in the most recent year for which information is available (1992), while the return to education was somewhat below the average OECD level (Figure 35, Panel C).

Two features of the Swedish education system would nevertheless still seem to impact negatively on long-term economic developments. The proportion of the relevant age cohort either not qualifying for or dropping out during upper-secondary education in the 1980s is estimated at around 20 per cent. In addition, there is an estimated non-completion rate for tertiary education of close to 20 per cent. While the high drop-out rates in the 1980s may not be fully representative of current circumstances, as they reflected the narrow wage differentials and favourable labour market of the time, an overall drop-out rate of 20 to 25 per cent has a significant effect on the bottom end of the skill distribution. A recently-implemented reform obliges local authorities to provide three years of schooling at upper-secondary level for all pupils who have finished compulsory education. There is a variety of programmes which are vocationally-oriented. All three-year programmes will meet with the general eligibility requirements for pursuing tertiary education while setting aside time for practical work. In these respects, the new system of upper-secondary education will have to cater to opposing

Figure 35. **FEATURES OF ADULT AND TERTIARY EDUCATION**



1. Registered students in tertiary education, fiscal years. Full-time equivalents in adult education and company training, calendar years.
2. Data for 1988, 1989, 1991 and 1992 are not available.
3. In 1993 for Sweden and Switzerland; in 1992 for France; in 1991 for Canada, Germany, Norway and the United States; in 1990 for Finland.
4. ISCED 3: Upper secondary education; ISCED 5: Tertiary education, short duration; ISCED 6/7: Tertiary education, long duration.

Source: OECD; *Education and Employment*; SCB.

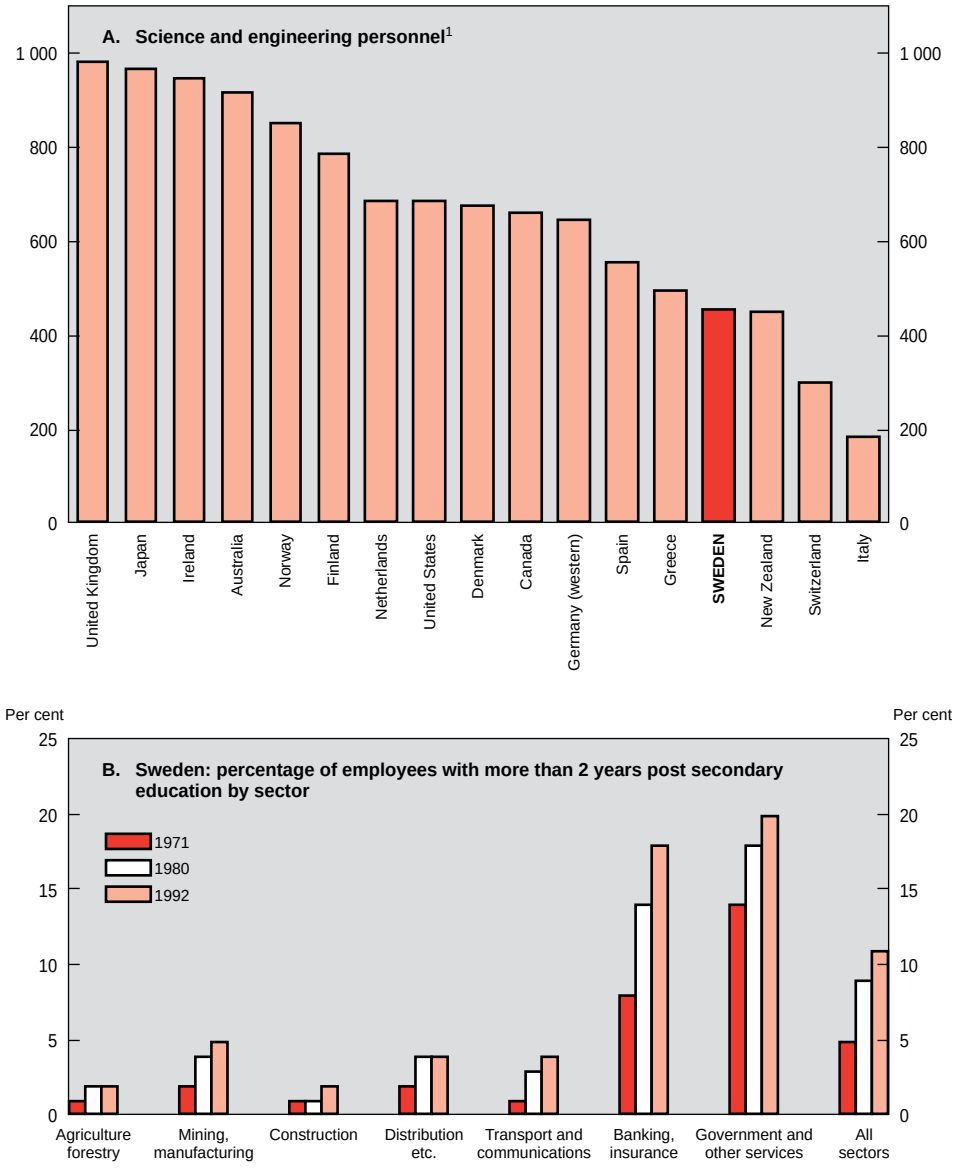
demands, and it remains to be seen whether the school-to-work transition can be made more effective for youths who are less theoretically inclined. Apprenticeship arrangements are being strengthened to cater for this group, but they are less important and have a far weaker tradition than in some continental European countries. The school-to-work transition may thus continue to be less smooth compared with such countries.

At the other end of the skill distribution, the continuing dominance of public sector employment among graduates from higher education may have slanted the education system towards the requirements of the public sector, restricting the supply of highly qualified university graduates with a scientific and engineering background (Figure 36). Moreover, tertiary education in Sweden tends to be of a shorter duration than in most other countries, thus magnifying the disadvantage of Swedish industries with respect to human capital supply and in particular with respect to research and innovation. The expansion of capacity in tertiary education already underway is projected to increase the share of the higher educated in the labour force from 22 to around 30 per cent over a 15-year horizon. But this increase is still projected to leave a substantial excess demand (5 per cent of the labour force) for this group, given the high knowledge-content of economic growth.[62] Closing the gap between the demand for and supply of higher-educated labour would seem to require a further increase in the capacity in tertiary education and in the returns to be expected from higher education, which has implications for the currently compressed distribution of income in Sweden.

Product market competition

Competition issues take on great significance in Sweden in view of the weak long-term performance of the economy relative to other OECD countries. For manufactured goods, the competitive regime which evolved during the thirty years of EFTA membership, prior to joining the European Union, has in general maintained competitive pressures in the face of dominant suppliers in a number of domestic markets. On the evidence presented in Chapter IV, it would appear that inadequate competition is most widespread in the markets for services, reflecting *inter alia* a host of government regulations and rules imposed by professional associations. The adoption in 1993 of the prohibition principle as the basis for the regulatory framework towards competition implies a stronger

Figure 36. **EDUCATIONAL PATTERNS**



1. Number of science graduates per 100 000 persons 25 to 34 years of age in the labour force in 1992.
Source: OECD, *Education at a Glance,* 1995; OECD, *Reviews of National Policies for Education*: Sweden.

pro-competitive stance in these sectors, underpinned by extended powers for the competition authorities to act against infringements of the law. But prices for services still contain a considerable premium relative to other European countries, indicating that incumbents are still in a position to exploit their market position to restrain the pace of new establishments, and that productivity levels in many service sectors are lagging best international practice.

Apart from the need to ensure that the new competition-law framework guarantees the market contestability in the service sector, the major outstanding issue is the privatisation and deregulation of public utilities and other public entities' participation in business activities (see Chapter IV for details). There has been considerable progress in the former area with the introduction of competition in the generation of electricity, while telecommunications, railways, and postal services are gradually exposed to more competition although the publicly-owned incorporated enterprises still maintain dominant positions in these areas. Otherwise public participation in business activities remains significant in the Swedish local government sector, which has controlling stakes in 1 500 companies, mainly concentrated in housing and in utilities. These activities which involve in-house participation in competitive tenders, present a range of problems. Foremost of these are predatory pricing strategies, often involving cross-subsidisation between exposed and monopolised activities and explicit or implicit public subsidies and guarantees, and conflict of interest as in-house bidders also act as in-house advisors. This would seem to call for *i)* a broad and internally consistent strategy, defining at national level the scope of the activities for such enterprises, bearing in mind the need for growth in the market-based sectors of the economy; *ii)* guidelines to secure a level playing field in those areas municipally-owned enterprises can operate; and *iii)* a framework for implementing and monitoring such guidelines.

Technology and innovation

Sweden spends more on R&D than any other OECD country, with the larger part taking place in the private sector and government spending mostly going to universities and colleges. The high level of private spending reflects the dominance of large multinational enterprises in the manufacturing sector and R&D is thus concentrated to a few areas. As noted in Chapter IV, these features may have reinforced the dualism of the Swedish economy, as small and medium-sized

enterprises in general have been unable to benefit from the large R&D effort. A variety of measures has been introduced to offset this, resulting in a proliferation of agencies combining support to R&D with general SME programmes and regional development support. The overall effect of these schemes have so far been below expectations. With respect to R&D, it would appear that the concentration of research at the university sphere has been associated with an inadequate diffusion of technology, the most important dimension of technological development for SMEs. An important dimension of this has been the overall low mobility of researchers, who rarely return to research institutions after a spell in business, thus depriving these institutions of valuable feedback. A shift in the balance of support from university-based institutions to institutions with stronger ties to business is thus warranted.

Nurturing an entrepreneurial climate

Based upon the evidence presented above and in Chapter IV, a host of government and private sector regulations add up to a fairly hostile climate to entrepreneurship and job creation in SMEs. Regulations in the area of employment protection and industrial relations, combined with relative wage rigidity, impose relatively strict impediments on their capacity to adjust. The extensive involvement of local government in the production of services noted above has distorted competitive conditions. Efforts to secure a similar tax treatment of incorporated and unincorporated enterprises and to prevent tax avoidance by shifting labour income to capital income have resulted in a complicated tax code and the long-standing double taxation of dividends may have hampered the emergence of a private placement market for risk capital. The overall effectiveness of the wide array of government support and lending instruments designed to stimulate the supply of risk capital may have been undermined by information and access problems. In sum, for SMEs to benefit further from the improvement in macroeconomic fundamentals of the Swedish economy, it would be highly beneficial to simplify both the tax code and support schemes.

Recent policy actions

The broader tax base created by the 1991 tax reform allowed a reduction in marginal income tax rates to near the OECD average, but the fiscal consolidation

measures adopted in 1994-95 have since lifted the highest marginal tax rate by 5 percentage points as a result of a surcharge on high incomes to be in effect to 1998. Moreover, employees' contributions to social insurance, introduced in 1993, are set to increase to 6 per cent by 1997, leading to a considerable increase in the overall tax on labour over the period from 1991 to 1997. Replacement rates in social insurance, which were generally 90 per cent for low and median-wage earners in the early 1990s, have been reduced and are now 75 per cent, both for unemployment and sickness benefits, while waiting periods of 6 days and 1 day have been introduced for the respective schemes. Access to part-time pensions for older workers has been curtailed by raising the minimum age and reducing the number of hours worked allowed. The replacement rate has also been reduced. Membership of a UI fund is now available only for youths above 20 years of age and subsidised jobs no longer qualify for first-time benefits. Concurrently, job availability requirements are being reinforced, with the benefit sanction being lengthened from 20 to 60 days if an unemployed person declines a job or training offer. ALMPs are gradually being concentrated on the long-term unemployed. Broadly, it would on balance appear that incentives to seek work have improved over the 1990s.

During 1994 and 1995, a wide set of issues in the fields of labour market policy, educational and regional policy were submitted to parliamentary-based or expert-based commissions for evaluation and recommendations. The reports cover issues related to the overall role of labour market policies, unemployment insurance, employment protection legislation, working time and the coverage of social insurance.[63] In the view of the Labour Market Policy Committee[64] which was given the broadest remit, the use of labour market policies as a stabilising policy instrument in the face of the economic shocks in the early 1990s reduced the efficiency of job broking and selection of participants for ALMPs; labour market policy should therefore focus on a narrower range of objectives than earlier, aiming primarily to lower the duration of vacancies and long-term unemployment and to reduce periods of joblessness. Within the area of job broking, the committee recommended a stronger focus on the need for individualised service, as expressed in ''individual action plans'', together with the need for an efficient division of work between public and private placement services. The wide array of ALMPs, each with their own statute, should give way to fewer and broader measures, while unemployment insurance should clearly serve as a

job-seeker's wage with limited duration. Two separate means of developing labour market training were identified: ordinary labour market training should be combined with work experience in enterprises, thus facilitating hiring into ordinary jobs, but it should also be possible for the unemployed to undertake formal education supported by UI benefits.

Broadly concurring with the recommendations of the committee, and following the preliminary assessment and guidelines given in the 1995 Growth Bill and the spring 1996 Fiscal Bill, the Swedish authorities tabled an Employment Bill in June 1996. This sets out a programme to stimulate labour demand in the short run, to strengthen small and medium-sized enterprises, to enhance incentives for human capital formation and to improve the functioning of the labour market, with the overall goal of reducing open unemployment to 4 per cent by 2000. In the fields of labour market policy and human capital formation, the main features of the programme, the first of which were adopted by the Riksdag (parliament) in July 1996, are:

- A reorientation of total resources from ALMPs to intensified efforts in the sphere of formal education. A substantial increase in the capacity of the formal education system, by 100 000 places in adult education aimed at low-skilled workers and 30 000 places in tertiary education, is to be financed partly by reallocating resources from ALMPs. This is to be complemented by a drive to upgrade the qualifications of unemployed persons between 25 and 55 years of age who may receive a one-year education grant equivalent to the UI benefit level. As a one-off measure, the unemployed above 55 years of age with more than two years of joblessness (estimated at 1 per cent of the labour force) will be offered public work at wage rates equal to UI benefits.
- An increase in the replacement rate in sickness and unemployment benefits from 75 to 80 per cent from 1 January 1998, which is to be partly financed by a reduction in the income base for these benefits, relaxation of indexing clauses and by increasing the employers' obligation to pay sickness benefits in the first part of a sickness period (''the sickness wage'') from two to four weeks.
- The introduction of an explicit upper limit for the duration of unemployment benefit. A given qualifying period with unsubsidised jobs can give

rise to a maximum of three years on benefit, but a work requirement – including ALMP participation – has to be satisfied at intermediate thresholds of 60 and 120 weeks. A fourth year may be considered if the job seeker has obtained relatively long periods of unsubsidised jobs or undertaken ordinary education. Further modifications to the unemployment benefit scheme, *inter alia* increases in the qualifying period and the number of waiting days, are pending as recent expert recommendations on its long-term structure are being assessed.

- A refocusing of the activities of employment offices from administering ALMPs towards their traditional role of improving the matching between unemployed and vacancies, the process of individualising job-broking being made more efficient by formulating individual action plans reflecting the job seeker's qualifications and opportunities and emphasising the responsibility of job seekers in finding ordinary jobs. Integrating educational efforts, ALMPs will serve to reduce joblessness rather than acting as a tool for reinstating UI benefit eligibility for job-seekers. Resources are to be moved to local employment offices and the ties with local authorities are to be strengthened, giving municipalities a more important role.
- A lowering of the tax burden on small and medium-sized enterprises, both through lower payroll taxes on the first three employees and a targeted relief of the double taxation of dividends.
- A reassessment of the balance between employment protection and enterprise adaptability, in particular with a view to facilitating the expansion and reorientation of activities. Attempts to bring about an agreement between the social partners on further changes in this framework have failed, and a government initiative has recently been taken signalling changes *inter alia* in the following areas: local unions may freely agree with their employers on recruitment and dismissal provisions (hitherto contingent upon approval at central level); enterprises may have five persons on fixed-term contracts of twelve months duration; the length of notice periods is to be differentiated by seniority and not by age; the rehiring obligation is to apply for nine instead of twelve months, and economic sanctions in case of breach of the rules are to be adapted to the enterprise's ability to pay. However, the last-in-first-out rule and the length of probationary periods will remain unchanged.

Assessment and scope for further action

While recent reforms of the instruments of Swedish labour market policy show a greater awareness of the longer-term implications of labour market policies on incentives, the underlying philosophy is adjusting rather gradually in response to external changes. The commitment to the ''work approach'' to labour market policy is still strong, but the links with education policy have gained recognition, making for a combined ''competence-work'' approach. This change in emphasis underlines the need for an upgrading of skills and is welcome, and together with the steps taken to improve the efficiency of active labour market policies should over time lead to a reduction of structural unemployment. However, given that the institutional framework of the labour market prevents wage differentials from reflecting relative labour demand, education policy alone may not be sufficient to match the supply and demand for skills. This is all the more pertinent in view of the ambitious goals for labour market developments set by the Swedish authorities for the medium term, aiming for employment growth in excess of 1½ per cent at an annual rate in order to cut open unemployment to 4 per cent and ''total'' unemployment to below 8 per cent by 2000. The areas of major concern are set out below, and further detailed proposals following from these considerations can be found in Box 5.

With respect to *wage and labour cost flexibility and incentives to job search,* it should be evident that the distribution of wages has not responded appropriately to the imbalances of the economy, either with respect to short-term changes in the balance between demand and supply in different segments of the labour market or with respect to the provision of incentives for investment in human capital. In particular, wages at the lower end of the income distribution have not adjusted sufficiently to enable the low-skilled unemployed to compete effectively for jobs. While the wage bargaining process which determines the effective minimum wage is institutionally independent of government, the introduction of a wider dispersion in the agreements of local and central government with their employees would constitute an important step towards greater wage flexibility. Government intervention in the labour market with generous benefit schemes and extensive use of ALMPs to prevent open unemployment also serves to underpin high individual reservation wages, thereby reducing collective resistance to wage structures that lead to unemployment. While replacement rates have been reduced, they are still among the highest in the OECD area and should be

Box 5. **The OECD Jobs Strategy: synopsis of recommendations for Sweden**

The *OECD Jobs Study* sets out a strategy for improving the ability of OECD economies to cope with structural change, through enhancing the ability to adjust and to adapt, and increasing the capacity to innovate and be creative. With respect to labour and product market flexibility it identified the need for initiatives in the following areas: working-time flexibility, the entrepreneurial climate, wage and labour cost flexibility, employment security provisions, active labour market policies, labour force skills and competences, and unemployment and related benefits systems.

Policy actions undertaken over the past few years have served to reduce replacement rates in social security schemes and also lowered marginal tax rates – although recently these have been on the rise once again. Moreover, access to transfer schemes has been curtailed, job availability requirements have been reinforced and steps have been taken to reduce the effective duration of unemployment benefits. The focus in labour market policy is shifting from labour market training and subsidised jobs to formal education, and from using ALMPs as an instrument to reinstate benefit eligibility to serving individual job-broking and reducing joblessness. While further efforts are warranted to reduce the generosity of transfer schemes, lower taxation, and make ALMPs more effective, the most pressing problem to be resolved remains wage formation where both aggregate and relative wage rigidity has prevented a reduction of unemployment.

Following from the detailed review of labour market and employment performance and the assessment of policy requirements, this chapter has identified the need for Sweden to move ahead in several areas. A common denominator of the proposals is the need to underpin the long-term strategy of raising the qualifications of the workforce with greater short-term flexibility that secure aggregate wage developments consistent with low inflation and underlying productivity trends and stimulates the hiring of low-skilled workers by enabling them to compete actively for available jobs and thereby improve life-time earnings. The effect will thus be to undermine insider-outsider mechanisms that at present preserve differences in unemployment risk and contribute to inequality in income distribution.

Increase wage and labour cost flexibility

High taxes on labour should be reduced. Given wage floors created by collective agreements and, ultimately, by high replacement rates in social insurance and assistance schemes, high taxes on labour act to depress job creation via increases in labour costs. These high taxes are an inevitable consequence of high public spending which needs to be brought down.

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The process towards lower replacement rates in social insurance should be reinforced. High replacement rates in transfer systems and ALMPs combine with high marginal tax rates to increase reservation wages and discourage job search, thus preventing low-skilled workers from pricing themselves into jobs and employees in general from seeking out higher-paid jobs. In the longer run this reduces human capital formation.

Greater wage flexibility with respect to skills, sectors and firms should be encouraged. Recruitment of the long-term unemployed and low-skilled youths will be facilitated if the social partners were to allow a widening of the income distribution at the lower end of the pay scale. The effective minimum wage is given by agreements reached in wage bargaining and is hence a central responsibility for the organisation of employees and employers, both in the private and public sector.

Wages may be made more responsive to labour market developments if financing of UI benefits and ALMPs were shifted from government to unemployment funds organised at industrial level. If this is not accomplished, the administration of UI benefits should be integrated with the other activities of the Public Employment Service, to make the enforcement of job availability provisions more effective.

The scope for secondary action should be reviewed. Such action may under present circumstances prevent the expansion of small and medium-sized enterprises as unions try to impose wage rates not fully consistent with local employment possibilities.

Reform unemployment and related benefit systems

Current replacement rates in unemployment benefits and sickness insurance are still high in an international context and should be reduced further.

ALMPs should not be allowed to prolong the effective benefit period beyond three years. The duration of benefits is high relative to other countries.

Increase working time flexibility and reform employment security provisions

Employment protection legislation should be reviewed to improve hiring prospects. Job security provisions may under present circumstances lock unemployment in at a high level, and discourage the recruitment of new entrants and groups with weak labour market attachment.

A greater decentralisation of decision-making with respect to terms and conditions of employment is required. The underlying structure of labour laws implies an excessive amount of regulations with respect to terms and conditions which is particularly onerous for small and medium-sized enterprises.

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Enhance the effectiveness of active labour market policies

A reduction in the scope of ALMPs is warranted together with a concentration on efforts to improve labour market re-entry. Active labour market policy has come under increasing strain, with the increase in unemployment reducing the effectiveness of matching jobs with vacancies, by diverting resources away from job broking and enforcement of job availability criteria.

Further targeting of ALMPs and job broking on the long-term unemployed and those at risk (inter alia social assistance recipients) is warranted.

The level of remuneration for subsidised jobs should be set to give incentives to seek out ordinary jobs. Although training measures with remuneration at UI benefit level now constitutes the major part of ALMPs and the work-for-benefit model has gained ground, some subsidised jobs are still at bargained wages.

Improve labour force skills and competences

The increase in enrolment rates over the past few years should be sustained, with the curriculum developing towards the needs of industry, and with emphasis on the scientific and engineering trades. Sweden has gradually lost its advantage relative to other countries with respect to human capital formation as younger cohorts have been discouraged from entering higher education.

Efforts should be devoted to securing an effective school-to-work transition. Sweden does not rely on apprenticeship schemes as comprehensive as in neighbouring countries and the practice of paying youths adult wages from the age of 18 has resulted in an overrepresentation of youths on ALMPs. The reform of upper-secondary education should be followed closely.

High drop-out rates from secondary and tertiary education, warrants further attention. Currently 20 to 25 per cent of the relevant age cohort does not achieve a vocationally qualifying education,

Enhance product market competition

The markets for services can only be fully contestable when a level playing field between private and municipally-owned enterprises when supplying local services, is secured. This should be a major point of departure in the forthcoming review of the conditions pertaining to this sector.

The privatisation process should continue, with due emphasis on the need to separate natural monopolies from competitive activities.

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The deregulation of the sheltered sectors should be continued on a broad front. Withdrawal of government support to the housing sector, already well under way, should have high priority.

Nurture an entrepreneurial climate

Given a favourable overall cost level and a low corporate taxation, small and medium-sized enterprises may be best served by simplifying both the tax code and the wide array of government support schemes and provisions regulating their activities. The Swedish industrial structure has developed differently from other countries, with small and medium-sized enterprises rarely entering into a phase of sustained expansion.

reduced further. The effective duration of unemployment benefits (including periods of ALMP participation) has proved to be very long and further efforts to reduce it are required. Moreover, wages would be made more responsive to labour market developments if the financing of unemployment insurance benefits and ALMPs were to fall to a greater degree on the social partners. Unless steps are taken in this direction, it would seem most logical to assign the administration of UI benefits to the Public Employment Service to strengthen job broking. Finally, high taxes on labour income, whether paid by employers or employees, are depressing job creation via their impact on labour cost. After the tax reform of 1991, marginal tax rates have again increased. Further fiscal restraint should be concentrated on bringing down the high level of public spending, allowing taxation of labour income to be lowered towards levels prevailing in other countries.

It would appear that there are several forces at work strengthening insider-outsider mechanisms in wage bargaining. *Employment protection legislation,* which ranks as relatively strict in a European context, is one such factor. It discourages hiring of youths and inexperienced workers who are in no position to adjust their wage demands to offset this potential cost insofar as the wage structure remains narrow and fairly rigid. Moreover, it imposes restrictions on the downsizing of firms but also on the subsequent ‘‘upsizing’’, and is particularly onerous for small and medium-sized enterprises. This legislation needs to be reviewed. More generally, as governments have enshrined regulations on terms and conditions of employment into law, the ability to vary terms and conditions

with local circumstances has been seriously eroded. While the Swedish model may be well adapted to an environment of large enterprises and comprehensive trade unions, which may jointly exercise a large degree of discretion at local level in applying rules and regulations set by government statutes, the present set-up severely detracts from the ability of firms to set terms and conditions with respect to job security, work environment and conflict rules better suited to local circumstances. These parameters are better determined through the bargaining process, where their costs and benefits can be assessed against improvements in earnings. It would seem that small and medium-sized enterprises are restricted rather severely by the extensive regulatory structure. Allowing more decentralised decision-making in this area is required.

Arguably, the present industry-based *wage bargaining mode,* which fragments responsibility for unemployment, cannot be maintained much longer if combined with the current emphasis on wage relativities. Whatever model may emerge in the future, it would seem that a return to the previous highly-centralised model, with a wage structure imposed on enterprises from above, is no longer sustainable. The ability to adapt to structural change can only be created and preserved at local level, and labour relations, including wage setting, are ultimately the responsibility of local management and unions. What could be obtained in a more co-ordinated mode is agreements on criteria for or an upper limit to wage growth at the local level. However, evidence on co-ordinated bargaining in the 1990s suggest that the willingness to abide by and enforce such restrictions is very limited. Against this background, further moves towards a more decentralised wage setting may be the only option available and, in view of the present problems, should be encouraged rather than delayed.

In the effort to contain the effects of joblessness too much emphasis has been put on *ALMPs,* with placement in such schemes becoming an end in itself. While unemployment spells can thus be broken up, underlying long-term joblessness has become a problem. ALMPs may have been fairly effective instruments in purely demand-management terms, but they may have contributed to the current unsustainably high overall wage growth and their effectiveness in preventing the decay of human capital has diminished. In dealing with long-term unemployment the balance has to shift back to intensified job counselling and placement and to test job availability. The measures contained in the Employment Bill of June 1996 are steps in this direction.

Human capital formation is dependent on the wage and tax structure allowing a sufficient return on the investment undertaken. Grants and credit on preferential terms may raise returns, but this is hardly an option with the strained fiscal position, and should be even less so over time as financial markets develop. The evidence presented so far clearly indicates that educational wage premia have been insufficient, and that high marginal tax rates have exacerbated this problem. While the 1991 tax reform was a major step forward, a trend towards higher tax levels has since reappeared and needs to be corrected. The potential payoff to improved incentives is evident in the differential in unemployment rates according to educational attainment, pointing to a levelling of unemployment risk and thereby of life-time incomes. As the education system adapts towards skill requirements – both formally and informally through company training that gives workers a basic qualification and allows a continued updating and upgrading of qualifications – the basis for low wage differentials in the future has been strengthened. However, this process is by nature long-term and in the short-to-medium term wage differentials have to be allowed to respond to imbalances between supply of and demand for different categories of workers. In particular, wages at the lower end of the income scale should be allowed to adjust to enable the low-skilled unemployed to compete effectively for jobs.

Major steps to improve product market competition would be to eliminate non-competitive practices and establish a level playing field between private and public production of services. For this purpose a comprehensive strategy should be developed which, apart from widening the scope for private production of services currently provided by the public sector, would include rules which ensure a uniform tendering process, specifying all dimensions of the tender with respect to prices, volumes, qualities and performance criteria, thus allowing full monitoring of the contract. In-house participants in public tenders should be subject to full cost accounting rules – including capital costs – to prevent cross-subsidisation. In addition, locally formulated long-term competition strategies would allow insight and oversight and should ensure that the government participation in business activities at local level is in accordance with overall national goals. Monitoring might be improved with an advisory body providing research and up-grading of local expertise. This framework would be conducive to small- and medium-sized enterprises, which would benefit further from a wide-ranging dismantling of targeted support schemes in favour of retaining and extending tax

expenditure instruments that are automatically available to all entrepreneurs. These comprise not only continued tax deductibility for R&D expenditures, but also modifications in the tax code that encourage equity investment and improve the return to entrepreneurial capital in unquoted enterprises. An example is the recently proposed relief of the double taxation of dividends, and a general removal of the double taxation of dividends, would serve to improve the supply and allocation of risk capital at all levels. Technology diffusion, arguably more important for enterprise and job creation than basic R&D, might benefit from transferring public R&D spending from universities to jointly funded science parks, the business-oriented environment stimulating the development of economically viable innovations.

IV. Policies for the business sector

Introduction

Starting from a very favourable position in the early 1970s, when overall productivity levels were substantially above the OECD average, Sweden has seen its per capita income relative to other OECD countries decline considerably over the past three decades (Table 15). Taking stock in the early 1990s, the government-appointed Productivity Commission[65] attributed this state of affairs to deficient human capital formation, insufficient fixed capital investment and inadequate competitive pressures. Among the explanations and advice offered,[66] the emphasis has been on the effects of centralised wage bargaining, extensive labour market regulations, tax policies and credit-market institutions in favouring established and (fixed-) capital-intensive enterprises at the expense of small and medium-sized human-capital intensive enterprises. Explanations have tended to focus on the relatively sparse small-enterprise sector catering to the domestic market, in contrast to the dynamic and thriving multinational enterprises in the manufacturing sector. The bias against small enterprises created by a narrow wage dispersion and rigid job protection legislation has already been noted in the previous Chapter. More generally, however, the effect of labour market rigidities on economic performance may be amplified or, in some cases, offset by the regulatory framework within which the enterprise sector operates.

This chapter reviews the institutional background to decision-making in Sweden, covering activities of incorporated and unincorporated enterprises and the self-employed. The issues involved relate to the impact of government regulation and competition-law enforcement in an economy with a rather large public sector; to the role of taxes and subsidies; to the support for research and development and the process of technological diffusion; to the availability of finance to risk-taking enterprises, and to the market for corporate control (‘‘corporate

Table 15. **Relative GDP per capita and labour productivity growth for selected OECD countries**

	Relative GDP per capita[1]				Labour productivity growth
	1970	1980	1990	1994	1970-95
	OECD average = 100				Per cent
Australia	104.1	102.1	99.6	98.6	1.5
Canada	108.0	117.9	114.4	109.4	1.2
Denmark	106.3	102.9	103.4	110.2	1.8
Finland	86.4	94.2	101.2	86.9	2.6
France	106.0	110.7	108.4	103.0	2.2
Germany	94.5	98.9	99.9	105.5	1.6
Italy	89.4	99.6	101.7	100.2	2.5
Japan	84.7	94.2	110.0	111.3	2.6
Netherlands	103.9	102.9	99.7	99.7	1.7
New Zealand	101.4	90.7	84.3	87.1	0.8
Spain	66.8	69.0	73.7	72.8	3.0
Sweden	**114.9**	**108.8**	**106.3**	**93.4**	**1.5**
United Kingdom	97.7	94.6	99.2	94.7	1.9
United States	147.7	139.9	137.3	136.8	0.9

1. At current prices using current purchasing power parities.
Source: OECD, *National Accounts.*

governance''). The structure of the chapter is as follows. In the first section, Sweden's economic performance and industrial structure are documented. The second section analyses the role of policies in the major areas of interest, while implications following from this are spelled out in the final section.

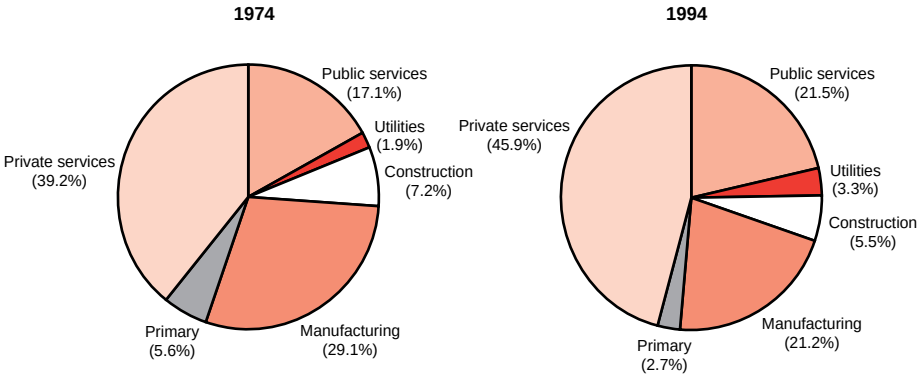
Structure and performance of the Swedish business sector

Overall and sectoral productivity performance

During the 1970s and 1980s, the growth of labour productivity in Sweden was among the lowest of OECD economies, at an annual average of 1.5 per cent (Table 15). This relatively poor productivity performance was associated, *inter alia*, with a change in the sectoral composition of the Swedish economy, as manufacturing and primary industries with high productivity growth gave way to private and public service sectors, with less measurable productivity growth (Figure 37). This development has been common to most OECD countries;[67] it has, however, been more accentuated in Sweden, where the public sector plays a

Figure 37. **SECTORAL COMPOSITION OF THE SWEDISH ECONOMY**

Per cent of GDP[1]



1. Net of unallocated items and net indirect taxes.

Source: SCB.

significantly larger role. Moreover, such output measures as are available for government-provided services show productivity levels within the government sector actually falling over the 30 years to 1990, with an annual average decline of 1.5 per cent occurring in the 1970s and 0.4 per cent in the 1980s, followed by a partial recovery in the early 1990s.[68]

Within the business sector total factor productivity (TFP) – the overall efficiency in the utilisation of labour and capital – also increased at a somewhat slower rate over the period 1970-1995 than in a number of other OECD countries (Table 16). With an investment rate somewhat below the OECD average, labour productivity growth in the business sector (which reflects both TFP growth and the increase in capital per employee) was likewise rather low. These results primarily reflect developments over the 1973-1990 period. Indeed, both measures of business-sector productivity picked up strongly at the beginning of the 1990s, as a considerable shake-out of labour occurred. Efficiency gains were particularly significant in manufacturing production, although the fact that profit margins have been under pressure has also induced enterprises in the sheltered sectors to minimise costs and boost productivity.

Table 16. **Productivity growth in the business sector**

Annual average percentage change

	A. Labour productivity				B. Total factor productivity			
	1960-73	1973-79	1979-90	1990-95[1]	1960-73	1973-79	1979-90	1990-95[1]
Sweden	**3.7**	**1.4**	**1.5**	**3.2**	**2.0**	**0.0**	**0.8**	**1.1**
United States	2.2	0.4	0.9	0.7	1.6	0.2	0.5	0.5
Japan	8.3	2.8	2.8	0.7	5.6	1.1	1.7	–0.2
Germany	4.5	3.1	1.7	0.1	2.6	1.8	1.0	0.0
France	5.3	2.9	2.4	1.7	3.7	1.6	1.6	0.7
Italy	6.3	2.9	1.9	2.5	4.4	2.0	1.2	1.2
United Kingdom	3.9	1.5	2.0	1.7	2.6	0.6	1.5	1.6
Canada	2.9	1.5	1.0	1.2	2.0	0.6	–0.2	0.1
Denmark	3.9	2.4	1.8	2.7	2.3	0.9	0.9	2.5
Finland	5.0	3.2	3.3	3.8	4.0	1.9	2.5	0.8
Netherlands	..	2.7	1.7	1.3	..	1.7	1.3	0.4

1. 1990-1994 for Germany, Italy and Sweden.
Source: OECD, *National Accounts*, Estimates.

Low productivity growth does not preclude high levels of efficiency. Indeed, Sweden has many manufacturing industries which have productivity levels comparable with best practice (Table 17, Panel A). This is particularly true of the paper and metal products sectors. Moreover, the productivity increase that Sweden has seen over the current business cycle has enabled a number of industries – including electrical machinery – to advance from being good performers to becoming the productivity leaders in the OECD area. A common feature for many of the ''productivity leaders'' in manufacturing would seem to be a high degree of concentration and specialisation together with a high export ratio. On the other hand, considerable scope for catching up to best practice remains in retail sales, construction and electricity generation (Panel B). In the early 1990s, prior to recent deregulations, revenue per employee in telecommunications was particularly low, reflecting the dominant position of the government-owned telecommunication company. As has been documented elsewhere,[69] a positive correlation would seem to exist between productivity indicators[70] for selected service industries and competition-enhancing regulatory reforms.

Prices and mark-ups

Estimates of relative prices in Sweden, as compared with the EU, suggest that as far as the goods market is concerned, the prices of durables and equipment

Table 17. **Labour productivity levels in selected OECD countries**

A. Manufacturing, 1993

	Value added per hour worked, leader country = 100									
	United States 1993	Japan 1993	Germany 1992	France 1993	United Kingdom 1993	Canada 1993	Australia 1993	Netherlands 1993	Sweden 1993	Sweden 1987
Food, beverages and tobacco	**100.0**	35.6	82.6	87.0	41.7	64.3	51.1	96.6	74.3	57.3
Textiles, clothing and footwear	78.3	41.9	70.3	67.1	51.5	46.3	32.3	**100.0**	69.8	60.8
Wood products and furniture	57.0	17.6	50.6	55.3	28.1	52.6	27.1	**100.0**	75.4	64.1
Paper products and printing	85.0	49.7	56.6	64.3	76.4	67.6	53.7	64.5	**100.0**	**100.0**
Chemical products	66.9	52.6	50.9	56.9	79.7	52.6	39.8	**100.0**	93.8	72.4
Non-metallic mineral products	81.8	62.9	73.9	99.4	70.6	78.4	77.4	**100.0**	85.0	75.5
Basic metal products	76.8	78.3	78.0	63.3	61.4	87.9	56.8	70.4	**100.0**	93.3
Metal products	68.9	67.6	67.2	46.4	42.5	54.8	35.9	54.0	**100.0**	**100.0**
Machinery and equipment	**100.0**	67.4	58.7	67.3	47.9	55.5	46.4	34.6	46.2	66.5
Electrical machinery	80.3	**89.0**	54.0	78.9	48.2	51.9	28.0	82.2	**100.0**	75.6
Transport equipment	88.4	**100.0**	82.6	85.0	47.8	71.9	45.5	41.8	56.3	55.8
Other manufacturing	**100.0**	41.4	39.6	31.4	43.5	33.5	22.1	27.0	48.4	67.0
Total manufacturing	**100.0**	76.6	81.3	84.2	64.1	71.3	52.0	95.6	93.7	82.0

B. Selected industries

	United States = 100				
	Manufacturing (value added per hour worked, 1993)	Retail trade (retail sales per employee 1990)	Construction (value added per employed 1990)	Electricity (gigawatthour per employed 1993)	Telecommunications (revenue per employee, 1992)
United States	100.0	100.0	100.0	100.0	100.0
Japan	76.6	70.7	79.7	77.5	80.6
Germany	81.3	100.7	75.5	26.9	63.1
France	84.2	94.8	83.3	46.8	68.3
Italy	n.a.	72.3	84.2	19.9	89.3
United Kingdom	64.1	77.6	62.0	26.7	68.9
Canada	71.3	n.a.	148.2	66.7	73.9
Australia	52.0	60.1	103.0	35.7	70.6
Belgium	101.2	94.1	90.1	38.5	58.6
Denmark	n.a.	68.6	65.8	40.5	53.4
Finland	96.6	85.9	94.4	37.4	48.0
Netherlands	95.6	54.8	69.9	37.7	88.0
New Zealand	n.a.	85.8	n.a.	41.4	65.2
Spain	67.6	45.7	86.8	40.1	74.2
Sweden	**91.8**	**86.9**	**75.2**	**68.5**	**50.4**

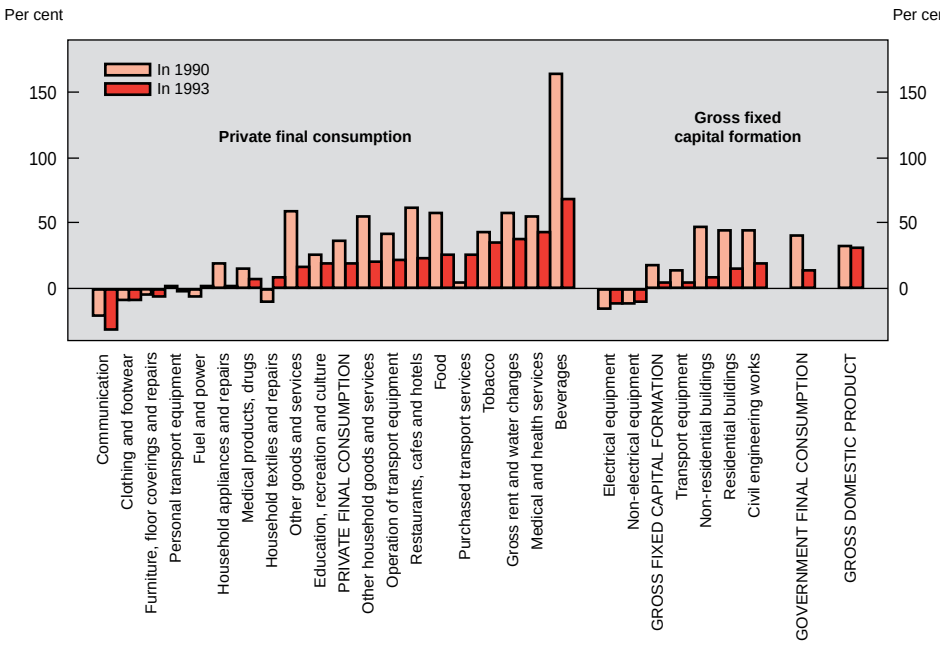
Source: D. Pilat (1997).

are near to European average levels. However, a considerable premium has existed on many products in the service sector. The squeeze on profit margins led to slow growth in prices of most components of final demand over the period 1990-93 (Figure 38, Panel A); in the process, the difference between Swedish price levels and those prevailing in the European Union generally narrowed. But it would appear that price levels in Sweden are still somewhat out of line with the normal relationship between per capita income and prices (Figure 38, Panel B).

Estimates for the mark-ups in Swedish manufacturing industry do not appear particularly high compared with other OECD countries[71] and the difference between the highest and lowest mark-up is among the smallest observed

Figure 38*a*. **RELATIVE PRICES AND REAL INCOME**

A. Relative price levels between Sweden and EU[1]

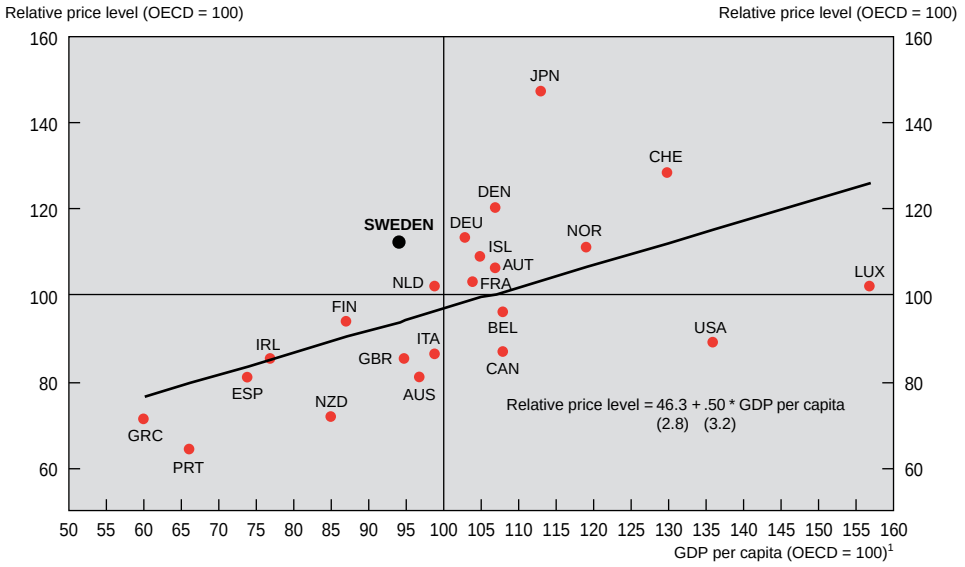


1. The percentage differences between price levels in Sweden and weighted averages of price levels in EU countries.

Source: OECD, *National Accounts* and *Purchasing Power Parities* and *Real Expenditures*.

Figure 38*b*. **RELATIVE PRICES AND REAL INCOME**

B. Real incomes and relative prices, in 1993 at market prices



1. GDP per capita, indices using current PPPs.
Source: OECD, *National Accounts* and *Purchasing Power Parities* and *Real Expenditures*.

(Figure 39). In some industries, such as the wood and paper industries and pharmaceuticals, mark-ups in Sweden are among the highest, but since these industries are internationally exposed, high profitability reflects to a large extent productivity levels above those of its competitors. Mark-up estimates for selected service industries are in general quite low (Table 18), with the exceptions of utilities and the financial industries. While low mark-ups might occur as a result of over-investment or high remuneration of capital, evidence from Chapter III indicates *inter alia* that capital costs have been high due to low capital working time. Moreover, wages in some sectors have been higher than productivity levels would warrant, and market rents would thus seem to have been appropriated by labour either as over-employment or above-market wages.

Table 18. **Mark-ups for a selected group of service sectors**[1]

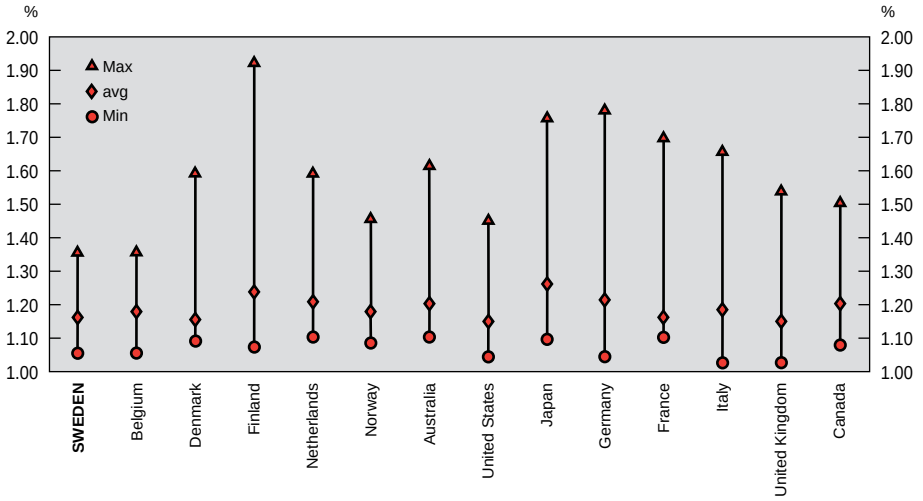
Per cent

	Sweden	Canada	Denmark	Finland	France	Germany	Japan	Netherlands	United States
Electricity, gas and water	**107**	154	36	29	50	39	58	25	34
Construction	**12**	16	9	19	19	23	26	6	17
Wholesale, retail trade	**16**	29	73	..	52	39	25	55	28
Restaurants, hotels	**13**	26	15	..	40	18	..	48	24
Transport, storage and communication	**20**	37	29	36	39	35	29	45	33
Communication	..	58	52	..	57	75	..	..	68
Finance, insurance and real estate	**86**	87	97	68	94	..	129	..	83
of which:									
Finance and insurance	**114**	25	74	45	..	70	..	..	33
Real estate	**92**	107	127	75	..	..	..	148	123
Community, social and personal services	**32**	59	50	42	88	57	37	27	20

1. 1980-1992. Mark-ups are calculated as the difference between prices and marginal costs. The estimates are adjusted for material inputs and are of net indirect taxes. (..) indicates that no data were available or that the estimated mark-up was not statistically significant.

Source: Martins *et al.* (1996).

Figure 39. **MARK-UP RATIOS IN MANUFACTURING 1980-1992**



Source: Martins *et al.* (1996).

Industrial structure, employment and enterprise creation

Sectoral trends and enterprise size

Sweden has been going through the same shift in industrial structure as other OECD economies, with the share of manufacturing in total employment falling (Table 19). Job creation in the private service sector has been modest in the last two decades, and whereas the distribution sector has been a major generator of jobs in other countries, it has been losing jobs in Sweden. Some of the weak employment growth in the private service sector, such as in private social services, can be explained by the wide scope of activities undertaken by the government sector in Sweden. In particular, reflecting the weak performance of private service sectors, self-employment is fairly low compared with other OECD countries.

Outside the public sector, employment patterns are influenced by the fact that manufacturing industry in Sweden is dominated by large enterprises (Figure 40), to a far larger degree than in other OECD countries; and

Table 19. **Sectoral employment**

A. Share of total dependent employment[1]

Sector	Per cent							
	Sweden	United States	Japan	Germany	France	Italy	United Kingdom	Canada
Agriculture	**3.8**	2.6	8.0	3.2	5.1	8.7	2.1	4.4
Mining and quarrying	**0.2**	0.5	0.1	0.6	0.4	n.a.	0.7	1.2
Manufacturing	**18.2**	15.6	23.1	30.2	18.9	20.5	20.4	14.5
Electricity gas and water	**0.7**	0.8	0.6	1.0	0.7	0.8	1.0	1.0
Construction	**5.9**	5.3	9.6	6.6	6.9	7.4	6.8	5.3
Wholesale and retail trade etc.	**13.9**	22.4	17.4	16.9	17.3	21.7	19.8	23.5
Transport and communication	**6.7**	4.2	5.6	5.6	5.8	6.5	5.6	6.2
Finance, insurance and business services	**8.9**	15.1	4.7	n.a.	11.0	n.a.	12.0	11.8
Community, social and personal services	**8.0**	19.0	22.6	n.a.	6.6	n.a.	12.0	9.0
Total industries	**66.3**	85.5	91.7	80.6	72.7	80.9	80.4	77.1
Government services	**32.6**	14.5	5.9	14.7	27.3	16.1	19.6	22.9
Total dependent employment	**100.0**	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Self-employment[2]	**12.6**	10.5	19.1	13.2	16.5	36.1	n.a.	15.9

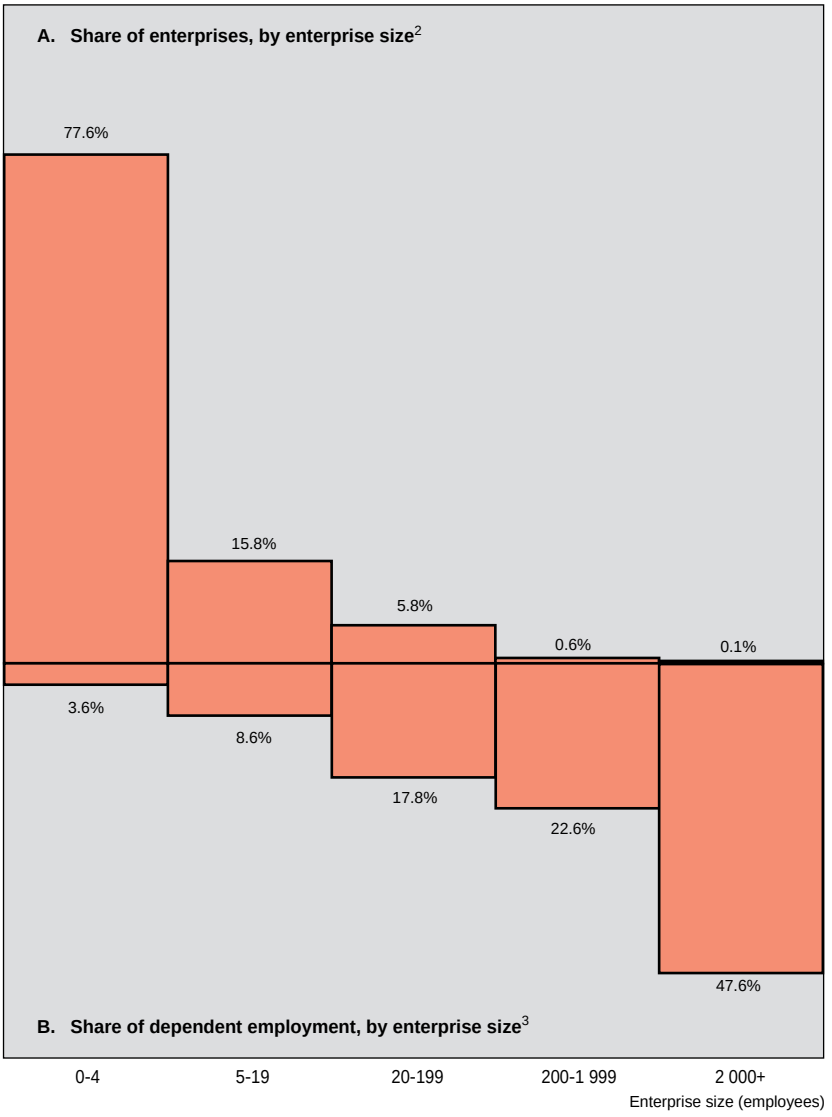
B. Growth rates 1974-94

Sector	Per cent per annum							
	Sweden	United States	Japan	Germany	France	Italy	United Kingdom	Canada
Agriculture	**–2.3**	–0.2	–2.2	–2.6	–2.6	–1.2	–1.2	–0.4
Mining and quarrying	**–2.3**	–0.1	–2.8	–2.1	–2.4	n.a.	–2.4	0.9
Manufacturing	**–1.5**	–0.2	0.4	–0.5	–1.1	–1.5	–1.5	0.1
Electricity gas and water	**0.3**	1.7	1.7	0.9	1.1	–1.3	–1.3	2.2
Construction	**–1.5**	2.0	2.1	–0.7	–1.1	0.5	0.5	1.8
Wholesale and retail trade etc.	**–0.1**	2.8	1.6	1.0	0.6	1.9	1.4	3.2
Transport and communication	**0.4**	1.2	1.2	0.8	1.0	1.8	–0.3	1.1
Finance, insurance and business services	**3.6**	7.3	3.9	n.a.	4.8	n.a.	5.0	6.2
Community, social and personal services	**0.3**	3.3	5.3	n.a.	4.6	n.a.	6.2	7.4
Total industries	**–0.5**	2.3	1.0	0.2	–0.3	0.3	0.3	2.2
Government services	**2.8**	1.2	1.2	2.0	2.4	2.3	0.8	3.5
Total dependent employment	**0.2**	2.1	1.1	0.5	0.2	0.6	0.4	2.5
Self-employment	**–0.5**	1.3	–1.4	–0.3	–1.9	0.7	2.5	2.3

1. In 1994.
2. The self-employment employment share refers to 1993.

Source: OECD, ISDB database.

Figure 40. **ENTERPRISE SIZE AND EMPLOYMENT IN THE MANUFACTURING SECTOR**[1]



1. 1994.
2. Per cent of total enterprises.
3. Per cent of total dependent employment.

Source: Federation of Swedish Industry.

Table 20. **Job creation by establishment size**

Number of employees

Employment size	Annual average increase in jobs 1985-1989 due to				
	New firms	Closures	Expansion	Contraction	Net contribution
0	0	0	14 700	0	14 700
1-4	11 900	10 700	12 800	9 000	5 000
5-19	7 100	5 300	13 000	13 100	1 600
20-49	3 100	2 400	7 600	8 000	400
50-200	2 900	2 400	8 400	11 200	–2 300
200-	2 400	2 500	7 300	13 800	–6 600
Total	27 400	23 300	63 700	55 100	12 800

Source: Davidsson (1995).

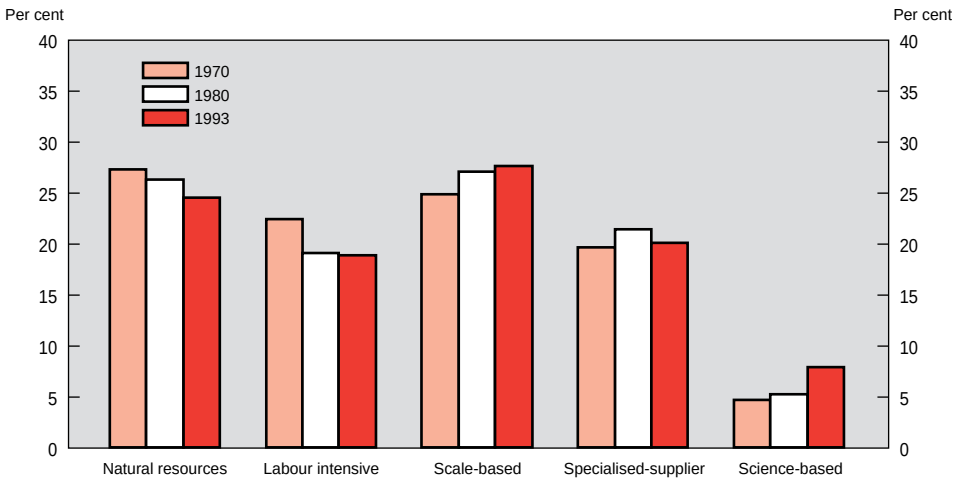
establishment size and net job creation have been inversely correlated (Table 20). In total, there are about 460 000 small and medium-sized enterprises (SMEs) in Sweden, employing close to 1.4 million, the majority in services, and it has been estimated that over the period 1985-89 60 per cent of the overall net job creation took place in SMEs.[72] Close to 80 per cent of net job creation in the service sector was in SMEs, with the majority in professional services, indicating that jobs to a large extent have been created in the high-skilled, high-paid segment of the labour market.[73] Gross job creation and losses were both much higher in the SME segment than in larger companies, indicating a more dynamic process. The survival rate of new firms after three years has been estimated at 66 per cent. After five years the survival rate is around 60 per cent and as many as 40 per cent survive after 10 years. However, the average employment in these enterprises after 10 years is typically fewer than three persons (including the owner), with very few employing more than 10 persons and around half employing no additional personnel.[74] Employment growth of SMEs is thus to a large degree based on a high start-up rate rather than expansion in already established companies. While, in most respects, the job creation process in Sweden is similar to other OECD countries, where 50 per cent of the net job creation has been in the 5 per cent fastest growing SMEs,[75] the failure of start-ups to expand is a negative feature of the Swedish case.

Factor-intensity, the technology base and R&D in manufacturing

As to factor intensity, Swedish industry is scale- rather than labour-intensive, with science-based industries having an above-average representation,

mostly as a result of a relatively large pharmaceutical industry and certain science-based metal industries. A few resource-based industries, such as paper and printing production, account for a very high share of the value-added in the manufacturing sector, but resource-intensive industries as a whole account for a smaller share than the average for OECD countries. The composition of manufacturing employment has remained relatively constant over the past couple of decades, although there has been a move away from natural-resource and labour-intensive industries towards scale- and science-based industries (Figure 41): science-based industries have almost doubled their employment share.

Figure 41. **EMPLOYMENT SHARES IN THE MANUFACTURING SECTOR BY TYPE OF INDUSTRY**

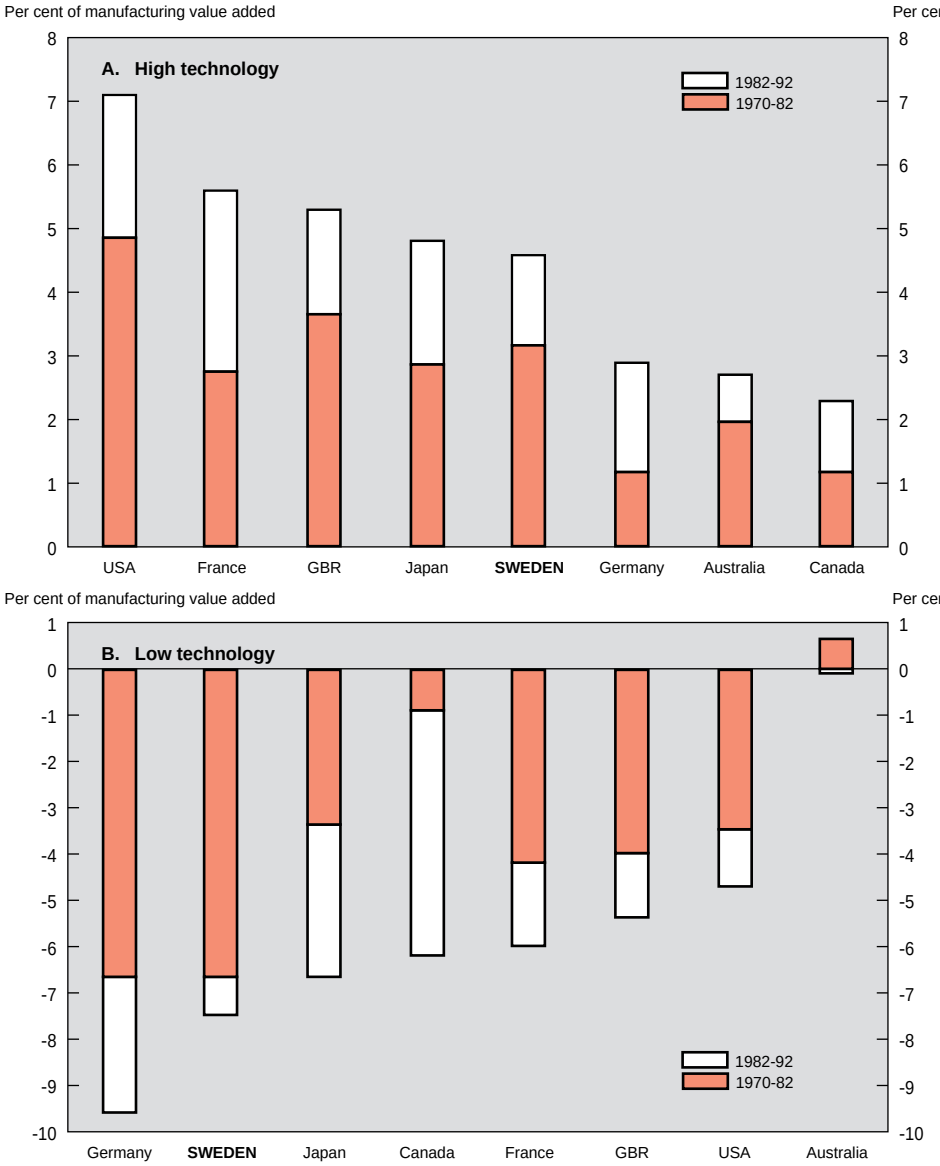


Source: OECD, STAN database.

When industries are classified by the technology level of their output – the point of reference being the R&D intensity in the OECD area as a whole – the weight of high-technology industries in total value added is seen to have been growing quite rapidly since 1970[76] (Figure 42). Moreover, the share of medium-technology based industries has also been increasing, contrasting with experience elsewhere in the OECD, while the fall in the low-technology share is relatively

Figure 42. **CHANGES IN OUTPUT SHARE BY LEVEL OF TECHNOLOGY**

Percentage points



Source: OECD, STAN database.

marked. High-technology industries have in general both higher employment and productivity growth than manufacturing as a whole. Thus, a change in industry structure towards high-technology based industries contributes not only to higher employment growth, but also to stronger growth in productivity. In this sense, structural change in Swedish manufacturing may be classed as relatively favourable. Nevertheless, a detailed decomposition of Swedish manufacturing output by technological level (Table 21) shows a continuing below-average weight for high-technology industries, such as electrical equipment and chemicals industries, with some notable exceptions such as motor vehicle and instruments production. Moreover, Sweden has a relatively high share of low-technology industries compared with other OECD countries: apart from natural-resource based economies such as Australia and Canada, only France has a similarly large share of low-technology based industries.

Despite the relatively low weight of high-tech industries, Sweden's expenditures on R&D, totalling about 3½ per cent of GDP in 1993, are the highest in the OECD area, and this holds for private as well as for overall spending (Figure 43, Panel A). It has also been growing faster than in other countries, boosted by higher private expenditures which increased from 64 to 71 per cent of total R&D

Table 21. **Share of manufacturing value added by industry**

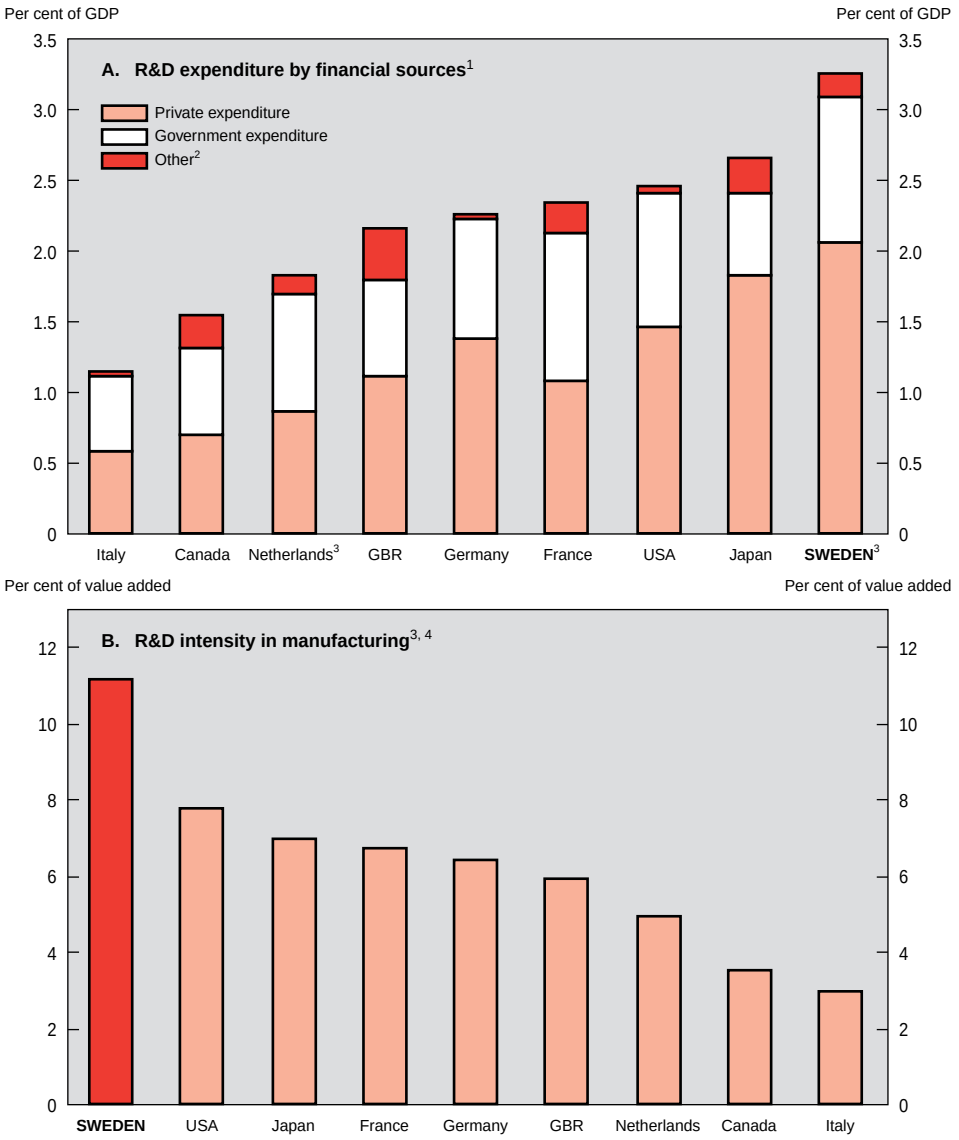
Current prices, per cent[1]

	Sweden	United States	Japan	Germany	France	Italy	United Kingdom	Canada	OECD[4]
Technology based[2]									
High-technology	**16.8**	24.0	21.4	20.1	18.6	13.7	22.0	12.3	20.6
Medium-technology	**29.7**	30.3	34.3	33.9	27.8	23.7	30.4	34.7	30.9
Low-technology	**53.5**	45.7	44.3	46.0	53.6	62.6	47.6	53.0	48.5
Factor based[3]									
Resource-intensive	**23.0**	21.0	20.9	25.6	31.1	26.5	23.0	29.7	24.0
Labour-intensive	**11.6**	13.5	15.6	14.1	14.9	27.7	12.2	12.5	14.9
Scale-intensive	**37.9**	33.8	33.4	31.3	29.6	23.8	33.3	39.0	32.5
Specialised supplier	**17.0**	15.9	23.1	22.0	16.3	15.3	18.4	12.7	18.0
Science-based	**10.6**	15.6	7.0	7.1	8.2	6.7	13.2	6.1	10.5

1. 1993.
2. Industries are grouped on the basis of their R&D intensity in the OECD area as a whole, defined as the ratio of business-enterprise R&D to production.
3. This classification is based on the primary factors believed to affect competitiveness.
4. Covers the 14 OECD countries for which data is available.

Source: OECD, STAN database.

Figure 43. **INTERNATIONAL COMPARISON OF R&D EXPENDITURE**



1. In 1994.
2. Higher education, private non-profit and foreign.
3. In 1993.
4. R&D expenditure relative to value added in manufacturing.

Source: OECD, STAN/ANBERD, STIU databases.

spending over the 1982-1993 period. In the process, the R&D intensity of Swedish manufacturing industry, taken as an aggregate, has become the highest in the OECD area (Panel B). This is because certain product groups are very R&D intensive, with 85 per cent of the R&D spending occurring in four product areas, all dominated by large firms: electro-technical products, transport equipment, pharmaceuticals and machinery.[77] Telecommunications and pharmaceuticals increased their share of total industrial R&D from 25 to 42 per cent between 1983 and 1991.

Export and import structure

The composition of merchandise exports and imports closely reflects the structural features of Swedish industry (Table 22, Panel A). Sweden is a net exporter of industrial goods and machinery and a net importer of primary goods, except for forest products. Swedish exports have been dominated by scale-intensive industries (Panel B), where price competitiveness plays an important role, making Swedish export performance sensitive to the external competitive position. But at the same time it has a relatively important specialised-supplier industry, which is dependent on non-price factors, such as quality, product innovation and differentiation, as well as marketing strategies. It also has an increasing export specialisation in science-based industries. However, Sweden continues to be more export-specialised in low-technology-based industries than other OECD countries except for Italy. In addition, Swedish import penetration is higher than most other OECD countries in the specialised supplier and science-based industries.

The trend has been for existing export specialisation characteristics to be reinforced. Shipbuilding, pharmaceuticals, petroleum refining, ferrous metals, wood products, electronics and plastic products had by 1989 a revealed comparative advantage, insofar as their share of exports were higher than for OECD countries on average and this advantage has become more pronounced (Figure 44, upper right hand quadrant). Export specialisation has also intensified in chemicals, other transport equipment, non-ferrous metals and electrical machinery, without yet constituting a revealed comparative advantage (upper left-hand quadrant). Many of these products are highly sensitive to price competition, although knowledge-based sectors have also gained ground. Moreover, Sweden has seen a shift towards importing goods with a high degree of human capital input with the reverse development in its exports.[78]

Table 22. **Foreign trade and industrial structure**

A. Composition of merchandise trade[1]

	Per cent of total	
	Exports	Imports
Agriculture and fish products	2.2	7.7
Energy products	2.5	7.6
Forest products	7.9	3.6
Industrial goods	33.7	27.7
Machines and equipment	45.0	37.8
Miscellaneous items[2]	8.5	15.4
Unclassified	0.2	0.3
	100.0	100.0
Total as per cent of GDP	28.8	24.3

1. 1993, SITC revision 3 data.
2. Includes clothing, furniture, photographic materials and clocks.

B. Export specialisation and import penetration by type of industry

	Per cent			
	Export specialisation[1]		Import penetration	
	1973	1993	1973	1993
High-technology	74.8	84.7	44.3	74.0
Medium-technology	84.5	88.4	37.3	64.2
Low-technology	126.1	127.1	21.6	30.6
High-wage	65.4	81.6	46.1	66.6
Medium-wage	139.7	137.0	26.3	45.2
Low-wage	72.1	65.0	21.7	37.3
Resource-intensive	79.6	89.6	20.5	26.7
Labour-intensive	60.7	58.2	29.5	51.4
Scale-intensive	131.3	121.7	29.0	47.9
Specialised supplier	111.1	106.9	37.6	68.4
Science-based	49.8	74.5	48.8	73.0

1. The index for export specialisation is above 100 if the export share of a particular industry in Sweden is above the average for total OECD.

Source: SCB; OECD, STAN database.

The last two decades have seen a decline in Sweden's merchandise export share, from 2¾ per cent to 2¼ per cent of the total OECD export market. The loss of market share was particularly pronounced in the 1980s. A detailed breakdown (Box 6) shows that Swedish exports benefited from a favourable

Figure 44. **MANUFACTURING EXPORT SPECIALISATION**[1]

1. Export specialisation (revealed comparative advantage) is the export of a particular good relative to the total export seen in relation to the same good's relative export performance for the whole of the OECD.

Source: OECD.

Box 6. **The five effects revealed in Constant Market Share Analysis**

Market composition effect – shows the effect of changes in the geographical distribution of trade on the exporting country's share of world imports. For Sweden this effect was negative over the period 1980-92, accounting for 2¼ percentage points of the total market loss of 15 per cent, as Swedish exports were concentrated on slower-growing markets.

Commodity composition effect – shows the effect of changes on the aggregate Swedish export share of changes in the commodity composition of world trade. This effect was slightly positive, reflecting a favourable commodity composition of Swedish exports in 1980, and added 1½ percentage points to the market share over the period.

Market adaptation effect – indicates the exporting country's success in adapting the geographical composition of its exports to the overall change in the geographical composition of world imports. In the Swedish case this effect was neutral over the period.

Commodity adaptation effect – indicating the exporting country's success in adapting the commodity composition of its exports to the change in the commodity composition of world imports. Reflecting the fact that the commodity structure of Swedish exports developed unfavourably, this effect was negative and of similar magnitude to the market composition effect.

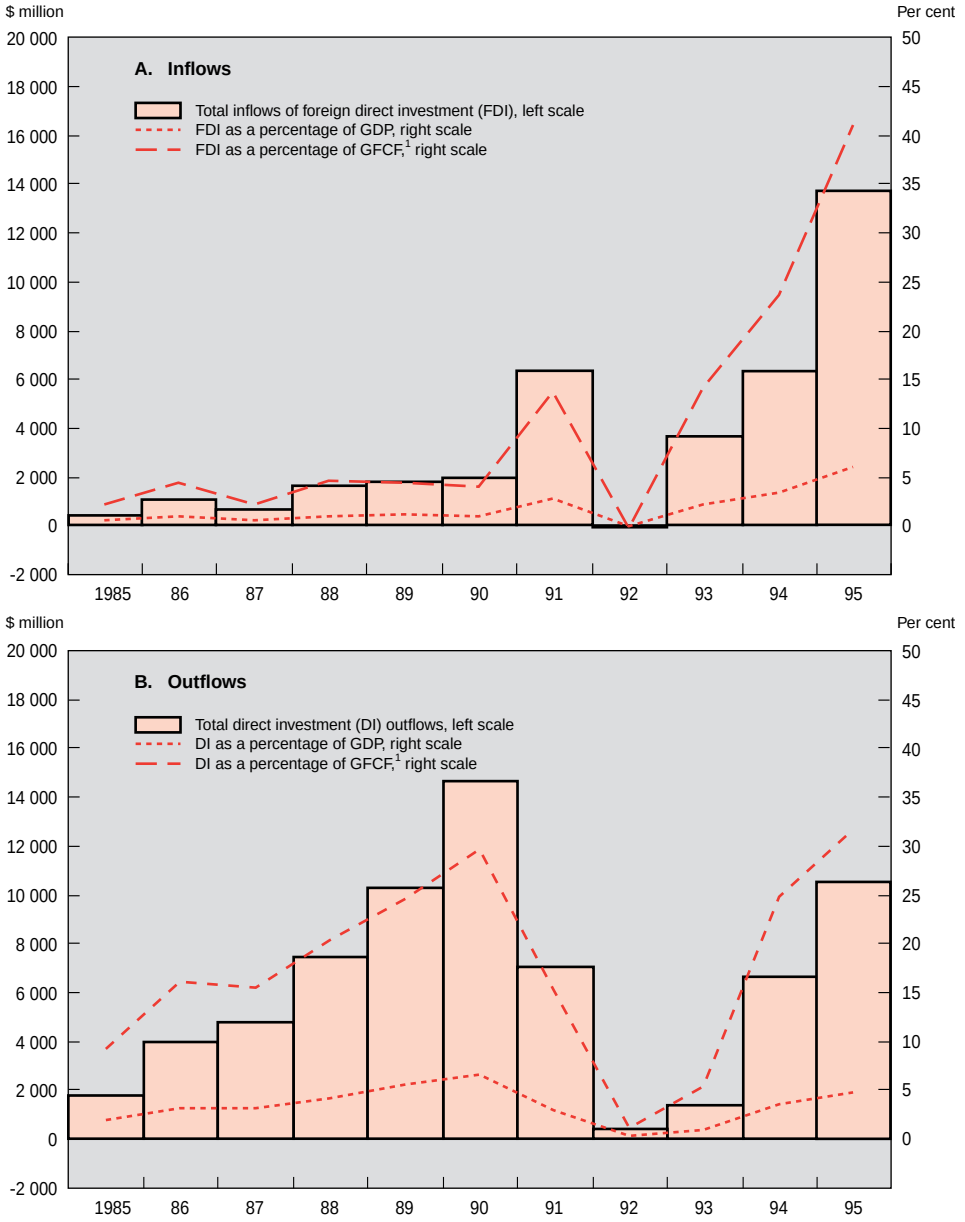
This leaves as a residual *the export performance effect* – indicating how well Swedish industry has performed on average in the different market segments. With an analysis based on values of exports and imports this effect catches both underlying trends in relative prices in world markets and the competitiveness of Swedish exports. This effect was strongly negative over the period 1980-92, explaining 12 percentage points of the total market loss.

commodity composition at the outset. This was offset by a failure to adapt to changes in the commodity composition of world trade over the period, even though the weight of faster-growing high technology exports increased.[79] Moreover, the market composition of Swedish exports has been disadvantageous. The loss of market shares partly reflects in part that Swedish companies moved production abroad[80] but also the weak external competitiveness of Swedish industry.

Foreign direct investment

While the gradual liberalisation of foreign exchange regulations that was taking place during the 1980's was fully implemented by 1989, restrictions on

Figure 45. **FOREIGN DIRECT INVESTMENT**



1. Gross fixed capital formation.
Source: OECD.

foreign shareholdings in Swedish enterprises were lifted only in 1991 as a part of adaptation to EU rules. FDI outflows, on the other hand, had been subject to a more liberal regime.[81] Sweden thus recorded a large net outflow of direct investment from the mid-1980s to 1991 (Figure 45), with a net inflow of FDI occurring from 1993 onwards,[82] when decisions as to the location of production can be seen to reflect more fully market forces such as fluctuations in the real exchange rate and international competitiveness, but also the new trade and investment regime following from Sweden's EU membership.

Between 1986 and 1990, 80 per cent of FDI outflows were invested in the EU, as Swedish companies positioned themselves for EU's internal market.[83] Swedish affiliates in the EU countries recorded solid gains (between 2 and 7 per cent) in value added per employee, while the corresponding affiliates in Sweden saw declines of around 1.5 per cent.[84] Employment in the foreign affiliates grew by 200 000, while employment in the part of the companies located in Sweden fell by around 80 000. As an offset, foreign-owned manufacturing firms increased their share of total manufacturing employment from 11 per cent in 1986 to 19 per cent in 1995.

The role of policies

The framework for competition

Sweden has traditionally used a liberal *trade regime* to promote domestic competition, as a counter-measure against tendencies to high concentration in many industries. In the thirty years prior to joining the European Union, Swedish trade was governed by membership of the EFTA. This framework ensured that by 1993 Sweden had among the lowest barriers to trade among the OECD countries.[85] Nevertheless, for a wide range of consumer categories, and for consumption in general, prices in Sweden have been higher, in PPP terms, than in the EU as a whole (Figure 38). EU membership should on the whole serve to reduce this differential. Sweden can no longer claim special rules for environmental, safety and security reasons. Moreover, as part of the EU harmonisation programme, new construction regulations have been implemented that increase both the range of possible technical solution to a given problem and the possibilities for entry of foreign enterprises. However, the EU trade regime is less

permissive than previous Swedish practices, particularly in the areas of agricultural policies, car imports, textile imports and anti-dumping rules. The state has also retained its monopoly on retail sale of alcoholic beverages, for which the former direct control of imports and wholesale trade has been replaced by a licensing system.

Since 1993, the Swedish regulatory framework concerning domestic competition has been in compliance with that of the European Union, with its emphasis on the prohibition principle (see Box 7). The responsibility for implementing the regulations rests with the Swedish Competition Authority (SCA), which also administers the EU-inspired merger rules and advises the government on competition-oriented measures to enhance public sector efficiency. The present framework contrasts with the previous emphasis on realising economies of scale and scope, and will increase competitive pressure. But incumbents are often successful in fighting off new entrants by exploiting their market power and by cross-subsidising sectors facing competition with profits from monopoly market segments. There thus has to be a learning process, in which the regulatory framework is gradually adapted to market structures and developments, as has also been seen in recent legislative reviews to promote competition within the food sector, both in production and distribution, and in the construction sector. Sweden has had very few court cases to establish a precedent and a basis for interpretation of the regulatory framework. A summary of recent rulings with respect to horizontal and vertical integration and abuse of dominant positions is attached in Annex III, indicating an activist stance with respect to opening markets to competition. But there is a considerable time-lag and effort involved in adjusting the competition laws and other relevant legislation, such as labour laws, rent controls, bankruptcy rules, environment legislation and regulation governing the financial sector.

The regulatory framework underpinning competition is particularly important for the development of the service sectors. These sectors tend to be either highly fragmented or concentrated due to natural monopolies or oligopolistic markets.[86] Retailing, restaurants, road transport and personal services may often be characterised by atomistic competition, but with the degree of competition determined by government regulations, such as planning and zoning restrictions, or by rules imposed by professional organisations and associations. For instance, the distribution sector in Sweden has been regulated with locally implemented

Box 7. **The Swedish regulatory framework to enhance competition**

The legal framework for competition policy changed on 1 July 1993 as the new Competition Act entered into force, abandoning the abuse principle in favour of the prohibition principle to comply with the EU regulatory framework. The Swedish Competition Authority (SCA) had been reorganised the year before with a mandate to supervise, implement and enforce the new law, with a staff of around 125. The Competition Act prohibits abuse of dominant positions and agreements restraining competition. Agreements are not in restraint of competition if market share is less than 10 per cent and if annual turnover for each of the involved firms is less than SKr 200 million (comparable EU thresholds are 5 per cent and ECU 300 million). In cases where the involved parties are unsure whether their agreement is in violation of the Act they can apply for a negative clearance, in which the SCA states that the agreement on ''known facts'' is not in conflict with the Act. If the SCA has not responded to a notification within three months an automatic clearance is granted for five years.

Certain provisions allowing restrictive agreements exist, namely if the benefits outweigh the negative effects and if available information does not warrant a conclusion that the agreement will infringe upon the rules of competition. Moreover, time-limited block exemptions exist for exclusive distribution and purchasing agreements, motor vehicle distribution and servicing agreements, specialisation agreements, franchising and retail distribution chain agreements (the only exemption not included in EU rules), and agreements concerning R&D. Moreover, special rules apply to the agricultural sector where co-operative activities are deemed to yield benefits outweighing the cost of impairing competition.

SCA also administers merger control rules, which stipulate mandatory notification of all mergers involving turnover of more than SKr 4 billion. Mergers that create or strengthen a dominant position may be prohibited if they have adverse negative long-term effects on competition. EU cross-border mergers fall under the rule stipulated by the EU, which comes into force when combined world-wide sales are more than ECU 5 billion and EU sales are at least ECU 250 million for each of at least two parties (unless each of the undertakings has more than two-thirds of its sales in one EU country).

The SCA can rule violating agreements null and void. Moreover, infringement can lead to penalty payments and fines of up to 10 per cent of the annual turnover of the undertaking varying with the seriousness and duration of infringements. The Act also opens up the possibility of legal action through civil courts by parties harmed by an infringement.

The Competition Act stipulates that in addition to the above tasks the SCA should initiate research and actively propose changes in administrative regulations restricting competition as well as propose competition-oriented measures to enhance government sector efficiency. In cases concerning competitive tendering the SCA is empowered to take improper practise to the Market Court. Such improprieties can either be discriminatory practises or other practises that distort conditions for competition.

building laws restricting the number of shops in local areas. In recognition of the detrimental effects on competition, this type of regulation have been repealed, even though change is delayed insofar as leading distribution chains enjoy exceptions from the new, stiffer competition law, and this sector continues to rank very low in international productivity comparisons. Industries such as public utilities, telecommunications, railways and air transport, on the other hand are generally characterised by oligopolistic (segmented) market structures as a result of high sunk costs. These may stem from infrastructure investments, economies of scale or network externalities, which require an appropriate regulatory framework to permit adequate competition.

Policies with respect to the regulatory environment and public ownership can thus have important effects on competition and economic performance. Furthermore, a large government involvement in the economy tends to lower the overall competitive pressure as the government sector is often the sole or the dominant agent in any given market. In the Swedish case this makes the interaction between the private and government sector an area of particular concern. Such interaction can take many different forms; for example, it can be private actors benefiting from special treatment by the government sector, such as long-term contracts with automatic renewal, or government units competing directly with private companies, but with the former often receiving preferential treatment of various forms, for instance as guarantees from the municipality.

Privatisation and competition

While public ownership of business sector companies has never been extensive in Sweden, over the 1991-1994 period privatisations amounting to SKr 23 billion were carried out, and the present Swedish government has embarked on a considerable privatisation programme, calculated to bring in revenues of SKr 40 to 50 billion.[87] A first step in privatisation is normally to turn the monopolies into incorporated companies, whereby the government can sell the entire company or retain a part of it in public ownership. Hence, by the beginning of 1996 the Swedish state remained the sixth largest shareholder on the Stockholm Stock exchange with holdings of SKr 30 billion out of a capitalised market value of SKr 1 210 billion. The holdings involve six companies with the state's stake in Nordbanken accounting for more than half its value.[88] The privatisation programme is focused on selling the state's interest in commercial

activities and no initiatives have been taken to persuade local governments to privatise locally provided services. Major state-owned enterprises are listed in Annex Table A2.

The main scope for further privatisation relates to the so-called network monopolies. As a first step, a number of them, including the public telecommunications company Telia, the postal service and the railways have been turned into companies. Because of ‘‘natural monopoly’’ elements associated with parts of these network industries, the transfer of ownership to private hands does not necessarily increase competition in these cases, although increased efficiency and higher returns can be expected as the incorporated company becomes more cost conscious. Hence, a separation of activities into those where competition can be introduced and those where it cannot is necessary. This has been done for the electricity market, where power generation has been deregulated to become more competitive, whereas distribution is kept within the public domain, and for railways, with transport activities separated from the maintenance of tracks. Complications arise, however, when networks are interconnected, as in telecommunications, making fair prices and free access to other networks of key importance to establish competition. An additional complication in introducing competition in industries previously dominated by a public monopoly is that the incumbent often has considerable market power, which it can use to prevent new entry.

During the 1990s, competitive forces have been given greater leverage in the electricity market. In 1992, the responsibility for electricity transmission – the national grid – was separated into a government-owned utility (Svenska kraftnät) while government-owned generation facilities was regrouped into an incorporated entity (Vattenfall AB). From 1 January 1996 a new electricity law entered into force, increasing freedom of choice both for consumers and suppliers and establishing a full separation between generation and sales activities on one hand and transmission on the other. Generators of electricity sell their product on a competitive market to distributors, who are responsible for re-sale to end-users, who are free to choose their distributor, thereby introducing competition at two levels. Reflecting the Swedish dependence on nuclear power (above 50 per cent), ownership in generation is dominated by the government, with the state owning around 50 per cent of capacity and local government another 30 per cent and only

about 20 per cent in private hands. Similar reforms have been taking place in Finland and Norway and a gradual integration of Nordic electricity markets should further competition in the sector.

Regulations protecting the government-owned railway company's monopoly of transport services have been altered, so that new entry has become possible in the intra-regional segment of the market and in inter-regional freight between certain cities, while Swedish Rail still has sole responsibility for inter-regional passenger transportation. The need to co-ordinate regional and national time schedules make inter-connectivity an important issue. With the inability of new entrants to establish themselves in the market, the competition authorities consider it unlikely that new entrants will be able to compete on equal terms with Swedish Rail.[89] It would thus seem that competition would have to be imposed on railways by developing inter-modal competition (long-distance bus transport). Following the full liberalisation of the *postal market* the incumbent (the government-owned Sweden Post) has been forced to refrain from engaging in exclusive contracts with customers and discriminatory pricing practices. The government has recognised that competition cannot be introduced in all segments of the market and it has established a range of productivity measures for Sweden Post to fulfil. These effectively rule out any price increases, but do not ensure price reductions in line with those registered in the more competitive segments of the postal market.

The state-owned *telecommunications* company (Telia) has, following the extensive liberalisation of the sector, been forced to allow competitors free and fair access to its infrastructure, ensuring that competitors cannot be forced to pay more for the use of infrastructure than Telia's customers. However, this leaves aside the whole question of Telia's pricing policy, which is not based on marginal cost pricing, since such costs are negligible.[90] Following the liberalisation of the telecommunication market Telia reorganised its price structure, so international calls – where it faces competition – became cheaper and local calls became 40 per cent more expensive, in the process reducing cross-subsidies. The government has felt compelled to regulate some of Telia services on a so-called RPI-X basis,[91] since competition can be and has been introduced in the areas of mobile-phones,[92] data-transmission and long-distance telephony, but Telia remains dominant in the local telephony segment of the market. The dominant position of Telia has moved the SCA to propose that Telia should divest itself of its customer

equipment division and mobile phone division, removing a possible source of cross-subsidising. Furthermore, the SCA has suggested that the sector's regulator be given larger powers to regulate the technical and economic aspects of the telecommunications market.

Government involvement in markets

Government-provided goods and services are generally free or heavily subsidised at the point of delivery and financed out of general tax revenues, as a result of which the link between costs and prices is removed. This reflects the nature of many publicly-provided goods and a policy of assuring universal access. However, the absence of competition in the government sector gives rise to problems similar to those applying to private sector monopolies, in the form of inflated cost levels. Given the large government sector in Sweden such cost inefficiencies have important economy-wide consequences. While the production of public services has been increasingly separated from their provision, the public sector retains a dominant role in producing a wide range of services, creating barriers to entry and hampering competition. Even where public provision is to be continued, the possibility of out-sourcing the actual delivery via competitive tendering may lead to substantial cost savings[93] and improved resource allocation. In addition, there are a number of advantages to be gained from having public companies compete on competitive markets, as they are forced to pursue cost minimisation, ensuring internal efficiency and competitive prices.

Government purchases of market-produced goods and services, which amount to more than 10 per cent of GDP, could in principle be opened to fair and equal access, with the winning bid going to the least cost producer. However, at the beginning of 1994 local governments had control (a minimum 50 per cent equity stake) of close to 1 500 companies with nearly 50 000 employees and an annual turnover of SKr 115 billion (Table 23). The strongest involvement is in housing and utilities. Swedish rules allow government providers to compete with private providers. But this access creates problems of ensuring that prices offered by government entities actually cover all costs, especially capital and rental costs; insofar as a government entity wins a contract on a price that does not cover all costs an element of tax-subsidy is involved. From the beginning of 1992 to March 1995 the SCA dealt with close to 200 cases concerning public procurement and the question of hidden price-subsidies in public providers' prices.[94]

Table 23. **Local government-owned companies**

	Number of companies	Number of employees (1 000)	Turnover (SKr billion)
Manpower and industry	36	0.2	4.9
Transport and communications	37	4.5	5.5
Parks and cultural activities	96	2.3	0.6
Energy and other utilities	223	12.5	49.8
Education	16	0.1	0.1
Social and health services	14	0.3	0.3
Property administration	673	16.5	50.0
Others	360	12.5	3.7
Total	1 455	48.9	114.8

Source: Svenska Kommunförbundet (1995).

When public entities participate in a competitive market in connection with out-sourcing of non-core activities, either through direct entry or in-house participation in competitive bidding, a range of problems arises. Given the dominant position of public entities they can pursue a predatory pricing strategy,[95] explicitly or implicitly supported by local government finances. However, to show that predatory pricing is taking place, it must first be demonstrated that the public entity in question is dominating the market. It must then be established that prices are below costs and that the purpose is to drive other participants out of the market. Additional competition problems arise with the practice of automatically renewing long-term contracts, as the market in question effectively becomes non-contestable. A further question arises as to which markets a public entity should be allowed to enter. For instance, the Swedish military services have considerable spare capacity in peace time in the area of transportation, which then has been offered to the commercial markets, such as helicopter transportation of sick people.[96] The persistence of public guarantees may carry a moral hazard problem, as the public purse functions as finance of last resort. For example, the rental market for property is dominated by housing co-operatives which have a market share of 55 per cent. These often have implicit or explicit guarantees and subsidies from municipalities. These co-operatives tend to base their rent on a cost-plus principle and not on demand and supply as is the case with private operators. The ongoing gradual withdrawal of subsidies from this segment of the housing market should thus improve competition and the overall efficiency of the economy.[97] Some issues in this area are covered in more detail in Box 8.

Box 8. **Issues arising from government involvement in competitive markets**

Predatory pricing occurs when a dominant market participant tries to eliminate a competitor or potential entrant through below cost pricing. The lowering of prices as a result of predatory pricing can be difficult to distinguish from effects of ''normal'' competition, which makes the issue of the dominant position an important element in assessing whether predatory pricing has occurred. The EU courts define this under normal circumstances as a minimum 30 per cent market share. Concerning the burden of proof, the EU practice is that if the price is below average costs but above average variable costs the EU has to prove the case. On the other hand, if the prices are below average variable costs the company has to prove that no intent of predatory pricing was involved.

Public subsidies often take place as a direct transfer to a private entity for providing a certain service, which insofar as the market is contestable follows the legal rules concerning predatory pricing. Transfers between two entities or from public administration to an independent, but publicly owned, entity are only in breach of the law if the purpose of the predatory prices is to eliminate another market participant. Cross-subsidisation – transfers within one entity – as such is illegal. Public entities that are active on several markets and/or have monopoly on one market but not others are required to keep separate accounts.

Additional general rules are in place for governing market activities of municipalities. Their activities must take place within their local area or serve the local population. In particular, local authorities are not allowed to enter traditional business activities. Furthermore, speculative activities are forbidden and activities, while required to be self-financed, are not allowed to generate profits. Moreover, all members of the local community must have equal access.

The possibility of conflict of interest exists when a public authority is offering a given service for competitive tender and its own in-house provider participates. Often the experts that evaluate the incoming bids are also participating in the calculation of the in-house provider's bid. This can lead to below-cost estimates of the in-house bid. This is a particular concern in cases where bidders have been allowed to re-issue their bids.

Public guarantees pose two set of problems. There is a moral hazard problem as inappropriate actions by the recipient are covered by the public. Moreover, they prevent more efficient private competitors from exposing an inefficient public producer to competition for two reasons: the public provider cannot go bankrupt and the public authority may change regulations and procedures to avoid bankruptcy of the public provider.

In sum, Sweden has been going through a transition similar to that experienced in other OECD countries with respect to privatisation and regulatory

reform. The privatisation process being relatively new, a full assessment is difficult. However, it appears that Sweden has addressed many of the issues involved with respect, *inter alia*, to inter-connection and abuse of dominant positions. The competition authorities have been actively implementing the competition laws, although given the lack of court decisions a full assessment of the impact of this is not possible. Moreover, the regulatory framework has a strong pro-competitive stance. For the competitive situation as such, the largest main remaining problems are found in the interaction between municipalities and private providers in market-oriented activities and in private sectors such as agriculture, food distribution and construction.

Government support for the business sector

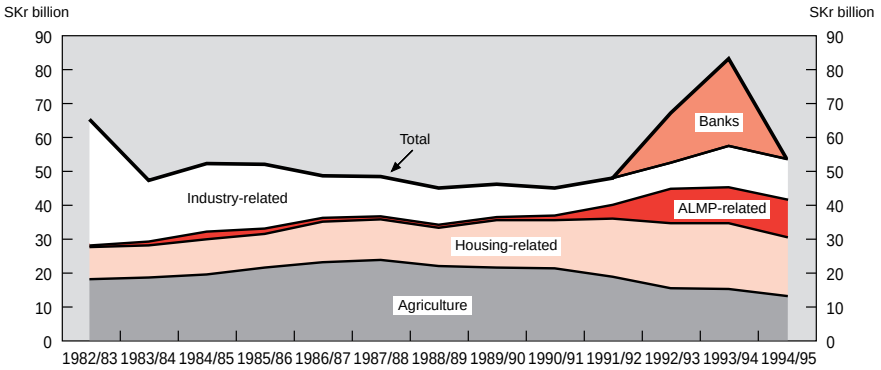
The range of government support

In Sweden, government support to the business sector has often been motivated by equity considerations, benefiting the housing sector and agriculture in particular, but regionally differentiated support can also be viewed in such a perspective. Business support has also been an integral feature of demand management policies, motivating not only transfers to industry in the face of negative demand shocks but also tax breaks for investment fund allocations which were released in cyclical downturns. With the reorientation of economic policies in the mid-1980s, the focus has turned to the promotion of structural change, and programmes directed towards small and medium-sized enterprises have, in general, been motivated by the failure of the capital market to make adequate financial provision for this segment of the economy.

Classification of support programmes according to objective is difficult in Sweden, since programmes tend to develop a life of their own and objectives change over time. In general, it would appear that the emphasis of support is still on maintaining profitability and employment in selected branches, rather than on fostering structural change, insofar as programmes directed towards R&D and SMEs constitute only 1 to 2 per cent of the total government support[98] (Figure 46). Moreover, the overall level of support has remained stubbornly high over the last ten to fifteen years, close to 5 per cent of GDP in 1993/94, and any variations seem primarily to reflect the business cycle. It should be noted, however, that while a half of business support in the early 1980s was given to manufacturing, direct support to this sector has since fallen continuously to about

Figure 46. **LEVEL AND COMPOSITION OF GOVERNMENT SUPPORT**[1]

SKr billion, 1993/94 prices



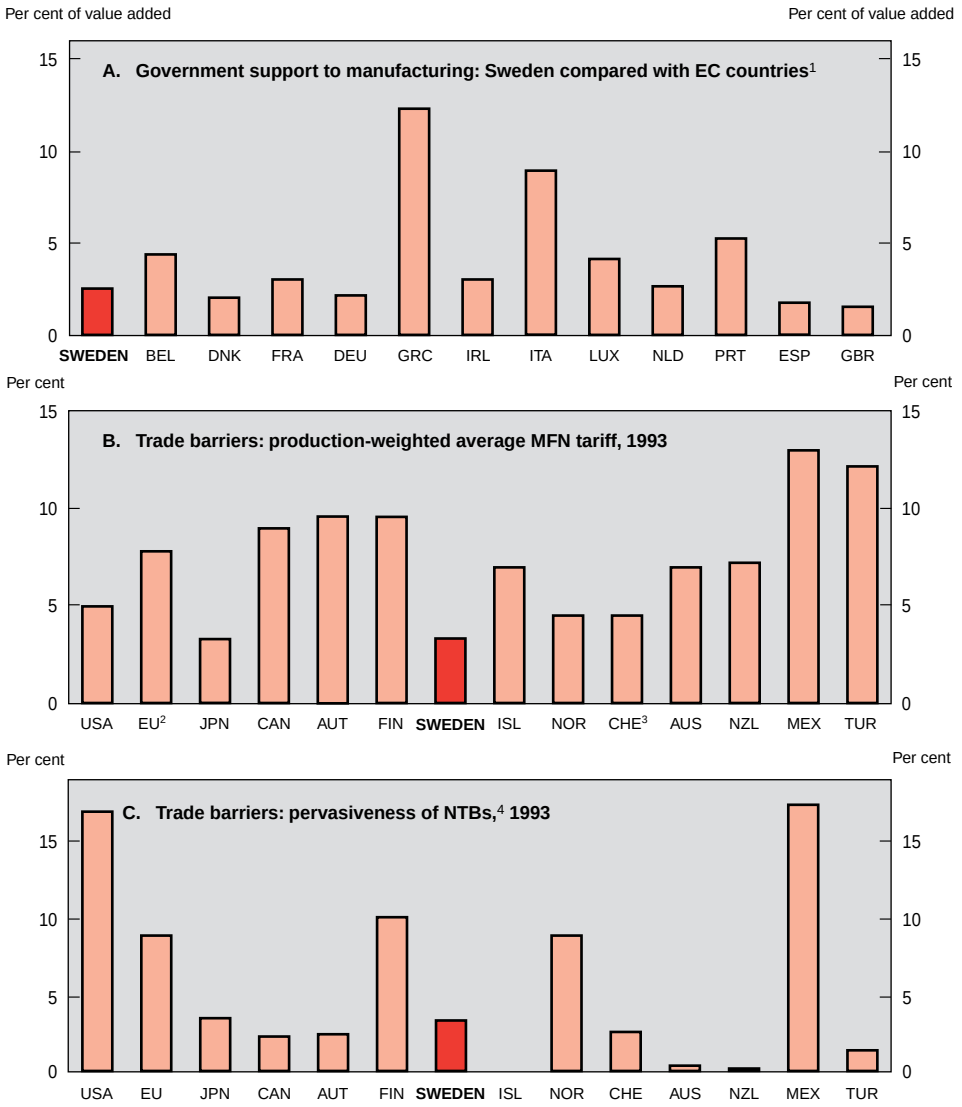
1. Includes subsidies and indirect support through market regulations. Tax expenditures are only included in a few cases. Agricultural support 1994/95 reflects the regime applying prior to EU membership.

Source: Barkman and Fölster (1995); OECD estimates for 1994/95.

10 per cent of the total. The increase in government assistance in the 1990s can be attributed to the support given to the banking system,[99] which was undertaken as a stabilisation measure, although a contributing factor to the crisis may have been the moral hazard generated by a system relying on implicit government guarantees rather than deposit insurance. Overall government support is set to decline as banking support has been dismantled with the removal of the government guarantee for banks and the establishment of a deposit insurance scheme.

On the whole, by international standards support to business in Sweden is modest. This applies not only to direct support to industry, but also to barriers to trade which, prior to Sweden joining the European Union, were among the lowest in the OECD area, both with respect to the level of tariffs and the pervasiveness of non-tariff barriers to trade (Figure 47). Indirect subsidies are, nevertheless, apparent at the local level, as noted above. The Swedish Competition Authority has pointed out problems in the areas of training and adult education, entrepreneurial support (starting own companies), industrial development consulting, outdoor public pools, within the hotel and restaurant sector, property management, translation services and adult dental care. Moreover, the wide array

Figure 47. **BUSINESS SUPPORT IN AN INTERNATIONAL PERSPECTIVE**



1. 1990-92. Supports on the basis of net costs to government. For Germany the western Länder only.
2. German weights.
3. Import-weighted.
4. Core NTBs import coverage ratio.

Source: EU Commission Services: Fourth Survey of State Aids; OECD.

of support schemes for a variety of purposes – in the early 1990s small and medium-sized enterprises could avail themselves of the services of 530 bodies, many often locally based, in the areas of advice, training and finance – could, in effect, have introduced distortions where only a few enterprises have access to these schemes.

Policies towards SMEs

In general, Swedish industrial policy has traditionally worked to consolidate and promote the expansion of large enterprises. Competition policy focused on promoting economies of scale, while the tax system, capital market regulations and the foreign direct investment regime have combined to favour large capital-intensive enterprises.[100] Both financial markets and tax policy are now in general more neutral with respect to enterprise size (see below), but as noted small and medium-sized enterprises rarely expand into larger enterprises. The full employment-generating power of small and medium-sized enterprises (SMEs) thus remains unexploited. Several issues may be involved here. First, the growth of SMEs is usually associated with a structural shift from manufacturing towards service sectors. Second, technological changes have been seen to favour small companies, lowering the minimum efficient scale of operation and placing a premium on flexibility and rapid adjustment. Increasing demand for customisation and specialisation should also increase the niche production opportunities for SMEs. The causes of the dearth of SMEs in Sweden could thus, in principle, be traceable to impediments to entry and expansion in the service sector, a weak private placement market, inadequate technological diffusion or a combination of these.

Sweden, like most other countries, offers SMEs a wide range of support with respect to investment, exports, technology, R&D, management and education, consultancies and environment issues. In all there exist 140 types of subsidies with an additional 110 available from the EU.[101] Financial support can also be obtained at county level. However, it should be noted that economic development depends neither on the dominance of large enterprises nor on SMEs, but rather the balanced development of both.[102] In this respect, it would appear that those Swedish regions with the soundest economic base, having a mixture of large and SMEs firms, have been the most successful in terms of new enterprise creation. Moreover, policies to support the creation of new SMEs appear to be

effective only in the manufacturing sector; service sector start-ups are more dependent on market conditions. Thus, while support to SMEs may be seen as a way of correcting the negative effect of policies pursued in the past on the size distribution of Swedish enterprises, such support play only a supporting role in the development of this sector in comparison with initiatives to remove the impediments contained in a biased public procurement process, restrictive labour market regulations and a tax code which may interact to discourage investment in both human capital and technological diffusion.

Technology diffusion and government support to R&D

As noted above, expenditure on R&D in Sweden is relatively high and has been growing rapidly. Private expenditures form the bulk, with government R&D spending rather low compared with other OECD countries (Table 24).[103] Government expenditures for business sector purposes are channelled to industrial research institutes, which have a 40 to 60 per cent financing split between government and industry. While tax allowances are less generous than the OECD average, the government provides indirect support to R&D spending through loans to industries. Various government funds have been created to support private research, the National Board of Technical Development being created in 1968 to promote and co-ordinate the Swedish research effort. A number of areas,

Table 24. **R&D expenditure by sector of performance**[1]

	Per cent				
	Total[2]	Business sector	Government	Higher education	Private non-profit
Sweden (1993)	3.3	71.1	4.0	24.7	0.2
Sweden (1981)	2.3	63.7	6.1	30.0	0.3
Canada	1.5	54.4	17.9	26.4	1.3
France	2.4	62.1	20.1	16.4	1.4
Germany	2.5	66.9	15.2	17.5	0.4
Japan	2.9	71.1	10.0	14.0	4.9
United Kingdom	2.2	65.9	13.8	16.5	3.8
United States	2.7	71.8	10.0	14.8	3.4
Total OECD	2.3	67.4	12.7	17.1	2.9

1. 1993.
2. Per cent of GDP.
Source: OECD, STIU database.

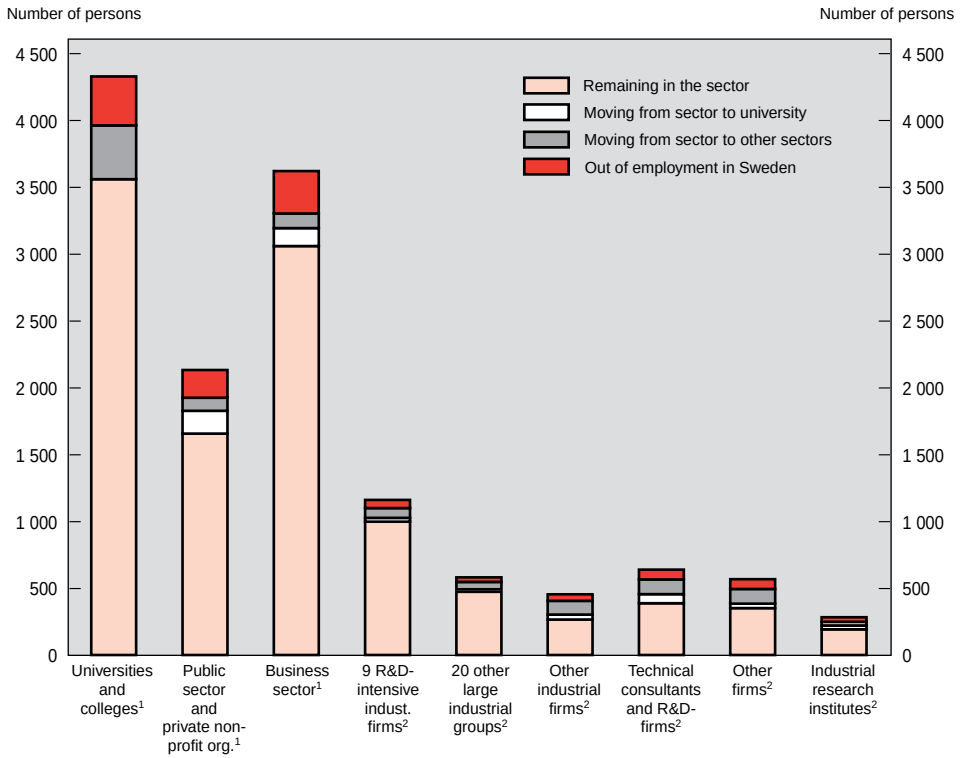
such as space technology, energy research, telecommunication services and micro-electronics, have received special support at different times.

The Board for the Development of Industry and Technology (NUTEK) is now the Swedish government's main channel for funding of R&D. In 1993/94 this amounted to SKr 1 150 million, of which 15 per cent went to direct support to private industries, either in the form of conditional repayment loans or capital in return for royalty with an expected long-term pay back rate of around 25 per cent. The majority (70 per cent) of the loans went to only three areas: information technology; minerals, energy and manufacturing; medicine and biotechnology. Moreover, a number of programmes have dual purposes, combining support for R&D with regional development and the growth of SMEs.[104]

The high R&D intensity in the manufacturing sector may thus be partly be explained by government promotional activities. However, around 85 per cent of the total private R&D spending was undertaken by companies with more than 500 employees, larger companies being in a favourable position to undertake R&D because of their ability to appropriate the benefits through economies of scale. In 1991 around half of the total was undertaken by just ten companies. Moreover, there has been a tendency in recent years to locate R&D spending abroad, the proportion reaching 19 per cent in 1990, albeit with great variations across industries. Foreign companies have increased their share of Swedish industrial R&D spending, from 8 to 17 per cent over the 1985-91 period. Overall, the impression is that R&D spending reflects the dualism of the Swedish economy, a dynamic and progressive multinational sector contrasting with a relatively stagnant small business sector, and government intervention has not served to modify this.

Impediments to the diffusion of R&D may be important in this respect. Government spending is mostly directed towards universities and defence-related research. But whereas the outward mobility from universities and colleges to other sectors is high (Figure 48), there is almost no inward mobility to these institutions. Moreover, there is little outward mobility from private firms to other sectors.[105] An additional aspect of human resource use in Sweden within the context of R&D is the very low share of researchers in government funded research institutions other than universities and colleges. Recent reforms have increased the number of such institutions to 15 with the active involvement of 600 private companies. Several government initiatives are now aimed at address-

Figure 48. **MOBILITY OF ENGINEERING PhDs LICENTIATES**
1990-93



1. Excluding "internal" flows.
2. Including "internal" flows.
Source: Stenberg *et al.* (1996).

ing these problems through joint public and private research and technology projects. These efforts may improve the diffusion of knowledge.

The role of financial markets

The process of financial market deregulation, first for internal and then for external transactions, was broadly completed in 1989. There is still, however, a legacy from the long period of financial market regulation, during which credit

was allocated to areas of high political priority, such as housing finance, whereas consumer credit and lending for business finance was restricted in order to maintain overall credit growth within ceilings deemed consistent with macroeconomic balance. An important vehicle for this policy was the National Pension funds (*AP-fonden*) with an investment policy subject to politically determined guidelines. These funds were accumulated in the 1960s and early 1970s, and by channelling them through ordinary credit intermediaries, the National Pension funds have played a significant role in the Swedish credit market up to the present day (Table 25). Though gradually being allowed to participate in equity financing through the establishment of additional funds, the inherent low-risk profile of their operations may have left a bias in favour of debt- rather than equity-financing. Moreover, the allocation of loans under credit rationing favoured established entities, generally larger companies, as these have had a long-standing relationship with financial institutions and have been able to furnish collateral for loans granted.

Table 25. **The domestic credit market**

Shares of total lending to the non-financial sector, per cent

	1980	1985	1990	1994
Banks	37.3	32.2	36.6	27.7
Mortgage institutions and credit companies	23.2	25.9	32.8	35.3
National insurance companies	8.0	9.9	7.5	11.3
Finance houses	3.2	4.6	4.5	2.7
Central government and Sveriges Riksbank	20.2	19.2	12.8	14.4
National pension insurance fund	8.2	8.2	5.7	8.6
Total	100.0	100.0	100.0	100.0
Relative to GDP (per cent)	157.7	175.2	207.8	209.7

Source: SCB.

Smaller enterprises are more dependent on outside financing in periods of weak earnings than larger ones (Table 26). For manufacturing, in particular, new equity issues played a far larger role in financing business investment in large enterprises than in small, with new equity issues amounting to about 5 per cent of net investment for medium-sized enterprises (20 to 49 employees) and more than

Table 26. **Financing of investment for different enterprise sizes**

Share of total, per cent

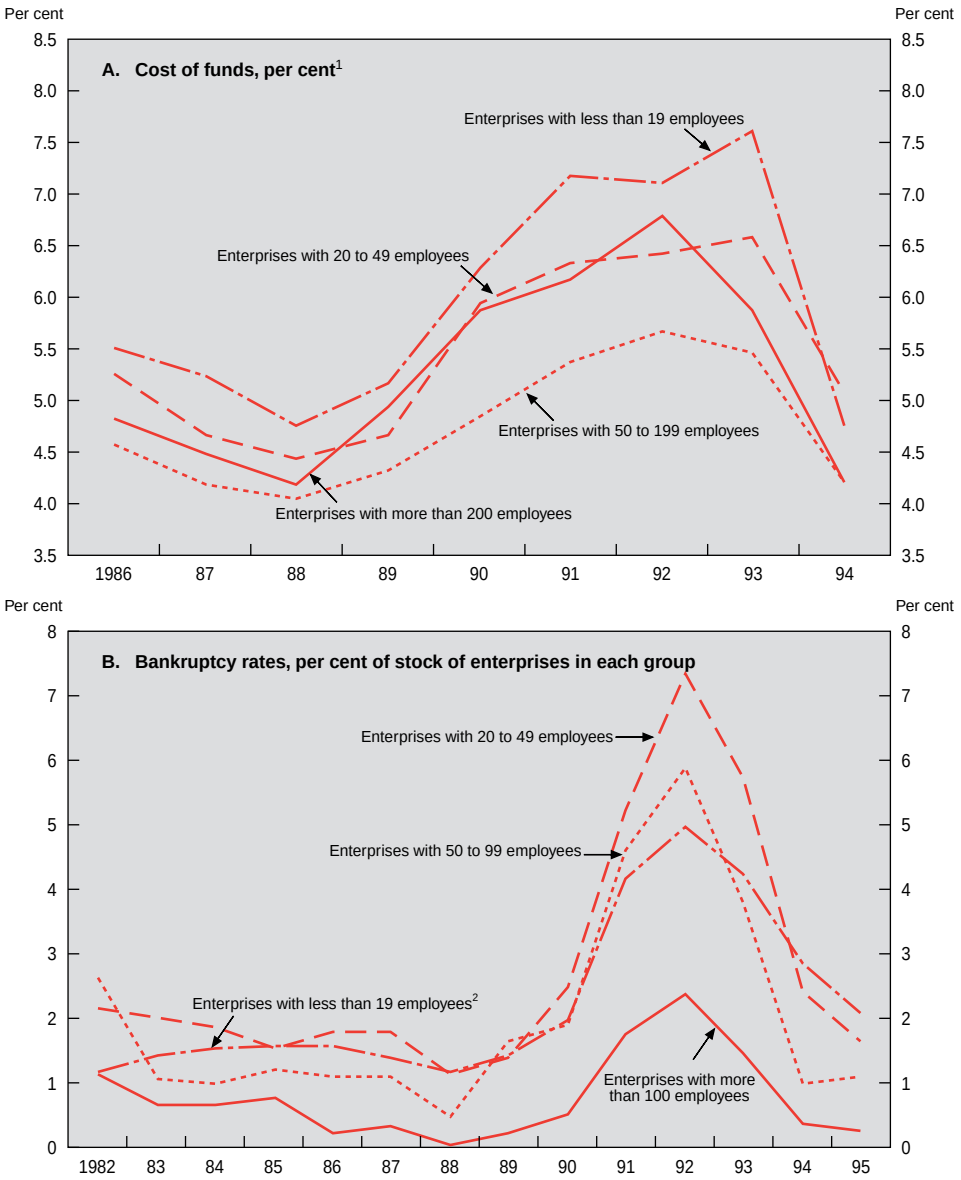
Source of funds	All companies		0-19 employees		20-49 employees		More than 200 employees	
	1980-83	1984-86	1980-83	1984-86	1980-83	1984-86	1980-83	1984-86
Retained earnings	61	69	49	76	58	61	66	67
New share issues	9	9	3	5	3	5	13	11
Loans	30	23	48	19	39	34	21	22
Total	100	100	100	100	100	100	100	100

Source: Oxelheim (1995).

20 per cent for enterprises with more than 200 employees. The dependence of small enterprises on debt finance reflects a host of factors restricting their access to securities markets. One is the direct cost associated with an equity issue, a cost that is generally independent of the size of the issue. Another is the lack of information investors face when small companies enter the market, resulting in a premium on their issues.

Evidence seems to bear out a change in the underlying riskiness of lending to smaller enterprises associated with a widening gap between the cost of funds for small and large firms (Figure 49). While overall bankruptcy rates remained virtually unchanged during the 1980s, they increased sharply with the economic downturn in the early 1990s and have yet to return to pre-crisis levels. Moreover, even during the 1980s there was a rise in bankruptcy rates, with smaller and medium-sized firms becoming relatively more bankruptcy-prone. This tendency has become even more conspicuous during the 1990s. With the fall in interest rates, the overall default risks should gradually fall, but a higher gearing ratio will probably ensure that smaller enterprises continue paying a higher risk premium than larger enterprises. At an aggregate level, it would appear that the macroeconomic imbalances in the first half of the 1990s have imposed a cost on the enterprise sector insofar as corporate bond rates have tended to follow government bond rates, and as the spread between borrowing and lending rates in banks has remained high (see Chapter II). As these imbalances are corrected, financing costs of enterprises will fall.

Figure 49. **FINANCING COST AND BANKRUPTCY RATES**



1. Interest payments. Currency gains/losses have been excluded.
2. Unweighted average of rates in five subgroups.

Source: Submission from Professor B. Gandemo, University of Lund; SCB.

Taxes and allocation of capital

Prior to 1991, the combination of high inflation and a variety of tax deductions and credits for specific purposes entailed significant distortions to business-sector rates of return according to the mode of financing (retained profits, equity issues or debt) and the mode of ownership (tax-exempt institutions, direct by households or indirect through insurance companies). With the tax reform of 1991 the most distorting features of the tax system were eliminated, with capital income, both for households and enterprises, initially being taxed at a uniform rate of 30 per cent (later reduced to 28 per cent for enterprises), and the tax base broadened as tax deductions and tax credits were cut back. With respect to the composition of household assets, the dispersion of marginal tax rates according to mode of saving has been reduced, although in an international context Swedish capital income taxation from the household perspective is relatively high (Table 27, Panel A). Moreover, although the 1991 reform and later modifications to the tax system alleviated the advantage previously attaching to debt finance relative to other modes of financing, and reduced the unfavourable treatment of households as investors, the effect of double taxation of dividends has been to encourage some forms of ownership at the expense of others. With the exception of 1994, when double taxation of dividends was temporarily abolished, the system has discriminated against holders of equity (Table 27, Panel B).

The leverage of Swedish industrial enterprises remains high in an international comparison and the role of households as purveyors of equity capital to Swedish enterprises is low. These biases may have particularly affected the small and medium-sized enterprise sector, as it may be argued that it suffered disproportionately under the old framework. Features of the tax system such as tax deductions for allocations to investment funds and accelerated depreciation schemes favoured established or fast-growing capital-intensive firms, thus reinforcing the capital market advantage that such firms have in offering collateral for loans. Special provisions have been made to offset some of the disadvantages of new and smaller enterprises, but such enterprises are often intensive in human capital and their profit base is often weak; moreover their capacity to take advantage of a detailed tax code is in general limited. It has thus been calculated that small family-owned enterprises with fewer than 20 employees in the mid-1980s faced an effective corporate tax rate 3 to 10 percentage points above other enterprises.[106]

Table 27. **Capital income taxation**

Per cent

A. Marginal effective tax rates on different types of capital income for medium income earner[1]

	1985	1990	1991	1994	1995
Current income					
Dividends	82.3	82.3	45.7	29.2	56.2
Government bonds	126.0	122.7	86.7	83.3	83.3
Interest payment deductions	–93.2	–84.0	–64.0	–62.0	–62.0
Non-current income					
Pension claims	0.0	0.0	21.9	21.9	35.0
Life-insurance claims	24.5	44.5	70.3	60.0	76.3
Capital gains on corporate shares[2]	57.9	58.5	41.4	36.6	43.3
Owner occupied houses[3]	86.2	75.8	52.4	44.6	44.6

1. Real interest rate = 3 per cent; inflation = 4 per cent; risk premia: on corporate bonds and owner-occupied houses = 2 per cent; on shares = 7 per cent.
2. 10 years holding period.
3. 15 years ownership, 100 per cent equity financed.

B. Overall effective tax rates[1]

	1985	1990	1991	1994	1995
Debt finance					
Households	88.2	84.2	49.1	48.0	43.5
Foreign investors	–73.4	–79.5	–41.5	–36.8	–48.8
Pension fund	–73.4	–79.5	–16.0	–19.6	–8.6
Life insurance fund	–15.6	–19.7	34.9	1.5	23.5
Retained profits					
Households	57.9	58.5	41.4	36.6	38.4
Foreign investor	35.9	37.4	18.5	25.9	14.9
Pension fund	35.9	37.4	24.3	27.8	24.4
Life insurance fund	42.0	44.4	39.0	29.8	33.8
New share issues					
Households	82.3	82.3	45.7	29.2	55.3
Foreign investor	36.8	37.8	22.8	40.7	35.9
Pension fund	20.3	21.6	16.2	30.0	37.0
Life insurance fund	38.6	39.5	43.5	33.4	50.4

1. Aggregate marginal effective tax rates on corporate income at investor level and with different modes of financing. Real interest rate = 3 per cent; inflation = 4 per cent; risk premia: on loans = 2 per cent; on shares = 7 per cent.

Source: Norrman (1996*a* and 1996*b*).

Investment incentives were far less generous for the self-employed than for incorporated enterprises until this difference was modified in 1991. A complete review of the taxation of self-employed, with the aim of neutralising differences

in tax treatment of self-employed and incorporated businesses, resulted in a new tax regime entering into effect in 1994. As a result, the real cost of capital is now lower for the self-employed than for a closely held corporation for all financing alternatives (Table 28). This has been achieved at the cost of introducing a more complex tax scheme for the self-employed. The capital income of the self-employed is imputed through a complicated formula and taxed at 30 per cent in line with other capital income. 25 per cent of the remaining income may be allocated to an income equalisation fund with a possible five-year tax credit, a rule also applying to incorporated enterprises. A parallel to the possibility for a corporation to expand through retained (single-taxed) profit is offered by allowing allocations to an expansion fund. The remaining income – if any – is taxed as labour income and subject to payroll taxes and the ordinary (progressive) income tax schedule.

Table 28. **Real cost of capital for personally-owned enterprises and closely-held corporations**

Per cent

	1991		1995	
	CC	PO	CC	PO
Source of finance				
Loan	3.1	1.4	3.0	1.2
Equity	3.8	4.5	5.8	2.6
Retained earnings	4.1	4.5	4.4	2.6

1. CC = closely held corporation, PO = personally owned. Real rate of interest 4 per cent, 4 per cent inflation.
Source: Agell *et al.* (1995).

In sum, the Swedish tax reform has contributed to a more efficient selection of investment projects, and although parts of the tax code still are complex, the simplification of the overall structure has focused the attention of enterprises on the need to obtain a satisfactory *pre-tax* level of return. However, although the overall rate of corporate taxation in Sweden is low compared with other countries, investor taxation is high, because of high personal income taxation, and the biases caused by the previous tax regime will take time to be eliminated. These

seem to have been particularly disadvantageous to small and medium-sized enterprises. A particular problem is still posed by small and medium-sized enterprises with active owner-managers, where the decisions to save and invest are integrated and where the high taxation and social charges may impinge directly on decisions with respect to expansion even though wages and required rates of return are in line with those elsewhere (Table 29). The recently proposed targeted reduction in double dividend taxation for such enterprises is a step in the right direction, but an extension to all dividends should have high priority when fiscal constraints allow.

Table 29. **Cost structure in small enterprises**[1]

Index: Total cost for average of six major trading partners = 100

	Sweden	Six trading partners
Wages (employer and employees)	72	73
Social costs (payroll taxes)	24	15
Debt service	7	6
Capital income taxation	5	2
Required return on own capital	4	4
Total cost	112	100

1. At exchange rates of April 1996. Countries covered are Germany, United Kingdom, France, Belgium, Denmark and Norway.

Source: SAF.

Ownership, control and decision-making in Swedish firms

In addition to the effects of an unfavourable non-wage cost structure, on the evidence presented in this and the preceding chapter, the regulatory environment has had a significant influence on the structure and performance of the Swedish business sector. In particular, the close link between the trade union movement and the Social Democratic party, the governing party for most of the post-war period, has ensured that the results of collective bargaining between nation-wide confederations of employers and employees often translated into national acts or regulations, leaving employers little discretion as to contractual and work-time regulations – to the particular detriment of small companies. The tax code has

given strong incentives to investment, and competition policies have emphasised the need to exploit economies of scale and scope. Towards this end, the Swedish unions adopted a favourable attitude towards outward foreign direct investment which served to facilitate the expansion abroad by Swedish multinationals, while at the same time allowing them to retain Sweden as their base.[107] The issue then at stake is whether the corporate governance system that has arisen in this environment – the institutional and behavioural relationships between managers, owners, employees, customers and creditors – ensures a fair and efficient allocation of capital for both established and new firms.

An active and concentrated ownership

The Swedish system of ownership, control and decision-making among publicly-held companies may be seen as occupying an intermediate position between the ''arms length'' or ''market-based'' Anglo-American model and the ''relations-based'' system of governance in Germany and Japan. The stock market is highly liquid, although stock market listing is not extensive compared with some major industrial countries: about 200 corporations are listed on the Stockholm Stock Exchange. The framework of company law shows a high degree of cohesion, insofar as it has been formulated to encourage active ownership, while possible conflicts due to asymmetric information between owners and workers have been minimised by mandating employee representation in corporate governing bodies. On the other hand, ownership of quoted companies has to a relatively large degree been retained in the hands of representatives of families of owner-managers or entrepreneurs.

In line with a number of other countries, the ownership structure has changed substantially, with individual ownership of listed enterprises falling from about 70 per cent of market value in the early 1960s to 30 per cent in the early 1980s and below 20 per cent in the 1990s which – in terms of numbers of direct owners – rank Sweden very high in a European comparison (Table 30). Institutionalised holdings thus amount to 80 per cent of the total, and of these one third are held by financial intermediaries with strong restrictions on their exposure to individual shares. The outstanding feature of the last few years has been the extension of foreign ownership, as foreign exchange regulations were dismantled and restrictions on foreign ownership in company regulations were disallowed. This might have been expected to result in more widespread and fragmented

Table 30. **Ownership of quoted shares 1983-1994**

Per cent of market value[1]

	Insurance companies	Mutual funds	NP - funds	Investment companies	Non-financial enterprises	Individuals	Other domestic owners[2]	Foreign owners
1983	11	5	2	16	16	30	12	8
1986	14	6	5	13	17	25	13	8
1989	15	10	5	8	20	20	15	7
1991	15	9	6	11	22	18	13	8
1992	15	9	6	10	20	16	12	12
1993	13	10	4	7	17	17	12	21
1994	13	10	4	6	12	17	12	28
1995	13	9	4	7	9	15	12	30
Memorandum item:								
Limits on stock ownership in companies[3]	5	5	10	–	–	–	–	–

1. 1983-1989 A-list only, 1991-1995 all lists. 1991-1992 in February, otherwise end of year.
2. Public sector, banks, foundations.
3. Percentage of voting rights.

Source: SCB.

ownership and, combined with the restrictions on financial intermediaries, to lead to a stronger position of management relative to shareholders. This outcome might have been all the more probable because of the extensive cross-ownership of shares between non-financial corporations, since such ownership may be used by management to reduce the influence of owners. Investigations covering the second half of the 1980s indicated cross-ownership of such pervasiveness that the ability of ''final'' owners to exercise influence could have been impaired. This structure was pronounced in Sweden and in many other countries such holdings are either limited by law or simply not in common use. On recent evidence, however, these cross-ownership structures have much become less pervasive over the 1990s.

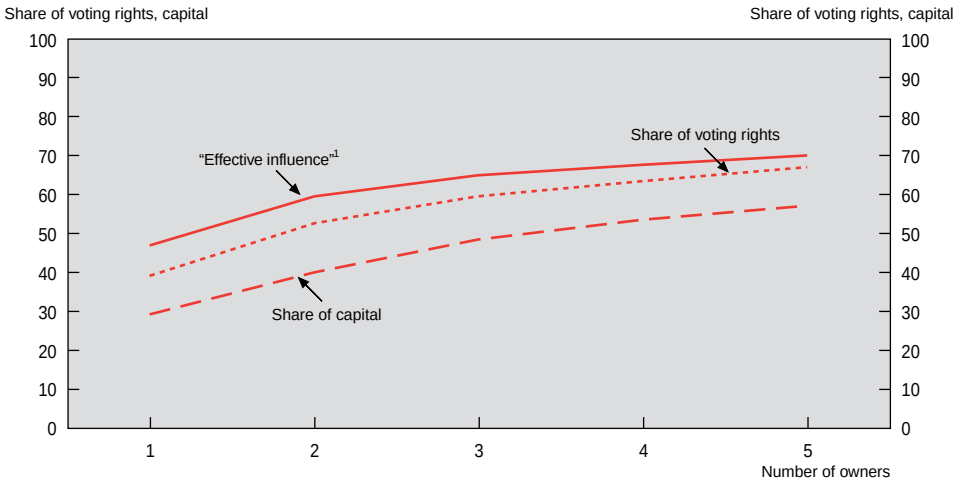
The private sector response to the diminishing influence of owners was the emergence of investment companies – also referred to as Managerial Strategic Investment Companies and themselves joint-stock companies – as financial intermediaries which undertake equity investment in selected companies, most often a minority position, and then actively monitor the performance of its portfolio companies. They may operate as venture capital funds towards closely-held start-up companies, but the main function is to establish strategic long-term minority positions in a few publicly-traded operating companies. The influence

of investment companies peaked in the first half of the 1980s, when they owned more than 15 per cent of the value of the companies quoted on the Stockholm Stock Exchange, and their influence has since then gradually diminished even though some new investment companies have been launched in recent years. This reflects their failure to attract new capital, a feature generally ascribed to the ''investment company rebate'' indicating that their market value as quoted on the stock exchange has been lower than the market value of their assets. This has discouraged investors from participating in the form of indirect ownership that investment companies represent.[108]

More important in preserving the influence of owners has been the Swedish system of dual class shares and voting restrictions, which facilitates effective minority participation and reinforces the role of active owners. Reflecting a long-standing tradition, Swedish corporations are allowed to issue shares with different voting rights, the legal restriction being a ratio of maximum 10:1 between A-class and B-class shares. (Higher ratios established prior to the present regulations may be carried forward (grandfathering) and ratios may therefore, in a few instances, reach 1 000:1.) As a result, a dominant position can be achieved with less than 50 per cent of shares issued. Evidence from the mid-1980s for the largest quoted companies indicated that a capital share of 30 per cent translated into a voting share of 40 per cent, and that in general agreements between shareholders added another 8 percentage points to the effective share of the largest shareholder (Figure 50). By 1992 the largest owner controlled on average 46 per cent of voting rights in the 62 largest Swedish corporations, and the five largest controlled on average 72 per cent. In corporations with a voting ratio of 10:1 the largest owner had a capital share of 35 per cent, but a voting power of 53 per cent.

The regulations regarding voting rights illustrate the emphasis Swedish authorities have placed on enabling an active ownership to be established in Swedish corporations. Strong minority voting rights can give leverage to arrangements between investment companies and larger shareholders in influencing corporate decisions. Conversely, they might also facilitate cross-enterprise shareholdings which reduce the influence of outside ownership. Regulations regarding voting rights could thus be seen as neutral in deciding the balance between owners and management, in the sense that any ownership constellation might develop at the expense of passive owners.[109] As part of an on-going assessment of

Figure 50. **OWNERSHIP INFLUENCE IN SWEDISH PRIVATE-OWNED CORPORATIONS**



1. Data refer to 1985. Includes the effect of shareholders' alliances/agreements.
Source: Isaksson and Skog (1988).

Swedish company law, a government-appointed commission has therefore recommended that this feature of the law should be retained; with a view to bolstering the ownership function even further it proposes that existing limitations on voting rights should be abolished and that voting rights can no longer be delegated to management.[110] More generally, however, dual-class shares and differences in voting rights raise issues with respect to the efficiency of the market for corporate control, the equality of treatment between different groups of equity holders, and the relationship between control of an enterprise and the bearing of residual risk.

The outcome of the corporate governance system is that ownership in Swedish business is highly concentrated and most quoted corporations are part of large industrial groupings or spheres of financial and non-financial enterprises. However, in Sweden control is centred on the major family names in industry, trade and finance, whereas in Germany and Japan bank-centred groups dominate[111] and other countries still have high government ownership. Insofar as dominant ownership still requires that a substantial amount of own capital is put at risk, shareholder interests, in the form of rates of return, are likely to dominate

over the possibly negative features of cross-ownership on competition. By ensuring a strong link between control and residual risk, the corporate governance system in Sweden may – for the listed segment of the business sector – fulfil one of the important objectives of a well-functioning corporate governance system in that it provides an environment where enterprises can secure financing at low cost. A system which enables owners to assess corporate profitability, monitor and influence corporate decision-making, will entail lower risk premia and thus less expensive financing than a system where information to investors is scarce and incomplete.

In some countries the rights of minority owners are protected, *inter alia*, by the so-called mandatory bid rule, which stipulates that an investor whose total of shares in a company exceeds a certain limit is forced to extend a take-over bid for the remaining shares. This rule is in force in the United Kingdom and is also integrated in the EU proposals for harmonising company law, but it is not part of Swedish company law. The argument for introducing such a rule is the possible adverse effect a change in company control may have for small investors. In respect of the Swedish case, however, it has been argued[112] that a mandatory bid rule would undermine the attachment to effective ownership of companies referred to above, because it increases the cost of obtaining or transferring effective control, and may possibly stimulate greater resort to de-listing to accomplish such changes, thereby reducing economic efficiency. In practice, major share acquisitions resulting in change of control have mostly been associated with an increase in share prices.[113] Moreover, average rates of return on equity in the largest listed Swedish companies have been relatively high by international standards. While dual-class shares and voting restrictions could, in some circumstances, give rise to conflicts between different groups of equity holders and reduce the incentive to hold minority equity shares (thus calling for offsetting protection of minority owners), these features have so far not been apparent in the Swedish case.

The market for corporate control

On the other hand, a consequence of the dual-class share system, where A-shares are not traded[114] actively in the stock market and are retained by the controlling owner or controlling coalition, is that the market for corporate control is to a large extent located outside the ordinary stock market. Takeover

negotiations then reflect the assessment of corporate prospects of two alternative majority owners, creating a possible asymmetry between the information available to the two classes of shareholders. In this respect, though, the Swedish system may not differ greatly from other countries, and it is difficult to demonstrate that take-overs have been associated with larger swings in share prices than in other countries as information hitherto extraneous to the market become embodied in prices. Moreover, premiums on A-shares seem to be in line with what has been observed in other stock markets,[115] and trade in shares with limited voting rights has in general remained brisk and the supply of new equity raised in the stock market above that of other small countries.

In sum, the Swedish corporate governance system contains a number of features that single it out from both the Anglo-American ''outsider'' model of corporate governance and the ''insider'' model that characterises Germany and Japan. There are latent conflicts with respect to the efficiency of the market for corporate control, and the relationship between different groups of shareholders and between control and residual risk. But in its present format, the system achieves a rather effective marriage of ownership and control, although at the cost of some concentration in ownership. However, one challenge to the Swedish system relates to the growing institutionalisation of savings and shareholdings, which is leaving a large part of the equity with owners who are not allowed to exercise active ownership and whose only option is the exit function. As the present pay-as-you-go system for old-age pensions gives way to a partially funded system, institutional shareholding may play an even larger role in the future as these funds may be expected to be placed as equity investment. The additional influx of capital through foreign portfolio investment in equity only adds to the importance of the issue. With the present structure of shareholdings, majority groupings of shareholders, even if underpinned by the advantage of the regulations of voting shares, are putting a substantial amount of their own capital at risk and thus have a vital interest in monitoring management and in securing a rapid reallocation of capital to more profitable uses. If the future outcome is to strengthen management at the expense of owners, a re-emergence of widespread cross-enterprise holdings of shares may be the result. In this respect, the recommendation given by the Stock Ownership and Efficiency Commission in 1988[116] to introduce regulations that limit the extent of cross-ownership may again take on greater significance. Investment companies in many ways represent an answer

to the governance problems involved, but these have had difficulties in raising risk capital since the early 1980s, and ways to address this problem should be explored together with a reappraisal of the role allocated to other large institutions with a view to allowing an active ownership function to a broader range of investors.

Implications for small enterprises

A broader range of investors may also facilitate the development of small and medium-sized companies, which are outside the corporate governance system. The failure of small Swedish companies to grow to a size where listing is practical is more a function of employment regulations than capital market inadequacies. (Although the German case shows that stock-market regulations can have important restraining effects when firms do reach this size.) The availability of capital to this segment of business may also be inhibited in a country with a relatively flat income distribution. Nevertheless, the absence of a venture capital and private placement market is a notable feature of the Swedish business scene. Governments' response to this has traditionally been either direct government participation in the risk capital market or an indirect risk sharing by implementing a tax code with a preferential treatment of particular forms of capital income. As documented above, the Swedish response has so far been a significant government intervention in the risk capital market through a wide array of instruments. The tax code has to a large extent been formulated to prevent income to be registered as capital income rather than as income from labour. This is a central concern in economies with a significant difference in tax rates for different types of income, but the consequence may have been a complex tax regime for small and newly established enterprises. A tax code that gave more generous treatment of capital income from equity investment in unquoted enterprises could both improve the supply of risk capital and pave the way for a government withdrawal from this market.[117] The proposed relief of double taxation of dividends, which has impeded the supply of equity capital is therefore a step in the right direction.

Policy implications

The Swedish economy is, in respect of the dynamism and rates of return of its larger enterprises, relatively well-performing. However, weaknesses in overall

productivity growth; falling export market shares and weak employment creation display an overall lack of vigour which extends beyond labour market rigidities *per se*. The economy is internationally very open, but it is evident that the capacity to adapt to changes in the international and domestic economy has been impaired by inadequate competition in the domestic market. More flexible domestic markets would give rise to positive market and commodity adaptation effects which would increase the diversity of its exports, making exports less sensitive to price and cost developments. In seeking greater flexibility and diversity, the general problems of a restrictive regulatory environment and heavy taxation need to be tackled. However, special problems exist at the interface of public and private sector activities and particular impediments have prevented the development of smaller firms. The public sector enjoys competitive privileges while larger enterprises have enjoyed a more favourable operating environment. To secure fully competitive domestic markets a range of issues must be addressed covering regulatory reform, further privatisation, a review of subsidies and taxes and adjustment to the system of corporate governance. A summary of OECD policy recommendations in these respects is given in Box 9.

Public procurement and in particular competitive tenders have to be (universally) opened up to all interested parties. Sweden has fully implemented EU directives, but further progress is desirable with the rules governing competitive tenders becoming more transparent and fully descriptive in terms of prices, volumes and qualities, allowing not only comparisons of offers but also the monitoring of performance and public oversight. When in-house providers participate in a competitive tender, clear accounting rules should be in place at the outset to prevent cross-subsidisation, particularly with respect to capital costs. Recognising that competitive tenders often only introduce competition in the market for time-limited local monopolies, renewable long-term contracts should as far as possible be avoided. Moreover, when a public provider competes in a given market along with private actors a problem arises because of the ability of government providers to pursue predatory pricing, as they have tax revenues as finance of last resort. The ability of a public entity to keep prices below costs magnifies competition problems during recessions, increasing the likelihood of the public entity gaining a dominant position.

While the privatisation of publicly-owned enterprises has gathered pace, incorporated enterprises fully owned by the government continue to dominate telecommunications, postal services and railroad transportation. Management are

Box 9. **Policy recommendations to improve business sector performance**

For a long time, industrial policy in Sweden worked to consolidate and promote the expansion of large enterprises. Competition policies focused on exploiting economies of scale, while the tax system, capital market regulations and the foreign direct investment regime combined to favour large capital-intensive enterprises. With the adoption of the prohibition principle competition policies have moved in an activist direction, and financial markets and the tax system are now in general neutral with respect to enterprise size. Insofar as small and medium-sized enterprises rarely expand into larger enterprises, it would nevertheless appear that the former policy regime has left behind a long-lasting legacy which is of particular importance with unemployment at unprecedented levels. The present chapter offers a wide set of policy recommendations which may act to trigger the employment-generating capacity of small and medium-sized enterprises.

I. Regulatory reform to improve competition

A uniform set of rules for competitive tendering should be established. The tendering rules should specify prices, volumes and qualities and stipulate performance measures to monitor the contract, both in terms of fulfilment of the contract and to establish rules for intervention and points of re-negotiation. Cancellation of tenders should be disallowed without compensation and automatic renewal of long-term contracts should as a rule not be allowed.

Full cost accounting rules should be mandatory for public entities participating in business activities. In particular, these rules should follow business sector standards and include capital costs. Public guarantees should be explicit and time-limited. Advisers should not be allowed to participate in tenders and in-house bidders should, preferably, establish themselves as independent (financially as well as legally) entities to reveal implicit cross-subsidies.

A longer term competition strategy for government entities engaged in business activities should be established. It should stipulate exact guidelines for what markets public entities are allowed to enter and whether they should be allowed to enter markets where effective competition exists already. In addition, it should contain rules for government entities' market behaviour.

A national independent advisory body could ensure nation-wide consistency in the implementation of rules for public procurement and competitive tenders. An additional benefit would be the centralisation of gained experience for the continuous updating of the regulatory framework. The body could co-ordinate the education at the local level. With appropriate legal powers such a body could also ensure a more uniform administration of regulation and be the first instance of appeal for negatively affected parties.

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Out-sourcing and competitive tenders provide competition for time-limited local monopolies. An alternative method for increasing competition is for the government to issue vouchers for recipients to purchase the publicly provided service on a market, allowing competition between the in-house and private producers. The recipients would search for the best combination of price and quality, thereby removing a considerable administrative burden in competitive tendering with its detailed specification. Another (less ambitious) method for assessing the competitive situation within a given market is to use yard-stick measures, comparing identical international or regional markets, to compare how well competition is functioning within a given market.

II. Privatisation

Natural monopolies should be separated from competitive activities wherever possible: A separation of the consumer equipment and mobile-phone division from other activities of the government-owned telecommunications company (Telia) should be implemented. Long-distance bus transport should be allowed in order to secure competition in inter-regional transport.

III. Subsidies and taxes

Subsidy schemes and tax expenditures should be reduced to pave the way for a reduction of the high tax level and eliminate distortions of the resource allocation. The ongoing withdrawal of the substantial housing support should be made as rapidly as macroeconomic considerations allow. Direct municipal participation in the provision of housing has served to reinforce market imbalances and competitive pricing should be adhered to.

Only those subsidies and programmes that foster structural change, such as support for R&D and SMEs, should be retained. To improve innovation and structural change in SMEs support should be streamlined, with respect both to the number of instruments and agencies involved and with respect to the linkages between public and private research efforts.

The tax code should be modified to encourage equity investment and improve the return to entrepreneurial capital in unquoted enterprises. This is a more general and more effective instrument for stimulating the supply of risk capital to SMEs than separate schemes. An important step has been taken with the proposed relief of double taxation of dividends from unquoted shares, but an elimination in full of the double taxation of dividends should have high priority when fiscal constraints start to ease. This would serve to increase the attractiveness of equity finance at all levels.

IV. Corporate governance structures

The ongoing review of company law should assess the possibility for institutional owners to exercise greater influence on corporate decisions and whether limits on cross-enterprise shareholdings should be introduced.

free to pursue business rather than social objectives, but it has nevertheless been found that as incumbents they exercise potential power to deter possible entrants. A separation of competitive and non-competitive parts of their activities seems the logical next step, particularly with respect to the telecommunications company Telia. In the case of railroad transport a wider scope for long-distance bus transport seems the most efficient way of increasing competition.

Insofar as the market structure characteristics are a hindrance for SMEs to establish themselves, regulatory reform which increases competition would facilitate the creation of new firms. Moreover, SMEs may face obstacles in other areas, such as lack of information about new technology and market opportunities, training of labour, capital and complicated legislation. Government support for these purposes, while secondary in importance to a less restrictive framework in the areas of labour relations and taxation, could be more effective if there was some rationalisation of the currently dispersed responsibilities.

In a competitive market producers will, in order to survive, be forced to acquire the most adequate technology either through their own R&D or by importing it through the capital equipment. Thus, a competitive market structure will typically ensure that the newest and best technology becomes embodied. Government support to R&D spending will result in an optimal outcome only if the market structure is sufficiently competitive; otherwise such programmes run the risk of pursuing non-commercially viable R&D. Hence, government support should be given on a non-discretionary basis, for example in the form of tax deduction for R&D spending, even if such support tends to favour large capital intensive organisations, which have the administrative capacity required to undertake research. It should also be kept in mind that the relatively low share of high-technology based industries in Sweden implies that the high R&D spending in the past was mainly exploited in foreign affiliates of Swedish corporations through FDI outflows.

Government support to manufacturing has been scaled down, and combined with a liberal external trade regime, ensures that this sector is exposed to competitive forces. However, overall support to business has remained high, particularly if tax expenditures are taken into account, and it is still geared towards maintaining production and profits in selected sectors, with only a minor share allocated to strengthen structural change. Sectors benefiting particularly from this have been housing, agriculture and the media. While EU membership sets a floor to support

in several areas, a reduction of business support would make it possible to lower the high level of taxation, with positive effects both for economic incentives and resource allocation. In the housing sector, direct local government participation in the provision of housing has added to market imbalances and efforts should be made to foster more competitive pricing.

The tax reform of 1990-91 has been important for the overall allocation of investment, in the sense that investment projects are now being selected mainly on the basis of their assumed pre-tax return. Corporate taxes are at a competitive level compared with other countries. However, high taxes at the personal level imply heavy investor taxation, which reduces investment opportunities, particularly for entrepreneurs for whom the decisions to save and invest are integrated. The double taxation of dividends has made equity finance less attractive than debt finance and contributed to the high indebtedness of Swedish enterprises. The planned elimination of double dividend taxation for equity investment in small and medium-sized enterprises is a step in the right direction, but an extension to all equity investment should have high priority when fiscal constraints ease. This would stimulate individual participation in the stock market. Increased familiarity with this kind of investment could have subsequent effects for the supply of risk capital also for SMEs. Insofar as a complex and fairly strict tax code acts as an impediment to the establishment and expansion of small and medium-sized enterprises, a more generous tax treatment of income from investment in unquoted enterprises both for outside investors and entrepreneurs themselves may be a better way of using public money than maintaining a wide array of schemes to supply risk capital to this sector. More generally, if the high taxation of labour were lowered, the tax code could be made simpler as less attention has to be paid to the need to close off windows for tax arbitrage. Finally, within the confines set by numerous labour and product market regulations, the focus of company law on active ownership has served to maintain a dynamic sector of large enterprises. Based on experience of mergers and acquisitions carried out in the past, this has not come at the expense of smaller shareholders, who on the contrary, have seen the value of their shares rise in such circumstances. At present, it would appear that no particular protection for these investors is warranted and would only make acquisition of controlling shareholdings more costly. The emphasis on active ownership might, however, imply, in certain circumstances, that limits on cross-enterprise holdings of shares would have to be enacted and that the possibility for institutional shareholders to exercise a more active ownership be strengthened.

Notes

1. Government expenditure cuts and tax increases have been concentrated on the household sector, combined with a front-loading of measures in 1995-96. This has led to an estimated reduction in household disposable income of 6 per cent over those two years.
2. If domestic demand had expanded at its trend-rate in the 1980s of (1.9 per cent) instead of the recorded (0.7 per cent) over the 1993-95 period, a conservative estimate suggests that imports would have had been almost SKr 20 billion higher.
3. There has been a considerable drop in sick leave from 8.9 per cent of hours worked in 1991 to 6.3 per cent in 1995, which accounts for about 115 000 full-time positions.
4. The monetary condition index is normalised with monetary conditions in a base period (1984-86) taken as a reference point, hence an interpretation of the index in level terms is not meaningful.
5. Real interest rates are based on yields on index-linked bonds.
6. Comparing long-term yields of Italy, Spain and Sweden with German yields from 1992 up to the third quarter of 1995, Favero, Giavazzi and Spaventa (1996), found that Swedish rates seemed to reflect purely exchange rate risk and no default risk. Moreover, on their calculations Swedish yield differentials were driven by different – although unidentified – factors from Italian and Spanish yields which were dominated by common factors (notwithstanding the formal differences in their exchange rate regime). Comparing long-term yields of European countries in the first quarter of 1996, De Grauwe (1996) concluded that markets attached a very small probability to Swedish participation in EMU from its inception.
7. The Swedish government commissioned an expert report to cover all the issues related to the effects of EMU membership for Sweden. The commission reported in late 1996.
8. It has been noted recently that the structure of the Swedish tradeables sector has little in common with the core member countries of the present ERM and that the correlation with GDP growth rates of these countries is below average, indicating that economic shocks will tend to be asymmetrical with other EMU partners. See Hansson and Sjöholm (1996). Pointing out that asymmetric demand shocks may be expected to diminish in significance with EMU, the authors also argue that supply shocks in Sweden are significantly correlated with only a few of the present EU members. Different assessments are provided by Assarsson and Olsson (1993) and Gros (1996).
9. A relevant case in point is the increased similarity of industrial structure in Austria and Germany following from the pegging of the Austrian schilling to the Deutschemark from the early 1980s. See *OECD Economic Survey of Austria* 1994, p. 34, ‘‘The two countries

initially fulfilled few of the necessary qualifications – the absence of asymmetric shocks and/or the presence of labour and capital mobility – but economic integration, increased wage flexibility and liberalised capital movements have created a situation where cyclical policy needs would normally coincide''. See also Frankel and Rose (1996).

10. The cost of capital effect of interest rate changes is thus more important than the concomitant income effect.

11. See Alexius and Lindberg (1996).

12. See *1995 OECD Economic Survey of Sweden*, pp. 27-28, for a full decomposition of the change in the budget balance from a surplus of 4.2 per cent of GDP in 1990 to the 1993 deficit of 12.3 per cent. The effect of supporting the banking sector is calculated at 3½ per cent of GDP in 1993.

13. The employment bill, presented in June 1996, containing the government's proposals for reducing open unemployment to 4 per cent by 2000, did not affect the budget balance, as educational measures were financed by diverting resources from active labour market programmes and by higher energy taxes.

14. See Molander (1992).

15. See *1994 OECD Economic Survey of Sweden*, pp. 53-55.

16. See von Hagen (1992).

17. See Ministry of Finance (1996*a*).

18. The Swedish tax expenditure accounts are based upon the outlay equivalent method whereby the tax expenditure is given by the corresponding gross transfer necessary if transfers were taxable. Amounts are calculated at the margin and adding them up, although not entirely meaningful, is done to assess their importance in different areas.

19. With the outlay equivalent method tax expenditures as calculated exceed the net budget balance effect. SKr 115 billion in tax expenditures thus correspond to a net budget balance effect of SKr 78 billion.

20. See Björklund, Palme and Svensson (1995).

21. See Hussénius and Selén (1994). Observations relate to the period 1987-91.

22. The effects on the income distribution of government-provided services in the area of health, child-care, elderly care and education are estimated in Fritzell (1994). However, no estimate on the effects on the life-cycle income distribution is given. *A priori*, one would expect the life-cycle redistribution effects to be quite pronounced in these areas.

23. See Roseveare *et al.* (1996).

24. The basic accounting rule underlying these calculations requires that the discounted value of the net tax payments of present and future generations equals the discounted value of government-provided services over a similar time horizon plus present net government wealth. The model requires that the real interest rate applied for discounting exceeds the productivity growth rate, and the size of the generational imbalance is critically dependent upon the relation between these two rates.

25. See Olsson (1995). The imbalances referred to reflect a differential between the real interest rate and the growth rate of 1 to 2 per cent. Generational accounts for Sweden published in

OECD (1995*c*) and in International Monetary Fund (1995), indicate far smaller imbalances (about 20 per cent) in disfavour of future generations.

26. Holmlund (1996*a*) has estimated that among job-seekers currently registered with employment offices, as many as 50 per cent may have been out of ordinary jobs for more than a year.

27. 1995 *OECD Economic Survey of Sweden*.

28. The recent outward shift in the Beveridge curve is far more pronounced when set out in terms of duration of unemployment and vacancy spells rather than in standard form with unemployment and vacancy rates.

29. Assarson and Jansson (1995).

30. See Calmfors (1993).

31. Responsibility for pay determination in the public sector has been delegated to agencies, with pay structure and employment to be contained within cash limits set by the government. See OECD (1996*g*).

32. See Edin and Holmlund (1994).

33. See Edin and Topel (1995).

34. See also further documentation in (OECD 1996*a*). Among low-paid, but full-time, workers in 1986, 37 per cent had moved on in the income distribution by 1991, 35 per cent remained low-paid and 28 per cent left full-time work.

35. OECD (1996*a*) concentrates on full-time workers and covers mainly the latter dimension of labour market outcomes.

36. An international comparison of labour market regulations is provided in Annex Table A1.

37. Shedding of labour in response to weak cyclical developments will normally be concentrated on low-productive workers. These will normally have problems in regaining employment, with unemployment spells acting as a negative signal to prospective employers. Stability of employment may induce firms to retain more less-productive workers.

38. The notice period is set purely according to the age of the worker and does not reflect seniority within a given firm. This leaves an older worker at a disadvantage relative to a younger worker. Older workers have in addition a shorter pay-off period for training expenses.

39. See Skedinger (1995) and Mosley and Kruppe (1992).

40. See Scarpetta (1996).

41. In a survey of Swedish employers, 70 per cent answered that they would not fire a detected shirker and 20 per cent would hardly ever dismiss such a worker. See Agell and Lundborg (1995). The authors note that in cases going to the labour court, smaller establishments have generally been allowed a greater discretion in dismissing shirking workers, due to the larger economic consequences to the enterprise.

42. Adjustment of the labour force to short-term variations in demand has, however, been strengthened by the deregulation of the activities of Temporary Work Agencies from 1993.

43. It should be noted that conditions for implementing temporary layoffs are already left entirely to the bargaining sphere, and the economic consequences of these are handled by the enterprises themselves.

44. See Aronsson and Walker (1995) and references cited therein.

45. See OECD (1995*a*).

46. See Ministry of Economic Affairs (1995). The year of comparison is 1994.

47. See OECD (1996*b*).

48. See Edin, Fredriksson and Holmlund (1994). On their stylised calculations, the tax system lowered the rate of return on undertaking tertiary education from 4.3 to 0.5 per cent in 1981, and from 6.0 to 4.5 per cent in 1991. The ''marginal'' tax on education was thus lowered from 88 to 25 per cent. It should be noted that the tax on education is zero when the tax system is fully proportional.

49. The suggestion is discussed in more detail in the 1996 *OECD Economic Survey of Denmark*, pp. 97-98. See also Holmlund (1996*b*).

50. The question remains to what extent long-term unemployment leads to a deterioration of human capital, see Pedersen and Westergård-Nielsen (1993).

51. This has been done in great detail in the OECD thematic review of ALMPs, see *The OECD Jobs Strategy: Enhancing the Effectiveness of Active Labour Market Policies* (OECD 1996), underpinned in the Swedish case by a review of the Public Employment Service, see OECD (1996*e*).

52. See Forslund and Kreuger (1995). See also OECD (1996*e*).

53. See Zetterberg (1996).

54. See Zetterberg (1996).

55. In a recent assessment, Forslund (1996) has estimated the short-term displacement effect to vary between 25 and 95 per cent, depending on type of programme and target group and with particularly high displacement for programmes targeting youth.

56. See references cited in Zetterberg (1996).

57. It has been estimated that ALMPs are the direct destination for 30 per cent of the total outflow from unemployment, see OECD (1996*e*), double the proportion in Austria and Western Germany.

58. See Carling *et al.* (1996). The effect is thus somewhat weaker in Sweden than in other countries.

59. A further discussion is provided in OECD (1996*e*). A broad set of requirements to ALMPs, based upon the experience of twelve OECD countries, is provided in OECD (1996*c*).

60. See OECD/Statistics Canada (1995). This study shows that Sweden outperforms the other countries surveyed in all dimensions of the literacy of the workforce. Estimates for 1994 give total expenses for university and adult education as SKr 35.6 billion for company training (measured as opportunity costs) at SKr 45.5 billion.

61. See Fredriksson (1994).

62. See SOU (1995*b*).

63. An intermediate report on the structure of the Unemployment Insurance scheme is given in SOU (1996*a*).
64. See SOU (1996*c*).
65. See SOU (1991).
66. See Freeman *et al.* (eds.) (1995); and Davis and Henrekson (1995).
67. In national accounts terms the public sector's productivity growth is conventionally set at zero in many countries, as – in the absence of a measurable output concept – output is measured by its input.
68. See Murray (1994).
69. See Høj *et al.* (1996).
70. Productivity in many service industries is difficult to measure as the output in such industries tends to be of a very intangible nature and with an emphasis on quality.
71. See Martins *et al.* (1996). Mark-ups are measured as the industry's income relative to cost of the three input factors: capital, labour and raw materials.
72. See Davidsson (1995). SMEs are defined as firms with less than 200 employees and accounted for two-thirds of private employment in the beginning of the 1990s. Close to 99 per cent of SMEs have less than 20 employees.
73. In addition, jobs in the professional services sector have a higher capacity for indirect job creation than is the case for lower-skilled jobs. See Davidsson (1994).
74. Federation of Swedish Industries (1996).
75. Definitions vary but SMEs generally have fewer than 500 employees. See OECD (1996*f*).
76. Like other OECD countries Sweden has experienced a growing share of high technology-based industries and a reduction in the share of low-technology based industries. Up to the second oil crisis Sweden showed the third largest gain in high technology manufacturing. It has since fallen behind France and Japan.
77. These are also areas with a large share of production abroad, thereby boosting domestic R&D intensity.
78. See Lundberg (1992).
79. The Swedish market share for high-technology commodities still fell from 2.2 per cent to 1.9 per cent, but Swedish performance in this market segment was relatively better than in others.
80. See Andersson *et al.* (1996).
81. See Blomström and Kokko (1995).
82. The bulk of the FDI flows has consisted of mergers and acquisitions. The most prominent are General Motors' 50 per cent stake in SAAB automobile, the merger between the Swedish ASEA and the Swiss Brown Boveri to form ABB, Tetra Pak's acquisition of Alfa Laval, Dutch Akzo's of Nobel Industries and the merger between Pharmacia and the American Upjohn.
83. See Andersson (1995).
84. See Andersson (1994).

85. The positive effects on competition of opening up for foreign trade were clearly demonstrated by the removal of quantitative restrictions on textiles from low-cost countries in mid-1991. Imported textiles exerted a downward pressure on prices, and over the following two years consumer prices on textiles declined 5.3 per cent while overall consumer prices increased by 6.5 per cent. However, when Sweden became a member of the EU, it was obliged to reinstate quantitative import restrictions on imports from non-EU countries, weakening competition in this particular field.

86. See Martins (1994) and European Commission (1993).

87. For Nordbanken, which became 100 per cent state-owned as a result of the banking crisis in the early 1990s, the first steps towards full privatisation were taken in 1995, while during the preceding year the pharmaceutical company ''Pharmacia'' was sold for SKr 9 billion and 50 per cent of the forestry company ''AssiDomän'' for SKr 8 billion.

88. The remainder of ''Assi-Domän'' accounts for an additional SKr 8 billion. The two other equity stakes involve a 25 per cent stake in the defence consortium ''Celsius Industries'' and 50 per cent ownership of ''Sila'' – one of the three mother companies of the Scandinavian Airliner ''SAS''. Under a recently implemented share buy-back scheme, the Government's stake in Nordbanken fell from 65 to 58 per cent.

89. See *Konkurrensverket* (1994). The only successful entrant in this market thus could not maintain its concession when it came up for renewal.

90. If Telia allows free and fair access for other competitors to transmit their calls over Telia's network, but does not pursue a marginal-cost pricing rule, then such inter-connections become a profit centre for Telia. This has led to the suggestion that any user of the telecommunication network should pay a lump sum and then have unrestricted use of the infrastructure.

91. In this case, Telia will be allowed price increases not exceeding the rise in the overall price level (RPI) minus an imposed efficiency gain on Telia (X). For a general overview of frameworks to regulate monopolised markets, see Høj *et al*. (1996).

92. Sweden has three providers of mobile-phone services, which has contributed to the very competitive market situation, unlike countries with duopolies.

93. See OECD (1992).

94. The present regulatory framework does not provide many possibilities for private companies to lodge formal complaints against government subsidies or other government activities. Often private companies find themselves selling to the main public entity in some markets and in direct competition in other markets. This may explain the small number of recorded complaints. See also *Konkurrensverket* (1995).

95. Cases presented to the SCA include the health sector, fire prevention, services concerning road cleaning, garbage collection, prepared meals, but also in financial services in connection with the pricing of retail over-night deposits at the postal service and the central bank.

96. The SCA has disapproved of this practice, since the competitive situation has been severely affected, and it insists that full cost pricing rules have to be obeyed.

97. See *Konkurrensverket* (1994). For the entire market to be fully competitive, a correct tax assessment of the income from owner-occupied housing is required since interest payments are fully tax-deductible.

98. Documentation is provided in Barkman and Fölster (1995).

99. An estimated recovery of SKr 30 billion of the initial outlay of SKr 65 billion has been taken into account.

100. See Blomström and Kokko (1995) and Davis and Henrekson (1995).

101. There is often an overlap between different agencies in the type of support offered, in terms of legal advice, commercial feasibility, technical possibilities and finance. The Swedish Industrial Development fund, for example, offers five types of financial support while the Innovation Centre offers financial support towards the cost of licences and entering markets. An overview of agencies and types of support are given in Annex Table A3.

102. See Davidsson *et al.* (1993) and (1994).

103. The effect of R&D on total factor productivity growth was examined in OECD (1996*d*) by applying a panel estimation across 10 OECD countries and across industries. In aggregate, the estimated rate of return of direct R&D was about 15 per cent, although there was a large variance between countries. Moreover, the rate of return of embodied R&D – purchased from other industries or countries – in the service sector was much higher. Information and Communication Technology was singled out as playing a major role in the generation and acquisition of new technologies, pointing to very high social returns of flows of capital-embodied technology. Indeed, it is not necessary to carry out R&D to obtain substantial returns if it is possible to acquire it through capital investment spending.

104. Government initiatives to promote research and innovation among small firms include the ''Regional Development Programmes'' and the venture capital initiatives of ''ALMI Företagspartner AB''.

105. See Stenberg (1995).

106. See Davis and Henrekson (1995).

107. See Blomström and Kokko (1995).

108. This ''rebate'' is thought to originate in the tax code which obliges investment companies to be taxed on capital gains of their share holdings. Mutual funds are exempted from this obligation which represent a double taxation from an individual investor's point of view. Other features of the tax code may, however, offset this to some degree, and the rebate has also reflected investment companies' high administrative costs, lack of diversification and exposure to capital losses when they wanted to divest themselves of shares in individual companies.

109. The increase in dual-class share companies in Sweden is a relatively recent phenomenon. It has been documented that the increase in dual-class share companies matched closely the number of take-over attempts through the decades from 1960 onwards. Moreover, 80 per cent of the new dual-class companies over the period 1960-90 issued only low-vote shares after going public. While the sharp increase in take-over activity in industrialised countries in the 1980s was followed by preventive defensive measures on part of management, the corresponding activity in Sweden – including incorporating as dual-class share companies –

must be seen as a response by majority owners. In the Swedish case the take-over negotiations would thus reflect the assessment of corporate prospects of two alternative majority owners. See Rydqvist (1992).

110. SOU (1995*a*). The proposal would allow the possibility of having limitations on voting rights retained in the company regulations, but not the trustee function of management. In early 1992 90 per cent of quoted companies had chosen not to use the option available under the present law to limit voting rights to one fifth of all shares represented at the shareholders' meeting.

111. See Annex Table A3.

112. See Skog (1995).

113. In the relatively few cases where both A- and B-shares are quoted, A-shares have been observed to trade at a premium of 15 per cent, a number that is dependent i.a. on the proportion of A-shares and the frequency of take-overs. In negotiated, non-partial take-over-bids an additional premium between 0 and several hundred per cent have been observed, giving an average of 23 per cent and in take-over contests the top marginal premium has averaged 95 per cent. Over the period 1980-88 take-overs paid shareholders in the target company an average return more than 17 per cent above normal, while the owners of the purchasing company got an average increase of 2.5 per cent. See Bergström *et al.* (1989), cited in Isaksson and Skog (1994).

114. Only 27 of the 202 companies quoted in early 1992 had no differentiation of voting rights of shares. Both A- and B-shares were traded in 65 of the remaining 175 quoted companies.

115. See Rydquist (1992) for a detailed discussion.

116. See SOU (1988).

117. It has been alleged that Swedish enterprises have a weaker position with respect to securing payment in trade relationships than enterprises elsewhere, making small enterprises with low equity highly vulnerable to financial problems in other enterprises. Adjustment in contract law may thus affect the capital base and survival rate of SMEs (see Företagarnas Riksorganisation (1995)). Tax arrears are strongly protected in bankruptcy proceedings, but this is in line with all other OECD countries with the exception of Germany.

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Annex I

Supporting material to Chapter III

Table A1. **Labour market regulations**

A. Indicators of job protection legislation

Values of the indicators[1]

Key[2]	Regular procedural inconveniences		Notice and severance pay for no-fault individual dismissals						Difficulty of dismissal			
	Procedures	Delay to start of notice	Notice period after			Severance pay after			Definition of unfair dismissal	Trial period	At 20 years	Reinstatement
			9 months	4 years	20 years	9 months	4 years	20 years				
	Scale 0 to 3	Days	Months						Scale 0 to 3	Months		Scale 0 to 3
EU												
Belgium	1.0	3.0	2.0	3.6	11.4	0.0	0.0	0.0	0.0	3.3	12.5	0.0
Denmark	0.5	0.0	1.6	2.8	5.0	0.0	0.0	1.5	0.0	3.0	9.0	1.0
France	1.5	12.0	1.0	2.0	2.0	0.0	0.4	2.7	0.0	1.2	15.0	0.0
Germany	3.0	10.0	1.0	1.0	4.5	0.0	0.0	0.0	2.0	6.0	18.0	2.0
Greece	2.0	1.0	0.6	1.7	9.0	0.3	0.9	4.6	1.0	2.0	9.0	2.0
Ireland	1.5	3.0	0.2	0.5	2.0	0.0	0.5	3.9	0.0	12.0	24.0	1.0
Italy	1.5	0.0	0.3	1.1	2.2	0.7	3.5	18.0	0.0	0.8	32.5	3.0
Netherlands	3.0	35.0	0.6	1.0	5.3	0.0	0.0	0.0	1.0	2.0	5.3	1.0
Portugal	2.0	17.0	0.8	2.0	9.1	0.2	1.7	9.3	3.0	1.0	20.0	3.0
Spain	2.25	40.0	1.0	3.0	3.0	0.2	1.3	6.0	2.0	1.7	35.0	0.0
United Kingdom	1.0	3.0	0.2	0.7	2.8	0.0	0.9	4.6	0.0	24.0	10.8	0.0
EFTA												
Austria	2.0	5.0	0.8	1.2	2.5	0.0	2.0	9.0	1.0	1.0	9.0	1.0
Finland	2.0	56.0	2.0	2.0	6.0	0.0	0.0	0.0	0.0	4.0	20.0	0.0
Norway	1.5	3.0	1.0	2.0	5.0	0.0	0.0	0.0	2.0	1.0	15.0	2.0
Sweden	**2.0**	**7.0**	**1.0**	**4.0**	**6.0**	**0.0**	**0.0**	**0.0**	**1.0**	**6.0**	**32.0**	**2.0**
Switzerland	0.5	1.0	1.0	2.0	3.0	0.0	0.0	0.0	0.0	3.0	3.0	0.0

1. Situation in the late 1980s. The definition of the indicators shown here are listed in Note 2, and some underlying assumptions are described in more detail in Grubb and Wells (1993). Information in this table relates to *individual* dismissals (provisions applying to large-scale plant closures may differ) and relates, where relevant, to an average across provisions for manual and non-manual workers and an average across provisions for technical (economic) reasons. In the Swedish case, the notice period was determined by age and not by tenure.
2. The variables tabulated under each key are as follows:
   Procedures: procedures to be followed when issuing a regular dismissal notice; 1 for a statement in writing to the employee of reasons for dismissal, 2 for notification to a third party (works council or local employment exchange), and 3 when prior permission for dismissal must be obtained from the third party.
   Delay to start of notice: the delay between a decision to dismiss and the time that notice can become effective after following required procedures in days (*e.g.* notification by registered letter is assumed to involve 3 days).
   Definition of unfair dismissal: scored 0 when worker capability or redundancy of the job are adequate grounds for dismissal, 1 when social considerations, age or job tenure must, when possible, influence the choice of which worker(s) to dismiss, 2 when retraining to adapt the worker to different work must be attempted prior to dismissal and 3 when worker capability can never be a basis for dismissal.
   Compensation at 20 years: the compensation payable to a worker who has been unfairly dismissed after 20 years with the employer.
   Reinstatement: scored 0 if, following a court judgement of unfair dismissal, reinstatement is never granted, 1 if reinstatement is ‘‘rare’’, 2 if reinstatement is possible, and 3 if the employee always has the option of reinstatement.

Table A1. **Labour market regulations** *(cont.)*

B. Regulation of temporary forms of employment contracts

Indicator values and country rankings [1]

Key [2]	Fixed-term contracts				Temporary work agency (TWA) employment				
	Valid cases other than the usual ''objective''	Maximum number of successive contracts	Maximum cumulated duration	Overall strictness	Types of work for which TWA is legal	Restrictions on number of renewels	Maximum cumulated duration	Can final user terminate at any moment?	Overall [1] strictness
	Scale 0-2 or all	Number	Months	Rank [3]	Scale 0-3 or general	Yes/No	Months	No = 0, Yes = 1	Rank [3]
EU									
Belgium	0.0	1.0	24.0	16.0	2.0	Yes	2.0	0.0	8.0
Denmark	All	No limit	No limit	2.5	2.0	Yes	3.0	1.0	4.0
France	1.0	3.0	24.0	13.0	3.0	No	24.0	0.0	3.0
Germany	1.0	1.0	18.0	14.5	2.0	Yes	6.0	0.5	6.0
Greece	0.0	3.0	No limit	10.0	0.0	n.a.	n.a.	n.a.	10.0
Ireland	All	No limit	No limit	2.5	General	No	No limit	1.0	1.5
Italy	0.5	1.5	4.5	14.5	0.0	n.a.	n.a.	n.a.	10.0
Netherlands	All	1.0	No limit	7.5	3.0 [4]	Yes	6.0	0.5	5.0
Portugal	2.0	3.0	30.0 [5]	9.0	1.0	Yes	9.0	0.0	7.0
Spain	2.0	6.0	36.0	7.5	0.0	n.a.	n.a.	n.a.	10.0
United Kingdom	All	No limit	No limit	2.5	General	No	No limit	1.0	1.5
EFTA									
Austria	All	1.5	No limit	5.0	General	No	No limit	Yes	
Finland	1.0	1.5	No limit	11.5	General	Not known	Not known	Not known	
Norway	1.0	1.5	No limit	11.5	1.5	Not known	Not known	Not known	
Sweden	**2.0**	**2.0**	**No limit**	**6.0**	**0.0**	**n.a.**	**n.a.**	**n.a.**	
Switzerland	All	No limit	No limit	2.5	General	No	No limit	Yes	

n.a.: Not applicable.

1. Situation in the late 1980s. Definitions of the indicators shown here are listed in the next note, and some assumptions underlying them are described in more detail in Grubb and Wells (1993).
2. The variables tabulated under each key are as follows:
   Valid cases other than the usual ''objective'': scored 0 if fixed-term contracts are permitted only in ''objective'' cases (*i.e.* to perform a task which itself is of fixed duration); 1 if specific exemptions apply to situations of employer need (*e.g.* launching a new activity) or employee need (*e.g.* workers in search of their first jobs); 2 when exemptions exist on both the employer and the employee side; ''all'' when there are no restrictions on the use of fixed-term contracts.
   Maximum number of successive contracts: refers to limits on the number of successive fixed-term contracts that an employee can have with a given employer (1 when no renewals are permitted).
   Maximum cumulated duration: the maximum cumulated duration of successive fixed-term contracts or TWA assignments with a given employer (and without a break in the employment spell).
   Types of work for which TWA is legal: scored on a scale with 0 where TWAs are never legal, 1 to 3 when legal only for certain work situations (*e.g.* not for permanent posts) or industries, and general where no work-situation or industry restrictions apply. Swedish TWA regulations were liberalised in 1993.
   Can final user terminate at any moment?: In some countries notice requirements apply to termination of the assignment of an employee hired through a TWA agency or TWA agency work takes the form of a fixed-term contract. Scored 0.5 if the TWA contract has a fixed term but can be terminated at a few days' notice.
   Overall strictness: see the following note.
3. The rankings for ''overall strictness'' are ranks of the unweighted average of the ranks of the variables in the preceding columns under the same heading. EFTA countries are not ranked for regulation of TWA employment because certain information for Finland and Norway was missing.
4. Netherlands: TWAs are forbidden in the building sector. Classification revised from Grubb and Wells (1993).
5. Portugal: maximum cumulated duration is 24 months for a new activity or enterprise, 36 months for other cases.

Source: The OECD Jobs Study, Part II; Grubb D. and W. Wells (1993): ''Employment Regulation and Patterns of Work in EC Countries'', *OECD Economic Studies* No. 21.

Annex II

Supporting material to Chapter IV

Table A2. **Major state-owned enterprises**

Company (part of company stock if less than 100)	Activity	Turnover 1995 (SKr million)	Employees 1995
A-Banan Projekt AB	Railway construction	5 912	2
AB Svensk Exportkredit (50%)	Exports credits	39 690[1]	86
AB Tipstjänst	Public betting	7 335	311
AmuGruppen AB	Training, competence development	3 089	4 892
Apoteksbolaget AB	Pharmacies	20 641	9 588
AssiDomän AB (50%)	Forestry	21 836	13 119
Celsius AB (25%)	Defence equipment, software	14 826	16 240
Luftfartsverket (agency)	Civil aviation	3 765	3 892
Luossavaara-Kiirunavaara AB, LKAB	Mining	4 557	3 432
Nordbanken (65%)	Bank	221 487[1]	7 124
Pharmacia and Upjohn Inc. (7%)	Pharmaceuticals	47 245	33 000
Posten AB	Postal services	22 355	46 048
Samhäll AB	Work programmes for the disabled	9 142	28 721
Securum AB	Liquidate uncertain credits	11 717	8 815
Statens Bostadsfinansieringsaktiebolag, SBAB	Housing loans	136 925	232
Statens Järnvägar (agency)	Railroads	17 737	24 847
Statliga Akademiska Hus AB	Administer real estate for universities	1 723	285
Svenska kraftnat Affersverket (agency)	National grid (transmission of power)	2 338	157
Svenska Penninglofteriet AB	Lottery	3 790	188
Systembolaget AB	Retailing of alcoholic beverages	20 078	2 658
Telia AB	Telecommunications	41 060	32 825
V&S Vin and Spirit AB	Production, alcoholic beverages	4 092	902
Vasakronan AB	Real estate	3 449	953
Vattenfall AB	Power generation and distribution	27 150	8 460
Venantius AB	Administer state granted housing loans	25 068	61

1. Total assets.
Source: Swedish Ministry of Industry and Trade.

Table A3. **Major industrial groups in Germany, Japan and Sweden**

A. Germany

Bank-centred	Industrial-centred	Family-centred	Other
Deutsche Bank	BASF	Quant	Aachen-Münchner[1]
Genosssenschaftsbank	Daimler	Klöckner	Allianz[1]
Hamburger Landesbank	Gute-Hoffnungs-Hütte	Krupp	Bank für Gemeinwirtschaft[2]
Landssparkassen	Höchst		Gerling[1]
	Mannessmann		VEBA[3]
	Thyssen		

B. Japan

Former Zaibatsu	New-bank-centred-groups	New manufacturer-centred groups
Mitsubishi	DKB	Nippon steel
Mitsui	Sanwa	Hitachi
Sumitomo	IBJ	Nissan
Fuyo	Tokai	Toyota
		Matsushita
		Toshiba-IHI
		Tokyu
		Seibu

C. Sweden

Bank-centred	Industrial-centred	Family-centred	Other
SHB	Volvo och Skanska	Bonnier	Custos
Sparbanks		Douglas	Industrivärden
		Hagströmer-Qviberg	
		Kempe	
		Lundberg	
		Norberg	
		Nordstjernan	
		Olsson	
		Paulsson	
		Stenbeck	
		Söderberg	
		Wallenberg	

1. Insurance company.
2. Hausbank for a cluster of companies owned by labour unions.
3. State-owned holding company.

Source: Kester (1992) and Sundqvist (1996).

Table A4. **Institutions engaged in industrial support**

Organisation [1]	Main financial instruments	Other targeted areas [2]	
		Technology and innovation, etc.	Business and human resources services
NUTEK (The National Board for Industrial and Technical Development) [3]	Loans and grants for regional development and SMEs.	Financial support for new technology in alternative energy and of new technology.	General financial advice. R&D support, regional advisors, training, information and network.
ALMI Business Partner Inc. [4]	Loans and credit guarantees for start-ups and existing firms.	Consulting and information in connection with technology projects.	Legal, technical and commercial consulting, financing and information for start-ups and SMEs.
The Swedish Industrial Development Fund	Loans for industrial projects, risk capital injections against royalty, project guarantees and credit guarantees.	Loans to SMEs in the areas of innovation and R&D.	General consulting.
The Innovation Centre	Financial support to innovation.	Technical support concerning the innovation process.	
County Administrative Boards	Loans and credit guarantees for investing in scarcely populated areas.		Grants to employees working from home.
County Labour Boards	Grants to unemployed and handicapped planning to start a new business.		Ten different programmes for developing human resources within firms.
Industrikredit, AB	Long term loans to SMEs.		
Nordic Investment bank	Investment loans in the Nordic countries.	International projects with Nordic interest.	
Bure Innovationskapital AB	Risk capital through minority investment in smaller firms.		
Fylkeinvest, KapN, Tuab, Företagskapital AB, Småföretagsinvest, Partnerinvest, Vencap	Different forms of risk capital for SMEs, totalling SKr 1.5 billion.		

1. In general, the public institutions listed below receive their funding from central or local government authorities, either as start-up capital or as recurrent appropriations.
2. In addition to this selection presented here SMEs can apply to another 130 Swedish and 110 EU programmes for support in the areas of investment, exports, technology, R&D, management and education, consulting and environmental issues.
3. An independent body under the Ministry of Industry and Commerce.
4. Corporation owned by a central government-owned parent company and by local governments via 22 Regional Development Companies.

Source: Ministry of Industry and Commerce (1996), ''Small Business in Sweden – Policies and Programmes promoting Small Enterprises in Sweden''; NUTEK (1995), ''Finansieringsmöjligheter i Sverige – et utdrag ur databasen ecuFinder''.

Annex III

Rulings of the Swedish competition authority with respect to horizontal and vertical integration and abuse of dominant positions

Mergers

The Swedish Competition Authority (SCA) has on a number of occasions acted to ensure that competition was not impaired by a proposed merger. In the cases of the brewery industry and lock-producing business, the SCA obtained voluntary commitment to reduce market domination, through reduced ownership and through divestment, respectively. In the food sector, the proposed merger of the two main slaughterhouses was only accepted when a non-competition clause was removed and two additional non-discriminatory clauses covering other market participants were introduced.

Vertical control and anti-competitive co-operation

Vertical control as such is not illegal,* but it is not allowed to lead to anti-competitive co-operation. A case in point is the SCA refusal to grant an exemption on exclusive distribution in the area of agricultural machinery in the south of Sweden.

It is fairly common among Swedish suppliers to issue price recommendation to retailers. Such practices are not considered as preventing competition insofar as the suggested prices are seen as recommendations by suppliers, and retailers remain free to set their prices.

In the area of information-collection and sharing there has been a number of decisions going against practices such as restrictions on the exchange of statistics and information in the timber-exporting industry and the Swedish Gramophone Suppliers' Association.

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* As an example, the largest electricity producer (the government-owned Vattenfall AB) was allowed to acquire two municipal distribution companies as it only controlled a very small part of the distribution system.

Horizontal price co-operation

Swedish trade organisations have traditionally distributed price-lists with recommended sales prices to their members. Such procedures are now considered as distorting competition, particularly in cases where the organisation covers a substantial part of the market. This has been demonstrated *inter alia* with respect to the Swedish Association for Motor Retail Trades and Repairs and the Trade Association of Electricians; and the agreement reached by the two largest television companies to regulate the sale of advertising time through a third, jointly owned, company. In the case of the Trade Association of Wooden House Manufactures, a price regulation clause in its standard terms of sale did not receive a negative clearance verdict.

Since 1 January 1993 there has in principle been free price-setting within the pharmaceuticals retail business. However, prices of medicines covered by the Public Health Service (*Riksforsäkringsverket*) are set by direct negotiations between the Public Health Service and the pharmaceutical industry. Generic products are priced through the so-called reference price system and pharmacies, which are state-owned in Sweden, have a monopoly on the retail sale of these products, but with parallel imports allowed. The development of a competitive market is thus severely distorted in this sector, and further efforts are required.

Abuse of dominant position

The SCA has made several rulings to curb abuses of dominant positions. Of particular importance have been:

- The Swedish Civil Aviation Administration used to provide the full range of airport ground handling services and through its dominant position as airport manager also controlled entry into the market. After an investigation by the SCA the two largest airports have now opened up for new entrants, with effect from 1 April 1995. The results remain to be seen, but, when ground handling services are provided on a competitive basis in other countries, prices have been found to be as much as 30 per cent lower than in airports with monopoly provision of such services.
- In air transport, the flag carrier (SAS) agreed to offer interlining (accepting another airline's tickets for the whole or part of a journey) to its sole competitor. However, in connection with the liberalisation of air transport on 1 July 1992, the two domestic airlines merged giving SAS a market share of 95 per cent with interlining no longer an issue. Since then a number of new operators have entered the domestic market, with the market share of SAS dropping to 70 per cent. Swedish evidence so far bears out that contested market segments have seen declining prices and a wider choice of departures while non-contested market segments have seen price increases and fewer departures.

- The postal sector was liberalised in 1993. However, several new entrants have complained that the incumbent, government-owned Sweden Post, has abused its dominant position by imposing sale arrangements with prices dependent on fidelity and exclusive agreements on customers, thereby effectively blocking new entry. In one case a ''cream-skimming'' clause imposed higher prices on all other services if customers bought one service from another company. In a number of cases the SCA found that Sweden Post was abusing its dominant position.
- In the telecommunications sector the government-owned incumbent was found to be charging competitors higher prices for interconnection than it charged its own customers. The SCA regarded this practise as price discrimination, thereby preventing new entry.

Annex IV

Methodology and assumptions involved in the calculation of effective tax rates on capital income

The required net rate of return – the cost of capital – of a marginal investment project depends on the corporate tax rate, investment grants and tax credits[1] in addition to the rate of inflation, the rate of economic depreciation and the discount rate. The cost of capital varies across assets insofar as the rate of economic depreciation and tax depreciation allowances vary between inventories, machinery and buildings. The discount rate reflects the return demanded by the owner – given by the return of an alternative financial investment such as bank deposits or bonds – suitably adjusted for the risk involved, which will be higher in cases of equity issues than for debt-financing. Insofar as the discount rate has to be adjusted for taxation of dividends and capital gains, financing through new equity issues or retained profits implies a higher required rate of return than in the case of debt-financing with interest payments fully deductible. The approach to calculating the effective tax rates on capital income applied in Chapter IV[2] departs from earlier studies for Sweden in two respects. In addition to explicitly incorporating risk compensation to lenders and owners it is assumed that the required rate of return net of corporate taxes is fixed, a possible interpretation being that enterprises face a given (world-market) rate of interest.[3]

The main parameters of the Swedish tax system are set out in Annex Table A5 below, while the effective marginal tax rates[4] are reproduced in Table 27, Panel B. For *households* as owners, the tax system acted as a subsidy to investment up to 1990 insofar as the required rate of return at enterprise level was lower than the return on alternative financial investments. With the 1991 tax reform this subsidy turned into an effective tax, of close to 50 per cent for debt-financed investment.[5] The removal of the double taxation of dividends in 1994 and its reinstatement in 1995 can be seen to affect the ranking of financing mode: equity issues had the lowest effective tax rate in 1994 while for 1995 they once again have the highest rates. *Foreign investors* by assumption are not liable to taxation of interest income and capital gains in Sweden and subject to only a 15 per cent withholding tax on dividends and continue to benefit significantly from debt-financed investments. Their effective tax rate is thus still negative in this case. Due to the withholding tax, financing with new share issues has consistently faced higher effective tax rates than when financing is undertaken with retained profits.

Table A5. **Main tax parameters for Sweden 1985-95**

Per cent

	1985	1990	1991	1994	1995
Corporate level					
Tax rate on distributed profits	38.5	40	30	28	28
Tax rate on retained profits	38.5	40	30	28	28
Investment grant buildings	0	0	0	0	50
Investment grant machinery	0	0	0	0	0
Households					
Marginal tax rate on dividends	59	59	30	0	30
Marginal tax rate on interest income	44	44	30	30	30
Marginal tax rate on equity capital gains [1]	25	25	30	12.5	30
Marginal tax rate on property capital gains [1]	0	0	30	15	15
Wealth tax rate on bonds	0.7	0.6	0.5	0.4	0.4
Wealth tax rate on equity	0.21	0.2	0.18	0.14	0.14
Tax assessment of property relative to market value [2]	65	40	75	75	75
Foreign investor					
Marginal tax on dividends	15	15	15	15	15
Marginal tax rate on interest income	0	0	0	0	0
Marginal tax rate on capital gains [1]	0	0	0	0	0
Pension funds					
Marginal tax on dividends	0	0	10	3.6	15
Marginal tax rate on interest income	0	0	10	7.0	15
Marginal tax rate on capital gains [1]	0	0	10	3.6	15
Life insurance funds					
Marginal tax on dividends	16.3	16.3	30	6.5	27
Marginal tax rate on interest income	18.5	18.5	30	15.56	27
Marginal tax rate on capital gains [1]	12.0	14.2	30	6.5	27
Depreciation rates machinery					
Tax-wise (depreciating balance)	30	30	30	30	30
Economic	11.39	11.39	11.39	11.39	11.39
Lifetimes buildings (years)					
Tax-wise (straight line)	30.4	30.4	30.4	30.4	30.4
Economic	41.8	41.8	41.8	41.8	41.8

1. It is assumed that stocks are held for 10 years.
2. Imputed income from owner-occupied housing was abolished in 1991 with the increase in property wealth tax offsetting the loss in tax revenues.

Source: Norrman (1996*a*) and (1996*b*).

The effective tax rates affecting households' choice between different assets are set out in Table 27, Panel B. It should be noted that borrowing for investing in pension and life insurance claims was highly profitable in the 1980s, but that the 1990-91 tax reform has acted to reduce substantially the differences in tax treatment of different assets.[6]

Notes

1. In the Swedish case, attention has to be paid to special provisions in the tax code, allowing *i)* a deduction at the corporate level if investment is financed by new share issues (the so-called *Annell* deduction, terminated in 1994); *ii)* a deduction from the tax base related to the size of the equity (the so-called *Surv*, introduced in 1991 and terminated in 1994); *iii)* a deduction related to the size of profits (*Periodiseringsfond*, replacing the Surv); and *iv)* a temporary investment grant for buildings at 50 per cent (effective in 1995).

2. See McLure and Norrman (1995).

3. See Norrman (1996*a*). The assumption used in other studies relating to Sweden, both King and Fullerton (1984) and a recent update in Södersten (1993), is that the required rate of return gross of corporate taxes is equalized.

4. Given by the difference between the required rate of return at enterprise level, net of depreciation, and the opportunity return of the owner (as per cent of the former).

5. The high and positive tax rates recorded for households prior to 1991 reflects that both the required rate of return at enterprise level and the rate of return on financial investments were negative under the assumptions applied.

6. See Norrman (1996*b*) for an elaboration of these effects.

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Annex V

Calendar of main economic events

1995

January

Additional fiscal measures aimed at improving the central government budget balance by around SKr 20 billion are announced. The main elements are: lower child allowances and compensation in family insurance; reduced pensions; reduced investment in roads and railways; and higher contributions and fees to government from labour market funds.

The Budget Bill for the fiscal year 1995-1996, extraordinarily covering the 18-month period from July 1995 to December 1996, is presented to Parliament. A deficit of SKr 243 billion is projected for the whole period.

February

The repo rate is increased from 7.60 to 7.83 per cent at the end of the month, and the deposit and lending rates by 50 basis points to 6.50 and 8.50 per cent respectively.

March

The repo rate is increased to 8.20 per cent.

April

To stimulate job creation employers' social contributions are lowered by SKr 6 000 per month when hiring an unemployed in the course of the first half of 1995.

The supplementary Budget Bill for the fiscal year 1995-1996 is presented to Parliament, showing a deficit of Skr 214 billion and including an additional net tightening of SKr 3½ billion.

The repo rate is increased to 8.41 per cent, the deposit and lending rates to 7.5 and 9.0 per cent respectively.

June

Measures to enhance employment among groups with high unemployment are approved by parliament, including a temporary 50 per cent wage subsidy (to be terminated by end-1997) for companies hiring long-term unemployed above 60 years of age. The number of ALMP schemes is reduced at the same time.

Indirect taxes are increased to finance the EU contribution. The increases primarily concern taxes on commercial property, energy, and private vehicles. Tax deductions for private pension contributions are lowered and government loans for housing are to carry market-related interest rates.

The repo rate is increased to 8.66 per cent.

July

The repo rate is increased to 8.91 per cent, and the deposit and lending rates are increased to 8.00 and 9.50 per cent respectively.

October

The government presents its Growth Bill, giving guidelines for labour market, education and industrial policies, together with the first update of the Convergence Programme. On the projections presented no additional savings are required to comply with overall fiscal objectives. Sanctions for not complying with the job availability criteria for unemployment benefits are tightened.

December

A new system for transfers to local government is decided by Parliament, aiming at narrowing income differentials between municipalities resulting from differences in economic and demographic structure.

1996

January

Following from the fiscal consolidation programme, replacement rates in social security schemes are lowered to from 80 to 75 per cent, and social security contributions are raised. A reduction in VAT on food from 21 to 12 per cent is more than offset by increases in excise taxes and property taxes.

The repo rate is lowered to 8.66 per cent and later to 8.45 per cent.

A new deposit insurance scheme comes into force, covering deposits up to SKr 250 000 per depositor and bank.

February

The repo rate is lowered to 8.05 per cent. The central bank's deposit and lending rates are lowered to 7.50 and 9.00 per cent respectively.

March

The repo rate is lowered several times to 7.40 per cent at the end of the month. The central bank's deposit and lending rates are lowered to 6.75 and 8.25 per cent respectively.

April

The Government tables its Spring Bill implementing a system of expenditure ceilings for 27 major expenditure categories and covering the years 1997-1999. Central government expenditures are set to fall from 41.0 per cent of GDP in 1997 to 37.8 per cent in 1999. On the projections presented, the budget deficit falls somewhat short of reaching 3 per cent of GDP in 1997 and balance by 1998, and offsetting measures are proposed. The permanent effect of the measures proposed is estimated to be around SKr 8 billion with additional temporary reinforcements of SKr 6 billion for 1997-98.

The proposal incorporates a programme to increase the capacity in adult and tertiary education, to be covered by lower spending on ALMPs and by higher energy taxes, and an agreement with municipalities which obliges them to keep tax levels unchanged while block grants from the government are kept unchanged in nominal terms.

The repo rate is lowered to 6.90 per cent at the end of the month. The central bank's deposit and lending rates are lowered to 6.00 and 7.50 per cent respectively.

May

The repo rate is lowered to 6.50 per cent at the end of the month.

June

The government puts forward an Employment Bill, aiming at bringing down open unemployment to 4 per cent by 2000. The length of spells on unemployment insurance is to be limited, the replacement rate increased to 80 per cent but financed with less generosity in other aspects. Payroll taxes for SMEs are lowered, and they also benefit from a targeted relief from the double taxation of dividends.

The repo rate is lowered to 6.10 per cent at the end of the month. The central bank's deposit and lending rates are lowered to 5.25 and 6.75 per cent respectively.

July

The elements of the employment package are approved by Parliament. They include an additional 130 000 places in adult and tertiary education and work-for-benefit for long-term unemployed persons over 55 years of age.

The repo rate is lowered to 5.55 per cent.

August

The repo rate is lowered to 5.25 per cent. The central bank's deposit and lending rates are lowered to 4.75 and 6.25 per cent respectively.

September

The Finance Bill, embodying the detailed appropriations following from the adoption of the expenditure ceilings, is presented together with the third update of the Convergence Programme. No additional savings are required to comply with the objective of budget balance in 1998.

The repo rate is lowered to 5.05 per cent.

October

The repo rate is lowered to 4.60 per cent. The central bank's deposit and lending rates are lowered to 4.25 and 5.75 per cent respectively.

STATISTICAL ANNEX AND STRUCTURAL INDICATORS

Table A. **Selected background statistics**

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
A. Percentage change from previous year at constant 1991 prices										
Private consumption	4.4	4.6	2.4	1.2	–0.4	0.9	–1.4	–3.1	1.8	0.8
Gross fixed capital formation	0.3	8.2	6.6	11.3	1.3	–8.9	–10.8	–17.2	2.0	10.9
Residential	–2.2	8.8	8.4	4.8	7.2	–2.4	–7.3	–32.8	–35.9	–28.0
Business	2.3	9.3	5.8	15.0	–1.1	–15.2	–15.5	–13.0	18.6	25.1
Public sector	–2.4	0.9	6.2	10.5	–2.0	8.3	0.6	6.8	11.1	–0.5
GDP	2.3	3.1	2.3	2.4	1.4	–1.1	–1.4	–2.2	3.3	3.6
GDP price deflator	6.9	4.8	6.5	8.0	8.8	7.6	1.1	2.6	2.5	3.7
Industrial production	0.5	2.5	1.3	3.7	1.1	–5.1	–1.6	–0.2	11.9	9.1
Employment (persons)	0.8	1.0	1.4	1.5	1.0	–1.9	–4.3	–5.8	–0.9	1.5
Compensation of employees (current prices)	9.5	8.3	9.2	12.8	12.5	5.1	–1.3	–1.8	4.1	4.3
Productivity (GDP/employment)	1.7	2.3	0.9	0.9	0.4	0.4	3.2	3.2	4.4	2.0
Unit labour costs (compensation/GDP) (current prices)	7.0	5.0	6.8	10.2	11.0	6.3	0.2	0.4	0.1	0.7
B. Percentage ratios										
Gross fixed capital formation as % GDP at constant prices	17.7	18.6	19.3	21.0	21.0	19.4	17.5	14.8	14.6	15.7
Stockbuilding as % GDP at constant prices	–0.8	–1.0	–0.8	–0.7	–0.3	–1.5	–0.4	–0.9	0.4	0.9
Current Account as % GDP at constant prices	3.2	2.4	1.7	0.6	0.8	1.5	1.9	4.9	5.7	6.8
Compensation of employees as % GDP at current prices	59.8	59.9	60.1	61.3	62.5	61.8	61.2	59.9	58.9	57.2
Direct taxes as % of household income	20.6	21.7	22.2	22.7	21.4	18.6	18.1	17.9	18.2	18.8
Household saving as % of disposable income	1.3	–2.8	–4.8	–4.9	–0.6	3.1	7.7	8.3	8.0	6.9
Unemployment as % of civilian labour force	2.5	2.1	1.7	1.5	1.6	3.0	5.3	8.2	8.0	7.7
C. Other indicator										
Current balance ($ billion)	–0.1	–0.2	–0.8	–3.4	–6.6	–4.8	–8.7	–3.7	0.8	4.7

Source: Statistics Sweden, *National Accounts;* OECD.

Table B. **Supply and use of resources**

SKr billion, current prices

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Consumers' expenditure on goods and services	487.3	537.9	584.4	632.7	692.7	771.3	777.3	796.4	834.5	861.4
General government expenditure on goods and services	260.2	273.4	290.3	322.6	372.1	394.4	402.5	406.1	416.2	424.9
Gross fixed capital formation	175.5	197.9	225.1	271.0	292.5	280.4	244.6	205.7	209.2	238.9
Change in stocks	–5.8	–4.8	–3.6	–0.5	–2.5	–21.2	–6.7	–13.7	7.2	15.9
Domestic expenditure	917.2	1 004.5	1 096.2	1 225.9	1 354.8	1 424.9	1 417.8	1 394.4	1 467.1	1 541.1
Exports of goods and services (non-factor)	311.1	332.4	359.7	394.5	406.8	404.2	401.6	473.3	557.8	673.0
Imports of goods and services (non-factor)	281.0	313.3	341.4	387.8	401.8	381.8	377.6	421.5	493.7	569.1
Gross domestic product in purchasers' values	947.3	1 023.6	1 114.5	1 232.6	1 359.9	1 447.3	1 441.7	1 446.2	1 531.1	1 645.0
Indirect taxes	159.0	176.9	182.8	199.8	233.4	256.8	235.1	226.0	228.8	237.2
Subsidies	46.1	48.4	48.9	55.7	63.3	72.7	78.6	84.9	80.8	84.2
Gross domestic product at factor cost	834.4	895.1	980.7	1 088.5	1 189.8	1 263.2	1 285.3	1 305.2	1 383.1	1 492.0
Depreciation and other operating provisions	118.3	128.1	142.1	160.2	178.9	191.6	193.3	203.5	. .	. .
Net domestic product at factor cost	716.1	767.0	838.5	928.4	1 010.9	1 071.7	1 092.0	1 101.7	. .	. .

Source: Statistics Sweden, *National Accounts.*

Table C. **Supply and use of resources**

SKr billion, 1991 prices

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
A. Gross domestic product by type of expenditure										
Consumers' expenditure on goods and services	708.2	740.5	758.5	767.3	764.3	771.3	760.5	737.2	750.4	756.2
General government expenditure on goods and services	360.6	364.1	366.2	374.0	338.8	394.4	394.2	395.0	392.4	388.4
Gross fixed capital formation	236.5	255.9	272.9	303.8	307.7	280.4	250.0	207.0	211.0	234.0
Change in stocks	–7.4	–5.8	–4.8	–1.9	–0.7	–21.2	–5.6	–13.0	6.4	13.0
Domestic expenditure	1 297.9	1 354.7	1 392.8	1 443.3	1 410.1	1 424.9	1 399.2	1 326.2	1 360.1	1 391.5
Exports of goods and services (non-factor)	369.5	385.3	394.9	407.1	413.8	404.2	413.7	445.3	507.4	571.3
Imports of goods and services (non-factor)	327.3	352.5	371.3	398.8	401.5	381.8	386.1	376.4	425.9	469.7
Gross domestic product in purchasers' values[1]	1 337.4	1 379.4	1 410.5	1 444.0	1 463.7	1 447.3	1 426.8	1 395.1	1 432.3	1 474.9
B. Gross domestic product by kind of activity										
Primary sector	31.3	30.0	29.4	32.6	34.0	31.2	30.3	30.8	31.4	33.3
Mining, manufacturing, gas, etc.	305.5	313.8	320.0	321.7	320.1	302.6	293.2	296.0	330.3	358.9
Construction	86.3	90.0	91.4	98.4	98.7	98.5	94.1	84.8	81.2	81.5
Services	783.0	809.2	830.3	851.0	870.9	875.2	873.4	853.7	871.1	888.5
Adjustment[2]	139.0	144.1	146.6	147.7	144.4	139.5	132.1	123.9	116.7	111.8
Statistical discrepancy	–7.8	–7.6	–7.3	–7.5	–4.6	0.2	3.6	5.9	1.6	0.9
Gross domestic product	1 337.4	1 379.4	1 410.5	1 444.0	1 463.7	1 447.3	1 426.8	1 395.1	1 432.3	1 474.9

1. On the basis of GDP by production.
2. Import duties, value added tax and other taxes net of subsidies, on sales and production of commodities less imputed bank service charge.

Source: Statistics Sweden, *National Accounts.*

Table D. **General government revenue and expenditure**[1]

SKr million, current prices

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Current revenue	571 844	635 099	686 050	785 348	860 747	870 804	857 210	849 736	887 179	957 130
Direct taxes	205 979	243 679	270 319	310 960	318 612	287 962	294 921	301 012	324 697	355 077
Indirect taxes	158 958	176 860	182 766	199 786	233 362	256 807	235 067	225 955	228 755	237 218
Social security contributions	134 427	140 836	156 757	186 883	211 889	223 768	213 145	207 560	220 701	242 922
Other	72 480	73 724	76 208	87 719	96 884	102 267	114 077	115 209	113 026	121 913
Current expenditure	575 935	600 358	640 887	710 565	797 372	876 511	939 213	985 485	1 023 274	1 066 547
Consumption	260 171	273 408	290 266	322 630	372 130	394 394	402 508	406 071	416 167	424 878
Property income payable	70 803	66 747	63 175	67 377	69 293	75 896	79 903	90 959	105 443	117 585
Subsidies	46 111	48 395	48 919	55 719	63 268	72 710	78 634	84 948	80 781	84 236
Current transfers	198 850	211 808	238 527	264 839	292 681	333 511	378 168	403 507	420 883	439 848
Net saving	–4 091	34 741	45 163	74 783	63 375	–5 707	–82 003	–135 749	–136 095	–109 417
Capital formation	25 057	26 321	26 195	30 277	32 787	32 683	39 144	15 481	45 706	48 139
Capital transfers net	0	16 284	115	–2 217	0	–3 337	–15 831	–51 828	–2 326	846
Consumption of fixed capital	17 321	18 734	19 988	24 054	26 376	25 404	24 993	25 566	25 977	27 422
Net lending	–11 827	43 438	39 071	66 343	56 964	–16 323	–111 985	–177 492	–158 150	–129 288
Per cent of GDP	–1.2	4.2	3.5	5.4	4.2	–1.1	–7.8	–12.3	–10.3	–7.9

1. According to OECD definition.

Source: Statistics Sweden, *National Accounts.*

Table E. **Balance of payments**

SKr billion

	1989	1990	1991	1992	1993	1994	1995
A. Trade balance	15.0	14.8	30.5	34.3	53.1	67.1	105.6
Exports of goods	329.0	336.0	328.8	322.4	384.2	463.1	564.4
Imports of goods	314.0	321.2	298.3	288.0	331.1	396.0	458.8
B. Services, net	–8.7	–14.0	–8.2	–11.6	–0.4	–1.0	–3.5
Transportation	11.3	11.2	12.1	9.4	10.0	10.2	9.4
Travel	–16.3	–20.0	–21.7	–22.8	–14.2	–15.8	–14.1
Other services	–3.8	–5.2	1.4	1.8	3.9	4.6	1.2
C. Trade in goods and services (A + B)	6.3	0.8	22.3	22.8	52.7	66.1	102.1
D. Wages and salaries	–1.0	–1.0	–1.3	–1.1	–1.1	–0.8	–1.2
E. Investment income	–14.0	–25.4	–37.5	–57.3	–66.9	–44.6	–46.0
F. Unrequited transfers	–13.2	–13.5	–12.4	–15.2	–13.9	–14.3	–21.1
G. Current account (C + D + E + F)	–21.9	–39.0	–28.9	–50.7	–29.2	6.4	33.8
H. Direct investments	–54.7	–75.6	–4.2	–2.7	18.1	–3.3	24.9
I. Trade in shares	–29.8	–18.7	–3.0	10.3	32.2	37.4	–37.7
J. Loans/bonds in foreign currency	113.9	114.6	–26.7	13.9	–35.5	–22.7	5.2
Government and the Central bank	–13.0	–14.0	–18.9	155.0	69.1	25.9	54.8
Other sectors	127.0	128.6	–7.8	–141.1	–104.7	–48.6	–49.7
K. Loans/bonds in SEK	36.8	64.5	64.7	37.3	42.7	2.6	44.4
Government and the Central bank	11.0	23.3	42.3	29.9	10.2	4.6	55.8
Other sectors	25.9	41.2	22.3	7.4	32.5	–2.1	–11.4
L. Financial derivatives	0.0	0.0	0.0	0.0	0.0	0.0	–6.0
M. Other capital transactions	–2.7	26.6	–36.9	0.6	25.6	34.0	–48.7
N. Official reserves	–9.3	–45.1	4.4	14.0	–75.5	–18.1	9.9
O. Net errors and omissions	–32.3	–27.3	30.6	–22.6	21.6	–36.2	–25.7
P. Financial account (H + I + J + K + L + M + N + O)	21.9	39.0	28.9	50.7	29.2	–6.4	–33.8

Source: Sveriges Riksbank.

Table F. **Labour**

	Labour force survey[1]					National labour market board					Participants in Labour Market Programmes[2, 3]		
	Labour force			Em-ployed	Unem-ployed	Total	Insured unemployed			Jobs vacant[4]	Subsided work	Training	Protected employ-ment[5]
	Total	Men	Women				Total	Mining, manu-facturing	Building, cons-truction				
	1 000 persons						Per cent of insured			Thousand	1 000 persons		
1986	4 388	2 284	2 103	4 278	110	84.2	2.5	3.0	5.5	39.1	29	34	74
1987	4 410	2 292	2 117	4 316	94	78.0	2.3	2.9	3.8	46.1	22	37	76
1988	4 452	2 312	2 141	4 375	77	61.1	1.7	2.1	2.7	52.1	19	47	79
1989	4 508	2 346	2 163	4 442	66	56.3	1.6	2.1	2.2	50.0	13	42	81
1990	4 560	2 372	2 188	4 485	75	66.4	2.2	2.5	2.5	41.5	10	36	81
1991	4 530	2 356	2 174	4 396	134	114.8	3.2	4.9	7.1	18.6	16	59	80
1992	4 442	2 309	2 134	4 209	233	214.5	5.9	9.7	17.5	10.6	29	86	77
1993	4 320	2 245	2 076	3 964	356	326.1	8.7	14.1	24.5	8.7	23	53	76
1994	4 268	2 219	2 049	3 928	340	331.7	8.8	12.9	25.3	38.3	42	60	80
1995	4 320	2 251	2 069	3 988	332	328.7	8.7	11.3	24.0	15.0	35	55	85

1. The employment concept covers all gainful employment, independently of the number of hours worked. Figures refer to persons aged between 16 and 64.
2. Averages of monthly observations.
3. The figure have been recalculated to conform with a new definition introduced in 1987.
4. From April 1989 only the vacancies with more than 10 days of duration are shown.
5. Employed in active work, sheltered workshops and semi-sheltered work.

Source: Statistics Sweden, *Monthly Digest of Swedish Statistics, Statistical Abstract of Sweden; OECD, Main Economic Indicators, Labour Force Statistics.*

Table G. **Production by sector and employment**

Mining and manufacturing

	Production, 1990 = 100									Employment[1] 1990 = 100
	Mining, manufacturing and electricity total	Mining and quarrying	Manufacturing							
			Total	Food, beverages, tobacco	Textiles, clothing, footwear	Wood and wood products	Chemicals, etc.	Basic metals	Metal products	
	ISIC 2-4	21-29	31-39	31	32	33	35	37	38	Hours worked
1986	92	107	92	97	115	90	91	100	86	102
1987	94	108	94	96	115	92	98	101	89	103
1988	95	103	95	97	106	93	100	109	91	103
1989	99	97	99	100	100	98	100	110	97	102
1990	100	100	100	100	100	100	100	100	100	100
1991	95	104	95	97	84	93	102	92	93	90
1992	93	99	93	98	78	82	110	96	90	81
1993	93	94	93	100	70	86	119	105	86	76
1994	103	99	103	99	76	96	125	112	101	81
1995	114	106	114	100	88	107	127	113	124	87

1. Monthly hours worked in mining and manufacturing.

Source: OECD, *Indicators of Industrial Activity; Main Economics Indicators.*

Table H. **Foreign trade, total and by area**[1]

$ million, monthly rates

	Total imports cif	Total exports fob	Imports by area						Exports by area					
			OECD			Non OECD			OECD			Non OECD		
			Total	OECD Europe		CEEC	OPEC	Other	Total	OECD Europe		CEEC	OPEC	Other
				EU[2]	Other					EU[2]	Other			
1986	2 725	3 105	2 417.0	1 582.4	445.6	78.5	40.7	188.4	2 689.2	1 553.6	652.2	72.4	75.0	268.6
1987	3 392	3 710	3 002.6	1 970.1	561.4	112.2	41.2	236.5	3 230.2	1 891.3	781.0	79.1	77.0	323.9
1988	3 809	4 142	3 374.2	2 161.3	637.3	113.4	39.0	281.9	3 608.6	2 163.7	837.6	90.1	99.1	344.6
1989	4 090	4 300	3 617.6	2 275.2	705.0	117.5	45.2	309.7	3 761.2	2 309.3	835.1	82.2	87.8	368.8
1990	4 569	4 792	4 041.2	2 526.4	839.7	125.6	56.2	346.2	4 192.5	2 604.1	933.6	91.8	106.9	400.8
1991	4 164	4 601	3 653.1	2 290.0	755.6	105.8	74.6	331.0	3 963.7	2 538.1	847.2	115.3	151.2	370.8
1992	4 169	4 676	3 621.6	2 314.6	691.5	132.0	77.5	338.3	4 071.4	2 610.6	831.8	117.7	117.8	369.4
1993	3 567	4 155	3 101.7	2 235.3	327.5	111.5	57.7	296.4	3 520.9	2 460.4	479.9	65.3	105.1	463.3
1994	4 308	5 091	3 805.1	3 015.5	406.9	100.7	62.5	339.8	4 302.5	3 020.3	582.3	95.2	95.9	597.0
1995	5 365	6 596	4 810.7	3 795.1	532.9	128.6	62.3	363.6	5 579.5	3 926.4	740.2	143.7	107.1	765.5

1. Break in data from 1994.
2. From 1993 EU-15.

Source: OECD, *Foreign Trade Statistics, Series A.*

Table I. **Prices and wages**

	Consumer prices, 1980 = 100				Net price index[2] 1980 = 100		Producer prices manufactured products 1990 = 100	Hourly earnings, Kronor[3] Manufacturing		
	Total		Food	Rent, fuel and electricity						
	Index	Annual growth rate[1]			Index	Annual growth rate[1]		Total	Males	Females
1986	160	4.2	185	154	156	4.2	81	53.35	53.82	48.31
1987	167	4.2	191	159	160	2.8	84	56.80	57.71	51.02
1988	177	6.1	202	170	171	6.8	89	61.33	62.30	55.62
1989	188	6.6	213	183	181	6.2	96	67.40	68.42	61.18
1990	208	10.4	229	211	194	7.3	100	73.70	75.48	67.01
1991	227	9.7	239	250	204	5.2	102	76.40	82.21	69.58
1992	232	2.6	227	269	213	2.9	101	80.67	82.73	73.40
1993	243	4.7	228	284	222	3.9	106	82.32	84.21	75.11
1994	249	2.4	232	289	225	1.7	111	85.32	87.20	78.02
1995	255	2.9	236	296	229	1.0	122	89.17	91.09	81.26

1. The 12-month growth rates take into account a revised December figure.
2. Consumer prices excluding indirect taxes.
3. Including direct wages for normal working hours, basic pay and supplements for overtime and shiftwork as well as supplements for staggered and inconvenient working hours.

Source: Statistics Sweden, *Monthly Digest of Swedish Statistics.*

Table J. **Money and credit**

	Interest rates							Central Bank		Commercial banks		Non-bank liquidity
				Bond yields				Holding of				
	Official discount rate[1]	Day-to-day money market rate	Treasury Bill 3 months	Government 5-year	Government 10-year[2]	Mortgage bonds	Industrial bonds	Foreign exchange reserves incl. gold	Domestic assets	Holding of domestic securities	*of which:* Government bonds	Money supply (M3)
	Per cent							SKr million				
								End of period				Average
1986	8.25	10.09	9.82	10.24	. .	. .	11.50	45 648	88 550	111 431	42 414	432 077
1987	7.50	9.20	9.39	11.44	11.68	11.99	12.21	50 288	95 697	102 841	41 521	467 767
1988	8.18	10.08	10.10	11.19	11.35	11.45	11.76	53 741	92 910	103 005	33 263	486 599
1989	9.25	11.57	11.50	11.30	11.18	11.64	12.16	60 704	58 461	117 759	41 247	520 231
1990	11.12	13.38	13.63	13.55	13.16	14.23	15.05	103 786	37 780	122 268	44 085	567 592
1991	9.39	11.76	11.56	10.90	10.69	11.84	12.47	99 751	131 721	118 385	49 327	623 864
1992	8.85	16.76	12.91	10.41	10.02	11.83	12.71	163 509	145 390	84 739	29 481	632 169
1993	6.78	9.05	8.33	7.65	8.51	8.95	8.97	179 651	98 397	155 656	56 196	676 627
1994	5.37	7.36	7.40	9.09	9.72	9.87	9.68	181 189	86 937	203 845	84 293	707 390
1995	7.13	8.54	8.75	9.87	10.24	10.27	10.51	171 320	71 025	195 050	78 399	698 362

1. Annual average.
2. Except 1994: 9-year government bonds.

Source: Statistics Sweden, *Monthly Digest of Swedish Statistics;* Sveriges Riksbank, *Statistical Yearbook;* OECD, *Financial Statistics Monthly.*

Table J. **Money and credit** *(cont.)*

SKr million

	Structure of banks deposits			Bank lending[1]			Bond, debentures, etc.[2]				Share issues (excluding bonus issues)
	Commercial banks	Savings banks	Total	Commercial banks	Savings banks	Total	Total	State	*of which:* Housing sector	Business enterprises	
	End of period										
1986	. .	. .	. .	. .	. .	. .	116 808	20 960	70 355	14 128	10 110
1987	. .	. .	. .	. .	. .	. .	56 640	–672	45 339	2 252	13 495
1988	342 840	147 153	489 993	458 628	162 558	621 186	86 512	–19 034	72 711	4 041	10 418
1989	393 837	156 657	550 494	577 463	203 503	780 966	77 343	–36 747	89 290	–6 303	14 498
1990	453 732	187 938	641 670	673 609	237 917	911 526	46 389	–32 910	71 733	–11 611	18 186
1991	495 585	191 838	687 423	677 060	220 764	897 824	83 545	13 121	74 383	3 240	21 261
1992	505 172	183 509	688 681	704 445	208 249	912 694	185 636	103 856	81 163	594	37 226
1993	532 124	191 206	723 330	566 059	171 601	737 660	296 261	196 036	77 349	4 588	14 980
1994	540 382	186 089	726 471	529 186	162 610	691 796	40 564	102 355	–39 171	–9 681	20 199
1995	566 962	186 801	753 763	519 627	165 723	685 350	126 969	212 294	–52 517	–16 790	51 426

1. Amounts utilised.
2. Net issued.

Source: Statistics Sweden, *Monthly Digest of Swedish Statistics;* Sveriges Riksbank, *Statistical Yearbook.*

Table K. **Production structure and performance indicators**

A. Production structure

	Share of value added at factor cost						Employment share (per cent of total)					
	1963	1970	1980	1990	1993	1994	1963	1970	1980	1990	1993	1994
Tradeables												
Agriculture	8.0	5.7	3.8	3.7	3.5	3.3	16.3	10.8	7.8	5.6	5.9	5.9
Mining and quarrying	1.6	1.3	0.7	0.4	0.4	0.5	0.7	0.6	0.5	0.3	0.3	0.3
Manufacturing	35.4	34.9	29.8	28.1	27.1	29.3	34.4	34.2	35.0	30.8	28.1	28.5
of which:												
Food, beverages and tobacco	5.7	3.8	3.5	2.9	3.0	3.0	3.2	3.0	3.0	2.8	2.8	2.7
Textile, wearing apparels and leather industries	3.1	2.1	1.1	0.6	0.5	0.5	4.6	3.4	1.7	1.0	0.7	0.7
Wood and wood products, including furniture	2.5	2.9	2.4	2.2	1.7	1.8	3.0	3.1	2.9	2.3	1.9	2.0
Paper and paper products, printing and publishing	4.0	4.8	4.8	4.3	4.5	4.7	4.3	4.1	4.8	4.3	4.1	4.2
Chemicals and chemicals petroleum, coal, rubber and plastic products	2.6	2.9	3.3	3.3	3.9	4.0	1.9	2.3	2.6	2.3	2.2	2.3
Non-metallic mineral products except products of petroleum and coal	1.6	1.5	1.1	0.9	0.6	0.6	1.6	1.4	1.0	0.9	0.7	0.7
Basic metal industries	2.1	2.8	1.5	1.3	1.5	1.7	2.3	2.2	2.2	1.4	1.2	1.2
Fabricated metal products, machinery and equipment	13.4	13.7	12.1	12.5	11.3	12.9	13.1	13.8	15.6	14.6	12.9	13.3
Other manufacturing industries	0.5	0.4	0.1	0.2	0.2	0.2	0.4	0.8	1.2	1.3	1.5	1.5
Non-tradeables												
Electricity, gas and water	3.0	2.5	3.0	3.8	3.9	3.6	0.9	0.9	1.1	1.1	1.2	1.1
Construction	11.7	11.6	10.4	9.9	8.9	8.0	10.7	12.2	10.2	9.9	8.8	8.1
Wholesale and retail trade, restaurants and hotels	14.3	15.2	14.5	14.6	14.7	14.5	17.8	19.1	19.8	21.1	21.7	21.5
Transport, storage and communication	8.1	8.7	7.2	8.5	8.4	8.3	8.2	8.6	9.4	9.9	10.4	10.3
Finance, insurance, real estate and business services	13.3	15.3	25.2	26.0	27.8	27.4	3.9	5.7	8.1	12.5	13.4	13.6
Community, social and personal services	4.6	4.7	5.5	5.0	5.3	5.0	7.0	7.8	8.1	8.8	10.3	10.7
Memorandum items:												
Business GDP	86.1	81.5	74.3	76.2	76.1	77.2	82.9	77.5	68.2	67.4	66.2	66.9
Public sector	12.9	17.6	24.3	22.6	22.6	21.5	14.3	20.6	30.7	31.6	32.6	32.0
Other producers	1.0	0.9	1.4	1.2	1.3	1.3	2.8	1.9	1.2	1.1	1.2	1.2

Table K. **Production structure and performance indicators** *(cont.)*

B. Sector performance – Constant prices

	Productivity growth (Sector GDP/sector employment)				Investment As a share of total business investment					
	1963-69	1970-79	1980-89	1990-94	1963	1970	1980	1990	1993	1994
Tradeables										
Agriculture	7.9	2.8	5.5	–0.3	4.2	3.6	4.9	3.2	3.2	3.4
Mining and quarrying	10.6	–0.3	2.4	4.9	1.3	0.9	1.0	0.6	0.8	0.9
Manufacturing	6.3	2.1	3.1	6.9	24.4	21.0	19.6	18.3	18.3	24.0
of which:										
Food, beverages and tobacco	3.6	1.2	1.6	5.5	. .	2.3	2.2	1.3	2.3	2.3
Textile, wearing apparels and leather industries	7.5	4.5	1.8	5.9	. .	0.6	0.6	0.3	0.2	0.2
Wood and wood products, including furniture	6.2	1.9	4.2	3.2	. .	1.2	1.6	1.5	1.0	1.4
Paper and paper products, printing and publishing	5.6	2.3	2.5	7.2	. .	4.5	3.7	4.1	3.6	5.2
Chemicals and chemicals petroleum, coal, rubber and plastic products	7.1	2.4	3.3	8.8	. .	2.4	2.3	2.2	2.6	3.7
Non-metallic mineral products except products of petroleum and coal	7.0	2.4	2.8	–0.1	. .	0.9	0.7	0.6	0.3	0.3
Basic metal industries	7.4	1.9	5.7	13.9	. .	2.7	2.2	1.0	0.8	1.1
Fabricated metal products, machinery and equipment	6.5	1.6	3.4	7.1	. .	6.3	6.2	7.2	7.6	9.7
Other manufacturing industries	8.5	2.1	10.1	1.2	. .	0.1	0.1	0.1	0.1	0.1
Non-tradeables										
Electricity, gas and water	6.9	5.1	5.2	1.9	9.5	9.9	9.6	5.2	6.5	6.5
Construction	1.5	3.6	2.4	3.5	3.4	2.1	2.3	3.1	1.9	2.0
Wholesale and retail trade, restaurants and hotels	2.8	1.7	1.9	3.3	7.5	5.8	6.5	9.2	8.7	8.8
Transport, storage and communication	4.2	3.2	2.8	2.1	12.5	10.3	11.3	12.4	14.6	16.8
Finance, insurance, real estate and business services										
of which:										
Dwellings	–2.4	0.3	–1.5	3.0	35.9	43.5	42.9	45.3	41.1	32.8
Community, social and personal services	–2.8	1.6	–1.3	3.5	. .	38.4	14.1	22.2	24.8	18.7
Discrepancy	0.2	3.0	1.0	–0.9	1.4	1.3	1.9	2.6	5.0	4.8
					As a share of total investment					
Business sector	4.5	2.5	2.1	3.5	83.5	80.1	83.8	88.7	80.2	78.1
Public sector	–0.9	–1.2	0.5	1.3	16.5	19.9	16.2	11.3	19.8	21.9
Other producers	1.2	2.8	2.7	2.5	. .	. .	. .	. .	. .	. .

Table K. **Production structure and performance indicators** *(cont.)*

C. Other indicators

	Size distribution of manufacturing enterprises in 1994				
	Under 10 employed	10-49 employed	50-99 employed	100-499 employed	500 employed and plus
Number of enterprises (per cent of total)	87.6	9.7	1.3	1.1	0.4
Number of employed (per cent of total)	7.5	11.6	5.6	14.0	61.4
	1987	1989	1991	1993	1995
Direct investment abroad, net (per cent of business investment)	–22.2	–38.0	–27.6	–9.0	–36.8
Total R&D expenditure as % of total GDP	3.0	2.9	2.9	3.3	3.1
Government funded R&D as % of total	36.9	38.1	34.0	31.4	. .

Source: Sveriges Riksbank, Statistisk Årsbok; *OECD, National Accounts* and *Main Science and Technology Indicators.*

Table L. **Labour market indicators**

	1986[1]	1990	1992	1993	1994	1995
A. Evolution of unemployment						
Unemployment rate (per cent of labour force)						
Total	2.5	1.7	5.3	8.2	8.0	7.7
Men	2.5	1.7	6.2	9.7	9.1	8.4
Women	2.5	1.6	4.2	6.6	6.7	6.9
Youth (age group 16-24 years)	5.3	3.7	11.4	18.5	16.7	15.4
Insured unemployed registered at the employment exchanges[2]	2.5	2.2	5.9	8.7	8.8	8.7
Unemployment average weeks[3]	16.6	14.7	21.1	23.5	27.6	26.6
Labour market policy						
Active measures (thousand persons)	92	57	165	189	227	191
Measures for handicapped, etc. (thousand persons)	74	81	77	76	80	85
B. Structural characteristics						
Participation rates						
Total	83.0	84.5	81.5	79.1	77.6	78.2
Men	85.4	86.6	83.4	80.9	79.5	80.2
Women	80.6	82.3	79.6	77.2	75.7	76.1
Employment, per cent of population between 16-64 years	80.9	83.1	77.3	72.6	71.5	72.2
Self-employed, per cent of total employed	9.8	9.2	9.8	10.8	11.1	11.2
Part-time employed, per cent of all employed	23.5	23.3	24.1	24.9	24.9	24.3
of which: Women (per cent of part-time employed)	86.6	83.4	82.3	81.3	80.1	80.1
Numbers of hours worked per week per person at work	36.2	36.8	36.5	36.4	36.9	36.8
Non-filled vacancies, per cent of employment	0.9	1.0	0.3	0.2	0.3	0.4
Overtime for employees, per cent of all employees	. .	11.5	7.9	8.3	9.8	9.4

1. These figures have been recalculated to conform with a new definition introduced in 1987.
2. In per cent of the members of the recognized unemployment benefit funds.
3. Average number of weeks, which persons still registered as unemployed, have been looking for job.

Source: Statistics Sweden, *Statistical Abstract of Sweden; OECD, Labour Force Statistics.*

Table M. **The public sector**

	1970	1980	1990	1992	1993	1994	1995
Budget indicators: General government (per cent of GDP)							
Current receipts	46.9	56.1	63.3	59.4	58.8	57.9	58.2
Non-interest expenditure	43.4	60.5	59.0	67.0	70.0	66.2	63.6
Primary budget balance	3.8	–4.4	4.3	–7.5	–11.3	–8.2	–5.4
Net interest expenses	–0.6	–0.4	0.1	0.2	1.0	2.1	2.5
General government budget balance	4.1	–4.0	4.2	–7.7	–12.3	–10.3	–7.9
Structure of expenditure and receipts (per cent of GDP)							
Expenditure							
Income transfers	14.5	19.2	21.4	25.5	27.3	27.2	. .
Interest payments	. .	4.1	5.0	5.4	6.2	6.8	
Subsidies	1.7	4.3	4.7	5.5	5.9	5.3	5.2
Consumption	21.6	29.3	27.4	27.9	28.1	27.4	25.9
Education	5.5	6.0	5.3	5.3	5.4	5.3	. .
Health	5.0	7.3	6.5	5.2	5.2	5.1	. .
Social welfare	2.4	4.8	5.2	6.4	6.1	6.3	. .
Housing	0.5	0.6	0.5	0.6	0.5	0.5	. .
Economic services	1.2	1.4	1.8	2.2	2.1	2.3	. .
Other	7.7	9.2	8.0	8.2	8.8	8.9	. .
Gross investment	6.6	4.2	2.4	2.7	1.1	3.1	3.0
Taxes and charges							
Property income	. .	5.1	5.9	6.6	6.7	5.9	
Taxes	32.9	35.0	40.6	36.8	36.4	36.3	. .
Indirect taxes	12.6	13.5	17.2	16.3	15.6	15.0	. .
Direct taxes	20.3	21.5	23.4	20.5	20.8	21.3	. .
Social security contributions	9.3	14.1	15.1	14.3	13.8	13.7	. .

	1988	1990	1992	1993	1994	1995	1996
Tax rates (per cent)							
Personal income tax							
State income tax							
Top marginal rate	45.0	35.0	35.0	20.0	20.0	25.0	25.0
Lowest marginal rate	5.0	3.0	0.0	0.0	0.0	0.0	0.0
Average local income tax	30.60	31.20	31.04	31.04	31.05	31.50	31.66
Capital income tax[1]	n.a.	n.a.	25.0/30.0	25.0/30.0	25.0/30.0	25.0/30.0	25.0/30.0
Social security fees	37.07	38.97	34.83	31.95	33.81	36.81	38.01
Value-added tax (basic rate)	23.5	25.0	25.0	25.0	25.0	25.0	25.0
Corporate tax	52.0	52.0	40.0	30.0	30.0	30.0	30.0
Wealth tax	1.5-3.0	1.5-3.0	1.5-2.5	1.5-2.5	1.5-2.5	1.5-2.5	1.5-2.5

1. Prior to 1991 capital income was added to earned income for tax purposes.

Source: Ministry of Finance, *The Swedish Budget*, various issues; The National Institute of Economic Research, *The Swedish Economy*, various issues; OECD.

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Denmark	Finland	France	Germany	Greece
Population											
Total	Thousands	1994	17 840	8 031	10 124	29 251	5 206	5 088	57 960	81 407	10 430
Inhabitants per sq. km	Number	1994	2	96	332	3	121	15	106	228	79
Net average annual increase over previous 10 years	%	1994	1.4	0.6	0.3	1.6	0.2	0.4	0.5	2.9	0.5
Employment											
Total civilian employment (TCE)[2]	Thousands	1994	7 680 (93)	3 737	3 724 (92)	13 292	2 508	2 015	21 781 (93)	35 894	3 790
of which: Agriculture	% of TCE	1994	5.3 (93)	7.2	2.6 (92)	4.1	5.1	8.3	5.1 (93)	3.3	20.8
Industry	% of TCE	1994	23.7 (93)	33.2	27.7 (92)	22.6	26.8	26.8	27.7 (93)	37.6	23.6
Services	% of TCE	1994	71 (93)	59.6	69.7 (92)	73.3	68.1	64.9	67.2 (93)	59.1	55.5
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1994	331.6	198.1	227.9	544	146.7	97.2	1 328.5	1 832.3	73.1 (93)
Per capita	US$	1994	18 588	24 670	22 515	18 598	28 181	19 106	22 944	27 826	7 051 (93)
At current prices using current PPP's[3]	Bill. US$	1994	327.9	162.3	204.2	596.7	107	82.5	1 111.8	1 601.7	118
Per capita	US$	1994	18 382	20 210	20 166	20 401	20 546	16 208	19 201	24 325	1 450
Average annual volume growth over previous 5 years	%	1994	2.2	2.5	1.6	1.1	1.9	–1.6	1.1	2.6	1.4 (93)
Gross fixed capital formation (GFCF)	% of GDP	1994	21.4	24.8	17.4	18.7	14.8	14.3	18.1	18.5	17.4 (93)
of which: Machinery and equipment	% of GDP	1993	9.8	8.7	7.8	6.2	7.2	5.90	8.10	7.5	7.8
Residential construction	% of GDP	1993	5.2	6.3	4.6	6.1	2.9	3.7	4.9	6.5	3.6
Average annual volume growth over previous 5 years	%	1994	0.8	3.7	0.4	–0.1	–2.8	–12.9	–1	0.8	2.7 (93)
Gross saving ratio[4]	% of GDP	1994	16.8	25.3	22	16	17	16.6	19	21	15.5 (93)
General government											
Current expenditure on goods and services	% of GDP	1994	17.5	18.8	15	20.2	25.3	22.4	19.6	17.7	19.1 (93)
Current disbursements[5]	% of GDP	1993	36.9	48.4	55.3	49	61.1	58.9	51.5	45.6	51.2
Current receipts	% of GDP	1993	33.5	48.6	50.1	43	58.3	52.5	46.8	45.7	40.2
Net official development assistance	% of GNP	1993	0.34	0.31	0.41	0.46	1.03	0.76	0.66	0.44	..
Indicators of living standards											
Private consumption per capita using current PPP's[3]	US$	1993	10 803	10 546	12 090	11 863	10 042	8 814	11 395	10 733	6 367
Passenger cars, per 1 000 inhabitants	Number	1990	430	382	387	469	311	386	413	480[8]	169
Telephones, per 1 000 inhabitants	Number	1991	464	432	410	586	577	544	511	420[8]	413
Television sets, per 1 000 inhabitants	Number	1991	480	478	451	639	536	501	407	556[8]	197
Doctors, per 1 000 inhabitants	Number	1993	2.2 (91)	2.3	3.7	2.2	2.8 (92)	2.6 (92)	2.8	3.2 (92)	3.8 (92)
Infant mortality per 1 000 live births	Number		6.1	6.5	8	6.8	5.4	4.4	6.5	5.8	8.5
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1994	3	5.5	3.7	3.3	3.5	4.8	3.5	5.2	14.6
Consumer prices	%	1994	3	3.4	2.8	2.8	2.1	3.3	2.5	3.3	16.2
Foreign trade											
Exports of goods, fob*	Mill. US$	1994	47 363	44 881	137 259[7]	165 358	41 850	29 514	235 337	422 243	8 958
As % of GDP	%	1994	14.3	22.7	60.2	30.4	28.5	30.4	17.7	23	11.5 (93)
Average annual increase over previous 5 years	%	1994	5	6.7	6.5	7.1	8.3	4.9	5.6	4.4	3.4
Imports of goods, cif*	Mill. US$	1994	49 731	55 071	126 006[7]	148 297	35 932	23 091	220 508	376 566	21 111
As % of GDP	%	1994	15	27.8	55.3	27.3	24.5	23.8	16.6	20.6	30.1 (93)
Average annual increase over previous 5 years	%	1994	4	7.2	5	5.4	6.1	–1.3	3.5	6.9	5.4
Total official reserves[6]	Mill. SDRs	1994	7 730	11 523	9 505[7]	8 416	6 203	7 304	17 986	52 994	9 924
As ratio of average monthly imports of goods	Ratio	1994	1.9	2.5	0.9	0.7	2.1	3.8	1	1.7	5.6

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Iceland	Ireland	Italy	Japan	Luxembourg	Mexico	Netherlands	New Zealand
Population										
Total	Thousands	1994	267	3 571	57 190	124 960	398	93 010	15 382	3 526
Inhabitants per sq. km	Number	1994	3	51	190	331	153	47	377	13
Net average annual increase over previous 10 years	%	1994	1.1	0.1	0	0.4	0.8	2	0.6	0.8
Employment										
Total civilian employment (TCE)[2]	Thousands	1994	138	1 168 (93)	20 152 (93)	64 530	162 (91)	32 439	6 631	1 560
of which: Agriculture	% of TCE	1994	9.4	12.7 (93)	7.5 (93)	5.8	3.7 (91)	25.8	4	10.4
Industry	% of TCE	1994	26.1	27.7 (93)	33 (93)	34	31.5 (91)	22.2	23	25
Services	% of TCE	1994	65.2	59.7 (93)	59.6 (93)	60.2	64.8 (91)	52.1	73	64.6
Gross domestic product (GDP)										
At current prices and current exchange rates	Bill. US$	1994	6.2	52	1 017.8	4 590	10.6 (92)	371.2	334.3	51.2
Per capita	US$	1994	23 199	14 550	17 796	36 732	27 073 (92)	3 991	21 733	14 513
At current prices using current PPP's[3]	Bill. US$	1994	5.1	54.3	1 068.4	2 593.7	11.7	673.3	285.9	57.3
Per capita	US$	1994	19 271	15 212	18 681	20 756	29 454	7 239	18 589	16 248
Average annual volume growth over previous 5 years	%	1994	0.6	4.7	1	2.1	4.1 (92)	3	2.3	2.5
Gross fixed capital formation (GFCF)	% of GDP	1994	15.2	15.1	16.4	28.6	20.4 (93)	20.7	19.3	20
of which: Machinery and equipment	% of GDP	1993	3.9	6.3	7.4	11.5	..	9.4	8.6	9.3
Residential construction	% of GDP	1993	4.4	4.1	5.3	5.4	..	4.9	5.1	4.9
Average annual volume growth over previous 5 years	%	1994	–4	1	–2.3	1.4	6.5 (92)	7.7	0.4	5.8
Gross saving ratio[4]	% of GDP	1994	16.9	19.5	18.8	31.2	60.2 (92)	15.1	24.4	20.7
General government										
Current expenditure on goods and services	% of GDP	1994	20.6	16	17.1	9.8	17.1 (92)	11.8[9]	14.2	14.7
Current disbursements[5]	% of GDP	1993	34.9	..	53.2	26.9	..	..	55.4	..
Current receipts	% of GDP	1993	35.9	..	47.1	32.9	..	..	54.5	..
Net official development assistance	% of GNP	1993	..	0.15	0.42	0.27	0.34 (92)	..	0.88	0.22
Indicators of living standards										
Private consumption per capita using current PPP's[3]	US$	1993	11 546	7 750	11 029	11 791	15 545	4 853	10 726	9 266
Passenger cars, per 1 000 inhabitants	Number	1990	464	228	478	282	470	85	356	440
Telephones, per 1 000 inhabitants	Number	1991	527	300	400	454	511	70	477	436
Television sets, per 1 000 inhabitants	Number	1991	319	276	421	613	267	148	485	443
Doctors, per 1 000 inhabitants	Number	1993	3	1.7 (92)	1.7 (91)	1.7 (92)	2.1 (92)	1	2.5 (90)	2
Infant mortality per 1 000 live births	Number		4.8	5.9	7.3	4.5 (92)	8.5 (92)	18	6.3	7.3
Wages and prices (average annual increase over previous 5 years)										
Wages (earnings or rates according to availability)	%	1994	..	4.6	5.9	2.4	..	5.3	3.2	2.1
Consumer prices	%	1994	6.3	2.7	5.2	2	3.1	16.1	2.8	2.5
Foreign trade										
Exports of goods, fob*	Mill. US$	1994	1 628	34 125	189 802	396 149	..	60 882	155 084	12 169
As % of GDP	%	1994	26.3	65.7	18.6	8.6	..	16.4	46.4	23.8
Average annual increase over previous 5 years	%	1994	2.7	10.5	6.2	7.6	..	21.7	7.6	6.5
Imports of goods, cif*	Mill. US$	1994	1 464	25 812	167 690	274 916	..	79 346	139 800	11 859
As % of GDP	%	1994	23.6	49.7	16.5	6	..	21.4	41.8	23.2
Average annual increase over previous 5 years	%	1994	0.7	8.1	1.9	5.5	..	25.5	6.1	6.1
Total official reserves[6]	Mill. SDRs	1994	201	4 189	22 102	86 214	..	4 301	23 655	2 540
As ratio of average monthly imports of goods	Ratio	1994	1.6	1.9	1.6	3.8	..	0.7	2	2.6

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population										
Total	Thousands	1994	4 337	9 900	39 150	8 781	6 994	60 573	58 375	260 651
Inhabitants per sq. km	Number	1994	13	107	78	20	169	78	238	28
Net average annual increase over previous 10 years	%	1994	0.5	–0.1	0.2	0.5	0.7	2.1	0.3	1
Employment										
Total civilian employment (TCE)[2]	Thousands	1994	1970 (93)	4 372	11 760	3 926	3 772	19 664	25 044 (93)	123 060
of which: Agriculture	% of TCE	1994	5.6 (93)	11.5	9.8	3.4	4	44.8	2.2 (93)	2.9
Industry	% of TCE	1994	23.1 (93)	32.8	30.1	25	28.8	22.2	26.2 (93)	24
Services	% of TCE	1994	71.3 (93)	55.7	60.2	71.6	67.2	33	71.6 (93)	73.1
Gross domestic product (GDP)										
At current prices and current exchange rates	Bill. US$	1994	103.4 (93)	87	482.4	196.6	257.3	130.7	1 019.5	6 649.8
Per capita	US$	1994	23 984 (93)	8 792	12 321	22 389	36 790	2 157	17 468	25 512
At current prices using current PPP's[3]	Bill. US$	1994	95.3	122	531.7	153	167.4	319.3	1 030.2	6 649.8
Per capita	US$	1994	21 968	12 335	13 581	17 422	23 942	5 271	17 650	25 512
Average annual volume growth over previous 5 years	%	1994	2.1 (93)	1.4	1.5	–0.3	0.5	3.6	0.8	2.1
Gross fixed capital formation (GFCF)	% of GDP	1994	22 (93)	25.7	19.8	13.7	22.8	24.5	15	17.2
of which: Machinery and equipment	% of GDP	1993	..	..	5.7	5.7	7.5	10.3	7.3	7.7
Residential construction	% of GDP	1993	..	..	4.1	4	14.9[10]	9.1	3.1	4
Average annual volume growth over previous 5 years	%	1994	–3.93	2.7	–1.2	–7.6	–0.4	5.1	–2.1	4.6
Gross saving ratio[4]	% of GDP	1994	21.9 (93)	24.2	18.8	13.7	29.3	22.5	13.5	16.2
General government										
Current expenditure on goods and services	% of GDP	1994	22.1 (93)	17.2	16.9	27.3	14.1	11.7	21.6	16.4
Current disbursements[5]	% of GDP	1993	..	..	43.7	67.3	36.7	..	42.7	35.8
Current disbursements[5]	% of GDP	1993	..	..	40.1	59	36	..	36.8	31.7
Current receipts	% of GDP	1993	1.23	0.36	0.32	1.33	0.49	..	0.34	0.19
Net official development assistance	% of GNP	1993	9 826	7 780	8 412	9 240	13 730	3 617	10 942	16 444
Indicators of living standards										
Private consumption per capita using current PPP's[3]	US$	1993	378	260	307	418	441	29	361	568
Passenger cars, per 1 000 inhabitants	Number	1990	515	273	340	687	603	143	445	553
Telephones, per 1 000 inhabitants	Number	1991	423	187	400	468	406	175	434	814
Television sets, per 1 000 inhabitants	Number	1991	3.2 (92)	2.9	4.1	3	3	0.9	1.5 (92)	2.3 (92)
Doctors, per 1 000 inhabitants	Number	1993	5	8.7	7.6	4.8	5.6	52.6	6.6	8.5 (92)
Infant mortality per 1 000 live births	Number		4	..	7.2	5.4	..	..	6.7	2.8
Wages and prices (average annual increase over previous 5 years)										
Wages (earnings or rates according to availability)	%	1994	2.7	9	5.6	5.7	3.9	73	4.6	3.6
Consumer prices	%	1994	34 645	17 072	73 129	61 122	70 467	18 456	205 170	512 627
Foreign trade										
Exports of goods, fob*	Mill. US$	1994	30.9 (93)	19.6	15.2	31.1	27.4	14.1	20.1	7.7
As % of GDP	%	1994	5	6.1	10.5	3.4	6.4	9.5	6.1	7.1
Average annual increase over previous 5 years	%	1994	27 345	25 967	92 182	51 730	68 126	22 976	227 026	663 256
Imports of goods, cif*	Mill. US$	1994	23.3 (93)	29.9	19.1	26.3	26.5	17.6	22.3	10
As % of GDP	%	1994	2.9	6.6	5.2	1	3.2	37.9	2.8	7
Average annual increase over previous 5 years	%	1994	13 033	10 627	28 475	15 929	23 790	4 911	28 094	43 350
Total official reserves[6]	Mill. SDRs	1994	5.7	4.9	3.7	3.7	4.2	2.6	1.5	0.8

* At current prices and exchange rates.
1. Unless otherwise stated.
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3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
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